

17. HUMAN AFRICAN TRYPANOSOMIASIS

The human African trypanosomiasis are endemio-epidemic diseases concerning intertropical Africa, and are caused by a flagellated parasite, the trypanosoma, which is transmitted by an infected human (or animal) to a healthy one via the sting of a specific vector, the glossina or tsetse fly.

The infection, progressive and fatal if appropriate treatment is not given, starts with an invasion of the blood and the lymphatic system by the parasite, and progresses, after a variable delay, to the nervous centres of the brain.

The term trypanosomiasis refers to the infection in receptive humans and animals. The popular name, sleeping sickness, recalls the typical symptoms of apathy and somnolence which appear at the advanced, chronic and nervous stage of the sickness.

The infection is caused by two similar parasites of the type T. brucei, which nevertheless differ in their host reservoir, their type of vector glossina, their virulence and their geographical distribution.

One of these parasites, T. brucei rhodesiense, manifests itself in an acute type of illness (in fact, a zoonosis transmitted occasionally to humans) peculiar to East Africa. Contamination occurs principally by xerophile and zoophile glossinae of the morsitans group.

The other, T. brucei gambiense, is responsible for the chronic type of sleeping sickness in West and Central Africa, and is transmitted by glossinae of the palpalis group.

The epidemiological chain of the trypanosomiasis is made up of three links: a vertebrate host, man (with the help of an animal as relay host); a

flagellate protozoan, the trypanosoma; and a vector insect, the glossina. All three are conditioned by the environment in which they develop.

The infection is transmitted by the bite of an infected tsetse fly. Not all types of glossina have the tendency to transmit the sickness and, in endemic centres, among the species with well-established tendencies, the percentage of flies actually affected is very low.

The tsetse flies cover a distribution area known in Africa as the fly-belt, with unevenly distributed species.

As the infection depends on favourable contacts between man and vector, particular sites in the fly-belt become foci of endemic infection, giving rise to outbreaks of epidemics.

The acute T. rhodesiense trypanosomiasis has a rapidly debilitating clinical evolution, which makes the diagnosis easy in an endemic environment.

This is not the case with the T. gambiense type, which has a chronic development with often a long incubation period. The first symptoms are similar to a bout of malaria or influenza, and the parasites are difficult to detect in blood.

The clinical consequences are such that, besides the relentless progression towards death in the absence of treatment (with the rare exceptions of a few healthy carriers), a late diagnosis makes the treatment more difficult, and a recovery without nervous after-effects, uncertain. From an epidemiological point of view, untreated patients constitute a source of parasites which may, in turn, infect the tsetse fly.

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HISTORICAL BACKGROUND

ORIGIN AND SPREAD OF AFRICAN TRYPANOSOMIASIS

1. *African lethargy in Arab chronicles*

As early as the fourteenth century, Arab chronicles referred to an *African lethargy* which occurred in the sultan states of Songhai or Sourai on the banks of the Niger and one of its tributaries, the Sankarani (Guinea and Mali). Some Arab authors, such as *Ibn Khaldun*, who died in Cairo in 1406, and his fellow countryman *Shihab ad-Din Ahmad al Qualgashandiy* related, according to the same sources, the death of the King of Melle, Sultan Maariy Djazaa, from a lethargy very common in that region which seems to have played an important part in the fall of kingdoms and sultanates in this part of tropical Africa.

The historian Baron de Slade obtained his information from *Abu Abdallah Ibn Mansul*, a judge in Koranic law, who translated the text in the following words: "He (the Sultan) was finally struck down with the lethargy, a very common illness in the country, which tends to affect people in high posts. This ailment starts with periodic bouts of lethargy reducing the patient to such a degree of inertia that it is almost impossible to make him stay awake, even for the shortest spell. Lethargy then completely takes over and causes the patient's death. Djazaa suffered from it for two years and finally died of it in the year 775" (by our calendar, in 1373-1374).

This lethargy could have been sleeping sickness, and this account has excited the interest of many specialists in tropical medicine. Although the objective value of the account may be rather poor, it offers those who attempt to retrace the progression of sleeping sickness the possibility of attributing its origins to a focus along the Niger. However, between the death of Maariy Djazaa and more recent documents, there is a gap of 360 years.

The identification of Loanda's disease described by *Alexeio de Abreu* (1623) as trypanosomiasis should be treated with caution.

2. *Indigenous tradition*

Many native affirmations testify to the fact that sleeping sickness existed before the whites came in. "Our fathers knew it", insisted Chief Madinga in the North of Mayumbe, in Niari, the focus in Congo Brazzaville. In Kwango, some small, old endemic foci of low viru-

lence were identified (Fr. Sadin, Gillet, De Vos). At Kwamouth-Tumbiri (see map 36, p. 1708), *Fr. De Vos* noted the existence of a sickness which induced sleep and killed many men among the Bobanzi (now Bayanzi); and in Lower Ubangi some cases occurred sporadically in centres known to the people (Rodhain, 1907). Information gathered in the Upper Congo and in the East, in Maniema, confirmed that sleeping sickness had always occurred sporadically. Hackars, the District Commissioner reports, before the coming of European settlers, that many groups attracted by the snow-capped Ruwenzori mountains were wiped out successively by a mysterious disease, in the southern plain of the Semliki which links Lake Edward to Lake Mobutu. The Babira, who settled in the Irumu plain at Bunia (in Ituri), are quite categorical: they lost more people during their stay along the Semliki river than in the fights they had sustained to get a passage there.

Similar statements have been heard in Lower and Middle Sangha (Congo-Brazzaville, now People's Republic of the Congo, and in the Central African Republic), where sleeping sickness is well known to the natives and is supposed always to have occurred (Heckenroth, 1912), so as to be seen on the map (p. 1686) published by *Dutton and Todd* in 1884; it is based partly on the statements of the people.

According to many reliable native sources, some places were *niches* of lethargy. This confirmed the accounts of epidemic outbreaks during forays and localized warfare. Moreover, some tribal elders have stressed that after the arrival of foreigners, these outbreaks had extended by leaps and bounds to wider areas.

The accounts given by natives should not be underestimated: they have been gleaned from various regions quite distant from one another, by many physicians, administrators and missionaries, all concerned that their research should be objective. The value of such findings depends on the quality of the observers (*Dutton and Todd, De Vos, Heckenroth, Rodhain*); and cross-checking is possible.

The oral history of some diseases naturally encounters difficulties which may be attributed to the multiple pathologies that prevail in tropical areas, thus limiting the possibility of identification. This is the case for many fevers. However, it was not so in the case of sleeping sickness: it was readily identified and well defined in the native language. Many a physician has discovered a disease he never suspected, by starting

with the name in the vernacular of a curious syndrome. Because they were verbal, the accounts should not be discarded.

It was only in 1902-1903 that *Castellani* identified the causal agent of human sleeping sickness, which has so far not been associated with the fever caused by trypanosomes. Clinical descriptions of sleeping sickness had become more and more accurate, as was the description of the related lesions of the central nervous system. However, all this had not prevented some physicians from advancing false hypotheses: Ziemann, for instance, still attributed sleeping sickness to cassava poisoning.

3. Early medical observations

3.1. In the coastal zone of West Africa

Observations by physicians were the first priority. Nevertheless, one had to wait until 1734 to find a work describing *sleeping distemper* or *restless sleep*, in the *Navy Surgeon*, written by John Atkins, a surgeon in the British Navy. He observed, in 1721, the existence of a sickness on the coast of Guinea which especially affected young people.

The same syndrome, observed by Winterbottom in the Gulf of Benin in 1793, and named by him *lethargus*, drew attention to the enlarged cervical glands. This feature bears his name; but it was already known to slave traders, who did not purchase any person who displayed such enlarged glands (lymph nodes). In some parts of the west coast, such as Casamance (Southern Senegal), Guinea, the Gold Coast (now Ghana) and the Ivory Coast, this led to the excision of such enlarged lymph nodes. This procedure was even considered to be therapeutic.

From the nineteenth century onwards, information became more abundant and detailed. After visiting the Gold Coast, among other places, *Robert Clarke* (1840), a missionary in Sierra Leone, gives a thorough description of the symptomatology of the disease: adenopathies, somnolence – thereby justifying the name of *sleeping dropsy* –, weight loss, loss of appetite, indifference, frequent amenorrhoea, oedema, and finally death, accompanied by convulsions. He observed that when carrying out post-mortems, an inflammation of the meninges was found, with an excess of fluid in the ventricles and at the base of the brain. Having travelled deeply into these two countries (Sierra Leone and Ghana), and after visiting Liberia (Cape Mesurado) and the island of Bioco (previously Fernando Po), Clarke drew attention to the fact that the incidence of *narcotic dropsy* was greater among individuals living

in the interior of the country than in those living along the coast, from where the slaves were shipped. In 1849, the naval surgeon *David*, according to a communication by Danielli, drew attention to the endemic state of lethargy in Guinea. This observation was to be confirmed by *Ferreira* on the island of São Tome. The British Expedition to the Niger in 1857 also noted the existence of the disease among the native population living in the interior.

From 1861, many French naval surgeons (Béranger-Feraud, Chassaniol, Corré, Dangeaix, Gaigneron, Grifon du Bellay, Guérin, Le Roy de Méricourt, Nicolas, Nielly, Santinelli), published their observations on sleeping sickness, and gave details of its symptoms and also of the lesions observed on post-mortem examinations. *Dangeaix* (1861) working in Gabon, confirmed Clarke's observations. He noticed, among the clinical signs, alternating anorexia and bulimia, rapid and slow pulse, and faltering steps. He also observed, at the autopsy, enlarged cerebral veins, hyperaemia of the central nervous system and the brain, hydropsy of the ventricles, softening of peduncles and optic nuclei; but he points out, very wisely, that the state of the organs could be due to the quick decomposition of the corpse.

3.2. In the Antilles (West Indies)

In 1806, *Moreau de Jonnes* drew attention to the existence of African lethargy among slaves sent out to the Antilles.

In the same archipelago, on the island of Martinique to be exact, *Nicolas* (1861) gave a complete clinical picture including the occurrence of a ptosis, a shrivelled skin, oedema of the face, urinary and faecal incontinence, awkwardness, involuntary movements and bouts of narcolepsy.

The dissertation of *Guerin* (Paris, 1869) was particularly important, based on observation of 148 African patients from Luanda (Angola) who had been transported to Martinique. He underlined cephalalgia and changes in behaviour as the first indicators of the disease, with oedema of the eyelids and ptosis, as useful additional signs, to recognize sleeping sickness. As the illness progressed, he noted an unstable gait, which forced the patient to alter his centre of gravity from a forward to a more backward position, followed by listlessness, convulsions and a lethargic sleep, with hypothermia in the final phase.

3.3. On the tropical African mainland

The sum of observations has shown that sleeping sickness is not a coastal disease as such. It affects as well the people coming from inland territories as those living at the coast.

If it had previously been considered a peculiarity of the coastal regions, this was because it had been spotted by naval surgeons, who limited their professional activity to people living in the harbour areas, fortified strongholds and the immediate hinterland. They never knew where their patients came from, and considered them indiscriminately as people from the coastal regions.

The first clue to modify this general opinion came from *Clarke* (1840). It was he who, as a result of his travels, broke the myth that the sickness was exclusively hitting coastal people. Further observations from other sources confirmed the validity of *Clarke's* opinions: *Dangeaix* (1861) and *Dumortier* (1868) in Gabon, which at that time was just a modest French base, pointed out that the sickness was prevalent inland, and that some of the infected people moved to the coast. This statement was confirmed by *Griffon du Bellay* (1869), and by *Ballay* (1880) regarding the lands bordering the banks of the Ogoe.

In Angola, where Europeans had until then settled in only some of the coastal enclaves, the first case was officially identified in 1871, in the Cuanza valley (see below, p. 1716). The existence of sleeping sickness had already been seen in Guinea-Bissau, as far back as 1855.

In the nineteenth century, numerous British and French physicians, travelling and practising along the coast from Senegal to Angola in the course of their ministrations, published their observations on the distribution of sleeping sickness, which allowed a better understanding of the epidemiology and nosogeny of African trypanosomiasis.

Corré (1877), a French naval surgeon, relates in his *Recherches sur la maladie du sommeil* the experience he acquired during the epidemics which decimated the small garrisons in Senegal. He was the first to realize that the sickness known as *Nelavane* in the Wolof dialect of Senegal was, in fact, sleeping sickness. His clinical description was outstanding. It brought forward cephalalgias, dry skin, pruritis, adenopathy, variable appetite, diarrhoea alternating with constipation, receded libido, apathy, sensitivity to cold (shivers), troubles of vision, and ptosis. Furthermore, he noted difficulties of speech in spite of a preserved intelligence, erratic somnolence, extreme apathy, choreal movements either widespread or localized, pseudoparalysis, convulsions and fits of narcolepsy. The disease was noted to progress slowly to incontinence and death, in convulsions.

Corré also indicated (1876) that, in Upper Senegal, villages were in ruins and deserted as a result of this scourge; he spoke also of epidemics localized in the

wooded swamps of the hinterland, and of an isolation village created at Sor, on the outskirts of Saint Louis.

Moreover, transportation by barges on the Black Volta, known today as the Mouhoun river, in Upper Volta (Burkina Faso), had been causing epidemics in this territory since 1880 and up to the north of the Ivory Coast. Observations gathered along the gulf of Guinea and later in Mali enlarged considerably the list of stricken countries.

The same facts apply to later investigations carried out in Nigeria. The results of these investigations, reported by *Duggan* (1962), confirm that foci of sleeping sickness were known since a long time in this vast country. Apart from coastal clusters close to Eket and Ahoada, and those scattered haphazardly further north, others stretch out to the area where the valley of the Cross river gives way to that of the Benue river. Beyond Lat. 8°N., foci are to be found in the valleys of the Niger and Benue and along the banks of the Kaduna. These foci spread as far as the frontier zone of the Sahel, the limit of the presence of glossinae. These regions were the starting point of virulent epidemics from 1850 to 1870 and from 1886 to 1906, a fact worth remembering.

The foci along the Benue tributary of the Niger and those in the Lake Chad territory go back to a long ago distant past. The Emirate of Bedde suffered a serious epidemic in 1846. In 1859, it was the turn of Digazon, a town on the Kaiawa, near Sherifari in the Lake Chad region. As a result, Digazon was deserted for 50 years. Dating from 1886, at the outbreak of a similar epidemic in Babuwari, the town was deserted for 20 years. The inhabitants were acquainted with the sickness, and associated it with glossinae. To protect themselves against tsetse fly bites, they rubbed their bodies with salt (*Duggan*).

3.4. Identification of sleeping sickness in Europe

Stephen MacKenzie (1891) was the first to observe, in London, a case of sleeping sickness in a black man. Two other cases hospitalized at the Charing Cross Hospital, where Manson was working in 1899, gave *Mott* the opportunity to make some anatomo-pathological observations still valid nowadays; he described the almost specific *morula cells*.

The same year, three Congolese men suffering from sleeping sickness were brought in from Africa by *Brumpt* and were examined in Paris.

In 1901-1902, an infection caused by trypanosoma was detected for the first time, in the Gambia, in a European by *Forde* and identified by *Dutton* in Liverpool. This sickness was called trypanosoma fever.

In 1903, sleeping sickness was diagnosed in an European woman in Manson's hospital wards. The results of the autopsy were published by *Low* and *Mott*. Her disease had in fact been recognized in 1871, in Angola.

But as these findings refuted the theory that Europeans were immune to the sickness, the Portuguese government appointed a team of experts to examine the accuracy of the diagnosis. A mission by Anibal de Bettencourt et al., as all of the other researchers engaged in the same search, focussed its attention on a bacteriological agent and found as several other physicians a streptococcus, which was named *hypnococcus* in the brain lesions. D. Zelia Botti, his laboratory technician observed, in a blood smear of a patient, a novel parasite, which was discarded by the master as a mere debris. Later on, with a relecture of the smear he was convinced that he failed to recognize a great discovery (Cambournac, 1882). When *Bruce* had the opportunity to re-examine some of the blood samples which were still in good condition, he had no difficulty in detecting trypanosoma.

It is only proper to recall that, at the end of the nineteenth century the Christian chief Mandombe, of Gombe Lutete, aware that he had been struck down with sleeping sickness, agreed to undergo an experimental treatment in London for the good of his people. In the London hospital where he died, a commemorative plaque was placed at the head of the bed he had occupied, bearing the inscription: "Greater love hath no man than this, that he lay down his life for his friends" (Chesterman, 1947).

4. Foundation of the Congo Free State (CFS)

In order to understand the endemo-epidemiology of sleeping sickness in the CFS, it is necessary to be aware of the complicated diplomatic negotiations which were at the origin of the Congo Free State.

The ambition of King Leopold II to endow Belgium with a colonial expansion in Central Africa was reflected in the Brussels Geographical Conference held in 1876, where he announced: "Let us implant the standard of civilization in the soil of Central Africa".

The aims and objectives were formulated to the international audience as follows:

1°) to organize a thorough exploration with the aid of people travelling singly and starting from different directions in the area bordered by the Ocean to the east and the west, by Egypt and Sudan to the north, and to the south by the Zambezi river;

2°) to build scientific and medical centres run by European residents, with the aim of assisting explorers and providing them with supplies, and extending their own protectorate to neighbouring regions.

Numerous explorations of the African continent in the previous years had paved the way for such a congress. In 1482 the Portuguese navigator *Diogo Cão* became the founder of this area by placing a *padrao* (a commemorative stone adorned with coats of arms) on the southern bank of the mouth of the Zaire river. In 1485 he sailed upstream, as far as the first rapids. In 1798, *Jose Larderda*, another Portuguese, discovered the copper mines of Katanga.

The main concern of geographical societies and of explorers, was to discover the course of the main rivers, their estuaries, and especially the source of the Nile, which was to lead to the discovery of the Great Lakes.

Stanley's arrival in Boma, in 1877, marked the start of feverish activity in the field of exploration. Many explorers, among them Stanley and his team, seized for their own benefit all the trade along the open channels; they systematically stopped all existing trade and every possible competition with all the means at their disposal including posts established as they sailed upstream, reaching as far as Iboko (Nouvelle Anvers, now Makanza) in 1881 (See map 32 p. 1688).

The first consequence of these explorations was the Berlin Conference convened by Bismarck in the Chancellery, at Wilhelmstrasse in Berlin, on November 15, 1884, after the Chancellor had withdrawn his veto on the (preferably peaceful) establishment of a German colonial territory, which was to have embraced Cameroon, Togo, and the East African states (Tanganyika, Rwanda, Burundi).

The Anglo-Portuguese treaty of February 26, 1884 gave Portugal free access to the sea all along the S.W. coast between parallels 5°12' and 15°18' lat. south, including the Lower Congo territory until then considered neutral. This treaty denied the International African Association any access to the sea. However, its application depended on the agreement of other powers. France, Germany and the USA were opposed to it, in the hope for free trade in the Congo basin. They suspected indeed that the Anglo-Portuguese treaty would bring with it the implementation of the Portuguese customs system.

The main object of the Berlin Conference was to safeguard free trade in Equatorial Africa, a consideration of the greatest importance. The discussions were painstaking and the General Act was not signed until February 1885.

Great Britain, as sovereign power of the lower part of the Niger, refused to allow free trade on the Niger. France requested a definite outline for the limits of the Congo basin, enabling to protect her Gabonese trading posts from the free trade. Portugal wanted detailed information on the territories it would be able to keep in the Congo basin.

In order to come to some agreement, negotiations alongside of the Conference started between France, Portugal, Germany, Great Britain and the International Association of the Congo, IAC (*Association Internationale du Congo*).

The treaty with Portugal stipulated that it would keep Cabinda, as this territory was specifically mentioned in the Portuguese Constitution, and it had acknowledged right along the left bank of the river from its mouth to Noqui, where the frontier veers to the east towards Kwango. The IAC was given the custody of a coast of 35km along the right bank of the river, from Banana Point to Cabo Lombo, as well as authority on the section of the river which did not belong to Portugal.

In 1884, the IAC granted preferential rights to France, to enable her to have direct access to the right bank of the river Congo between Brazzaville and the Atlantic, a decision which irritated Great Britain. The way in which the frontiers were drawn up compelled the IAC to give up Niari-Kouilou, without receiving any compensation, although it had already established several posts. The IAC considered the dividing line Congo-Niari-Kouilou as far as Manyanga to be the accepted limit.

The frontier limits were set from that point along the river Congo up to its confluence with the Ubangi, then to the Lower Ubangi. The frontiers of the Upper Ubangi and the Bomu had not been defined and would be the subject of many explorations and later treaties between France and Britain.

Germany and Britain acceded to a request from the IAC, to extend the hinterland of approximately 450 small areas along the river into vast, unoccupied and unclaimed territories. The boundaries were as follows:

- in the east, the mountain separating the Congo from the Nile basin, the median line of Lake Albert (Mobutu), the Semliki, the median lines of Lakes Kivu and Tanganyika, the line stretching from Moliro to Lake Moero, the line from the Luapula river to Lake Bangwelo and the ridge between the rivers Congo and Zambezi (see map 33, p. 1694 and 35, p. 1698);

- in the south, the same ridge up to Lake Dilolo, then the course of the river Kasai from Luao towards the north, then the parallel 7°11' lat. south towards the Kwango, and west towards Noqui (see map 36, p. 1708).

When acknowledged as a *Constitutional State*, by the nations present at the Conference, the IAC was able to bring the decisions to the notice of the Conference and to sign the General Act in its own right. IAC became the Congo Free State.

The General Act of the Conference granted freedom of trade and navigation, to the basin of the Congo, as agreed by the Convention but demanded the abolition of the slave trade and an effective occupation.

The last request resulted in an intensive inspection of the Upper Ubangi, Bomu and the neighbouring regions (see map 33, p. 1694). In Chari, Ouaddai, Darfour and other unoccupied territories where the IAC had contemplated establishing scientific centres and hospitals for emulators of Heinrich Barth and Gustav Nachtigal. This brought about competition and protests from France; the Congo Free State had to withdraw to the Bomu, but attempted a push towards the Nile. A major expedition led by *Van Kerckhoven* reached Wadelai (East of Aru, in Uganda). The leader of the expedition died from an accident at Jabal Wattî on August the 6th, 1892.

Great Britain agreed to the occupation of Southern Sudan, and in 1894 offered the lease of the territory covering the south of the Nile basin to latitude 10°N. (that is as far as Fachoda). Renewed protests from the French compelled the Congo Free State to repudiate the treaty with the British, to limit the territory leased to the Lado enclave (see p. 1695) and accept the left bank of the Bomu as its northern frontier. The establishment of this frontier, which cut across homogeneous ethnic groups, proved to be the cause of migrations of populations swayed by the unpredictable and capricious allegiances of their sultans.

4.1. Inland exploration

In the south, many explorations were undertaken, from Luanda (Angola) by German explorers, among them *Hermann Von Wissman* and *Pogge*, who surveyed the Kasai and the Sankuru before reaching Nyangwe. The baptist missionary *Grenfell* came upon a considerable number of tributaries in the basin of the river Congo, and was the first to publish a map of the region. He also explored the Ubangi, and the Lulonga, Marina, Ruki, Busira, Tshuapa, Kasai and Kwango rivers.

The real thrust to the heart of Central Africa, an area already, to some extent, known as a result of several other explorations, was undertaken by the International African Association (IAA). Following the example of previous explorers, the Association set off from the east coast. This enabled them to follow the caravan trails used by Afro-Arab dealers from Zanzibar, Bagamayo and Sadani.

The five expeditions that took this route were able to establish themselves on both shores of Lake Tanganyika at Karema and at Mpala (see map 35, p. 1698), but faced the usual difficulties encountered by all inexperienced explorers. By this route there was however no risk to contract sleeping sickness. Some glossinae did live in these territories, but not *T. gambiense*; an acute and fatal infection of *T. rhodesiense* though very unlikely, would more often than not be diagnosed as pernicious malaria; this trypanosoma was only discovered three decades later.

The push towards the west was to become a reality through the odyssey of Stanley who, from 1874 to 1877 and despite numerous obstacles, succeeded in establishing the link between the Indian Ocean and the Atlantic by sailing the river Lualaba-Congo (Zaire) downstream from Nyangwe. He started with 347 men and lost two-thirds of his contingent on the way, many of them stricken with disease. However, although the journey took the expedition through infected areas, no case of lethargy was ever detected, and the actual causes of death remained unknown.

The IAA decided, from then on, to adopt the western route as well as the Congo (Zaire) river, which would be used for expeditions into the interior. On November 25th 1878, when the Upper Congo Study Committee, UCSC (*Comité d'Etudes du Haut Congo*) was formed, it was decided to examine the best possible way of using the river Congo and its basin. Stanley was to be appointed by the UCSC to carry out the study. He landed at Banana on August 14th 1879, sailed upstream to Vivi, and established a post on the right bank of the river. Later the town of Matadi was erected on the left bank. Vivi became the first capital of the Congo Free State. From that point, sailing up-river became impossible because of its successive cataracts. In December 1881 in order to reach the Malebo Pool, Stanley worked unceasingly at building two road stretches totalling 245 km divided by a 120 km navigable portion of river, between Isangila and Manyanga. The infrastructure was hereby prepared for the Congo Railway, started in 1889, which by 1898 would make it possible to cover the 400 km between Matadi and Leopoldville.

In 1882, the Upper Congo Study Committee became the IAC (International Association of the Congo). In 1883, heading a new expedition in the boat *En Avant*, Stanley explored the river as far as the Stanley Falls, discovered Lake Tumba and the Lomami and Lukunga rivers, and set up the posts of Bolobo, Equator, Bangala and Kisangani. A few years later, epidemics of sleeping sickness were rife all along that route.

The importance of the river and some of its tributaries was quite obvious; but it was not the only priority.

During that quarter of the century, the International African Association and the Congo Free State were both busy, organizing more explorations into the hinterland and army expeditions aiming at suppressing the slave trade. Up-to-date administrative and commercial offices but also missionary posts were set up in rapid succession. The building of a wide network of modern means of communications and the setting up of other infrastructures were initiated.

Other access routes were used by military expeditions, organized in the north-east of the country, which came under the Sovereignty of King Leopold II in 1885. These expeditions had a dual objective:

- to repress the tendency of the Afro-Arabs to extend their territories: in 1860, they had established the post of Nyangwe, from where they proceeded gradually to the region of Lomami in the west, then up-river to the north as far as the Stanley Falls, thus expanding into the territories where they could find the ivory which was carried by long lines of black slaves to Zanzibar;
- to keep in check the attacks from Mahdist troops, members of a Muslim sect, whose chief claimed to be a descendant of Mohammed and who was determined either to exterminate or convert the infidels. These troops, which had already advanced deep into the province of Equatoria, proved a great obstacle to the plans of King Leopold II, who was eager to reach the Nile, the gateway to the Mediterranean.

4.2. *Occupation of the territories – Movements of soldiers, traders and missionaries*

To fulfil the obligations prescribed by the General Act of the Conference of Berlin (1885) and achieve among others an effective occupation, the Congo Free State had to create an entire administrative structure across its territory. This called for the country-wide movement of explorers, of military and of civil personnel, which in turn opened the interior up to traders and missionaries.

The movement of large military groups was likely to spread disease. Soldiers, particularly at the first stage of the explorations, were recruited from various areas, mostly West Africa and Zanzibar, but also locally. To these contingents were added auxiliary soldiers and irregulars, porters, oarsmen, servants and women.

As soon as the administrative quarters were set up in the inland posts, soldiers and policemen were appointed to travel from village to village, to maintain law and order. The suspect and the guilty were brought over to the administrative quarters. Moreover, messengers travelled ceaselessly to and fro, even on long distances.

The freedom of trade and navigation guaranteed by the Acts of Berlin (1885) and Brussels (1890) was abolished in 1919 by the Convention of St. Germain-en-Laye. For the time being this freedom generated the establishment of a wide network of trading posts which were set up at the junction of waterways, where inland tracks and roads converged. Traffickers and dealers of rubber, copal and ivory brought with them their domestic help, as well as various other people, all of whom added to the mobile population on the move. Furthermore, transport by porters expanded

considerably. Commercial firms had no monopoly of trade; the State conducted parallel activities in its vast privately owned territories. Local small-scale merchants dealt also in copal, rubber-latex and ivory.

Caravan trails, travelling through glossina-infested territories, became one of the main means of the dissemination and increase in virulence of the trypanosomes. Herds of cattle and their drivers moving from place to place added to the risk. All these factors were responsible for the sudden increase in the spreading of pathogens and accelerated the transmission of diseases. In such a situation, there is no need to search for a special factor to favour epidemics.

5. Health conditions in the nineteenth century

Promoters of this colonial venture in Central Africa, and the early participants in this project, were fully aware that they were embarking into an unhealthy continent; they also knew that it was the main reason why, for centuries, Arab, British, Danish, Spanish, French, Dutch and Portuguese dealers kept to the coastal zones and waited in their fortified posts for their coveted goods to be brought to them.

5.1. First explorations

5.1.1. Captain Tuckey's expedition

The Lords of the Admiralty had instructed *Captain Tuckey*, to sail the river upstream, to ascertain the exact site where the mouth of the river Niger (whose estuary was discovered only in 1830) took its source. In 1803, *Captain Maxwell* had been able to sail the river Congo only as far up as Noqui, a few kilometres downstream from Matadi.

Tuckey's two sailing-ships, the *Congo* and the *Dorothea*, with their 56 members, reached the estuary on July 5th 1816. The explorers' rowing boat was blocked by rapids on the 9th of August, and 30 of its members landed at Noqui.

They proceeded over land with great difficulty; by September 9th half the members of the expedition (those who were still able-bodied) managed to reach the navigable stretch of the Isangila river at Manyanga, a distance of about 210 kilometres. However, sickness and exhaustion forced them to return. By the time they arrived back, on 4 October, their numbers were disastrously depleted: commander Tuckey, his second-in-command, the four naturalists and eight crew members were dead. To these casualties had to be added the deaths of four more members of the crew left behind in the river's estuary, totalling

47% of the staff. The survivors returned to Britain forthwith.

The assistant surgeon *Mackerow* wrote an excellent report, which asserted unhesitatingly that the deaths were due to pernicious malaria – although at a later stage it was thought that yellow fever had been the cause. The meagre information gleaned by the eminent botanist *Professor Smith*, on diseases that afflicted the native population, suggested evidence of smallpox scars, scabies, leprosy, haemorrhage, fever, elephantiasis, soft tumours and wounds.

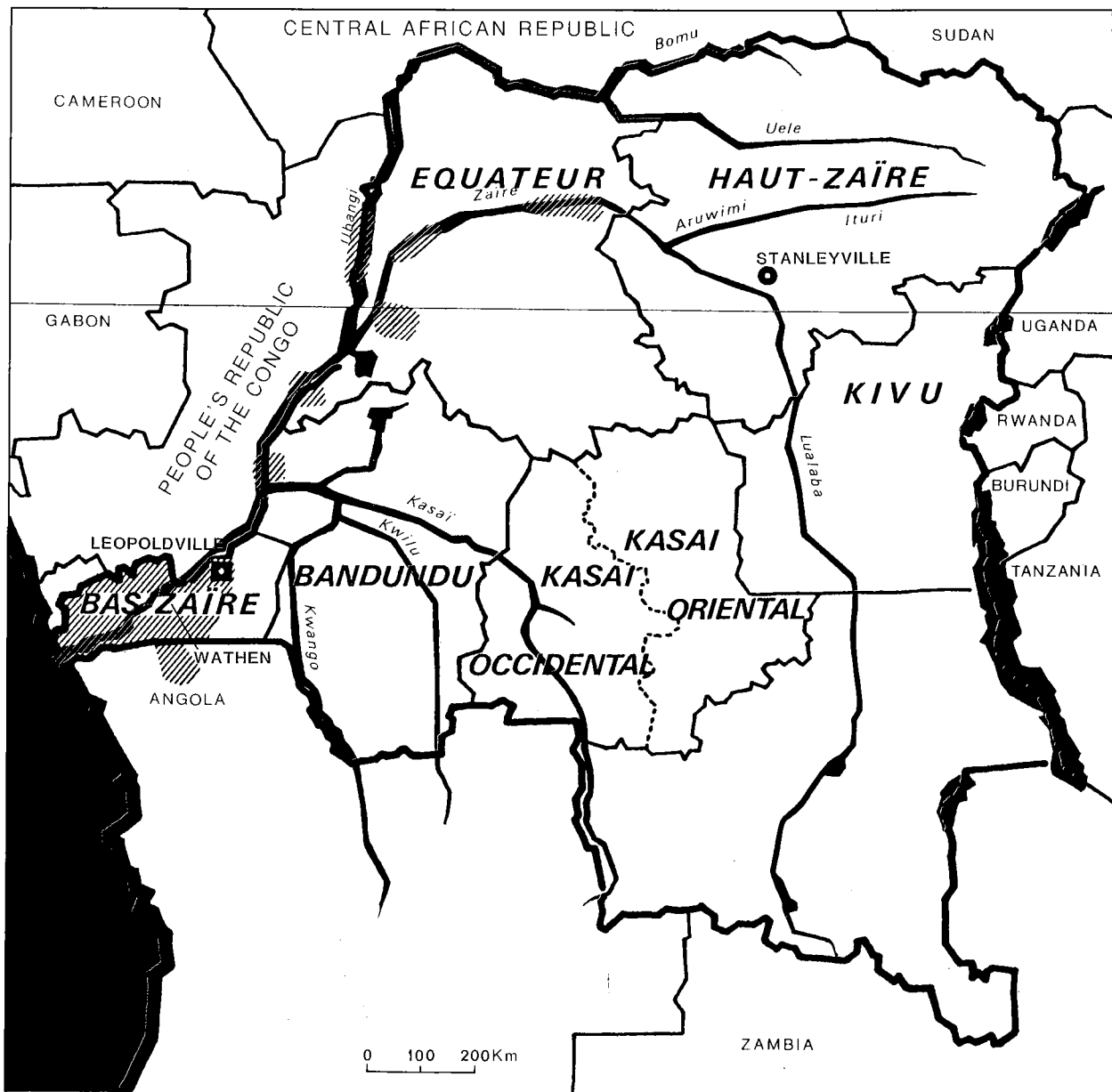
The failure of the British expedition led by Tuckey in 1816, was a bad omen.

5.1.2. Livingstone, Cameron, Stanley

Livingstone, Cameron and Stanley, pioneers in the art of exploration in the very heart of the continent, had also given some indications of the health risks they were running.

Stanley had undertaken his first trip to East and Central Africa in 1871. As a reporter with the *New York Herald* assigned to Europe, he had been instructed to find Livingstone, a protestant missionary of Scottish descent, who had been exploring Africa at the time, without giving news of his whereabouts. Stanley found him on the 10th of November, at Ujiji on the Eastern shore of Lake Tanganyika, struck down with a severe attack of malaria. Stanley was well acquainted with the signs of the recurrence of the illness, and knew how to treat it adequately with quinine (3 x 5 grains, equivalent to 0.35gm), a drug he had come across and used in Arkansas. In his travel notes, he mentioned several health problems he observed among the natives, such as smallpox, malaria, pneumonia, dysentery, ophthalmia, ulcers and other cutaneous infections.

V.L. Cameron, a British traveller, crossed Africa from east to west between 1873 and 1875. After reaching Nyangwe on the river Lualaba, he was stopped from proceeding down-river by Tippu Tip (whose real name was Hamed-ben-Mohammed), a rich and energetic young Zanzibar merchant who acted as if he were the ruler of the whole region. Cameron decided to go west, reached Lake Kisele, continued towards the region of Upper Lomami, travelled upstream, and finally, by following one of the routes used by the Portuguese caravan trails, reached the west coast of Angola at Benguela, south of Lobito. Disease was rife: malaria for the white population and smallpox for the natives, a cruel disease which spreads like wildfire, devouring vast regions of Africa. He also reported seeing lepers isolated, relegated to an island in Lake Tanganyika.



Map 31 – Foci of sleeping sickness in the Congo, 1884
(adapted from *Dutton and Todd*, 1906, on modern administrative and political boundaries)

Stanley, at his second attempt to link the east coast to the west (1874-1877), did not take with him any physicians, naturalists or soldiers, but only three auxiliaries: the two Pocock brothers, who were fishermen; also Barker, an employee at the hotel where he had been staying. Moreover, in Zanzibar he recruited about 360 African auxiliaries. His voyage on the *Lady Alice*, a collapsible boat, was meant to be a careful reconnaissance of Lake Victoria.

On May 27th 1876, he signed a pact with Tippu Tip, who had obstructed Cameron's exploration, and paid him a sum of 5,000 dollars. Stanley was then allowed to embark at Nyangwe on the *Lady Alice* and, with a flotilla of canoes, he aimed to sail down-stream on the river Lualaba. He reached the estuary of the Congo on 9 August 1877. On his way, he noticed that in the small town of Ujiji, with about 3,000 inhabitants, an average of 75 persons were dying of smallpox daily. Stanley himself lost 173 members of his party out of 363: a total of 48%. Among them, the three white auxiliaries and 45 of the 173 blacks died of smallpox, 21 of dysentery, 58 by wounds and accidents and 14 from drowning.

On August 14 1879, Stanley and his expedition reached the mouth of the Congo. In this mysterious continent they wanted to explore the possibility of establishing a regular link between the Lower Congo and the navigable upper basin on behalf of the Upper Congo Study Committee.

At that time, the Dutch (*Afrikaansche Handels Vennootschap*) and the French trading posts (*Dau-mas, Beraud & Co.*), which assembled the goods carried by river from the hinterland, were nearly all based at Banana, the extreme point on the Congo River. Only eight trading companies were set up in Boma and were run by about 20 Europeans.

The European personnel appointed to serve in the hinterland were aware of the risks of disease and death in the space of only a few days: a threat hanging over one like the sword of Damocles.

5.2. Distribution of sleeping sickness and accounts of health conditions

5.2.1. The *spread of sleeping sickness* was one of the most serious evils that tropical Africa had to endure. Manifestations of its gravity followed relentless, one after another, like waves beating on the shore. Outbreaks were observed in several foci but the source of the sickness was not known.

In 1904, the *Dutton and Todd* mission published retrospective details gathered from the local population, and drew up a map of the distribution of sleeping sickness for the year 1884 (see map 31 on p. 1686).

5.2.2. Accounts of health conditions

Fever, dysentery, skin diseases, even the African sun for the white man and smallpox for the natives were the disasters that gave the west coast of Africa the name of White man's grave. A saying, with a typical touch of British humour, summed up the situation in these words: "the Gold Coast needs two Governors, one on the spot and the other on his way, to replace the first one, who is on the way to die".

Towards the end of the nineteenth century, much progress had been made in all scientific disciplines enabling to discover the agents responsible for an increasing number of diseases, and the specific drugs needed to control their effects.

Very little was known about the sleeping sickness described as a mysterious lethargy thought to be peculiar to the black population. The initial fever, its transmission through a vector no-one suspected, and its pathogenous agent were still unknown. This lethargy had been noticed to a slight degree in confined foci, a fact which had not escaped the natives. Great was the surprise of the Europeans when they realized that this lethargy was not a mere curiosity, but could become a major problem.

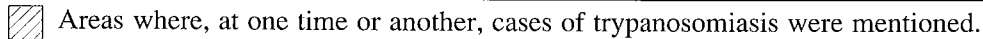
Year	No. of Europeans employed	Deaths Number	%o
1879	20	1	50
1880	32	5	156
1881	34	1	30
1882	74	2	27
1883	106	17	169
1884	187	16	86

The *mortality rate* among Europeans was assessed through the figures supplied by various employers. The mortality rate of the staff of the IAA and the UCSC was high, and continued to rise.

Of the 1,136 whites employed in the building of the railway, 113 (that is 99 %o) were to die during their stay of nine to 16 months: 17 as a result of accidents, explosions or drowning, and 96 from illness.

Statistics on deaths among the personnel of the Free State showed that:

- between the years 1885 and 1896, the mortality rate among the staff of the Department of the Interior reached 56%o of whom 46%o died of sickness. Deaths among the staff of the Department of Justice were obviously much lower because of their more settled lives;
- in 1905, the rate still reached 54 %o; whereas in Belgium, by comparison, the mortality rate among soldiers doing their national service barely reached 4 %o.



Map 32 – Main rivers and towns or sites of medical importance in western and north-western Zaire and neighbouring countries

The main cause of fever was malaria with its black-water or haemoglobinuric fever; it had acquired a sad reputation, and was proving fatal in some 10 % of the cases. It was, however, rarer in certain regions. From 1903 onwards, sleeping sickness was to be added to the fevers which affected Europeans.

The figures showing the mortality rates should not be a surprise, taking in account the low standards of hygiene, the few basic preventive methods which existed at that time, and the ludicrously small number of physicians employed by the government: in 1885 there were two physicians for 45 posts, in 1891 eight out of which six were in the centres; 14 physicians in 1897, 25 in 1899 to cater for 215 posts, 27 in 1903, 59 in 1910 and 81 in 1914. There was also an unknown but undoubtedly small number of physicians appointed by private firms and missions.

Nevertheless, sleeping sickness appeared as a major threat. *H. Ward* (1910) wrote in 1895 that sleeping sickness was one of the major diseases to which the Congolese were exposed. The disease was widespread among Babangui and Bangala. Its nature was unknown, but it led to death.

5.3. Endemic foci of sleeping sickness

5.3.1. Western Congo Free State

A menacing increase in the cases of African lethargy occurred towards the end of the nineteenth century.

The particulars given by *Mense* (1888) confirmed the first reliable information on African lethargy, given by *Dryepondt* in the *Guide pratique, hygiénique et médical des voyageurs au Congo* (1895; a practical guide to hygiene and medicine for travellers in the Congo), which mentions the existence, in the Lower Congo and in the Equateur Province, of a sickness which only affects black people born in Africa. This observation is repeated by *Van Campenhout* in the *Rapport du laboratoire de Léopoldville* (1901). It was added that sporadic cases had been observed in Uele at Basoko, in the Bangala district, in Leopoldville and at Boma; it also mentions that the sickness had existed many years earlier, endemically in the Cataracts region (Banza Manteke) and along the left bank of the river Congo from Kwamouth to Bolobo.

The major interest of this report lies in its study of the epidemic which spread in the Berghe-Ste-Marie Mission, on the promontory formed by the left bank of the river Congo and the right bank of the Kwa-Kasai in Bandundu.

Van Campenhout went out to examine the situation on the spot. The focus was a school complex opened

in 1890. By 1900, out of a total of 1,134 children brought to the school, only 250 of them were left. Among the survivors, some had been born there.

From 1890 to 1894, deaths were due essentially to dysentery; 114 deaths were registered (26% of the 546 boarders). In 1895, most of the fatalities had been caused by dysentery, at a mortality rate of 37%, but cases of lethargy were becoming more frequent. From 1896 onwards, about 100 deaths from lethargy alone occurred among an average population of 420 inhabitants. *Van Campenhout* managed to examine 82 patients, but only seven were actually natives of the region.

The Berghe-Ste-Marie Mission was abandoned in 1898. The transfer of survivors caused the dissemination of trypanosomes in hardly supervised areas, in previously disease-free zones and in foci already identified by the presence of enlarged lymph nodes, where the transfer may have introduced new strains.

The Nouvelle Anvers Mission (Makanza) took in 136 children rescued from the disaster at Berghe-Ste-Marie; of these, only 13 survived. However, their role in spreading the disease would have been negligible at such mission centres, with extremely strictly enforced rules of hygiene.

From 1900 to 1905, similar disasters hit other missions, namely at Tumba, built along the railway line linking Matadi to Leopoldville. The 70% mortality rate was due to the fact that Tumba was the only mission where seriously sick people could find refuge. In 1903, the St Trudo mission near Lusambo was the scene of a similar tragedy reported by *Broden*. St. Donat in Maniema was to be yet another example. Further east, on the western shore of Lake Tanganyika, the Mpala mission avoided such a disaster by systematic clearance of shrubs, trees and undergrowth, a successful procedure already used in 1905 by *Rodhain* in Ubangi. The Lusenda mission, near Baraka, also on the shore of Lake Tanganyika, had to be evacuated.

The disaster at the Catholic Mission of Berghe-Ste-Marie, at Kwamouth, emphasized the risks already referred to prudently by *Dr Walfridson* of the Swedish mission of Mukimbungu (see also p. 1717). A number of other physicians working for protestant missions, and *Dr Dye* in Bolenge in particular, also confirmed that the disease had a tendency to spread widely.

The existence of sleeping sickness was known in various places: Wathen, Lukolela, Banania, the Cataracts (Banza Manteke), the Kwango (see map 32). For more than fifty years, and certainly before the colonial period, the disease had moreover been known in several areas of Lower Congo, Kwango and Equator but as a sickness which only affected Africans.

Several new cases were observed, particularly along the Congo river which was travelled by an increasing number of personnel, – individuals, officials, missionaries, merchants and others. The extension of the disease occurred in river areas presenting as very homogenous. Although its appearance was sporadic, it gave an impression of continuity along the river banks.

As early as 1890, *Dr. Paul Briart* drew attention in his reports to the danger of tsetse flies along the Lubudi, Lomami, Lovoi and Lualaba rivers. The Delcommune expedition had seen cattle and dogs dying after being bitten by tsetse flies suspected of injecting a toxin.

Starting from 1894, men were recruited to work on the Lower Congo railway in densely populated regions such as Pania-Mutombo. By 1896, the first case of sleeping sickness had been observed in this area, a staging-post for caravan trails. The disease spread rapidly along the Lusambo-Kasongo caravan route. In many locations, the people themselves were able to supply detailed information on the time the first cases occurred, the names of the patients and even their origin. By 1890, sleeping sickness was rampant along the Upper Congo river and the Lualaba.

In 1903, *Dutton*, and a year later, in 1904, *Todd and Christy*, found at Boma 11% positive cases among 253 soldiers and prisoners drawn from various regions. In contrast, among the 103 examined locals, the rate of positives was only 0.9%. At Matadi, 2.9% out of 138 people examined were infected; in the Cataracts area (Wathen, Lutete), the rate soared to 7.5% of the 212 people examined. In Leopoldville, where 465 blood tests were carried out, the rate was 11.6%. The blood test was the only technique used, and limited to those who showed some clinical signs of sleeping sickness.

5.3.2. *The foci in Equatorial Central Congo* (see also map 32, p. 1688)

The vast territories crossed by the tributaries of the equatorial section of the river Congo were important foci of sleeping sickness. Close attention was given to territories north of the river, but only along its banks and at the mouth of the Mongala and the Itimbiri.

However, this was not the case with the inland of the central basin of the river on which only fragmented information was found in general administrative reports, particularly those of the Health Commission, or in the medical reports of the few physicians and in the writings of missionaries or of commercial firms, when available. Globally, all

these accounts convey a picture worth considering, although the data are too sparse to be of scientific value.

The Equator region showed, on the whole, that the infection followed the course of navigable rivers and decimated the people who lived along their banks. The progress of the disease stopped where the river trading ended, remaining isolated.

It was also observed that, broadly speaking hospitals caring for patients infected with trypanosomiasis and isolation *lazarettos*, were foci spreading the disease when they were built in locations where the tsetse fly thrived. It was also discovered, incidentally, that in hospitals, as well as in isolation units, patients with sleeping sickness came in contact with smallpox patients, to the extent that the importance attached to trypanosomiasis tended to overshadow the possibility of other deadly epidemics of different aetiology.

It was later proved that in army encampments (Irebu, Lisala) where soldiers were billeted, apparently healthy recruits often carried the disease without actually being a source of infection. Regular cleaning of the camps, measures to control tsetse flies and much less the immediate transfer to isolation lazarets of soldiers in the initial phase of the illness, helped to keep the disease in check.

On the river Mongala and its many branches such as the Ebola and the Due, there was a chain of little villages. It was impracticable to enforce any rules of hygiene or to attempt any drainage, as the chiefs of the various tribes living along the river had no influence over the population of fisherfolk. The situation inland was just as bad. It is as well to remember that the medical mission set up to control sleeping sickness in 1920 in the territory of the Bangala, was to meet the greatest opposition.

Mouchet (1927) confirmed that it was impossible to carry out a systematic investigation. A rate of 1.08%, representing 3,306 sick patients out of a population of 180,523 inhabitants, was showing only a small proportion of the actual problem. The estimate of 2.3% (0.18% to 7.3%) seemed more realistic.

Upoto, a post a few kilometres downstream of Lisala, was visited by all the regular and casual control teams on the river. In 1890, the area consisted of an uninterrupted chain of fishing villages, which were to be reduced within a few years to a few dilapidated huts.

In the vast territories south of the North-western loop of the river, four major foci of sleeping sickness developed. In 1903, the region around the river

Lulonga, confluent of the Lopori and Maringa rivers, was the first to be affected (see map 32 p. 1688):

- in 1908, at Lulonga post, built at the meeting point of the Lulonga river with its tributary, 19% of the deaths were attributed to sleeping sickness;
- that same year the State Inspector, *Gerard*, reported that the banks of the rivers were deserted, particularly those of the Maringa river, and that many villages had been wiped away; the same situation was to be observed along the river Ikelemba;
- the third and best known focus was the Ruki-Busira-(Salonga)-Tshuapa focus. On the Lower Ruki, the infection had certainly existed since 1901. As everywhere else, it was the cause of a worrying depopulation which was difficult to stop in spite of all the vigorous measures taken to combat the disease. Even the building of an isolation lazaret on an island overrun with tsetse flies seemed futile, at a time when their role was still obscure. The spread of the disease stopped abruptly (and for no apparent reason) at Mondombe;
- the fourth focus was located at Lake Tumba.

5.3.3. Northern Congo Free State

The course of events in the border areas and in the hinterland is particularly well known. However, actual political frontiers of these controversial regions have to be disregarded, as these were continuously crossed by the people.

In the regions watered by the Ubangi river and its tributaries, many observations were carefully and repeatedly noted by French physicians of renown. The Lower Ubangi area represented an important focus which joined that of the Lower Sangha (see map 32, p. 1688).

The basin of the Sangha, a tributary of the river Congo, was the centre of active foci of sleeping sickness, and had probably been there at least since 1886.

The focus of Lower Sangha was located at the same level than the one on the Lukolela, on the other river bank. The Sangha crosses the equatorial forest as far as Bania, running through densely wooded banks up to Carnot.

The Upper Sangha is in the Central African Republic. From Carnot, it flows across the Sudanese savannah. The Upper Sangha embraces the Nola focus of great historical importance. This post was at first a relegation village for political deportees (Hausa), erected at the point where the Kadei and the Sangha merged. Its inhabitants were wiped out within two years. The 1885 Sangha epidemic was due, according to *Ouzilleau* (1911), to raids by the Foulbes and to contamination from the Loango focus. In the valley of the Lower

Kadei, there were several small foci which, from time to time, flared up and then subsided. The same peculiar evolution was observed in the Congo Free State where, in 1894, a large village on the Lower Kadei was reduced to about 20 huts.

Ubangi had been hit by the African lethargy before the arrival of Europeans. Statements from the involved population attested to this and were confirmed in 1903 by *Brumpt*. The foci on the left bank of the river, opposite those identified on its right bank, were just as rife.

The Lower Ubangi, which starts at the confluence with the river Congo, that is at the level of Liranga, was mainly in the later People's Republic of the Congo. The Ubangi-Chari territory has been divided over the People's Republic of the Congo and the Central African Republic. In Central African Republic, the Ubangi river started upwards from the mouth of the river Lobaye. The foci of the Lower Ubangi were opposite those of the Congo Free State, from Imese to Mokwangai, via Libenge and Zongo on the bank opposite Bangui.

A mild endemic situation prevailed in the Bangui area, with one southern focus at M'baka and another at the confluence of the Ombella with the Congo river. Observations on the river dwellers of the Ubangi and its three big tributaries, the Motaba, the Ibenga and the Lobaye, were further collected by scientific expeditions. The scientific data confirmed the disastrous findings recorded on the left bank in the Congo Free State. At the township of Etimba, near Liranga, the population was decimated. At the end of the nineteenth century, three large villages close to Irebu, with a population totalling more than 5,000, were reduced to 600 people, and eventually to 500 only. Lipeite (Ibenga) was to suffer a harsh epidemic which killed three-quarters of its people. At Impfondo, a population of about 5,000 to 6,000 inhabitants was reduced to a mere 800.

Observations recorded by *Rodhain* in 1907, after his first visit to the Ubangi (from 1904-1906), were confirmed at his second visit to the region in 1914: many villages which had been hit by sleeping sickness had now disappeared.

In the Lower Ubangi, the sleeping sickness had been noticed before the coming of white men; however, cases of the disease were few and far between. *Rodhain* maintained, nevertheless, that the new and more virulent sources of infection might have been introduced by infected military and civilian personnel, as well as by sick labourers returning to the region.

An examination of ganglionic fluid revealed the existence of trypanosomes in patients who were recognized as having caught the sickness during their

journey upstream (Imese, Ekuba, Libenge, Bamon-dema, Mokwange). In this last post, opposite to the Fort de Possel (Djoukou), Rodhain drew attention to a more serious outbreak of the disease. He was also convinced that there were as many, but more virulent cases on the French bank of the Lower Ubangi.

The Upper Ubangi region, which stretches from Fort de Possel (Djoukou) to the source of the Ubangi river, namely where the rivers Bomu, Uele, and Bili meet at the level of Yakoma, was free of infection, or only slightly endemic, except around Mobaye (Banzville), on the bank of the Bangui and the Lower Kotto. Mobaye-Bongo is on the opposite bank to Mobaye. All cases detected, apart from those ascertained by Rodhain in the village of Guerloget, situated between Yasanyania and Yakoma, were those of soldiers or their wives. The greatest concern of all was the danger of the disease spreading to Monga, an important commercial centre on the Bili.

In 1891, *Ponthier* identified the presence of sleeping sickness among the people who had become affected by Arab culture and lived in Bomokandi, an area at that time considered to mark the end of the spread of the sickness in Uele, as reported in 1902-1903 by *Bertrand*.

Rodhain (1916), in a fact-finding mission carried out in 1913-1914, pointed out first of all that *Brumpt* had proved the existence of glossinae along the Uele river in 1903. He also recalled the findings of *Dr Russo*, who had diagnosed some clinical cases near the banks of the Asa, a left tributary of the Bomu, downstream from Zemio, when he was part of an expedition sent to fight against the Azande chief, Sasa. Rodhain suggested a classification of the foci in a series of groups. In the south-west, in the forest, a few cases were discovered by *Heiberg* in 1907, in the isolation lazaretto of Ibembo; cases on the river Itimbiri, especially among the oarsmen (paddlers), and some isolated foci in the heart of the forest at Zobia, Buta, and Libokwa were also to be found. The focus on the river between Dungu and Surongo, in the swamps, was also known. In the north-east, there was a focus at Monga on the river Bili (see map 32, p. 1688), where *Dr Fauconnier* reported the rate of infected subjects to vary between 0.2% and 4%; however, Rodhain had noted that at Biateke the rate was even higher.

Since 1905, according to *Dr Bomstein's* findings, a focus existed at Yakoma near the Ubangi-Uele confluence, where the rate of infected subjects reached 21% to 35%. The whole of the north-west was affected, and

there was a definite extension towards the east. The Zande focus of Rafai, on the Bomu, showed a rate of 0.3% to 11% and the disease was spreading towards the south.

The Yakoma area stretches along the right bank of the Ubangi river, beyond the Kotto river, between the Sango in the west and the mouth of the M'bari in the east. Villages, with huts built closely together, were set up high on the river bank on the spot which separated the river from the depression behind it. When the level of the river was low, water from the swamp flowed into it and dwellers took advantage of this opportunity to catch fish in their nets or in wattle traps, set in the overflow for this purpose.

The Yakoma people are not farmers, but fishermen or itinerant traders. Their trading trips by canoe often lasted several months and were undertaken in difficult conditions, which left them exhausted and a prey to whatever infection might be around. A very dense population lived in often poor conditions, in clusters of huts built at regular intervals. Sleeping sickness was widespread among these boatmen, who were constantly liable to infections due, perhaps, to new sources. As early as 1904, a state of epidemic was noticed.

The N'zakara country is separated from the Yakoma region by a desert zone of 20 to 30 kilometres, and is centred on Bangassou. In this fertile area villages were scattered in forest glades surrounded by thick foliage which protected them from the heat. The dwelling places (*zeribas*) were of exceptional quality for tropical Africa. They were surrounded by hedges of flowering shrubs with abundant pink and white flowers, interspersed with candle euphorbia, clusters of banana trees and castor oil plants. The huts were very large (8 to 10 metres wide), with the floor at a height of 30 centimetres above ground level. The *zeribas* were spaced out along the creek, each with its own private access to the water. The Government decided to relocate the dwellings along the roads, in order to regroup the people, who then lost interest in maintaining their dwellings, which became dilapidated. The political organization was in the hands of a small nucleus of Zande families, Bandia conquerors from the south, and was quite remarkable. These rulers were also judges, held in high esteem by the litigious little world they governed; in addition they controlled local trade. The Nzakara had conquered and ruled over different small ethnic native groups belonging to the large family of the Badi, who were mainly farmers but also small scale cattle-breeders. Unfortunately, these ethnic groups endured a great depopulation: only

40,000 remained of the 120,000 estimated when the area was first surveyed.

Uele was partly safe, but certainly had a focus in the area of Yambio-Tambura-Doruma, and possibly another in the area Dungu-Kibali-Nzoro (see map 33 on p. 1694).

The 1920 mission against sleeping sickness, organized in Uele and headed by *Dr 's Heeren*, published the following year a detailed account of the situation in Lower Uele; its findings were supported by points already observed by Rodhain. Another interesting document, produced by *Dr Mouchet*, who was at that time (1923) the provincial physician, dealt with the history of the disease in the whole Eastern Province between 1921 and 1923. Also worth mentioning is the report submitted in 1928 by *Lyndhurst Duke* and *Van Hoof*, two members of the International Commission of the League of Nations for the study of human trypanosomiasis; the report dealt with the epidemiology of the disease in the Upper Uele.

The Bomu rapidly became the stake of rivalries between the Congo Free State and the French Congo. When several Belgian explorers arrived peacefully in the Ouaddai, the Kanem and the Darfour, in order to set up scientific and medical stations according to the statutes of the IAA, they met great opposition from the French, who were categorical in their insistence that the Bomu was the northern boundary of the Congo Free State. A further legal dispute, in 1894, established the left bank of the Bomu as the border.

The Bomu flows into the Ubangi; and the focus of trypanosomiasis which was to develop there was, in fact, only an extension of a single focus whose eastern point was at the demarcation line between the Congo and the Nile basins.

Medical and administrative problems which were to occur all along the Ubangi and the Bomu proved that it was a great mistake to link the geopolitical borders to those of the river, because in so doing homogenous ethnic groups living on both sides of the river found themselves split up artificially. Furthermore, as the groups were known by the name of their sultan, the same name was quite often used on both banks, causing additional confusion.

The Zande country was much larger. It embraced territories above the Middle Uele and the Bomu region; it also stretched out into the Central African Republic as far as Zemio and reached the Sudan through the valley of the Tore, a secondary tributary of the Nile, and through the river Yei. Zande is situated in the transition zone between the equatorial forest and the Sudanese savannah, with only sparse

bush but criss-crossed by forest corridors. The country straddles the rather flattened watershed between the Congo and the Nile basins.

The Zande people came south with the last influx of migrants, which started two centuries ago. In the course of their migrations, they managed to subdue a great number and variety of ethnic groups, who were to be absorbed among the Zande people in less than three generations. The eastern Zande group was known by the name of Makraka. While ensuring a *Pax Zandea*, they were also planning further expansion, but their plans were thwarted by the creation of the Congo Free State.

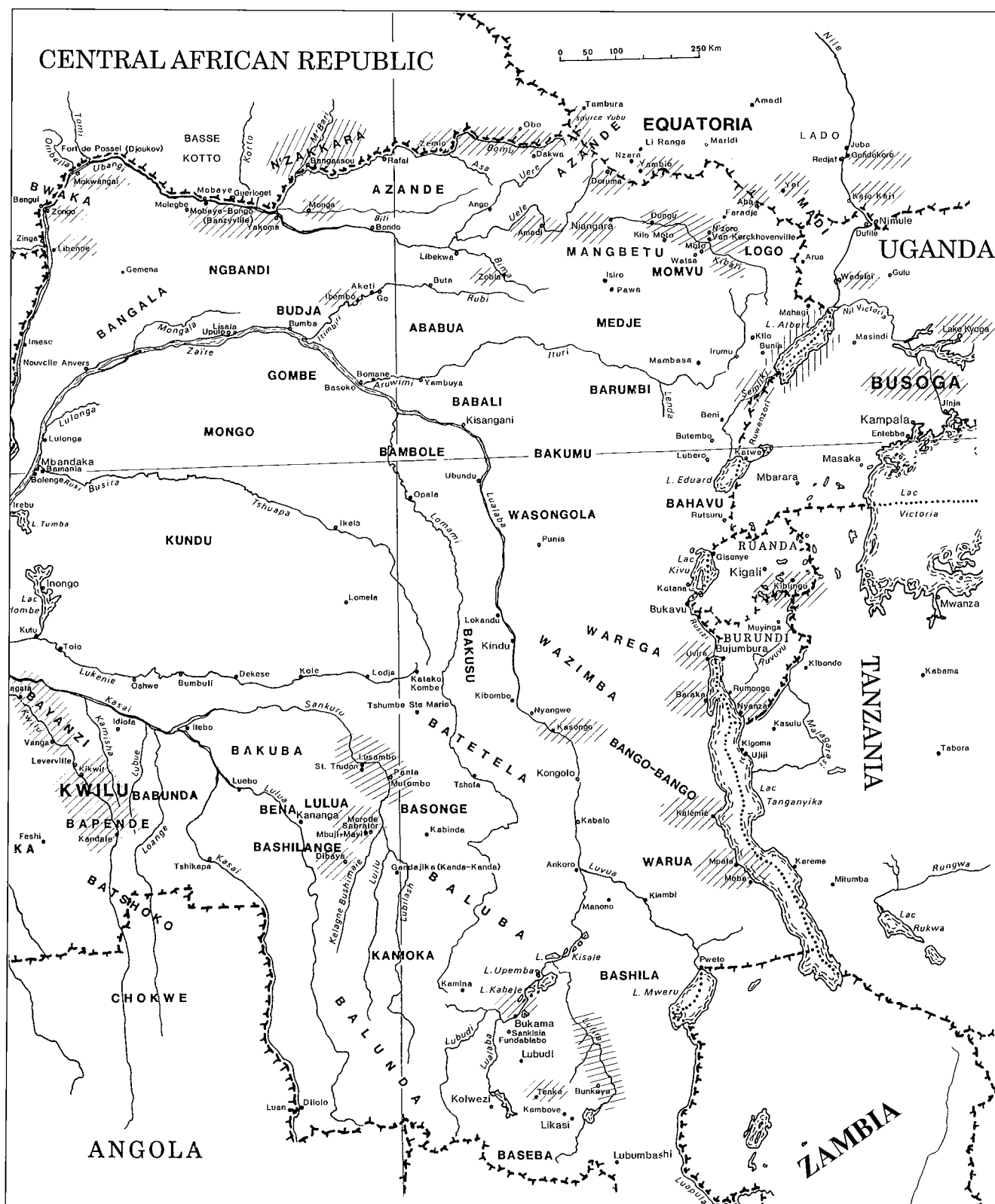
The Zande people, heterogenous but unified, numbered between 600,000 and one million scattered over a vast but sparsely populated area (two to three inhabitants per square kilometre). Half of these people lived in Zaire, the other half were spread out over the Central African Republic and the Sudan.

They were known to be mobile people often travelling great distances, which involved a serious risk for the spread of diseases, including sleeping sickness.

The Uele and the Bomu territories with the scarcely discernable watershed between the basins of the Congo and the Nile (at the Dirbi hills) were carefully investigated by Ouzilleau, Heckenroth and Rodhain. All three gathered information from the best possible sources on the history of these territories in earlier days, and described their ecological and socio-cultural characteristics.

The dividing line between the Congo and the Nile basin can hardly be distinguished as there are no mountain ridges. Indeed, the transition is such that the source of the river Yebu, a tributary of the Nile, and the springs of the Bomu tributaries are only 200 metres apart; hence, there was no obstacle to stop the passage of men, animals, or glossinae from one basin to the other.

According to *Lyndhurst Duke* and *Van Hoof* (1928), the river Bomu with its tributaries, starting in Bahr-el-Ghazal and covering an area common to the Congo and the Nile, were the cradle of sleeping sickness. The disease spread to the west along the Bomu, and also to the south along the Dakwa, to Amadi and to Niangara, and to the east in the territories watered by the Kibali, the Yebu and the Nzoro. The sleeping sickness epidemic was observed to progress slowly and did not spread significantly; the cases remained sporadic, in spite of the great number of tsetse flies present in these areas. It looked like an endemic condition in slow motion. According to the mentioned authors, the foci of Aba and Faradje were false foci because they



▨ Areas where, at one time or another, cases of trypanosomiasis were mentioned.

Map 33 – Main rivers and towns or sites of medical importance in north-eastern Zaire and neighbouring countries

were fed by imported cases. Nowadays, these foci have disappeared.

The valley of the Yei (see map 33, p. 1694) was an endemic zone (Aba-Faradje); several cases appeared at the isolation lazaret of Aba: 15 soldiers, nine Baganda porters and nine State personnel from other regions, a fact which emphasized the absence of infected local inhabitants.

Rodhain also stressed that the mobility of man as a carrier was an important means of spreading the disease; he examined also many other elements which contributed to the progression of the sickness.

Finally, it is worth mentioning that, according to Maillot (1962), the Upper Bomu focus in Central African Republic was imported from Sudan in 1897, and had reached Rafai and Zemio by the year 1912-1913. Maillot also emphasized the fact that, if the infected cases were rare, it had to be borne in mind that the population itself was very sparse (0.7 inhabitants per km²).

5.3.4. *Foci in the north-east and along the equatorial stretch of the Nile at the crossroads of the Upper Uele, Upper Nile, Equatoria and the Lado Enclave*

Although the findings of Rodhain on the north-west regions of the Congo Free State had provided a great deal of information on the foci and the progression of sleeping sickness, for many years our knowledge about the north-east region remained very patchy. Much of the reason for this were the repeated military operations and constant fighting in the eastern corner of the Congo Free State, a never beneficial situation for medical or epidemiological research.

Because of its political and military implications, reaching the Nile became the aim of many states. Access to the river was imperative, if Central Africa was to have an outlet to the Mediterranean. The need to achieve this aim contributed to spread the sleeping sickness in this region coveted by all concerned colonial powers, as there were many movements of missions and soldiers.

The importance ascribed to southern Sudan by the statesmen involved in those regions was highlighted in 1864 by the annexation of the province of Equatoria by Egypt, a member of the Ottoman Empire. The province lived through stormy times, first under the governorship of Emin Pacha after the routing of the Egyptian army, followed by the occupation by the Ansars from Mahdiya and finally in 1897, the Mahdists' defeat by Chaltin at the battle of Redjaf. The army Chaltin commanded was made up of Haussa, Congolese and Zande soldiers who continued

to occupy the area, as ordered by the Congo Free State. These events brought about a continuous movement of troops, messengers, porters and others.

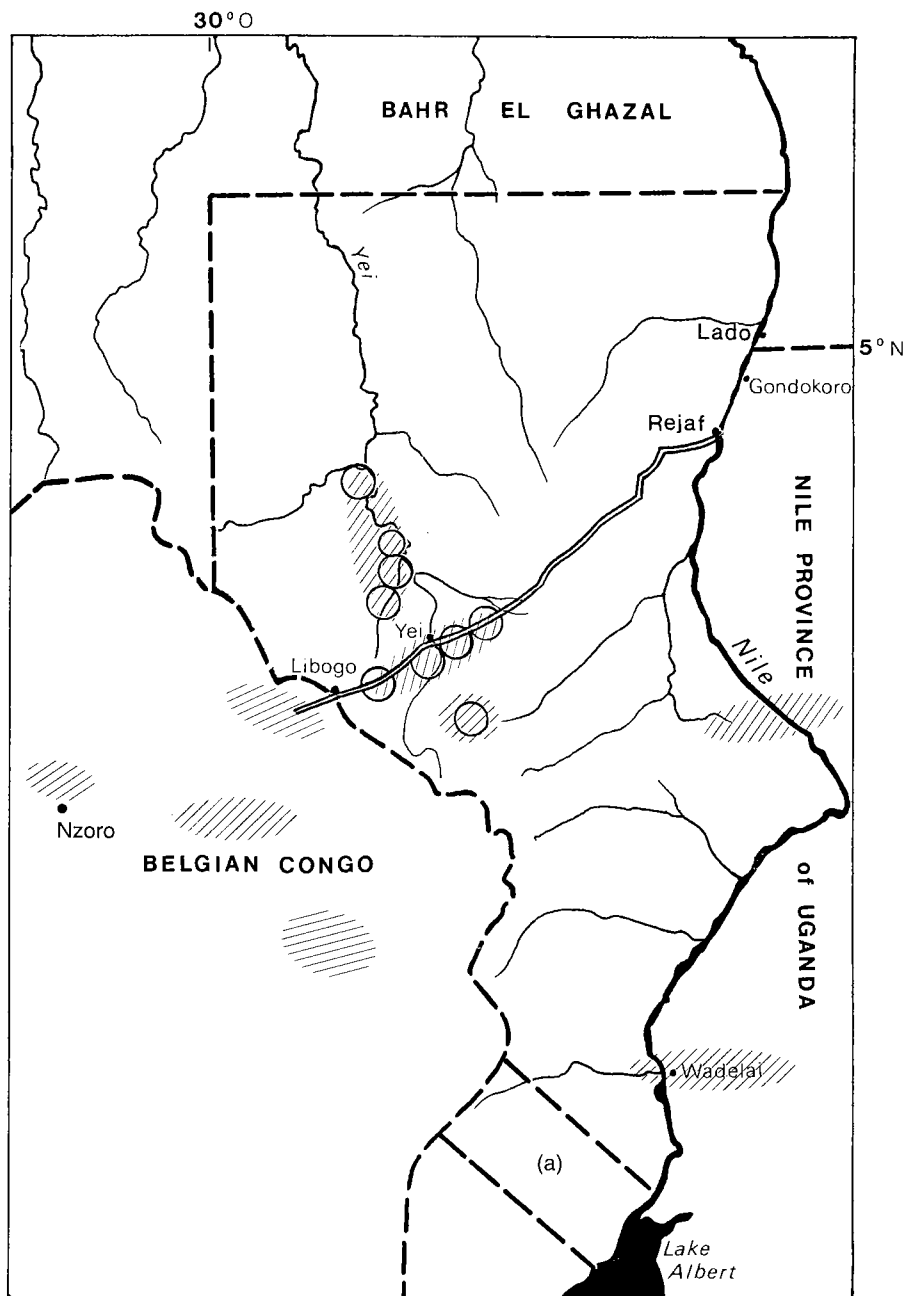
This North-eastern area, close to Ethiopia and Uganda, was an important crossroads to the north and the river Nile. The left bank of the Nile is comparatively flat and inhabited by many different ethnic groups: Madi, Sudanese, farmers, fishermen, hunters akin to the Zande, Makraka (warriors by tradition), peaceful people like the Logo, Mando, Kaliko, etc. Traders' caravans, coming from the north or the east and heading for the Bomu and the Uele, often crossed the territory.

The right bank of the Nile is hilly and extends in the north east towards Ethiopia, and in the south west, to Uganda and to Kenya. Its people are Bari shepherds with patrilinear origins, living on the bank of the river, feeding traditionally on milk and blood. Similar populations extend southwards up to the Masai. The Kakwa, Lutuka and other tribes also belong to the same group.

In this strategic zone, a chain of well-organized trading posts had been set up by Arab, Afro-Arab or Swahili (Wagwana) traders, who had chosen the coast alongside the Indian Ocean as their base of operations. These traders were duly recognized by the sultans and the native chiefs thanks to signed agreements. This system of trading came to be adopted by chartered companies and other organizations set up by the colonial powers. Many of their porters came from Uganda. As a consequence, if a disease present in Uganda could have been caused by pathogens imported from the Congo, the same could have happened in reverse, from Uganda; this fact was proved for sleeping sickness at several observation posts, particularly at Ibe-mbo (Dr Heiberg). Furthermore, some of the Ugandan porters left their caravan to take up permanent residence in the Congo (Zaire); others ran away to go and live in what they considered to be more favourable places such as Busoga. Many of these Ugandan run-aways and settlers were involved in contraband, fraud, and illegal trade, at some time or other.

This north-eastern crossroads had been partly spared; however, as will be shown later, there was most certainly a focus in the Yambio-Tambura-Li Ranga area. A second may have existed also, at Dungu and along the river Kibali as far as Nzoro.

A very noteworthy incident, known as that of the **Lado enclave**, occurred during this turbulent period and was often associated with the spread of sleeping sickness (see map 34, next p.). The Lado enclave was a place where the populations of Sudan, Uganda and Zaire were meeting, because it was an important



- (a) Corridor conceded to Great Britain
 [Shaded area] Areas where, at one time to another, cases of trypanosomiasis were mentioned

Map 34 - The Lado enclave
 (adapted from Mackenzie 1910)

crossing area for caravans. It was also a point where many roads converged, especially the so-called *royal road*, used by motor vehicles from 1910 onwards; it was linking Redjaf to Aba through Yei. The enclave was also connected to a network of navigable waterways on the Uele and the Ubangi-Bomu. After the murder of General Gordon in Khartoum, the British, who had to do a great deal of work to win back the Upper Nile, looked with an approving eye on the campaign of the Congo Free State forces against the Mahdists.

A Franco-Congolese convention forced the Free State to give up its occupation of the eastern part of Bahr-el-Ghazal and to evacuate the right bank of the Bomu. The valley of the Bomu as far as its source (or valley of the Ali) had been acknowledged as being the frontier between the French Congo and the Free State (see p. 1683). This latter arrangement split some important Zandian chieftains whose leaders had signed agreements with the Free State. On the other hand the right to occupy the Lado enclave was acknowledged and also the use of the port of Redjaf (previously Gondok) on the Nile, easy to reach through Yei.

The first large-scale expedition of the Congo Free State to the Nile was organized in 1890 and the Bahr-el-Djebel was reached at Wadelai in 1892. The occupation of the territory proceeded according to a well-tried method, by setting up a succession of posts, particularly the one at Dufile. A second expedition was organized in 1896, based on the simultaneous advance of two columns converging towards Redjaf. The Dhanis column crossed the forest following the Aruwimi and the Ituri, but was blocked by the mutiny of the vanguard, followed by the rebellion of the bulk of the army. The other column, led by Chaltin, reached the Nile at Redjaf in 1897.

This main crossroads in the south of the Sudan has known a period of instability in the late nineteenth and early twentieth century, culminating in the sudden flare-up of active foci of sleeping sickness.

The history of sleeping sickness in the Sudan is a typical example of the way its presence was discovered. The existence of animal trypanosomiasis, which was diagnosed in 1904, led a commission to investigate the possibility of the presence of the human sickness in Bahr-el-Ghazal. Great efforts were made to detect the sickness by a small medical team who carried out on the spot thousands of tests which turned out to be negative. As there were no boundary indications, a vast area was mapped out, having as its northern border a line drawn from Raga to Shambe in Sudan, on the White Nile (see map 33, p. 1694). Raga, in Bahr El Ghazal, was a frontier post close to the French Congo

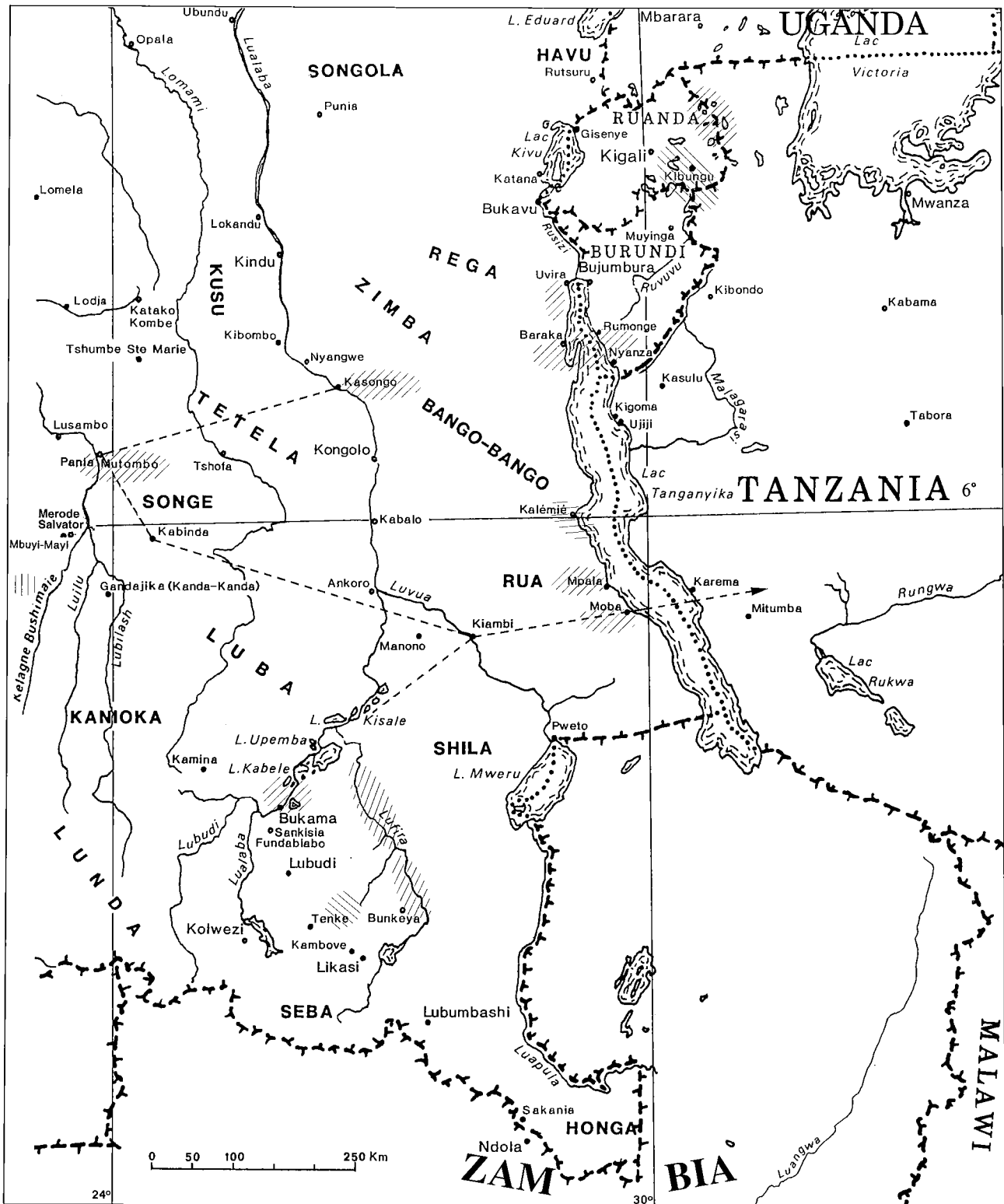
(Central African Republic), where the surveillance post had put an end to a single case of sleeping sickness which could have imported the disease. From such case it may be readily deduced that it would be almost impossible to stop the infiltration of a sickness which had been thriving for ages in its natural focus, which was straddling the political frontiers of Ubangi-Chari, the Congo Free State and Sudan, three regions with no natural boundary. According to *Maillot* (1962), as already mentioned the disease was introduced from Sudan into the Central African Republic in 1897.

Health conditions in this enclave had been carefully studied by medical officers of the Congo Free State. Cases of sleeping sickness had already been diagnosed by *Drs Polidori* (1903) and *Neri* (A.A. 840.3., 3rd July 1906); the latter described his visit to the enclave and to a few villages on the right bank of the Nile near Wadelai. Sleeping sickness had appeared there soon after the raids of Egyptians, Mahdists and Zande, and at the same time as a great famine struck the region; however, it never developed into a real epidemic. It was therefore not surprising that, after the return of the enclave, cases of sleeping sickness were found in Yei.

Four foci were discovered between 1910 and 1920:

1) in 1910, the first focus was positively identified at Yei. Out of 14,000 subjects examined, 238 were found positive. This number increased to 547 within three years, and finally settled at about 30 cases yearly. This first focus seems to have developed in the Lado enclave, where sleeping sickness had already been detected in 1903 by practitioners of the Congo Free State.

The report drawn up by *Dr Mackenzie* (1910), soon after the Sudan had repossessed the enclave on June 10th 1910, clearly explained that he had tried to determine the probable limits of the disease from the information given to him by *Dr Errera*, an Italian physician then in charge of the isolation lazaretto at Aba and previously assigned to the enclave. Mackenzie had first of all confirmed that 15 patients from a village close to Yei suffered from trypanosomiasis, and then delimited a zone near the Belgian Congo frontier. *Thompson* (1911) agreed that this focus was very localized, and that it was the result of travels by Congolese workers and porters of itinerant traders of various nationalities who had evaded the control. He ordered that the brushwood along the river, the contact point with glossinae, be completely cleared, likewise along the isles and in the *Khors* (villages); he also drew attention to the role of the banana plantations. *Mathias* (1911), the President of the Commission for



 Areas where, at one time to another, cases of trypanosomiasis were mentioned

Map 35 – Main rivers and towns or sites of medical importance in south-eastern Zaire and neighbouring countries
(adapted from *Times Atlas of the world*, 1984)

Sleeping sickness, approved the limits of the infested zone and visited the isolation camp with its 120 patients. It was now obvious that the southern limit of the focus had been drawn too high: Wadelai had indeed to be abandoned due to sleeping sickness in 1906 and the administrative quarters moved to Kobe, 30 kilometres further south. This place had in turn to be evacuated in 1910, after sustaining over 1,000 deaths at Gulu, 75 kilometres to the north-east.

Drew (1911), who had examined the situation between Wau and Raga, found two cases in the quarantine quarters of this frontier post. One was an Arab in the final stage of the sickness; the other came from Zemio, the Zande focus, which was an even more alarming event.

In 1914, *Ranken* was supervising the Yei camp, which hosted an average of 383 patients, suffering from sleeping sickness. He drew attention to the fact that the number of admissions fell in 1913, from 140 to 24.

2) in 1914, a second focus was discovered in the Kanjo-Kanji (Sudan): most cases came from 18 villages further south, opposite Nimule. The focus was attributed to an individual imported from Uganda. At that time, the focus ceased to be Sudanese and became Ugandan, because the enclave had just been made part of the west Nile district of Uganda, a fact unknown to the local practitioner. Once again, all this emphasizes how difficult it must have been to keep track of epidemiological happenings, in a climate of constant administrative and geographical changes. *Gray* noted that during his term of office at Nimule, in the course of 1901-1902, he had looked after an old Sudanese soldier in the nervous phase of his illness which, later on, was diagnosed as sleeping sickness.

3) in 1918, a third focus presented the problem of the origin of the Tambura focus. Indeed its origin was not easy to trace, because those involved were Zande people, a mobile population who had taken shelter in the French territory and had been helped to settle there by a contingent of Sudanese troops. These Zande had been exposed to the infection, and thousands of them had also taken refuge in Sudan, when they decided to oppose the rules of the Congo Free State. Some were already suffering from sleeping sickness. In 1919, Syrian medical officers diagnosed together around 621 cases. This first epidemic, which lasted from 1919 to 1929, decided the authorities to build a reception camp at Yuba; out of 3,630 hospitalized cases, 1,661 died, a mortality of 45.6%. The number of new cases dwindled rapidly and, from 1930 onwards, became sporadic. The years 1946-1952 witnessed a new outbreak in the territory, as well as in its neighbouring

state, the Belgian Congo; a total of more than 100 cases were observed.

4) between 1910 and 1920, seven foci of trypanosomiasis were identified within a narrow band at the southern border of the Equatoria province. They developed subreptitiously like all the other foci in the Upper Uele, with only an occasional case to remind one that the trypanosomes were still lurking about. Thus a false sense of security prevailed until the sickness cropped up once more in 1943, in particular at Yei, after a quiescence of 20 years.

There were other outbreaks during the forties and fifties, including the second serious epidemic between the years 1953 and 1961. The civil war, which had already lasted 17 years, had prevented control measures. The third outbreak was discovered by a travelling Norwegian physician coming from Uganda: towards the end of 1971, he detected by thick-drop examination, a number of cases with sleeping sickness near the river Sue (West Equatoria).

In 1972, when a peace treaty was signed, control of the sickness resumed by fits and starts and a volunteer physician (F. Denis), who was located at Li-Rangu, subsequently detected thousands of cases of trypanosomiasis, from which it was deduced that the current epidemic had started around 1971.

The **Southern Sudan** experimented at least two epidemics of sleeping sickness. The first was from 1920 to 1929, and the second from 1953 to 1961, after a lull of more than 20 years, and developing since 1974 to the present day situation, which was very similar to the difficult times present at the beginning of the century. Refugees from several different endemic foci of Zaire and the West Nile area of Uganda, as well as the endless comings and goings to and from south-western Ethiopia, were added difficulties to the problem.

These general details show how difficult it is to assess the distribution of the sickness, and its epidemiological links.

It is now believed that Lake Albert (Mobutu) was struck by sleeping sickness in 1898. However, Greig (1906) did not meet any cases during his time in Uganda in 1903, but in 1904 he did encounter some suspect subjects along the Nile at Wadelai. On that occasion, he found *G. palpalis* as high as Gondokoro (Sudan), which led him to believe that the northern limit of *G. palpalis* should coincide with the progression of the sleeping sickness.

The medical facilities in the neighbourhood of Lake Albert on the Congo (Zaire side) were minimal. In 1905, there was only one physician for the whole of

Ituri district. Unyoro was affected by sleeping sickness in 1904, although its focus did not stretch beyond Nimule. On the west bank, and particularly in Mahagi territory, trypanosomiasis appeared in 1905, imported from Uganda and Lado by soldiers and by porters of trading caravans. The death rate reached a very high level among the Alur and Walendu populations. This focus gradually died out by 1915. In 1927, according to Van Hoof (1928), the degree of endemicity was again rather low. *Van Hoof* noted the presence of *G. palpalis* and *G. pallidipes* in the forest of Kawa, but he observed that the nearby escarpment stopped any possible progression towards the high plateau. Although it cannot be ruled out that the sources may have been Yei and the Zande focus, it seems more likely that the disease started from the Bagandan focus.

5.3.5. *The Busoga epidemic, the region of the Great Lakes, and the Semliki focus in Uganda*

It is difficult to ignore the disastrous sleeping sickness epidemic which swept Uganda from 1900 onwards. It raised the question about its origin and the factors that triggered it off. These questions will be reported further (see pp. 1752-1753), but remain unanswered.

a) The year 1896 marked the start of the Busoga epidemic. The first case was observed by *Jack Cook*, a physician at the mission hospital of Mengo. As the numbers of infected subjects escalated, the medical personnel were compelled to turn the sick away, because of the lack of therapeutic means. The *Rev. Baskerville* noted, in 1900, that the dying were deserted by everyone.

Four cases came from Busoga, a refuge of undesirables, whose continuous uprisings had required repeated military interventions. Busoga was a few hundred kilometres away from the active foci of sleeping sickness in Central Africa, but the year 1899 had been one of severe famine. The location of Busoga, between Lake Victoria and Lake Kyoga in the proximity of the Teso focus, this fact is not without interest, since the latter may have been the source of the epidemic.

The scientific fact-finding commission sent out from London by the Royal Society were not made welcome by the local administrative and medical authorities. During their search for the aetiological agent, the research-workers became interested in *Fil. perstans* (whose role was never proved by their analysis), then in streptococci, which were noted as being present in the terminal phase of sleeping sickness in Angola (*hypnococcus*, see p. 1682) and the Congo (Brodén, 1901). However, their absence in the lymph nodes weakened the assumption that they could be the agent

of the disease. The bacteriologist *Aldo Castellani*, attracted by this aetiology, advocated the production of a vaccine which was never tried out. Finally, on November 12th 1902, *Castellani* discovered a trypanosoma in the Cerebrospinal fluid of a patient.

He waited four months (March 1903) before adopting a definite position, attempting, in the meantime, to produce a vaccine from cocci which were often found in the CSF of dying patients.

C.J. Baker captured the first tsetse flies. He was also the first to discover trypanosomes in the blood of a patient, a policeman from Busoga with a high fever, and later on in the blood of two other patients. This discovery was to prove the existence, in Uganda, of a fever caused by these parasites. Their presence was confirmed by *Castellani*, but no one seemed to link them to sleeping sickness. *Baker* suggested the possibility that the tsetse flies might have transmitted the trypanosomes. This assumption was not original:

Livingstone in his day had already drawn attention to a fly disease, which killed horses and bovines. The fly had also been identified by *Bruce* during his research at Ubombo in Zululand, but he considered that the disease transmission was mechanical.

R. Koch, who had made good use of his retirement by going to Tanganyika, put forward the idea that the cyclical development of trypanosomes in tsetse flies was the most plausible explanation. This was to be proved in 1909 by *Kleine* and *Taute*.

In the meantime, *Hodges* drew a map showing the distribution of cases of sleeping sickness in Busoga and build a laboratory at Jinja. In order to satisfy the commission, a makeshift laboratory was set up at Entebbe under the unfriendly eye of Moffat, the Chief Medical Officer.

Adenopathies were for some time the source of confusion with the plague which was raging in the Wandri Hills (Kenya). Disorientation was not surprising, considering the series of discoveries made since the beginning of the twentieth century: in 1901 *Dutton* identified *T. gambiense* in the blood of a feverish man in Senegambia; in 1902 *Castellani* discovered this trypanosoma in the cerebrospinal fluid of patients infected with sleeping sickness in Uganda; in 1903 *Dutton* and *Todd* established the epidemiology of the disease in the Congo Free State; in 1905 *Thomas* manufactured the first active drug, *Atoxyl*®, in Liverpool.

The origin of sleeping sickness in Uganda

It is generally accepted that the origin of sleeping sickness in Uganda was closely linked to Stanley's expedition, organised in 1888 to rescue Emin Pasha. This expedition, which travelled through the great

Ituri forest in which there was no sleeping sickness, took him to Lake Albert and to the great western depression. The difficulties entailed in the journey were thought to have inevitably triggered off the sickness in the carriers of trypanosomes and thus lead them to their death.

This presumption ignores the observations and facts reported by *Cutberth Christy* (1903) a member of the Commission sent out by the Royal Society at the instigation of the Foreign Office, to investigate the severe epidemic of sleeping sickness in Uganda on the Victoria-Nyanza (lake) and in progress in the Province of Busoga. He made an epidemiological reconnaissance of the affected region and proved to be an outstanding naturalist and field-man.

He underlined the peculiar distribution of the disease, confined to the margins of the lake and all the islands but without having a tendency to spread inland, neither along any of its rivers nor along caravan routes. Christy was of the opinion that the sickness had been endemic in some parts of the Busoga Province for some years before it was first recognized. The cattle grazing around were free from the disease but in the wooded regions the inhabitants were affected to the extent of between 30 or 40 per cent.

Busoga is separated from the Kavirondo region by a strip of uninhabited land before the road enters into a vast cool forest of magnificent trees, matted creepers, and tangled undergrowth of tropical luxuriance, where forest and banana groves take the place of the rolling downs and fields of millet of Kavirondo. J. Ford will later on underline the importance of this scenery (see p. 1753). Christy found the majority of the banana plantations untenanted and signs whereby the once large population had disappeared. Along his route he came across sickness and death. He marched into Bukoba, in the German East Africa, saw the snow range of the *Mountains of the Moon*, the smoking volcano beyond Lake Kivu, proceeded to the lakes Albert, Edward and Nyanza, the Unioro, the Murchinson falls, then along the Nile to Gondokoro, continued by steamer to Fachoda and Khartum and by rail to Cairo. By that time he had learned that the trypanosome had been identified as the pathogen.

Christy underlined that the way and manner by which the disease first reached the country was a matter of conjecture. He suggested that its introduction may have been made possible when the representative of the British East African Company, Captain (later General Sir) Francis Lugard, decided that he needed an armed force at his command in order to maintain peace between the Protestant, Catholic and Mohammedan communities, which were not made up of Ugandan

natives. As it was impossible to recruit trained soldiers from the coast (a four-months journey), he decided to use the remnants of Emin Pasha's army. They had established themselves at Kavali, West of the Albert Nyanza after the relief of Emin by Stanley.

He succeeded in recruiting 400 or 500 of these Sudanese soldiers, and brought them, with a rabble of 7,000 wives, followers and children, into Toro, where he established them in the newly created Kingdom over which he placed Kasagama, their King. Owing to their lawless conduct, they ultimately marched further eastward, in 1892, through the province of Uganda, and settled in Busoga, where they found themselves subject to a better control. These Sudanese recruited by Emin, from all parts of the Egyptian and Western Sudan, now made up a section of the armed forces of the Uganda Protectorate. From 1892 to 1896 more Sudanese with their followers were brought through to the Victoria Nyanza (lake), and four years afterwards sleeping sickness first appeared in Busoga, where the majority of them had been quartered. To this day, however, the Sudanese themselves are free from the disease, so that supposingly it had originated in this manner, it must have been brought by their slaves and followers.

These facts give a hint to another origin than that of Stanley's soldiers. These assumptions have ignored furthermore the constant traffic of people between the Bomu, the Nile, Lake Kyoga and the coast.

The focus in Uganda was due to *T. gambiense*, transmitted by the hygrophile glossinae, *G. palpalis*, *G. fuscipes*, etc. As such, this focus was similar to the endemic situations in southern Sudan, the Central African Republic and north-eastern Zaire. Nevertheless a *T. gambiense* trypanosomiasis, which was believed to have caused some 200,000 deaths in a population of 300,000 in such a short period of time, it is most unusual and raises many questions.

b) The region of Lake Edward, nowadays known as Rutanzinge, had not yet, in 1906, been attached to a political power, and was still without any administrative body. Its trypanosomiasis focus became a great worry to both Congolese and Bagandan authorities. The entomologist *Van Someren* had observed a great number of cases at Katwe (Uganda), where saltworks were operating commercially, extracting salt by evaporation from Lake Edward and the smaller neighbouring lakes. The post attracted many traders, whose Congolese and Bagandan porters often carried not only salt but also trypanosomes. In 1907, a Koro porter who had been infected at Katwe was identified; this observation led to the identification at Mbarara among the porters of caravans of numerous infected cases.

c) The **Semliki focus** is important in several ways. The Semliki river links Lake Albert (Mobutu) and Lake Edward (Rutanzinge), which are 120 kilometres apart. The river valley straddles the Equator, but at an altitude of 910 metres; and the river flows between the Mitumba Mountains and the Ruwenzori massif, on a gravel bed. The northern section of the Semliki, with neither humans nor glossinae in sight, is a paradise for hippopotamuses and crocodiles. In its southern section, its banks are wooded and rich in glossinae. Torrential rivers and numerous streams flow into its steep, wooded ravines, rich in *G. palpalis*. The valley is fertile and many tracks run through it, used by caravans to transport salt from Katwe. The region is also a rich hunting ground and fishing area (see map 33, p. 1694).

The District Commissioner, *Henri-Martin Hackars*, had been responsible for these territories since 1918. When nominated to take charge of the Park Albert (Virunga) he partially relinquished his administrative charges in 1938.

Inspired by Hackars' experience, Van Hoof (1926) describes this historic focus. On the basis of information he had gathered from notables of different ethnic origins, Hackars was ready to proclaim that sleeping sickness had existed in the region before the arrival of the white man. Attracted by the snows of the Ruwenzori, travellers who hoped to settle in the Semliki plain walked to their death. So, two centuries ago, did the Mabudu, who settled in Nepoko, when crossing the plain on their way. The Babira, who live in the Irumu-Bunia plain, affirm that they lost more people during their brief stop in the valley than in all the battles fought to gain passage during their last migration.

The region was occupied by the Congo Free State in 1896. The civil servant in charge was surprised to see that the largest part of the population lived in the Mitumba foothills, safe from glossinae, and that on the contrary, the fertile Semliki plain had been left to a very sparse population. He moved the people from the mountains to the plain; from his point of view this was logical. But the transfer caused a fearsome epidemic: in the valley of Kavirondo, whole villages disappeared, and in one particular village, 27 inhabitants out of 175 died. Canoeists from Lake Edward and the Semliki river lost five or six of their men yearly. The use of *Atoxyl*® at the isolation lazaret in Vieux-Beni (the prime localization) brought no relief.

In the course of 1923-1924, it was decided to evacuate the plain, and in preparation the cultivation of crops was started higher up, on the plateau of Bugula, 300 metres higher and vector-free. This move proved to be a lengthy but rewarding undertaking, because in the plain the prevalence of the disease and the mortality

rate were high. Traders remained in Vieux-Beni and Kasindi; people continued coming and going to and from the plain, with its fields and banana plantations, and where game was in great supply and fish in abundance in the river. The Wanande and the Bahema are related to the people of Unyoro, Ankole and Toro, where their hereditary chiefs lived on the other side of the frontier, out of reach of the administration of the Belgian Congo.

Only the Bahema shepherds stayed behind. The drug *Bayer 205*® and the shift of the population overcame the epidemic.

5.3.6. *The Rusizi focus*

The Rusizi river, after careering downhill for a considerable distance links Lake Kivu (already declared in 1908 by *Kleine* and *Breuer* to be free of glossinae) to lake Tanganyika. The northern section of the river flows through Bugarama in Rwanda (see map 35 p. 1698), but the greater part of its course flows through the frontier valley between Zaire and Burundi. *Breuer* (1909) found some *G. palpalis* on the eastern bank of the river at each ford, while for the most part the river remains impassable. The limit of the endemicity was to be found, according to *Marshall* (1909), at the military post of Tschiwitoka (now Cibitoke), half-way between the lakes, where the man in charge, a German *unter-offizier*, had contracted trypanosomiasis and where some of the population had become infected. Where the Rusizi flows into the lake, seven out of the 473 persons examined were carrying trypanosomes. *Breuer* (1909) reported the presence of hordes of glossinae in the banana plantations on the banks of the Rusizi's tributaries. At Usumbura and in the outskirts of the town, no glossinae were to be found.

As to the foci along the Rusizi and the eastern shore of Lake Tanganyika (Burundi had been a Belgian protectorate since the end of the 1914-1918 war), FOREAMI was asked in 1932 to detect and to treat the patients infected with sleeping sickness (see pp. 126 and 1779). *Dr V. Prati* (1933) was put in charge of the operation. His full-time appointment, together with FOREAMI's strategy, enabled him after several prospections and systematic treatment of the patients, to collate figures on the infection for the years 1932 and 1933. The infection rates were as follows: from 6% to 2.5% for the Bugarama; at Usumbura, which had a sizeable floating and foreign population, from 1.6% to 1.2%; at Rumonge, from 1.2% to 0.3%; and at Nyanza from 1.1% to 0.3%. It was moreover decided to relocate clusters of well-maintained dwellings along the Cyangugu-Uvira road, and at the same time to undertake a limited clearance of brushwood at fords

and along essential pathways, at canoe moorings and at water holes. Attempts to capture glossinae with traps proved disappointing, probably because their type was not suitable. In 1939, the INC (Index of New Cases, *Indice de Contamination Nouvelle*) was 0.15%. In 1942 a recrudescence was noted, and in 1946 the INC reached a figure of 1%. Treatment with pentadimine brought down the indices: between 1960 and 1965 only seven cases were observed and the focus has been extinct since then.

It is worth stressing that it was at this focus that *Leleup* (1952), drew attention to the banana plantations as a habitat for *G. fuscipes martinii*, especially at Bujumbura. While discussing this report, *Rodhain* pointed out in 1947, that while passing through Bujumbura, he discovered a vast banana plantation harbouring many *G. palpalis*; this plantation had to be destroyed. Even today, *G. fuscipes martinii* are still to be found in the banana plantations of Bujumbura (*Coosemans*, personal communication).

5.3.7. The foci of lake Tanganyika and Katanga (Shaba)

a. The Lake Tanganyika focus

The sleeping-sickness focus which developed around Lake Tanganyika requires a good understanding of the geographical position of the lake.

Lake Tanganyika is located in the same depression as Lake Albert (Mobutu) and Lake Edward (Rutanzinge), but it belongs to the Congo basin.

Topped by Bujumbura, the capital of Burundi, the eastern shore of the lake lies in Burundi and in Tanzania, formerly German East Africa. Its western shore belongs to Zaire, (which in 1909 was still called the Congo Free State) from the Uvira territory, and in its upper section embraces Fizi and Kabambare. This western shore waters the North of Katanga (now Shaba), along Moba and Kalemie. In this province, the depression proceeds to Lake Moero (Mweru). The northern territories are named Lower Katanga, in contrast with the Upper Katanga mining area. The whole area borders to Zambia (formerly Northern Rhodesia) and Angola. The administrative situation has been rather special in this region, once the *Comité Spécial du Katanga* was set up (see below, p. 1705).

In the middle zone of its western shore, Lake Tanganyika became the scene of a deadly epidemic in the early days of 1900. The infection came from the west and was probably imported by infected individuals. The first case was detected in 1896, at Pania-Mutombo on the Sankuru river in the heart of Zaire, in a labourer who had been recruited in 1894 for large scale works in Lower Congo. The number of sick people escalated rapidly along the caravan trails to

Kasongo, where the first cases were diagnosed in 1901. From the Lualaba, the infection advanced eastward, along the caravan trails. The banks of the Elila river escaped the infection.

According to the official reports of 1905, the shores of the lake were infected. The infection may even have occurred earlier, since in 1904 *Dutton and Todd* heard, in Leopoldville, that a sleeping sickness epidemic was raging in the region of the lake. They would both have liked to observe the progression of the disease, but they were headed for Kasongo, where they were able to witness a similar spread of the epidemic. *Marshall* (1909) mentioned the occurrence of the sickness in Tanganyika in 1901.

b. The foci of Lower Katanga (Northern part)

For the section of Lower Katanga (presently the northern part of Shaba) adjacent to the mining area, recorded information on the start and propagation of sleeping sickness is scanty or hard to obtain.

Data on sleeping sickness were scarce, although Lower Katanga was to become a major disaster area, particularly in zones bordering Lake Tanganyika and the Lukuga and Lualaba rivers. The lack of data was the result of a particular administrative system signed between the Congo Free State and the Company of Katanga.

The report from the Dutton-Todd mission had shown that sleeping sickness was detected at Kabinda in 1904. This area was remote, but represented an important staging post for caravans carrying salt and dried fish from Kiambi and the lakes.

The posts (Kalemie, Mpala, Moba) on the shores of Lake Tanganyika, and also the territories of Lower Katanga, were under the authority of the officials in Elisabethville (Lubumbashi), but without much being their own concern.

However, sleeping sickness caused real havoc on the western shore of the lake. This tragedy was described by Mgr. Roelens. *Dr Polidori*, in charge of the Urua, was to bring help and treatment with *Atoxyl*®, trypan-rot, and with other arsenical compounds.

Dr Schwetz related his experiences in 1925 as director of the isolation lazaret of Baudouinvill (Moba) in his publication of 1946. He realized that lazarets were inadequate to control sleeping sickness. The detection of the disease in villages encountered great difficulties: a single examination was insufficient, so that only a few hundred sick people could be treated, when there were thousands who needed care. Moreover, a course of 10 injections of *Atoxyl*® over a period of 10 weeks was not always effective. With the help of Fr. *J. Kindt*, he ordered to clear the vegetation on the banks of the Lufuku and the shores of the lake, an area

of about 20 hectares around the Mpala Mission; the bullrushes had roots up to two metres deep.

The administrative measures recommended by circulars the relocation of villages to higher areas or the prohibition of fishing, but all went unheeded. In other regions, like Lomami, such measures simply could not be applied. Health department teams, formed to ensure the clearing of bushes and scrubland round the villages, often became maintenance teams for the administration posts.

In his memoirs, Schwetz (1946) described the Belgian shore of the lake as having scattered villages at water's edge or perched on hill slopes. In fact, these villages were simply hamlets, with huts spread in the banana plantations, often close to the tributaries flowing into the lake. Contact with the glossinae was direct and permanent and all the efforts made to destroy their habitat were ineffective.

The epidemic on the Belgian shore of the lake led to the belief that a similar situation was impending on the German eastern shore. The most critical point was at the level of the Ubwari peninsula, which forms Burton Bay. At that point the distance by boat across the lake to Burundi is very short. The infection of the German shore, thought probable, became a certainty in 1906 at Urumbi and Kiguena, south of Rumonge, where *G. palpalis* were found. The Urumbi focus came to be infected at 60% to 75%, the Kiguena focus at 90%. Ujiji was hit and also the banks of Malagarasi river. In the Rushuggi focus out of 200 examined individuals 14 were found infected (7%). In the Kirundu focus, the rate was 15 out of 200 (7.5%). When a caravan at Usumbura was checked, 10 out of 16 porters were found to be positive, a rate of 62%.

How serious the 1911 sleeping sickness epidemic by *T. gambiense* may have been, the figure of 20,000 deaths put forward by the State official in residence, Hans Meyer, seems grossly exaggerated. In any case, such an assessment, without any population census, is most unreliable, to say the least.

Marshall's remarks (1909) can not be ignored on the futility to control the epidemic on the German shore, when nothing was done on the Belgian side. To realize how significant Marshall's comments were, one should only compare the medical provisions on both sides: in Burundi, there were five German Physicians and eight *Sanitäts-unteroffiziere* (medical orderlies), while on the Belgian shore there was only one physician, at Rutshuru.

Schwetz had put his stay to good use by studying the methods applied by neighbouring colonies. As the British shore of the lake was sparsely populated, especially in the extreme south, relocation of villages from

the lake shore to a place free of glossinae presented no great problem. The German shore, from Ujumbura to Ujiji, was very densely populated and the higher land, free of glossinae, was at a great distance. The people in charge then decided simply to relocate clusters of villages and to carry on with clearance of brushwood. It should also be wise to add that the German physicians were convinced that the caravans played an important part in the spreading of sleeping sickness; they also thought it unfortunate that no preventive measure had been taken in the territories of the Belgian Congo which bordered that part of the lake.

c. *The Upper Katanga foci (southern part of Shaba)*

Four expeditions were launched, almost simultaneously, in the southern part of Katanga (now Shaba): only one was organized by the Congo Free State while the other three were commissioned by the Company of Katanga. These late attempts had been enforced by the British South Africa Company who, under the direction of Cecil Rhodes, had been trying since 1889 to take possession of the whole mining basin. This undertaking could have been facilitated by the pro-British feelings of Mwami Mwende Mushidi (Msiri to Europeans), founder and absolute ruler of the Muyeke Empire (Garenganze). The occupation of Upper-Katanga followed different attempts and expeditions:

- *Paul le Marinel*, organizer of the fortified camp of Lusambo, was sent on a mission to Katanga in 1889. He surveyed the rivers Lubi, Bushimaie, Luilu, Lubilash (Upper Sankuru) and Lubudi (tributary of the Lualaba); but he failed in his mission when he visited Msiri in Bunkeya. He then ordered an observation post to be built on the Lofoi, tributary of the Lufira, at a place pointed out by Msiri, and then returned to Lusambo.

- An expedition was undertaken in 1891 by the Company of Katanga, a chartered company and subsidiary of the CCCI (Congo Company of Commerce and Industry). It was headed by *W. Stairs*, a British explorer who had been with Stanley in the Emin Pacha Relief expedition and who made, at the time, the first ascent of the Ruwenzori to an altitude of 3,260 metres. The mission reached Bunkeya from the east and erected Fort Boden. Stairs died of haemoglobinuria (at the moment the *Bia-Francqui* expedition, from the same Company of Katanga, arrived at Bunkeya).

- The mission headed by *Alexandre Delcommune* (who in 1886 had joined the CCCI and from there was seconded to the Company of Katanga) had *Dr Briart* among its members. He drew attention to the danger of glossinae. The expedition reached Bunkeya in 1891, where it met up with the *Bia* expedition in 1892.

The mission went on to explore the regions of Kundelungu and Marungu before returning to Lusambo.

- The *Bia-Francqui expedition* included a geologist, J. Cornet, and was also heading for Bunkeya through Upper Lomami and Lufira. En route it encountered the string of lakes of the Lualaba river, lakes Kisale, Upemba and Kabele. Cornet discovered enormous deposits of copper ore and the mission negotiated the submission of many local chiefs; it also managed to fix the boundaries of this still unoccupied region.

A newly appointed Committee, named the *Comité Spécial du Katanga*, CSK (Katanga Special Committee), was empowered to exercise the authority of the State and the Company of Katanga, as well as to ensure the follow-up of statutory health measures. This authority on behalf of the State was not withdrawn until 1910. Documents are not available for the period ruled by the CSK.

The *Comité Spécial du Katanga* was not particularly interested in the area adjoining Lake Tanganyika and the Lukuga river. Their attitude changed when Dr Yale Massey, company physician to the Tanganyika Concessions Limited, a sub-contracting company of the *Union Minière du Haut-Katanga*, UMHK (Mining Union of Upper Katanga), sounded the alarm. In the course of 1905-1906, he pointed out that the death rate due to sleeping sickness among the riverside dwellers of the Lualaba, was very high upstream towards Bukama and at Kayembe on the Lufira. On a journey to Tengelouzi, a Lualaba riverside village on the bank opposite Bunkeya, he observed that 80% of the population were carriers of cervical lymph nodes. From then onwards, physicians of the CSK and of the UMHK (*Union Minière du Haut-Katanga*), started to deal with this threat. Also the administration started to be interested because of the anxiety and comments expressed by the Rhodesian authorities and, in a lower key, by those of German East Africa.

In 1907, Dr S. Neave explored the region and, with the help of his son, an entomologist, he drew the first map showing the distribution of the glossinae along the Lualaba towards Kongolo and Fundabiabo, as well as along the tributaries Fungwe, Kalule and Lubudi. Dr Pearson, the first physician of the *Union Minière du Haut-Katanga*, studied the upper course of the river Dikulwe, tributary of the Lufira, north of Kambove. Dr Stohr, in charge of the Bunkeya isolation lazaret explored the shores of the Lufira and the Fungwe. Dr Goessens established that *G. palpalis* progressed along the main watercourses (the rivers Lualaba, Lufira, Luapula, Luvua), but that no tsetse flies were to be found above an altitude of 1,200 metres.

Alarmed by the observations made by Dr Massey, the CSK decided to take steps in response to complaints

made by the authorities of Northern Rhodesia, who feared that the link of the Elisabethville to the Cape railway might have to be postponed; they also feared the potential risks that the building of a railway towards Bukuma and the extension of the navigation on the river between Bukuma and Kongolo might present.

For the year 1908, Pearson (1909) summarized the results of the relentless control of trypanosomiasis by the medical officers of the UMHK. In first instance, with the cooperation of the Employment Office, every African was thoroughly examined by a physician, when taken on or dismissed, and was also submitted to a quarterly check-up. This plan was not easy to implement, because the mine workers did not like it. These medical examinations would not only reduce the risk for trypanosomiasis to spread, but also help to record its geographical distribution.

Cases of sleeping sickness among the workmen of the enterprise were rare: Kambove 28 out of 517, or 0.3%; Ruwe 2 out of 1,476, or 0.15%; Etoile 6 out of 1,783, or 0.34%. In the supply centre of Kapiri, outside the copper mining zone, in the Dikulwe valley and the neighbouring region, the situation was different: 274 out of 3,540, or 7.7% of infected subjects.

An expedition sent to the neighbouring Portuguese territory did not find any infected subjects among 1,519 natives, nor any glossinae. In the Mutendele basin, no infection was found. *G. palpalis* was present neither along the Lufira, downstream of the Kaparowe, nor along the Kafubu.

The conclusions drawn were that the disease could only have come from the north, with the limit of its distribution corresponding to that of *G. palpalis*, and that *G. morsitans* was not a vector of the disease in Upper Katanga.

In October 1909, Dr Polidori of CSK, Dr Pearson and Dr Goebel of UMHK, held a meeting to discuss a common plan of action against sleeping sickness: each of their zones was mapped out, zones with *G. palpalis* were defined, caravan trails were identified and ways and means to examine porters were determined. The health measures to be adopted were based on the recommendations of Brussels and Leopoldville: distancing the villages from glossina areas and provision for building isolation lazarets. Moreover, it was decided to open a joint office, to collate all the information received and to publish the figures.

In 1909 and 1910, meetings were held with responsible German and British authorities from the bordering countries. The lack of action on the Belgian side was once more clearly underlined. The measures devised to restrict the movements of carriers of trypanosomes and to prevent contacts between men and glossinae were not realistic and even harmful: to ban

fishing, confiscate and destroy canoes, move villages away from the shores of Lake Moero and from the rivers, where the people had been settled for generations. The German measures were more realistic. One of these allowed fishing at night. A similar proposal had been made by the White Fr. of Mpala: "Go fishing when the glossinae are asleep". But it was also decided to forbid fishing at night with a gaff, a method used on the river banks. Moving villages to hilly heights was too distant and not considered.

In July 1910 a conference was arranged between N.E. Rhodesia and Katanga, to discuss problems concerning Lake Moero and the frontier river Luapula and was attended by L. P. Beaufort, acting Administrator in N.E. Rhodesia, Administrator Gheur, chief of the Luapula zone and on the medical side by Drs Polidori, Storrs, Master and May. It was decided that access to the river banks would be limited to specific spots where effective measures against the glossinae had been taken, that frontier crossings would be limited to two places, Pweto (Lake Moero) and Kasompi (on the Luapula), both to be kept under medical supervision, and that the sick should be sent back home. The trypanosomes of the Luapula area were said not to be very virulent, and *G. palpalis* had been detected near Kambove.

In this general context, when the Belgian Congo took its responsibilities for the implementation of good health conditions, the **Scientific Mission of Katanga** made up by Rodhain, Pons, Van den Branden and Bequaert was active from October 1910 to the end of September 1912, to determine how far south, in this region of abundant waterways, the trypanosomiasis had extended towards the mining zone of Upper Katanga.

Parasitological examinations proved that the Bukama region, abandoned by the *Comité Spécial du Katanga*, had been badly hit, and that the wooded shores were swarming with glossinae. *Raynaud* of the medical services had estimated that 10% of the people living in Kongolo and Ankoro were infected, and he also noted that the population in both places was declining. *G. palpalis* was to be found along the river banks and *G. morsitans* lived in abundance in the plain.

The Rodhain mission had settled at Shankisia, along the Bukama-Fundabiabo road, close to the Bukama-Kambove railway.

The significant increase in the number of cases and the infection rate, was noted respectively by *Neave* (1906), *Pearson* (1908) and *Goessens* (1910), in villages established close to three rivers: close to the Dikulwe the four cases reported in 1906, increased to 5.7% and 37.6% of the population; along the Kabanga, cases increased from zero to 2.1% and then

to 4.7%; around the Lufira, infected cases numbered 13.1% and thereafter 30%.

The distribution of the glossinae was uneven and there may even have been migrations of *G. palpalis*. None were ever found at an altitude above 1,140 metres. *Goebel* (1911) raised this limit to 1,400-1,500 metres.

On the other hand, the ubiquitous *G. morsitans* geared the expedition's main work to the possibility of a transmission of *T. gambiense* by this type of glossina. Possible with laboratory bred *G. morsitans* and may be occasional in natural environment, the possibility remained without practical consequences. Indeed, *G. palpalis* uses man as its main source of blood, whereas *G. morsitans* prefers to feed on animals.

In the Fungwe valley with glossinae present throughout the year, the rates of infection rose from 7.4% to 15%; the Mwanza basin, where glossinae were scarce, showed similar rates (6.8% to 15%). In swampy lakes with papyrus along the shore, there were no glossinae, but in waterside villages, surrounded with eleis palm trees, trypanosomiasis was responsible for a great number of deaths.

The real danger was thought to be the introduction of *T. rhodesiense* in these regions; however, this was an unlikely occurrence because of its acute clinical evolution. One suspected case was observed by *Dr Conzemius* at Sakania, but it turned out to be an isolated case.

In 1912, *Stohr* covered the valley of the Dikulwe, a tributary of the Lufira, and was of the opinion that sleeping sickness was occurring there for quite some time. In the Bunkeya isolation lazaret 81 patients had been registered: 30 had died, 20 had escaped and the 31 people left continued to receive treatment. "The isolation lazaret is a charitable, expensive and useless refuge", said *Stohr*. He worked out an experimental treatment for the villagers of the Dikulwe valley, which proved very encouraging.

He drew attention to the importance of indigenous commerce: traders came from Uganda for copper, from the east for ivory and slaves. Salt, fish and palm oil were highly valued and conveyed by caravans from Kiambi to Kabinda, and from there towards Sankuru and Kasai, or from Kiambi to Tanganyika through the southern route (see the map p. 1698). Caravans from Angola exchanged firearms and powder for rubber and ivory, and in earlier years for slaves too. In Katanga, the importance of the river traffic should not be underestimated.

The copperbelt was said to be free of *G. palpalis*, but according to *Stohr* this fact was worth checking out during the rainy season, when the glossinae tend

to appear in greater numbers. Unless they are boggy, river banks below 1,200 metres tend to be infested. In places with only *G. morsitans*, there were no cases of sleeping sickness.

Stohr noted that, since 1907, the Belgian authorities had allowed sleeping sickness to spread, without bothering about it. Theoretically, 16 physicians had been appointed, of whom 12 were to deal with sleeping sickness, but in fact there were only six (one Belgian and five Italians), and only four were assigned to the sleeping sickness problem.

In 1911, specific campaigns were organized; and it was in this region, long neglected, that new strategic approaches to control sleeping sickness were first undertaken. The change in strategy, not any longer based on isolation lazarets, was undoubtedly the achievement of *Dr E. Lejeune*. He was the first to find a practical solution to the problem by visiting the villages, carrying out a systematic census of the population and examining every individual living in the area; moreover he advocated a methodical treatment in local centres to be given by auxiliary nurses, specially trained for the job. This endeavour was interrupted by mobilization at the outbreak of the 1914-1918 war, but in 1919 the scheme was resumed with success, in 54 villages in Bulungu, and became the source of the health system implemented by FOREAMI (see the chapter Evolution of Health Services, p. 112).

In these villages, the rate of infection in 1918 was 6.4% (682 out of 10,592). In 1921, it was 2.4% (305 out of 12,615); and in 1922-23 the figure was 2.15% (299 out of 13,883). In 1927, Mouchet reported the rate in Kiambi as 1%, and communicated other figures: for Upper Luapula district, Bukama: 0.53%; Kisale: 0.38%; Kibara (random sampling): 1.1%; Lukafu valley: 1.92% (72 out of 3,749); North Lufira: 0.53% and South Lufira: 0.65% to 1.15%; Tanganyika-Moero district: 2.5% (1,339 out of 47,225); Baudouinville (Moba): 0.95%; Albertville (Kalemie): 1.6%; Ankoro: 2.5%; Kabalo: 2.2%; Kongolo: 5% (for the region of South Niunzu, the percentage varied from 6% to 7%, with rates reaching 25.9% and even 39.4% in some villages). In Bukama district, out of 25,000 people, 150 infected individuals were receiving treatment.

Reading these figures, one cannot but wonder why the people in charge of this province, prosperous both economically and financially, did not set out to discover, at least in a pilot sector, the natural epidemiological history of the spread of human trypanosomiasis. Research which could have been carried out in a particular section of the province might have yielded the possible determinants of the variations in time and space which, hitherto, had puzzled everyone. Investi-

gation of the rates of infection carried out by the Rodhain mission, and those of a group of physicians in Katanga, were completed in 1913. This helped in the practical organization of the treatment of infected persons but did not add to the understanding of the why and wherefore of the random distribution of trypanosomiasis, of the movements of healthy carriers of parasites, nor of the conditions producing the link between glossinae and infective parasites, nor even of the non-propagation of the disease in some river zones, etc.

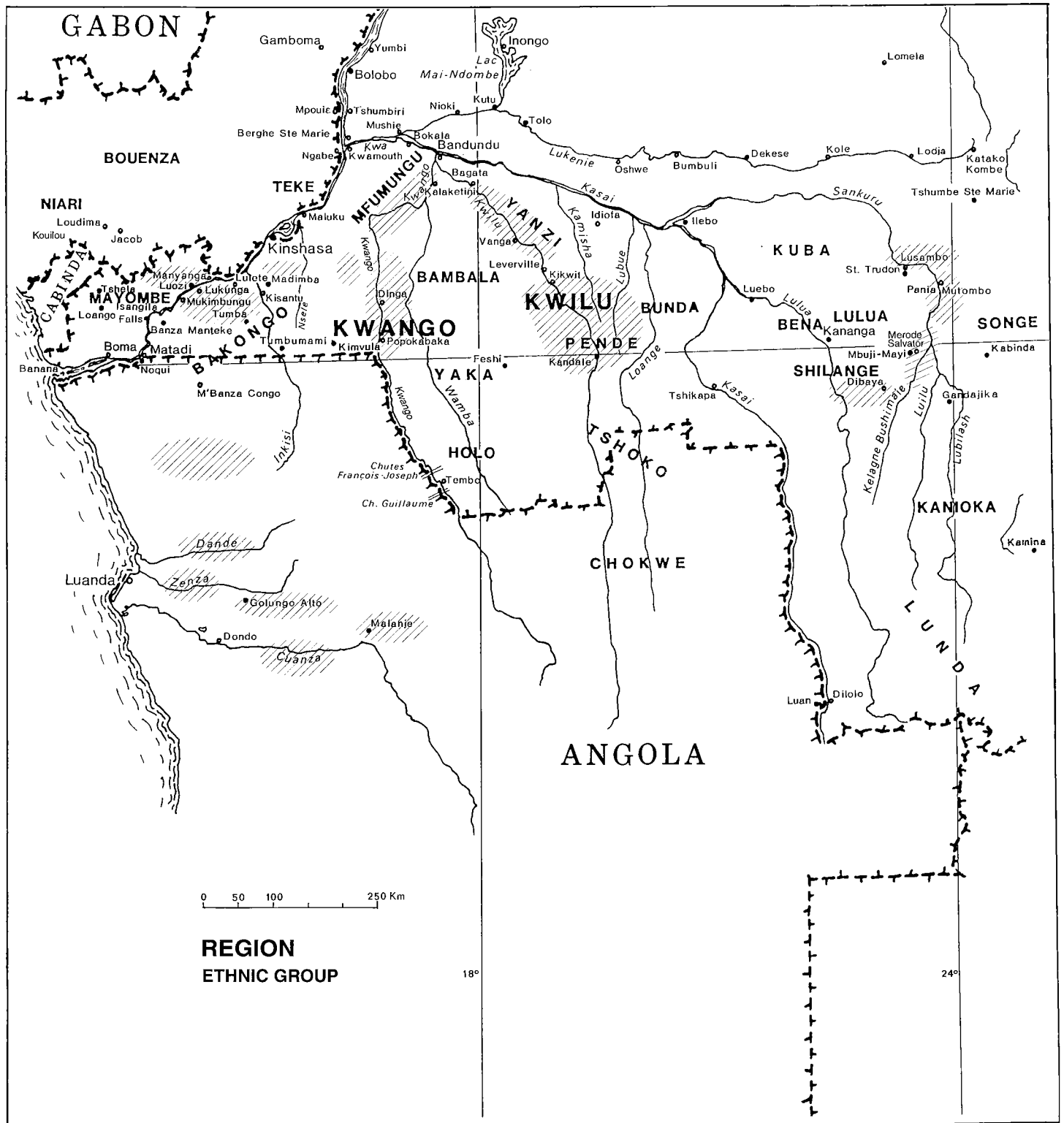
5.3.8. *Foci in the Kasai, Sankuru, and Lake Mai Ndombe*

Until the uprising of the Batetelas (1897-1899), the Kasai-Sankuru had been the route for illegal trade and official transport of goods from Malebo Pool (Stanley Pool) to the Lualaba river and to Tanganyika. This route was then moved towards Kisangani and further upstream on the Lualaba towards Kasongo: in this way, sleeping sickness was able to proceed along new paths.

The Kasai, the Sankuru and other tributaries were explored by *H. Von Wissmann*. The first expedition set off in 1881, and with it was *Dr Paul Pogge*, a lawyer and naturalist, who had lived with the Bashilange for two years. The group landed at St Paolo de Luanda, assembled at Dondo, then was joined by traders at Malanje (see map 36, p. 1708), from where it headed for Kimbundu and along the Upper Tshikapa, a tributary of the Kasai, in the Lunda kingdom. They followed the Kasai valley upstream, crossed the Bashilange area, headed north-east, crossed the Sankuru, then the Lomami, to reach Nyangwe. Pogge turned back towards the Atlantic coast and Wissmann headed for Lake Tanganyika, crossed it near Ujiji and reached the east coast.

Wissmann hoped to return to Kasai, but the *Deutsche Geographische Gesellschaft* (German Geographical Society) was unable to finance his new journey; however, King Leopold II subsidized his venture. Wissmann conducted his assignment most conscientiously, and planted the flag of the Congo Free State at every opportunity. In his reports to the King, he never mentioned sleeping sickness. On the contrary, he described the populations as being numerous and vigorous.

An occasional commercial circuit of clandestine traders from Angola was already in place but such exchanges were of little importance compared to the human trade and cargo handled by steamers, barges and caravans brought into use by the Congo Free State. For the important infrastructural works and for building the railway in Lower Congo large contingents of manpower were brought in by these routes.



Areas where, at one time or another, cases of trypanosomiasis were mentioned.

Map 36 – Main rivers and towns or sites of medical importance in south-western Zaire and neighbouring countries.
(adapted from *Times Atlas of the World*, 1984)

Soldiers were dispatched all over the country. Inevitably, some of them might have caught sleeping sickness, either at the building sites or camps or during their journeys to and from their homes, for example when they crossed the Kwamouth focus. It was therefore no surprise to anyone when, at Pania Mutombo one of the crossroad centres, as already mentioned the first patient with sleeping sickness was identified by name. It was also normal that the population would identify the typical cases of lethargy, the terminal stage of the sickness, while those who spread the disease were the far more numerous patients with fever or the apparently healthy carriers. The transmission chain had hereby been established.

The situation at Pania Mutombo could evidently be repeated at other crossroads, like Kabinda and Lusambo, and this explains the tragedy at the St Trudo Mission.

As at Pania Mutombo, sleeping sickness appeared in 1899 in its terminal stage at the St Trudo mission, which had just been opened at Muteba, on the Lubi river, 15 kilometres from Lusambo. Its first case was a slave who had been freed and had come from Pania-Mutombo.

In 1902, cases multiplied rapidly, and by 1903 villages on the river banks were devastated, along a distance of six days' walk from Lusambo, and as early as 1904, five large villages had been wiped off from the map.

Broden went to study the situation on the spot. The medical personnel was absolutely minimal, and it was not until 1911 that the only medical officer in charge of the vast district of Lualaba-Kasai was joined by another practitioner, to give assistance at the isolation lazaret of Lusambo and carry out control of sleeping sickness. The team was supplemented by a third physician with residency at Luluabourg (Kananga). Accounts from missionaries bring some additional information on the subject, and from the year 1910, some of them started to attend courses at the School of Tropical Medicine of Brussels or attended a practical course at the Leopoldville laboratory.

From 1900, the condition at Merode-St Salvator, a missionary centre, was just as serious. There were one hundred and sixty people in a somnolent state, and the village population was dwindling steadily.

Thurstan, the British Vice-Consul reported (1911) on a tour in Kasai district, that sleeping sickness was causing real distress throughout large sections of the district. Villages were wiped out along the lower course of the Kasai, but the disease did not spread across the lower Loange towards Bashilile. Close to Bushimai (Mbuji Mayi), thousands of natives must

have perished. South of Kanda Kanda, it ravaged a vast number of Kanyoka people, who concealed their sick and denied the existence of the sickness in their zone. In one Baluba village the chief admitted that he lost 30 out of his 50 wives in the space of five years. The large village of Bene Lulua mourned more than 1,000 victims.

In the Mbuji Mayi region, the death rate showed a steady decline. In this area there was a very extensive salt-making industry with frequent contacts with tsetse flies but with a remarkable insight, the salt of the European company was blamed for all the diseases and deaths that occurred.

In 1922, *Dr Gillet* carried out an investigation within some sections of Tshikapa and Dibaya territories, where he discovered some important foci of sleeping sickness, with rates of infection varying from 1% to 5%, reaching at times 10%. This situation led to an agreement with the **Medical Mission of Kwango-Kasai** (headed by Dr David), to enable the *Forminière* Company to organize an independent prophylactic mission against sleeping sickness, and to be entrusted to *Dr Fourche*. His field of action was at first extended to the territory of Luluabourg, where *Forminière* and its subsidiary companies recruited their labour force; and later, for a limited period, it was to spread to most of Kasai district, until the Government services of the **Dibaya Medical Mission** were organized.

The campaign and its findings were reported in the account of *Haveaux* (1945), who drew attention to the most worrying focus of all, the one among the Batshok, close to the Angolan frontier. The dramatic drop in the rate of infection to 0.06% in 1943 was the result of the systematic control methods applied.

Mouchet (1927) summed up the situation for Kasai district as follows: an average rate of 2.6%, with a minimum of 2.3% but a maximum of 25% in the worst hit foci.

Prati (1926), in charge of the **Dibaya Medical Mission** for the control of sleeping sickness, reported that, in the second half of the year, the distribution of sleeping sickness between the Lubi and Lukula rivers was very uneven. Some villages had a low level of contamination or even none; but, around the post of Dibaya, and in the core of the epidemic in Tshibabala, it reached a rate of 28%. Among the labourers recruited from Katanga for earth works, 1.6% showed to be positive, a rate that was also found among people examined at the frontier, and among foreigners.

For Sankuru district, *Mouchet* (1927) pointed out that a thorough investigation had been carried out (67,605 people): the average rate of infection was

1.2%, with a minimum of 0.1%. He also said that Katako Kombe, Ilebo (Port Francqui), Lomela, Lodja and Kole had been practically spared. In 1930, *Fr. Hemeryckx* wrote in his first report that the isolation lazaret of the Tshombe-St-Marie mission sheltered 30 to 35 patients; and that during his rounds in the inland territories he had found 29 people with trypanosomiasis in a population of 7,949 (0.4%). However, at the dispensary, he diagnosed 121 cases out of 3,359 patients, who had come on their own to be examined.

The Kwa and the tributaries of the Kasai had a series of posts: Mushie, Kutu (the principal district town), Tolo, Oshwe, Bumbuli, Dekese. Trypanosomiasis developed there since 1900. It was believed to have come from the focus of Berghe-St-Marie, but at Kutu (1900) and Dekese (1901) soldiers coming from the Lower Congo had been accused of having transmitted the disease to the local population. In 1908, Oshwe was infected at 12% and the disease progressed to Tolo and Nioki. Reports from the local public health services (*Commissions d'hygiène*) stated that 14% to 16% of the people in the villages were infected. These villages were located at the river banks; it were the boat crews, the canoe paddlers and the wood storage places which were spreading the disease.

In the interior of the country, where the population lived far away from river banks, the territory was free of infection, particularly the villages between Dekese and Oshwe. At Inongo, cleared of all brushwood and covered with plantations, there were no infected persons among the inhabitants in 1905, but the disease was recognized among soldiers and foreigners.

Mouchet, in 1927, reported that around Lake Leopold II (Mai Ndombe) the rate of infection reached an average of 3.3%, with a maximum of 4.9% and a minimum of 0.16%.

5.3.9. Bas-Zaire, Kwango and middle Congo

a) The Jesuits active in the region were a valuable source of information. At the mission of Kimuenza (to become later the site of Lovanium University, now the National University of Zaire), the fathers themselves had a first case of sleeping sickness in 1900, and were badly hit in 1902 and had finally to abandon this missionary post in 1905.

The account of the situation in the Yungu St Ernest Mission at Tumba Mani (now Tumbumami), near the Inkisi, is the first report including statistical parasitological information. Out of 913 subjects examined, 468 carried suspect lymph nodes; in 44 of them (9.47% of lymph nodes carriers and in 4.8% of the overall population), the lumbar puncture revealed the presence of trypanosomes in the cerebro-spinal fluid.

These figures were underrated, as examinations repeated at different intervals would certainly have yielded higher rates.

b) A survey of the Kwango region in the Congo Free State was carried out in 1886 by the missionary *G. Grenfell*, who travelled up to the Sirka Tembo falls. These falls had been reached previously by Ivens and Capello via the Southern route, coming from Angola. Grenfell reported the existence of the junction of the Kwango with a large tributary, the Djuma, 10 kilometres before it unites with the Kwa. The Kwango follows a south-north direction, then turns east to join the river Kasai. At about the same time, *Alexandre Delcommune* explored possible commercial prospects on behalf of the CCCI.

Dhanis, who had first explored the Bambala area, was ordered to investigate and take possession of the Eastern shore of the Kwango in 1890. He set off with 80 soldiers and 150 porters, and negotiated treaties of alliance during a one year peaceful mission. On the same occasion he established the post of Popokabaka. Although a focus of sleeping sickness did exist in Western Kwango, and had been known since at least 1870 in the region east of Malanje, this risk was not mentioned. However some cases were observed at Popokabaka in 1898 (*Boulanger et al.*).

The Kasai Company had settled in Kwilu before the State administration. From 1902 onwards, a series of trading posts were established. A commercial network on such a scale would naturally result in a continuous traffic of caravans and porters to the trading posts along the river, promoting hereby the extension of sleeping sickness.

In 1911, the *Huilleries du Congo Belge*, a subsidiary of the Lever Brothers group of oil factories, set up oil-palm plantations in the area of Lusenge, and used river transport to carry the goods to their centre, Levertville; this of course would also accelerate the spread of the disease.

Although Kwango district had been incorporated into Middle Congo in 1906, the administrative quarters were set up only in the year 1911-1912, at Popokabaka. At that time, already 60% of the prisoners were infected, but the first official investigation in 1914, by *Dr David*, reported that 20% to 30% of the people in villages across the territory were carrying suspicious lymph nodes.

In order to understand these epidemiological figures, it must be borne in mind that the occupation of eastern Kwango was for a long time limited to the area between the rivers Inkisi, the Nsele and the bend of the Kwango towards its mouth in the Kasai.

The ravages caused by sleeping sickness in that region, where people were also starving to death, were really impressive. Fr. *de Vos* (1908) wrote that the zone between the Nsele and the Kwango river resembled a desert, and he added that famine opened the door wide to sleeping sickness.

Fr. *Sadin* (1910) and Fr. *Greggio* (1917) had heard from reliable sources that sleeping sickness had existed in the area before the coming of foreign settlers, although it appeared only in isolated cases. This statement confirmed the information gathered by Fr. *Baltus*, a Scheutist (1898), on the existence of the disease in some main villages of Bobangi and Bayanzi population groups.

A point made by Fr. Allard was that those concerned were easily persuaded to submit to an examination, but that infected patients were reluctant to be treated: indeed, only 20 out of the 44 agreed to undergo treatment, of 24 others, four died rapidly. Fr. Allard also noted that, in villages, the conditions were extremely favourable for the transmission of the disease, because the sick lived in valleys covered with a profusion of *eleis* palms, a favourite breeding ground for tsetse flies and his attempts to induce the villagers to move towards higher grounds, away from the glossinae, proved almost impossible to achieve.

The information supplied by J. Schwetz (1919), after his visit in 1918 to Kwango-Kwilu, which he restricted to Middle Kwilu, is very interesting. He described a country of grassy swamps, with small wooded valleys watered by the tributaries of the Kwilu, and he stressed particularly the abundance of *eleis* palm groves, which at times extended as parks mile upon mile; forests of *raphia* palms covered the bottom of the valleys.

Schwetz was particularly struck by the fact the glossinae were rare during the dry season and very abundant in the region south of Kikwit, where many hogs were to be seen in copse. These copses, typical of the region, contain *eleis* and *raphia* palm trees, bamboos, climbing plants and bushes. Far from rivers and rich in *G. palpalis*, there were remains of deserted villages. Schwetz could not say if these glossina havens were temporary or permanent, because no time had been available to look for pupae. The attraction of tsetse flies for both domestic and wild pigs is such that the natives named these flies: the pig fly.

Schwetz observed sleeping sickness in all the villages he visited, and noted its unevenly spread, also when villages were very close to one another. He based his diagnosis on the palpation of lymph nodes. In villages along the Gobari river, he examined 10,910 people, of whom 7% were found to be sick (10% of men, 8% of women, 7% of boys, and 3.3% of girls),

but no children under three years. In the villages along the Kwilu, out of 8,469 people examined, 8.5% were positive (9.5% of men, 9% of women, 12% of boys, 8% of girls and 0.3% of children under three years). These average figures did not take into consideration other findings, according to which in 18 Gobari villages and in 30 of Kwilu, the rate of infection was over 10%; this rate reached even 25% in respectively four and seven villages of these regions and more than 30% in one of the villages of Gobari, where 30% of men and 50% of women were infected.

Fr. *Ivan de Pierpont*, trained at the Leopoldville laboratory and working thereafter in Kwilu, reported in 1915, that 22 villages had disappeared; in a large village of 636 huts only 20 were still standing. At that time there was only one physician in Bandundu for the whole district.

In 1929, the situation was very much the same, except that the *Huileries du Congo Belge* (HCB) had appointed two physicians, one in residence at Leversville and Tangu, and the other assigned to the protestant mission of Vanga.

The **Kwango medical mission** had to carry out the census of a comparatively large but very scattered population, and had to draw its own maps. Although these medical and administrative assignments were time-consuming, the medical mission managed to examine a considerable part of the population.

It is as well to mention that Schwetz based his detection of the extension limit of the disease on the palpation of swollen cervical ganglia, and used the microscope only for special cases. His quick examination technique gave rise to many controversies. Finally, the microscopic diagnosis became general practice.

Schwetz justified his method restricted to palpation by saying "the natives do not like the needle". The ganglionic puncture for microscopic parasitological detection was accepted at first, but rejected later, when rumours spread that the needle injected the disease. On the other hand, *Atoxyl*® injections were accepted, because the treated subjects appreciated the stimulating effect of the arsenic.

The rates of infection observed by Mouchet in 1920-1923 and 1927 respectively, noted a considerable improvement: at Kikwit, 12.8% and 0.6%; at Bulungu, 12.2% and 0.38%; at Kandale, 5.3% and 1.1%; at Niadi, 5.8% and 0.45%; at Lower Kwilu, 4.7% and 1.2%; on the Lukula, 3.7% and 4.3%; on the Kamisha and the Lubue, 9.8% and 0.65%; among the Bapende, 7.5% and 2.5% (Mouchet, 1927).

c) There was a focus close to Kwamouth (Fr. *de Vos*, 1903) and a second in the southern part of

Middle Congo district, between the falls of Kingani and Mwene Dinga (Fr. Greggio, 1917), adjacent to the West Kwango focus of Angola. Fr. F. Allard, who had been in the Congo since 1905, went to the Leopoldville Laboratory, under Dr Mouchet and Dr Dubois, to learn techniques enabling to recognize and treat sleeping sickness; whereby the missionaries would contribute to control this scourge which killed so many people in the region.

6. The obstacles to the discovery of sleeping sickness

Albeit incomplete, this brief survey shows the ways and means in which sleeping sickness had extended in very different regions, by the end of the existence of the Congo Free State and during the first decades of the Belgian Congo. It underscores that the control of the disease was carried out with tenacity in spite of very few administrative personnel and understaffed medical services. With the increase in personnel to control trypanosomiasis from the year 1910-1911, and in spite of the gap caused by the 1914-1918 war, *ad hoc*, appropriate and accurate methods began to take shape and to benefit from discovered effective trypanocides (*Atoxyl*®, tryparsamide, Bayer 205®). The steadily encouraging improvements continued up to 1959, when *T. gambiense* trypanosomiasis ceased to be a major public health problem. Details relating the evolution of sleeping sickness in the Belgian Congo (Zaire) can be found in publications by Gabba (1938), Trolli (1941), Van Hoof (1948) or Gillet (1954).

The abrupt interruption of the procedure in 1960 proved the vulnerability of the scheme, which dealt solely with the human link in the transmission of trypanosomiasis and did not dispose of a drug capable of eradicating the disease.

The resumption of sleeping sickness control, by Belgians and Zaireans working closely together (see p. 1784), has been recognized as a model of co-operation. The 30 teams assigned to different sections of the field work proved, nevertheless, that the foci called historical continue to be centres from which the sickness tends to expand.

During the early colonial times, useful but fragmented data had been collated; unfortunately, erroneous conjectures had arisen, because at that time little was known about the essential factors which fostered the development of the trypanosomiasis. Some of these misconceptions have biased the epidemiological philosophy, as well as the methods of control.

6.1. *The theses of isolation by lack of communication*

Disregard of information available in Central Africa had seriously reduced the awareness on the danger of sleeping for the population and the way it could spread.

The fact that endemic foci existed in the hinterland of tropical Africa was not perceived immediately. The current belief rested on the theory that inland areas of tropical Africa had remained isolated until the second half of the nineteenth century, and that its communities had always lived there, confined in themselves and out of touch with their neighbours. This ethnic compartmentalization was attributed to constant insecurity, and to permanent threat of tribal wars, incursions and forays.

However, this theses disregarded the migrations, whether peaceful, aggressive or even warlike, of the Bantu, Sudanese, Fulani and other populations and also the seasonal migrations of herdsmen, the long trips of fishermen, the visits to markets or the trade stimulated by Portuguese, Dutch, English and French commercial agencies. These agencies, with a network of head offices on the Atlantic coast, had a monopoly over goods highly praised by Europeans which were purchased or bartered from the inland territories. Among all these transactions, the purchase of slaves, brought in from afar along the caravan trails, should not be forgotten.

The wheel was unknown to the people. The use of burden or draught animals was almost impossible, due to the many fatal diseases killing them at the moment of their introduction into the country. Only two choices remained to the inhabitants: the canoe and portage. The latter system was slowing down the travels, and would discourage many explorers and traders from venturing into the hinterland.

The **thesis of isolation** also stemmed from unawareness of the many communication trails in tropical Africa and the rest of the continent.

6.1.1. *Communications between northern and tropical Africa*

North-south connections towards the centre of Africa had been existing for centuries without interruption, though on a somewhat limited scale. They started from Maghreb and Libya, using caravan trails across the Sahara; some went down to Timbuktu (now in Mali) and Kano (in Nigeria), others linked the black mountains of Fezzan (in the south of Libya) to Bornou (region of Niger and north Cameroon), to Kanem (region of west Chad north of Lake Chad), to Ouadai and even to Darfour, through the Tibesti massif. Others

started from the south of Egypt via the royal route of the Nile to reach the Sudan, Bahr-el-Ghazal and the other territories of the vast Nile basin stretching out into East Africa.

North-south traffic by land had been operating by Hausa caravans coming down regularly from Kano to Mobaye on the Ubangi river, taking three months to make the journey. Just like the Bornou itinerant peddlers, they bartered and traded salt from Bilma (Sahara), clothing materials, leather, metal objects, hardware, silk, perfume and other products in great demand, such as arms and powder. The Fulani for their part came down from the Chad to Bangui. Finally, the regular incursions by Nubians from Darfour (western Sudan) and from Kordofan (central Sudan), penetrated right into central Africa. The mentioned forms of trading continue even today, although some transport is mechanized and bartering has been replaced by cash payments.

The importance of the Nile was assessed in a very special way during the events in Bahr-el-Ghazal and southern Sudan, towards the end of the nineteenth century. Through annexation in 1864 by Egypt, at that time still a part of the Ottoman Empire, this region became the province of Equatoria. This particular network of inter-African communications was emphasized by several events: the exploits of Emin Pacha, a key figure in that stormy period, the Mahdists' invasion, the persistence of the British Empire in its efforts to achieve the north-south Cape-Cairo link, and the attempts of France to establish an east-west Libreville to Obok connection.

Arab and Afro-Arab traders, Nubians and Copts, had built up a chain of trading posts which were extremely well run. In order to back smoothly their business, they signed agreements with the local chiefs and thus were allowed to set up trading points in their respective lands. Moreover this type of agreements was used later by chartered companies and other European organizations.

6.1.2. *Communications between central Africa and the east coast*

Other means of supply and export were being operated from the east African coast. The trade movement started from some 400 different small ports scattered over 3,000 kilometres along the coast of Kenya, on the Indian Ocean. These ports were used by Arab traders, Afro-Arabs (Swahili) and Indians who dealt in the traffic of ivory, rock-crystal (quartz), gold, copal and slaves. Buying and bartering took place along the trading routes: utilitarian goods were exchanged for shells, pearls, copper wire and so on. African goods were bound for Arabia, the Persian Gulf and the Coast of India.

a. *From ports of east Kenya*

Starting from small ports like Shanga and Manga, the routes served the hinterland. They crossed Kenya and Uganda, reached Lake Victoria and Lake Mobutu (previously Lake Albert) through Jinja, Masindi, Lake Kyoga, Butiaba, over the West-Nile district to Mahagi, Nimule, the Nile, and then continued westward in the Equatoria province, the territories of the Bomu, Uele, Ubangi and many other tributaries of the Congo basin. These routes were epidemiologically speaking not so much in the limelight of the concern, because the east coast was spared from sleeping sickness, but human exchanges occurred in both directions and many of the porters came from Uganda. Some of them must have played a part in the dissemination of sleeping sickness from west to east.

b. *From ports further south*

The means of communication from the southern ports towards the hinterland were better known. A network of links with the deep interior had been set up from Zanzibar, Bagamoyo and Sadani. From 1830 onwards, it had grown to such an extent as to reach, via Tabora, the Lake Tanganyika and the trading post of Nyangwe near Kasongo on the Lualaba. From the trading post of Nyangwe, the route was extending along the river towards Kisangani. This access route from the lower east coast had become familiar, as it had been used by the explorers Cameron and Stanley and by the five expeditions of the International African Association.

6.1.3. *Communications between central Africa and the west coast*

When they settled on the Atlantic coast at the mouth of the Zaire river, in 1455, the Portuguese, who reached Matadi, had no intention of establishing a colony but they simply wanted to secure a source of goods (gold, spices, etc.). Trading relations with the hinterland were started by the Portuguese, but were maintained mainly by a group of educated Africans, known as Luso-Africans or *Ambaquistas* (meaning traders because Ambaca was a major trading centre; those traders came from Luanda, Ambaca and Malenje. The *pombeiros*, mistakenly identified as Ambaquistas, ran the Portuguese business between various trading posts and depots, and the *soases*, native peddlers crisscrossed the territories, pushing deep into the interior.

These Luso-African traders had developed a commercial network directed mainly to the south of Central Africa. Pania-Mutombo and Kabinda in the south-east were centres of inward-coming and outward-going

trade, catering for Lake Moero, Shaba and Kasai; they provided a link with Maniema and the East Africa trade.

Wathen, situated in the south-west Lutete territory (Lower Congo), was a relay post for all kinds of products such as ivory, spices, salt, dried fish, wrought iron, arms and slaves, which were being transported to Angola from the Ubangi, the Tshuapa, the Kwa-Kasai and the banks along the whole river Congo. Wathen was also a centre for goods coming from Loanda and Mossamedes.

Commercial caravans bearing the same kind of goods were formed on the coast of Angola, heading for Kasai and Shaba.

It is worth noting that these vast territories, thought to be populated only by communities living in total isolation, had always been crossed by caravan trails (whose travellers may have been trypanosome carriers), as well as by rivers, where glossinae could have been or become infected and transmit sleeping sickness to or from any passing-by person.

When considering the small local trading branches, it becomes clear that the central African people were far from living totally isolated. To all these occasional and trade contacts could also be added the fact that the numerous, often large-scale migrations of communities seeking new fertile lands and pastures may also have contributed to potential infectious diseases' transmission, including sleeping sickness.

Any critical analysis of the propagation of sleeping sickness shows that these numerous ways of communication were multiplying the contacts with vectors, and endangered the lives not only of the travellers but also those of the local inhabitants.

Although the above mentioned obstacles may have limited incursions into the hinterland by Europeans for several centuries, they did not stop raids and incursions, nor the slave trade or the export of gold, ivory and rock-crystal.

6.1.4. Depopulation along the River Congo and in areas of endemic sleeping sickness

The account by *Alexandre Delcommune* of his three journeys along the river is particularly striking. On each trip following his first one in 1887, he noticed that the number of villages on the banks of the river kept on decreasing to such an extent that, in 1909, he found only the remains of small villages where once large communities of the powerful Bayanzi, N'Gombe and Bangala tribespeople had been living.

Observations made by *Rodhain* in Ubangi (1904-1906) confirm *Delcommune's* findings. In 1914, on the site where large thriving villages used to stand

when he was in charge of the local medical services in 1904 he saw only ruins, which indicated that the villages had not been moved but had simply been wiped out by sleeping sickness.

Similar situations were described in accounts and discussions during the successive meetings of the National Colonial Congress, particularly at the session on December 1920, which was completed by a report of *Broden*, on the survival and increase of the Congolese population.

The attitude of the authorities and their executives in Brussels has certainly concurred to implement realistic infrastructures. However, it may also have been responsible for the over-simplified belief that occurring calamities, such as the depopulation of villages, was solely due to sleeping sickness.

Anyone acquainted with the daily occurrences in the field of health problems knows the great number of morbidity factors with an epidemic potential. A careful reading of documents on outbreaks of epidemics shows that other causes, such as famine, dysentery epidemics or smallpox are frequently mentioned. Furthermore, sleeping sickness caused by *T. gambiense* is essentially a chronic disease which may be quiescent but is liable to flare up when some biological and ecological changes occur.

Sleeping sickness has been accountable to a large extent for depopulation, but it would be too feeble-minded to put all the blame on it. Some affected communities completely regained their vitality, while others went on declining, even though they had been freed of human trypanosomiasis.

The turmoil of colonization had considerable repercussions on a population already unstable: expeditions, occupation of territories, recruitment, and uprooting of the labour force from its traditional surroundings. These innovations occurred at a previously unknown pace and rhythm, and caused important ecological changes which included changes in the epidemiology.

6.2. Presumed resistance of Europeans to sleeping sickness

It was believed for a long time that Europeans had a special resistance to sleeping sickness. The fact that they were small in number in any particular region, and probably not often exposed to the risk of infection, bolstered this belief. Furthermore, it was thought that the symptoms of the disease were limited to lethargy, and as a result its initial fever phase was overlooked as being a trivial symptom. The discovery of human trypanosomes will clarify this matter.

In Gambia in 1901, *Forde* discovered in the blood of an European some vermicular micro-organisms to

which he ascribed the cause of a fever which failed to respond to quinine. A year later, in Liverpool, *Dutton* identified these parasites as trypanosomes. The deadly fever was to be called *trypanosoma fever*.

Broden crushed for good the theory of resistance to the disease, when he discovered between 1903 and 1905 many Europeans were suffering from trypanosomiasis. It had also been detected, in London, in 1904 by *Manson*, in an European missionary lady infected in Angola. It should be added that *H. Vanderyst* remembered to have treated two missionaries from Scheut at Moanda in 1891; they suffered from intermittent fevers resistant to quinine and died the following year. With hindsight, the displayed disorders were those of trypanosomiasis.

6.3. Attitude of the authorities

The administrative authorities, physicians included, were naturally baffled when faced for the first time with a disease about which they knew very little: fever without characteristic symptoms, lengthy latent periods, terminal phase manifested by strange neuropathology, apathy, stupor and occasional somnolence. All these physical signs brought very little help in the search for the origin of the symptoms observed during the acute stage of the illness or at its terminal phase.

It is hardly surprising therefore that the authorities gladly accepted any reassuring explanation given, even if it did not take into account the magnitude of the disease, its extension and its seriousness. But it was regrettable that these same authorities continued stubbornly to stick to misleading opinions, even when new facts proved how inadequate and insufficient were their measures to fight the disease.

Some even wished to minimize the existence of the problem.

a) The authorities in former **French West Africa** have negated the reports of *Bouffard*, *Thiroux*, *G. Martin* (1876; 1877) and the observations of *Corré* and his publications (1906; 1909; 1910). They all demonstrated the existence of foci of sleeping sickness in these territories. These authors drew attention to the decline in the population that had occurred on the banks of the Senegal river and the Black Volta, once barges had been used to transport goods. Although some people went as far as to deny the existence of sleeping sickness, others acknowledged that it was to be found in the hinterland, and just an imported sickness which had spread along the coastline between the Congo and Ivory Coast during the twenties and thirties.

It was due to *Jamot* who happened to be in Dakar at the time, that the government officials were finally duly informed on the *so-called* epidemic. In thirty months, *Jamot* identified 68,000 cases, a figure which increased to 180,000 in 1938.

Had it not been for *Jamot*, the unfounded assumptions made by health officials could have impeded all progress in the much needed research on trypanosomiasis.

b) It was unfortunate that the **Congo Free State** (*Etat Indépendant du Congo, EIC*) authorities did not learn from the experience gathered in Angola. The situation there was a model for what was to hit the Free State, whose officials were totally unaware of what was happening in a neighbouring country. This state of affairs was all the more surprising because, on the occasion of the Berlin Conference (1885) and in subsequent years representatives from both countries had held long discussions on boundaries, particularly those of Kongo and Lunda territories. These discussions culminated in the agreement of May 25th 1891.

The Portuguese had learned that ecological and epidemiological disasters were the price to be paid for the transition to an export policy. The factors involved were numerous and complicated. In addition to famines caused by climatic instability, aggravated by deforestation and invasions of locusts, there was also depopulation, due mainly to the transfer of a considerable labour force for general infrastructures and the building of the railway (from Luanda to Ambaza 1873-1888). Both export of goods and military expeditions were made easier by steam navigation on the river Cuanza from 1886 onwards, and played a part in depleting the population while it also contributed to an economic instability (see map 36 p. 1708).

The disruption of the traditional way of life of an undernourished population, ignoring the simplest rules of hygiene, affected by fever, dysentery and acute respiratory troubles or pulmonary tuberculosis, covered by skin ulcers and often exposed to new pathogens, increased the havoc of epidemics such as smallpox.

The presence of sleeping sickness had been established with certainty in 1869, when it was diagnosed in Martinique among slaves whose origin was the Kongo district (slave trade had been officially abolished in Angola in 1836, but the application decree was proclaimed in 1858; with a transition period of 20 years ending in 1869).

The year 1870 has been marked by Portuguese military campaigns to occupy territories. The Portuguese soldiers were not suspected of being carriers of pathogens but sleeping sickness raged in the district of Malanje (Malange), a busy crossroads for caravan trails to neighbouring West Kwango (Cuango).

The **Malanje epidemic** sounded the alarm and disclosed the expansion of sleeping sickness towards the Lower Cuanza, the banks of the Dande and the Nzenza. Although the first official accounts did not appear until 1880, the disease had taken its toll there since 1860. The many trading caravans could include trypanosoma carriers from endemic zones in western-central regions still unoccupied.

These epidemic outbreaks were not the result of an unmediated generation. Concessions of arable land for coffee growing had attracted, particularly Portuguese settlers coming from Brazil. Within 20 years, they increased from 600 to 6,000 and the plantations multiplied from 1,000 to 16,000. The establishment of farms and estates involved serious deforestation, hunting on a large scale, and recruitment of agricultural workers for the regions of Cazengo and Golungo Alto, and the hinterland of Luanda. The thickly wooded region of Cazengo-Golungo, cleared by the *coffee barons*, became an ideal site for the glossinae, which were ready to infest the coffee plantations.

The notorious epidemic of 1895, in the plantations of the Cuanza, where the banks of all the tributaries were very fertile, caused a general exodus from the villages (Cazengo, Cambamba, Kisenga, Massangao); the inhabitants fled, settled in other villages or built new ones. The region between the Dande and Congo rivers (see map 36, p. 1708) lost half its population, which had dwindled from 228,000 inhabitants in 1876 to 115,000 by 1898. The same situation was witnessed in the region between the Dande and the river Congo, but no figures are available for the Kongo district.

All these facts were foretelling the situation which was to occur throughout the Congo Free State, but they remained local history. The Free State was to live its own experience by bearing the heavy burden of a dramatic development of sleeping sickness, similar to the developments in Angola. However, Angola failed to make useful contributions to decree control measures; the governing body simply drew attention to the existence of the sickness, but did not enact effective laws.

In the same way as it was supposed, in 1966, that only one focus of sleeping sickness remained at Abengourou, in the south east of Ivory Coast, it was decided in the Congo (Zaire) to use every possible means to eradicate the disease. Simultaneously with a temporary but unbreachable *cordon sanitaire*, the health authorities decreed that the entire population be submitted to a medical examination, including parasitological and serological detection, treatment of all infected and suspected subjects. Finally, sound ento-

mological research and measures to contain and eliminate the vector were planned and organized.

In spite of all these control methods, the results did not meet the expectations. Some important facts in need of a careful analysis, were neglected, on the fallacious belief that, as later on in Abengourou, there was only one focus in the territory. The flare-up in Ivory Coast of the focus (unknown until then) at Daloa, some 400 kilometres from Abengourou, proved how precarious such conclusions were. Daloa has since become the WHO research centre for trypanosomiasis in West Africa.

7. Actions taken

7.1. Data gathering in the Congo Free State (Etat Indépendant du Congo)

For decades information on the state of health of the Congolese population remained fragmentary. However, it would be grossly unfair not to mention the efforts that were made to improve the situation.

In 1897, for instance *Bourguignon et al.* presented, at the National Congress on Hygiene and Medical Climatology of Belgium and the Congo, a report drawn up from the findings collected with the help of a very detailed questionnaire submitted to district commissioners and administrators. One of the topics referred to sleeping sickness.

Some opinions linking sleeping sickness to beri-beri (Corré, 1877), or to filaria (Manson) were recalled, but the authors of the report drew attention to the haphazard dissemination of the disease in the Lower Congo, particularly in the Cataracts region, as well as to the fact that the sickness prevailed upstream from the Malebo (Stanley Pool), where some districts had been decimated.

The questionnaire posited six degrees of prevalence for concerned localities. The results were:

- 1) rare and mild cases: Popokabaka on the left bank of the Kwango;
- 2) numerous and mild cases: Basoko on the river between Stanleyville (Kisangani) and Bumba;
- 3) serious but rare cases: Katola (situated in one of the arms of the Congo estuary), Temvo (between Boma and Tshela), Bolobo on the left bank of the river between Leopoldville (Kinshasa) and Coquilhatville (Mbandaka), Banzville (Mobaye-Bongo) in the north-east, on the Ubangi;
- 4) serious and numerous cases: Kwamouth where the River Kwa meets the River Congo (Zaire);
- 5) serious but rare cases: Boma, Matadi and Palabala (on the heights, a few kilometres from Matadi),

Tumba (on kilometre 156 of the Lower Congo railway), Leopoldville, New Antwerp (Makanza) on the river, halfway between Coquilhatville and Lisala, Luluabourg, Lusambo (on the right bank of the Sankuru); 6) numerous and serious cases: Isangila, Kibanzi, North Kingila, Mukimbungu, Lukungu (State post and American Baptist Mission), Lutete (Wathen), Manyanga (on the right bank of the river, 102 kilometres from Thysville), all these localities are in the Cataracts territory of Lower Congo.

It may be concluded from the above that the Cataracts region, the banks of the Inkisi and Kwamouth were badly affected.

The report on the Mukimbungu district of the Protestant Svenska Mission Forbundet (see p. 1719) supervised by *Dr Walfridson*, emphasized the havoc brought by sleeping sickness; a mortality rate of 20% was forecasting a future depopulation. A similar situation in secondary posts as Kibunzi, Diadia and N'Ganda led the authors of the report to observe, no doubt influenced by the prevailing ideas on the situation, that the infected territories were not very large. A great disaster was impending, and would erupt during the first decade of the twentieth century in the Cataracts and Inkisi regions.

It was only when several disasters had occurred, such as the numerous deaths among the students in the boarding school of Kwamouth, the depopulation of the river banks and the forced abandonment of the depleted mission stations, that the authorities realized the extension and the seriousness of the problem.

7.1.1. Medical, parasitological and epidemiological missions

The *Liverpool School of Tropical Medicine* was called in to assess the gravity of the problem. The first rough estimate of the virulence of the disease in the Lower Congo and in Leopoldville was given by the **Dutton-Todd-Christy expedition**, in their report (see also pp. 100 and 1689). Blood samples taken from individuals who looked healthy showed that in Boma 6.3% were infected (18 out of 284), in Matadi 0.9% (1 out of 116), in the Cataracts region 7.5% (16 out of 213), in Leopoldville 2.4% (6 out of 255). On the other hand, some healthy carriers were found. The rates observed in hospitals were obviously higher: in Boma 15.3% of the sick were infected with trypanosomes (11 out of 72), in Matadi 13.6% (3 out of 22), in Leopoldville 28.7%, and in the dispensary of the Baptist mission of Wathen (Lutete), visited by *Dutton* and *Christy*, 13.9% (11 out of 79). The care they took over their clinical observations was quite

remarkable. *Christy* reported that out of 54 examined cerebro-spinal fluids of patients with trypanosomiasis, 24 (44.4%) were found to contain trypanosomes. One of their major concerns was to establish the identity and distribution of the trypanosome and of the glossina. Their parasitological and entomological investigations confirmed the identification of *T. gambiense* as the pathogenic agent and *G. palpalis* as the vector. It is as well to remember that at the time, because no proof of any development cycle in the body of the tsetse fly had been produced, the glossina could be thought only to play a mechanical role in the transmission.

Dr Heiberg, whom *Dutton* and *Todd* had known since the time he attended courses at the Liverpool School of Tropical Medicine, was seconded to the expedition. When they learned at Coquilhatville (Mbandaka) how important it was to look for the trypanosomes in the juice of enlarged cervical lymph nodes (*Greig* and *Gray*, 1904), the team pursued their observations, making use of this new technique.

The prevalence of trypanosoma infection was ascertained in various places. At the same time, the diagnostic value was compared of a lymph node juice examination with that of the traditional system of examining a blood sample, which had been used systematically in the first stage of the expedition's work. These examinations drew attention to the difference of risk for the riverside dwellers who were exposed along the wooded banks, as compared with the inhabitants living in villages in the interior. In Upoto, slightly westward of Lisala but still along the Congo (Zaire) river, the rate of infection was 12% (4 out of 33), whereas in the neighbouring interior villages there was no infection (0 out of 225); in Bumba, the rate was 2.1% (9 out of 426) and in Tubila 1.5% (2 out of 136); in the village of Katanga near Kisangani it was 0.5% (1 out of 196); in Ponthierville (Ubundu), at an altitude of 470 metres and in the villages in the interior, there was not one case (0 out of 124); at Lokandu 18.7% (19 out of 96) and an hour's journey inland 0.9% (2 out of 224, who were imported cases); in Tshofa 5.4% (10 out of 185); in Kabinda 5% (5 out of 99); in Lusambo 1.5% (3 out of 197); none at Manghay. However, not all of the villages a distance away from rivers or built on high ground escaped the infection; in Palabola (Matadi), 0.9% of the people had contracted the disease and in Kabinda (altitude of 868 metres) 5%.

The expedition decided to investigate the various means whereby the disease had spread and to identify its pathogenic agent and vector. The west shore of Lake Tanganyika, which was then passing from an endemic situation to an epidemic upsurge, seemed the

ideal spot for the purpose. They did not get to the west shore, but found a similar spot in Kasongo, where Dutton identified the spirillum responsible for the recurrent fever.

Meanwhile the sick suffering from sleeping sickness were admitted to hospitals. Careful physicians, like Rodhain (1907) in the sector of Libenge (Ubangi), produced very revealing reports. By examining the ganglionic fluid, he proved in 1905 the presence of trypanosomes in patients from the Lower Ubangi (Imese, Ekuba, Libenge, Bamondema, Mokwange). He confirmed that there were people suffering from sleeping sickness on the French bank of the Lower Ubangi. In spite of the positive parasitological diagnosis made on the inhabitants of Guelorget village, between Banzville (Mobaye-Bango) and Yakoma, he believed that the Upper Ubangi had not yet been affected, and indeed all cases discovered there came from soldiers or soldiers' wives.

7.1.2. First specific treatment trials

Pretty soon after the trypanosome had been identified as the pathogenic agent, a first specific treatment (Atoxyl®) had been found. Even more information became available and concrete facts were gathered from patients suffering from sleeping sickness or being treated, in either isolation lazarets or hospitals.

The laboratory in Leopoldville spent a great deal of its time on trypanosomiasis research and pooled all the known relevant epidemiological information. There, Broden stated that sleeping sickness was endemic in most of the territories of the Congo Free State (EIC) and that the more the occupation of the country extended, the greater risk also was the spread of the trypanosomiasis. To observations of the sick and post-mortem examinations therapeutic trials with Fowler's *potio* and especially with Atoxyl® were added.

Atoxyl® had been discovered by Bechamps in 1863 and was found to act on *T. gambiense*, as demonstrated experimentally by Harold Wolferstone Thomas (1905), in Liverpool. This medicine was the first one to bring a glimmer of hope.

Broden (1907), who had already noted the satisfactory results obtained with the use of high doses of Fowler's *potio* for the treatment of the disease, had also acquired some experience in handling several arsenical substances. He consequently felt quite confident when he came to use Atoxyl®, and followed the method prescribed by Thomas: two to three injections weekly. However, he expressed the opinion that other ways of administering the drug might produce quicker results. He was particularly concerned about the side-

effects of the drug, and the criteria of recovery. Broden had also examined the situation in the interior area, especially in the observation centre of Stanleyville and in the isolation lazaret of the St Trudon mission, near Lusambo.

Rodhain, who had already shown his concern for sleeping sickness in the Lower Ubangi since 1905, became Broden's assistant at the laboratory of Leopoldville in 1907, with additional responsibility for the Leopoldville lazaret. This double task was to afford him the opportunity of developing therapeutic trials, and studying them in depth. Atoxyl® which could be administered orally, by intramuscular injection and also intravenously, required a lengthy treatment which did not preclude relapses. Arsenious acid (advocated by Loeffler), antimony salts (emetic), colouring substances (trypanroth, parafuchsine, Afridol), and combinations of various medicines were all tried out; these were constantly monitored as well clinically as parasitologically. Serious attention was given to the criteria of recovery, based on the control of changes in the CSF.

Rodhain had noticed among the personnel of firewood-storage depots for the river steamers, infection rates as high as 50%. On his way to Lower Katanga, he was able to observe the prevalence of trypanosomiasis along the river, as far as Basoko, about 200 kilometres downstream from Stanleyville (Kisangani), and also in a number of posts between Ponthierville (Ubundu) and Kasongo.

At Maluku, upstream from Malebo (Stanley Pool), 13 out of 54 men (5.2%) were infected, but no women; at Kinzulu, 5 out of 32 men only were infected (15.6%); at Kwamouth 12 out of 33 men (36.3%), and 3 out of 18 women (16.6%); at Sandy Beach (between Bolobo and Kwamouth) 4 out of 23 men (18.1%); at Balobo 2 out of 20 men (10%); at Yumbi 2 out of 34 men (5.5%); at Lukolela 6 out of 163 men (3.8%), and 2 out of 28 women (7.1%); a total of 60 out of 562 men (16.7%), 12 out of 166 women (7.2%), and 1 out of 46 (2.3%) of children were infected; an overall average of 82 out of 837 (9.7%).

Rodhain noted the presence of *G. palpalis* in all the stations. In the few places where the statutory rules had been obeyed, efforts were of no avail because deforestation and the clearing of the brushwood had been either inadequately or not at all applied, especially near water-holes, a favourite but overlooked abode of the glossinae. Upstream, from Lisala to Basoko, nearly all the villages were in the same state of neglect, and downstream, on the island of Ukaturaka, 9.5% of the people were infected, not an exceptional rate for that area.

Upstream from Basoko, glossinae were rarely found, as was the case as far as Ponthierville (Ubundu); the only person to be found infected in that post was a soldier's wife, who was already infected when she arrived. In the area 210 subjects were examined, workers at the post, residents and even riverside people; all were free from infection.

In 1911-1912 *Mouchet* and *Dubois* (1914) were put in charge of the trypanosomiasis control in Middle Congo district, stretching 500 kilometres along the left bank of the river. They found that the average rate of infection in the district was about 4% to 5% although, in some villages, it was found to reach between 10% and 12%, after only one examination. In the Sandy Beach area, a little way downstream from Bolobo, they discovered the record rate of 30% (6 out of 20). They decided to organize treatment on the spot, because to evacuate the trypanosed subjects to the isolation lazaret would have exposed a great number of workers to the serious risk of contracting sleeping sickness through contamination. Treatment consisted of sterilizing all infected subjects with *Salvarsan*®. This arsphenamine (or 606®) had, in monitored trials, given a peripheral clearing of parasites in the blood, lasting three months. At the same time, the multiplicity of small clusters of dwellings made the sweeping of brushwood extremely difficult, and a campaign to eliminate the glossinae almost impossible.

In 1911, *Dubois* explored some posts on the Kasai, as far as Bokala, a little way upstream from Mushie, on the Wissmann Pool. He observed a global infection rate of 3.7% (9 out of 237), that is 3% men (5 out of 168), and 5.8% women (4 out of 69); all the children were diagnosed as being negative. In 1914, he carried out an investigation in the Bokala region, which revealed an overall rate of 2.4% (16 out of 646). However, among a group of 115 volunteers at a market he found three women and one child positive or an overall rate of 3.5%.

Accounts from diplomatic services came to be quite an unexpected source of information on health conditions. After a visit to Kasai in 1911, the British Vice-Consul Thurstan reported the very serious situation in the southern region near the river Bushimai (Mbuj-Mayi) and gave other important details already reported (p. 1709). The American Vice-Consul, Armstrong, noted that sleeping sickness was unknown in Uele and described his visit to the isolation *lazarettos* of Ibembo on the Itimbiri and of Aba on the Sudanese border.

7.1.3. Role of religious missions

The missionaries played an active part in the control of trypanosomiasis. The many facts they gathered

from the end of the nineteenth century represent some valuable information, the more so as they report that some decline in the population had been observed around the missionary posts even before the eruption of sleeping sickness, or of any other epidemic.

a. Protestant missions

The protestant missions gave professional help from the outset. The *Svenska Mission Forbundet* (SMF) settled in Bas-Zaïre, at Mukimbungu in 1886 (see map p. 1708). It had the help of Dr Walfridson, who had probably, in the year 1892-93, observed trypanosomes in the blood of some of his patients. His notes, left unpublished, would certainly not have invalidated the discoveries made by Forde and Dutton in Gambia, 10 years later. Dr Walfridson died in 1893. His successor, Dr G. Palmer, was infected with sleeping sickness, from which he very slowly recovered.

The *Livingstone Inland Mission* settled in Palabala (Matadi) and from 1882 received help from Dr Aaron Sims, a Scotsman who was to remain active in the region for 40 years. The *American Baptist Mission* (ABM) later took over this mission.

Other missionary societies were also sending physicians. At the *Baptist Missionary Society* (BMS), established near the Stanley Pool in 1883, Sidney Comber was appointed as physician but died a year later. Dr Girling from the BMS was appointed to Bolobo in 1888. Dr Royal Dye of the *Livingstone Inland Mission* was sent to Bolenge in 1883.

Dr W. H. Lesley and Dr Catherine Mabie of the *American Baptist Mission* (ABM) were sent to Sona Bata (Kimpese), to which a succession of seven physicians followed one after the other from 1882 to 1891. The *Balolo Mission* settled in Lulonga in 1884.

b. Catholic missions

– In Bas-Zaïre and Kwango

On their arrival in 1893, the Jesuits discovered the Inkisi-Nsele and Kwango-East foci of sleeping sickness. They were also to witness a depopulation far more serious than any which had occurred during an outbreak of smallpox. For want of a physician they were compelled to deal with the problem themselves.

Their colleague, *Hyacinthe Vanderyst*, a priest and agricultural engineer based at Kisantu, was encouraged to familiarize himself with diagnostic problems and treatment, in conjunction with Drs *Broden* and *Rodhain*. In the meantime, many other missionaries working in the regions infected by trypanosomiasis were given a technical training at the Leopoldville laboratory. This project was to inspire Dr A. Broden to start a course for medical auxiliaries to be orga-

nized at the School of Tropical Medicine in Brussels, once he was appointed to this School. This course was to benefit a large number of missionaries (see p. 113).

The Inkisi-Nsele region lost 67% of its inhabitants between 1899 and 1907. *Fr. Butaye* wrote that, within a succession of ten villages, he came across no natives, no fields, no agriculture. Along the Inkisi and the railway, the village of Mbo had disappeared and the 200 huts in Kilumbi housed only four to five people. Research undertaken by *Fr. P. De Meulemeester* and the *Brothers Gillet* and *Van Houtte* in 42 villages in the chieftainies of Kisantu, Kinturi and Kibanzu, showed that in 13 years, 85% of the people had succumbed to the disease.

The situation described in the Kwango focus was similar. In 1908, *Fr. De Vos* described the area between the Nsele and the Kwango, where the Banfunguma people lived, as a desert.

Having learned the techniques to recognize and control the disease with Drs Mouchet and Dubois, *Fr. Fernand Allard* took over the health and hygiene of the inhabitants served by the missionary posts of Yungu St Ernest and Tambu-Mani. In his 1913 report, he recorded that the average rate of infection in 14 villages amounted to 4.8% (44 out of 913), but he added that this figure was well below the true one. *Fr. Greggio* (1917), in charge of the lazaret of Kisantu, analysed very carefully the consequences of trypanosomiasis on mortality in general, and on children in particular. He also pointed out the beneficial effects that Atoxyl® had on the sick, and how well it had served the population. *Fr. Greggio* also made some interesting suggestions to standardize the methods to be followed and to issue reports from the various missionary posts. To this end he advocated annual meetings to collate, discuss and assess the work achieved by all those committed to the struggle against sleeping sickness in their diocese.

– Along the Tanganyika lake

Two physician catechists appointed to the White Fathers Missions, (see the chapter Medical Education, the Lavigerie project p. 200) practised in the Lake Tanganyika region: *Adrien Atiman* at Karema, on the east shore, and *Joseph Gatchi* at Mpala and Kibanga on the west shore. The latter post was particularly unhealthy (the mortality rate had reached 20% among the residents), and it was later moved to Kirungu (near Baudouinville, now Moba). From 1894, Gatchi served magnificently the mission St Louis of Mrumbi, and collaborated with Leopold Joubert, captain of the papal zouaves. In 1904 Gatchi observed cases of sleeping sickness and was amazed to notice its worrying and rapid expansion, which coincided with the

installation of a telegraph network. His two youngest children died of sleeping sickness. The St Louis mission of Mrumbi was moved in 1909 to Moba, where Gatchi himself died of trypanosomiasis in 1912.

Mgr. V. Roelens was to be an invaluable witness to the way in which sleeping sickness invaded the Upper Congo, Maniema and the shores of Lake Tanganyika. During his regular visits to the different missionary posts in his vast diocese, he was able to follow the disastrous extension of the sickness from Kasongo, and to observe the results of the invasion by a highly virulent trypanosomiasis which wiped out most of the population of the Ubwaru peninsula. He was obliged to abandon three of his missions: Luisi St Lambert, Lusenda to the north-east of the lake and Bruges-St-Donat in Maniema.

According to instructions given by Cardinal Lavigerie, all missionaries had to set up in every missionary post an out-patients centre and a pharmacy to nurse the sick with self-denial and to grow medicinal herbs. *Fr. J. Kindt* from Bruges returned to his post in 1908, after having attended a course at the Pasteur Institute in Paris where he acquired the medical expertise needed to control sleeping sickness. At that time, the course for medical auxiliaries at the School of Tropical Medicine in Brussels had not yet been instituted. *Fr. Kindt* brought with him his own microscope and a supply of Atoxyl®. He was an energetic and able assistant to the Drs Quartella and Schwetz in charge of the region. Correct diagnosis and specific treatment coupled with careful clearing of scrubland, marked the beginning of the decline of sleeping sickness in the region.

– In Northern Congo (Zaire)

The Norbertin Fathers of Tongerlo (Belgium) had participated actively and systematically in the control of sleeping sickness since 1908, along the Itimbiri, near its mouth at Bumba, on the way to Aketi and as far as Buta. Eight of their missionaries had been trained in Brussels, at the School of Tropical Medicine. In 1913, they spread the gospel among the living-dead; the rate of infection had reached 11.3% (997 out of 8,882). In 1914-1918, the rate had fallen to 2.1%, in 1919 to 0.6% and in 1930 to 0.005%. @The treatment incorporated the services, in the same village, of a male nurse, whose job was to give weekly arsenical injections to the sick, for a period of three months, to ensure a complete treatment.

Most of the missionary orders settled in endemic zones, made use of auxiliary staff who had received additional training and practical experience in the field; these qualifications helped considerably in the increasing success of the campaign against trypanosomiasis, at a time when professional staff were still too scarce.

7.1.4. Collaboration with private societies

Another variant in means of collaboration was negotiated in 1922 between the Forminière Company and the **Kwango Medical Mission**. An investigation carried out by *Dr Gillet*, in the territories of Tshikapa and Dibaya, where the company held mining concessions, discovered the existence of important foci of sleeping sickness with an endemic rate at Tshikapa of between 1% and 5%, reaching 10% in a focus at Dibaya. Faced with this situation, the Forminière Company decided to form its own **Medical Prophylactic Mission against Sleeping Sickness (MPMS)**. Its field of operation extended up to the region of Luluabourg (now Kananga), an area of manpower recruitment where there was not even a mobile medical service. The work method was the same as in the Kwango Medical Mission.

In 1925, an agreement between the Ministry of Colonies and the Forminière group was to allow the activities of its MPMS to stretch to the greatest part of the Kasai district. However, these activities were restricted, at first, to the years 1926 and 1927, then extended to the year 1928, when the **Dibaya Medical Mission** was able to resume its own programme.

Haveaux (1945), who reviewed these events, divided the 20 years of activity of the MPMS into three periods:

- 1923 to 1925, when all efforts were concentrated on the focus in the Batshok region near the frontier with Angola, where the global rate was found to be 1.9%, peaking at 10%;
- 1926 to 1931, when a sudden upsurge appeared, including the so-called *black zones*, reaching a rate of 10% of infected subjects in the area going from Kanda-Kanda, passing through Merode and Dibaya, and stretching along the railway as far as Luluabourg (see map 36, p. 1708);
- and the period 1932 to 1943 of well-ordered systematic activities, as described in the 1943 report, which also mentioned that out of 234,657 examined individuals, 136 were contaminated, a rate of 0.06%.

It was in the course of this activity that *Fourche* and *Ricklin* (1928), then *Fourche* and *Haveaux* (1930), tried out a preventive treatment with *Germanine*® or *Bayer 205*® (*Suramine*®), monitoring the infection rate. These trials led to the conclusion that the drug had a preventive effect, lasting six months. As a result of its application it became possible to advise other prophylactic measures, in a zone of high endemicity.

7.2. Legislation and recommendations to control the diseases

The Congo Free State (EIC) never had a thoroughly organized medical service. Physicians were appointed for specific assignments: to join expeditions and scientific missions or explorations, to provide medical care at major centres, to care for the labour force enrolled to build the railway, and to run the services of the laboratory at Leopoldville (which was a private enterprise). For instance *Ralph Leslie*, after a short stay with the AIC in 1883, came back in 1884 as physician for *Sir Francis de Winton* the Governor-General and was also head of the medical service of Nouveau-Vivi (capital).

As the medical service was poorly staffed, the administrators of the Congo Free State (EIC) had to solve the problem of trypanosomiasis, legislative measures and recommendations as the sole available means.

The year 1903 was to be a crucial one. As soon as the alarm had sounded, the Governor-General promulgated a decree, on May 5th, specifying that sleeping sickness was to be listed as one of the notifiable infectious epidemic diseases. This decree came into effect by law on August 26th. A circular prescribed all public health and hygiene measures to be taken in the event of an epidemic: isolation of the sick in places distant from population settings, a ban on sending back the sick to a non-infected areas, or to transport them in State-owned boats, and the obligation to clear thickets round the stations and villages and maintain them clean.

It was also in 1903, that *Castellani* discovered the trypanosome as the pathogenic agent of human sleeping sickness. Its identification would help to draw up a reliable diagnosis for the sick under observation, and to ascertain the prevalence of the disease in the communities. In the same year, *Broden* reported the first cases of sleeping sickness in Europeans.

The prevailing belief at the time was that sleeping sickness spread through infected subjects and thus their isolation was imperative. At the time the diagnosis was based solely on clinical observation of lethargy, every affected person had to be isolated at a distant place far from the post and the sick were not allowed to move around. At that time also, all bed-ridden, abandoned and dying persons were admitted into isolation lazarets of the religious missions.

In 1905 the Vice-Governor-General, relying on the report from the Dutton-Todd mission (1903), which emphasized the urgent need to take strong measures to stop the extension of sleeping sickness, had sent out a

circular strictly forbidding all infected Africans to travel, with the exception of any suspect to be transferred to the neighbouring medical centre and the sick on their way towards the lazaret or isolation lazaret. The prohibitory order of river transport effected also the passage of steamers.

On March 24th 1906, a circular was issued as a reminder that it was imperative to maintain a high standard of cleanliness in the villages. However, experience showed that clearance of scrub and undergrowth was practically impossible, even in administrative stations, through lack of manual labourers. Only the military camps and some of the missions were well kept.

In his letter dated June 3rd 1906, addressed to his general secretaries, the king stated that "no effort must be spared to control sleeping sickness", and he urged to apply without delay the recommendations put forward in the Dutton-Todd report. These were integrated in the directives of May 1903, and in the circulars of March 24th and August 24th 1906, and finally repeated in the rules edicted on December the 5th 1906.

They specified the most important measures:

- isolation of the sick in *lazarettos* (isolation hospitals) to be set up in infected regions (see p. 1723);
- organization of posts for observation to prevent further travel of infected cases on the edges of regions presumed uninfected;
- drainage, other health improvements and clearance of undergrowth around dwellings and huts in villages and around wood-depots for steamers;
- relocation of villages on a healthier place (hills or plateaus);
- a ban on maritime travel for the sick, unless their destination was a medical centre; in that case they were to be isolated on deck, under a mosquito net.

These mandatory measures showed that the people were aware of the different factors which caused the sickness: by that time it had become possible to recognize the infection during the initial phase, by detecting the parasites in blood or swollen cervical lymph nodes.

The role of glossinae in the transmission of the disease was well known, but not yet the transmission mechanism, hence the need to protect the population from the vectors by clearing the undergrowth of 50 metres around each dwelling, or by destroying the nests of pupae.

As *Fr. De Vos* rightly remarked in 1907, it was not enough for the Governor to prescribe immediate short-term measures, they had to be applied; but the work-

force available for scrub-clearing was too small, as was the number of tools at their disposal.

Moving villages from glossinae areas to places further away met with opposition from the population, whose habits were deeply rooted and who showed a marked preference for setting up their huts in low-lying ground or on swampy land, where palm trees grew in great profusion. It was also often quite difficult to find a suitable site for an isolation lazaret away from the glossinae: a place thought to be suitable in the dry season could well prove unsuitable in the rainy season.

On October 15th 1906 the General Secretary of the Department of Interior of the *EIC* (Congo Free State) at Brussels, *Mr Liebrechts*, officially opened in Belgium the School of Tropical Medicine, on which occasion he drew the attention of the public to sleeping sickness.

On October 18th 1908 the Belgian government took over the administration of the Congo Free State, and the *Colonial Charter* became the basic law for the country. On October 30th a new branch of the Government was formed: the Colonial Office. The Minister of Colonies enjoyed the same status and the same rights as those of his colleagues. On April 30th 1909 *Jules Renkin*, the first Minister of Colonies, set sail for the Congo to visit every region and take stock of the situation. In the medical sector, he decided to increase the manpower, but the recruitment of Belgian physicians remained low in number.

On April 29th 1909, an administrative order was issued at Leopoldville (Kinshasa), stating the limits of the infected and suspected zones and defining compulsory entry stations at every frontier; portage was forbidden in the Ubangi-Uele zone.

On September 9th 1909 a circular for the benefit of missionaries and other volunteers was sent out, giving technical instructions to help them in treating the sick. These instructions also dealt with the use of *Atoxyl*®. The circular stressed that strict isolation of the sick was to be observed in the first fortnight of the treatment; and that every infected subject should be moved to an isolation lazaret during the second stage.

On September 8th 1910 another statutory order promulgated a set of rules coordinating the measures to stop sleeping sickness with penalties for offenders. This was supplemented by a set of detailed rules calling for the full cooperation of the heads of missions, supervisors of trading posts and agricultural holdings, colonial staff and even native chiefs.

After specifying in detail the role of physicians and station supervisors to identify trypanosed subjects with the relevant actions, particularly the issue of a

medical travel permit, specific regulations were established: the running of lazarets, the precautions to exterminate the number of tsetse flies, the role of supervisors at the entry posts bordering infected regions, the transport of the sick by boat, and the treatment of trypanosomiasis by missionaries and others non-professionals.

The decree of March 22nd 1910 withdrew from the Katanga Special Committee (CSK) the executive power which had earlier been delegated to it and gave supreme authority for the government of Katanga territory (Shaba) to a senior official, a Vice-Governor-General. On September 21st 1910, this Vice-Governor-General issued an edict listing all the previous measures prescribed in the general order of September 1910 in Leopoldville, and appended to it a long list of specific measures to be applied in Katanga.

It goes without saying that the enforcement of these measures aimed to check the expansion of sleeping sickness had to face many difficulties throughout the territory. In the local context, most of the measures advocated remained a dead letter. The British Vice-Consul, G. B. Beach, fearing that the sickness might extend to Northern Rhodesia, remarked that the suggested measures were too complicated to be applicable and that they would not stop clandestine recruitment of workers from uncontrolled neighbouring territories. Dr Stohr claimed that illegal dealers were disregarding all rules.

More edicts followed in 1913 and 1914. A positive fact however was the appointment of physicians by the Government starting from 1911; they would have to develop an active campaign against sleeping sickness in the infected territories.

The ordinance of December 31st 1910 repeated and supplemented the earlier list of measures against sleeping sickness. It spelled out the role of the medical officers and their auxiliaries, and obliged them to follow a training in tropical diseases. It is remarkable that this ordinance called also upon non-medical organizations such as missions, trade companies and local chiefs, who would be provided with informations to detect disease carriers by their enlarged cervical lymph glands. Detailed and specific instructions were supplied to them on lazarets, observation centres, clearing undergrowth, control of travellers and suspects, sterilization of peripheral blood of infected subjects by injections of 0.5g of *Atoxyl*® (5ml of a 10% solution) and on issuing and checking medical passports.

In 1910, a map of the geographical distribution of sleeping sickness in the Belgian Congo was published and distributed at the occasion of the Brussels International Exhibition.

It highlighted the following facts:

- the new foci of sleeping sickness scattered from north to south in the eastern regions, especially in Kilo, Irumu, Beni, Upper Semliki, on the western bank of Lake Moero, and on the upper course of the Lukuga and Lufira;
- the existence of *many* cases at Aba in the Lado enclave, at Lake Edward, at the northern part of Lake Moero and on the west bank of the Luapula;
- the existence of *innumerable* cases along the western shore of Lake Tanganyika.

The magnitude of the work achieved may easily be drawn from the annual reports published in the medical journals of 1911 and 1912. These show the existence of 27 isolation lazarets with a total of about 3,500 sick, with an average of 130 in each lazaret, and there were seventeen observation centres with two physicians (see p. 1703) and 15 heads of centres *agents sanitaires* (sanitary officers), assisted by clerks. The clearing of the scrub was entrusted to 32 health teams, totalling 1,700 workers. However, many of the measures prescribed could not be fulfilled.

7.3. Organization of health services

The organization of health services was divided into three sections: isolation lazarets, observation centres and mobile treatment.

7.3.1. Isolation lazaretos

These lazarets have often been criticized, even recently (Lyons, 1988, Worboys, 1979). Some of the arguments pertained particularly to the definition of these refuges. According to Littré, they were "places where persons with all objects originating from places where epidemic contagious diseases prevail are held in isolation". There are different etymological meanings of the word *lazaretto*; it could be associated either with *lazzaretti* (hospices for leprosy patients) or with *lazzaro* (a beggar). Often they were places of quarantine. Anyway, their origin was unquestionably linked to places of shelter set up in missions, where abandoned and dying people were given refuge, out of humanitarian or evangelical mercy. The term was given a broader meaning when it became imperative to isolate people suffering from sleeping sickness as the only means to avoid contamination of healthy individuals and expansion of the disease.

Isolation lazaretos had been prescribed in the circular of August 24th 1906, specifying that they should be built in every infected area; the first ones were set up at Leopoldville (Kinshasa), Nouvelle Anvers (Makanza), and Lusambo. Observation and travel-control centres,

backed up by *lazarettos*, were established on the edges of unaffected areas, especially at Stanleyville (Kisangani) and Ibembo.

The idea of building lazarets was inspired by the desire to offer shelter, aid and assistance to the victims of this scourge, and more particularly to State-employed personnel. A first mistake however was not to make any distinction between the various stages of the development of this essentially chronic illness. Inmates who were in the first stage and still able-bodied were expected to undertake ordinary maintenance and cleaning jobs, on a voluntary basis. This arrangement was never accepted, as the patients felt unjustly interned. A second error was to allow only a short stay for the sick, on the erroneous premise that the drug *Atoxyl*® would achieve either total recovery, or at least a satisfactory outcome within a period of three, four, or maximum six weeks. Finally, these lazarets were not only unattractive for the sick, but also frightening, and ended up by being shelters for incurable, demented, abandoned or dying people.

The isolation of the sick in these *lazarettos* (also known by some as *hypnosaria*, isolation camps or hospitals, etc.), resulted from the widespread terror of becoming infected. This fear led the British authorities to contemplate isolating those infected with sleeping sickness on an island in Lake Victoria. This procedure was similar to the *quarantine* system introduced in 1377 at Ragusa, aimed at stopping plague from being brought in by ships which had come from infected places; they did not offer treatment at all.

The isolation lazarets which resulted from this historical experiment offered an approach which at best only allowed patients to be isolated, as already mentioned, until a peripheral sterilization of their blood and lymph had made them harmless to others. However, the people in charge were soon to be disillusioned, because the number of infected individuals was much larger than expected and the *Atoxyl*® did not guarantee the neutralization of the disease within three, four or six weeks as expected.

In the year 1907 *Rodhain*, who was in charge of the Leopoldville lazaret, was faced with the problem of overcrowding. He decided to let the patients move about freely, after an initial treatment with *Atoxyl*®, which ensured a peripheral sterilization, followed by treatment on fixed days. In 1908, *Trolli* observed that the number of admissions at the isolation lazaret at Stanleyville was higher than the discharges.

From 1911 and 1912 the Leopoldville lazaret became a centre of international reputation for its experiments with treatments and evaluation of their efficacy. It was and remained for many years the

place, *par excellence* to define the treatment schemes with *Atoxyl*®, tryparsamide, suramine, pentamidine and melarsoprol. The aim of isolation changed rapidly in havens for advanced, bed-ridden and demented cases, who were often abandoned by their families. These *death camps*, denigrated by many, nevertheless rendered great services to the patients with trypanosomiasis.

7.3.2. Observation, detection and containment centres

Observation centres were set up as sentinel posts at strategic points on the edges of infected zones. The most important ones were placed under the supervision of experienced physicians. Ibembo and Stanleyville (Kisangani) were assigned *Heiberg* and *Trolli* respectively both to be involved in the overall organization of the health services (see p. 106).

These posts came to provide some very useful information of valuable epidemiological interest, but they could not form an efficient barrier. Movement-control measures, so favoured by army physicians, were to prove useless in the African bush. Nevertheless these detection posts highlighted the risks entailed in moving personnel to new places or soldiers and their families to other garrisons, but also when demobilized the military personnel were sent back to their place of origin. The role of the porters in general, and especially of professional portage in commercial caravans was given particular attention.

Medical activities were supplemented by attempts to group small villages together into larger units, but this endeavour was often not followed by the people. The relocation of villages away from the glossinae-infected areas was recommended, but often not easy to apply.

7.3.3. Development of mobile services

Physicians, disheartened by the lack of success of the isolation lazarets, tried their best to come up with practical treatment procedures. The first solution was the development of a mobile system of treatment, which was tried out by *Rodhain* in Leopoldville in 1907, and by *Mouchet* and *Dubois*, at the wood depots for steamers on the Congo river in 1911.

Several physicians did their utmost in the field to organize mobile treatment units, some of them in conjunction with population censuses, others without. In 1913, two interesting projects were set in motion: one at Ibembo, on the river Itimbiri, downstream from Aketi (province of Upper Zaire), where *Dr Olivier*, through a systematic medical census of the populations in his sector, detected the number of infected subjects, and treated them either at the isolation

lazaret or at the general hospital. The other was at Kiambi, on the river Luvua (province of Katanga, Shaba), where *Dr. Lejeune* also detected his patients through a medical census but organized their treatment in the villages, according to a specific scheme whereby he entrusted to auxiliary medical aids the job of giving regularly the prescribed injections to the sick, under his supervision (see pp. 112 and 1707). In 1914, the war put an early end to the trials.

The year 1913 started with a special grant of 1,250,000 francs from the King of the Belgians, to be repeated in 1914 as an additional contribution to both ordinary and extraordinary budgets. The Ghent International Exhibition in 1913 allocated a substantial space to tropical diseases. The British section distributed a pamphlet published by HMSO (His Majesty's Stationery Office), in which Professor *J.W.W. Stephens* had written an article on sleeping sickness. The financial and advertisement efforts were a proof of the general determination to intensify the control of the disease. These good intentions and plans were also to be frozen by the 1914-1918 war, which saw a large number of the medical staff leave to join the forces.

In 1919-1920, activities began to resume systematically as soon as adequate medical personnel was again available and could be enrolled.

7.4. Medical missions for the control of sleeping sickness (see also pp. 1709 and 1711, 1719 to 1721)

The first medical mission was entrusted to the Liverpool School of Tropical Medicine (see pp. 100, 1689 and 1717). On July 8th 1920 two prophylactic missions were formed by ordinance, with a twofold purpose: first, to resume systematically the census and medical examinations to apply once more a mobile detection and treatment procedure with male medical auxiliaries trained in microscopy and also to administer injections; secondly, to define clearly the tasks of *agents sanitaires* (see p. 113) and of voluntary auxiliaries from religious missions.

The **Medical Mission of Kwango** was set up in this area where, on a preliminary visit, *Schwetz* (1919) had noticed an alarming infection rate of 6 to 10%. The procedure applied was palpation of the cervical lymph nodes, followed by treatment with *Atoxyl*®. *Schwetz*, assisted by five physicians and five *agents sanitaires*, managed to assemble, identify and examine more than 500,000 people.

The **Medical Mission of Uele** was set up to cater for the Uele region, under the direction of the physician *s'Heeren*, and examined 503,000 individuals. In those with enlarged lymph nodes, trypanosomiasis was

looked for by microscopic examination of the puncture fluid of the lymph nodes. The 12,305 positive cases (2.4%) were treated. The value of *s'Heeren's* report was enhanced by the constructive comments made by *Rodhain*, who was well aware of the problems in Uele.

In 1923 similar missions were set up among the Bangala, in Mayumbe and in four other regions: along the Bukama-Kindu stretch of the river, around Dibaya (East Kasai), in Kwango and the Kasai regions (Bandundu province) and around Faradje, the chief town of the territory on the river Dunga (north-eastern Congo, near Sudan).

The control of sleeping sickness was now well organized, well planned and in full operation. In 1930, FOREAMI was to improve the administrative procedure by making it more flexible. Finally this system was at the root of the future health organization of the Colony, which was to be called Zaire (see the Chapter Evolution of Health Services p. 118 et sqq.). *Mouchet* urged, in 1912, the importance of developing an independent vertically-run service.

Apart from the work of FOREAMI in zones where activities were integrated, the role played by this body in the control of *T. gambiense* trypanosomiasis in the Trustee territories of Rwanda and Burundi should also be mentioned. This assignment was committed to *Dr Prati* in 1932, and the mission covered the contaminated areas of Cyangugu, Bugarama, Rusizi, Usumbura, Rumonge and Nyanza. By this mission 2,319 trypanosed subjects, including 101 who had been infected outside the zone, were treated. After one year 683 cases were pronounced as being cured, 796 cases remained under observation without treatment, 219 cases continued to show positive infection and 78 cases had disappeared. The mission had moreover discovered 850 new cases. In 1933, 1,494 infected patients were treated, the average incidence being 1.33%. The inhabitants were resettled far from the glossinae and the focus died out (see also pp. 126 and 1779).

7.5. Treatment and prophylaxis

Specific drugs were gradually improved, without reaching however the required degree of efficacy or the desired speed of action. Furthermore, these drugs did not fulfill the requirements of a simple administration, nor did they provide the security required for large scale treatment.

The *Ehrlich* era contributed to develop substances acting on trypanosomes. The experimental model *mouse-trypanosoma* was a great help in the search for quick acting substances on the *Treponema pallidum* agent of syphilis. Syphilis was still, at the beginning

of the century, a major concern of the medical profession and *Treponema pallidum* too elusive to be used in routine investigations, quite the reverse of the trypanosoma, which was easily inoculated in mice and providing a steady infection curve. At the same time this model was useful for the selection of trypanocides.

Since Livingstone, via Broden, Mesnil, Laveran and others, arsenic had gained the reputation of being an active agent against trypanosomes. Indeed, the first active trypanocides were pentavalent arsenicates. *Atoxyl*® was to become the first in line after the experiments of Thomas (1903).

The exact formula of the product synthesized by Béchamp in 1863 was ascertained in 1907 by Ehrlich and Bertheim. *Atoxyl*®, which could be administered orally, but was preferably injected parenterally, had been the sole trypanocide available for two decades. In practice its efficacy was limited to the first phase of the sickness; Broden and Rodhain wrote in 1907 and confirmed in 1920 the success of the treatment reaching 73% in the initial stage but having 100% failures during the meningo-encephalitic period. The toxicity of *Atoxyl*® had a very deleterious effect on the optic nerve, provoking more than 3.5% cases of blindness.

Atoxyl® followed by Salvarsan and Neo-salvarsan which were active against the treponema, proved for a time to be useful on account of their quick peripheral trypanocide activity.

Trypanroth was also found active. If the doses needed were too large to be commonly used, the product became the starting point to the synthesis of suramine.

Tartar emetic, like various other organic antimonial derivatives which were later synthesized, showed a moderate activity and could be used as a back-up medicine.

Glyphenarsine or *tryparsamide*, a derivative of *Atoxyl*®, which had been synthesized by Jacobs and Heidelberg in 1917, was tested by Pearce and Van den Branden at the Leopoldville laboratory and at the isolation lazaret attached to it. This derivative penetrates into the meningeal barrier, modifies the nervous tissues, keeps the incurable cases in a satisfactory state for years, and presents a chance of recovery for the sick, detected in the second phase of their illness. Tryparsamide gave the possibility of not only working out a plan of intravenous weekly treatments, but also to design techniques to evaluate the treatment and to determine the criteria for a complete cure.

In the thirties, a problem of resistance to tryparsamide appeared gradually. This loss of activity was to be expected, as no medicine is 100% active on

parasites. In a population of parasites having different sensitivities, the drug destroys the most sensitive clones but not the less sensitive ones thus selecting the long run resistant strains. Furthermore, insufficient doses and/or irregular treatment may induce a state of chemo-resistance.

At about the same time, *Suramine*, or *Bayer 205*®, a complex derivative synthesized from urea and trypanoth by Dressel and Kothe (1917), was to prove an active trypanocide. This activity was observed in humans in Africa, by Kleine and Walraevens in Katanga. Suramine does not penetrate through the meningeal barrier and its use was limited to the early stages of the disease, but it had also a limited prophylactic activity when administered intravenously, protection lasting for only three to six months.

A new arsenious derivative, this time trivalent, *Melarsen oxyde*, was synthesized by Friedheim in 1947 from melaminy and was transformed into *Melarsoprol* by a combination with dimercaprol or BAL (British Anti-Lewisite), in order to counterbalance the toxicity of the trivalent arsenical derivative: it was named *Arsobal*®. As was to be expected, this arsenical was very active during the second phase, but also very toxic. *Arsobal*® was easier to handle and is still the main drug in the campaigns to control trypanosomiasis. This drug carries the risk of some rather unpleasant side-effects, such as the *arsobalian storm* in the cerebro-spinal fluid, which complicates any therapeutic evaluation. Unfortunately, resistance to the product has appeared and yet other complications as encephalopathies can lead to a fatal outcome. In advanced cases one can only take the risk and trust in Providence.

Diamidines became available at the same time as the previous arsenical drug. Almost all the hypoglycaemic effect had been eliminated but the trypanocide power of synthaline from which they derived was retained. Pentamidine or *Lomidine*® was the derivative to remain in use. It was effective only during the sanguino-lymphatic phase, but was long-acting because it was retained in the tissues.

This property was used as an advantage in promising prophylactic campaigns, which had however to be stopped because they tended to camouflage the disease and thereby delayed the diagnosis until the meningo-encephalitic stage when the prognosis was very unfavourable. Furthermore, prophylactic campaigns tended to bring about a primary resistance (Kayembe and Wery, 1972). Henrard and Peel (1950) had already observed a primary resistance of strains which had never been under the influence of this drug.

Diminazene aceturate, or *Berenil*®, turned out to be a very active trypanocide as well as against *T. gambiense* as *T. rhodesiense* trypanosomiasis. However, the firm that produced it refused to supply it for medical purposes, because it had produced cerebral haemorrhage in dogs.

Such a decision caused two problems:

- on the one hand, *Berenil*® had stopped any further research on trypanocides as it successfully fulfilled the needs of veterinary medicine, by ensuring the economic profitability of cattle raising. As in other fields in need of new parasiticides, the interest of veterinarians accounts for the decision-making of the expensive research as cattle herds do have an economic value, while this does not apply for human beings.

- on the other hand the question must be raised if it is necessary to apply the very strict current criteria to approve medicines treating deadly diseases as for those relieving minor ailments. The only drugs available to cure trypanosomiasis are all dangerous ones; the active and less dangerous *Berenil*®, which could have been used to great advantage, was classified as not acceptable by the authorities.

The success expected from prophylactic aromatic diamidines in the fifties also slowed down research of pharmaceutical industries, not only because it was becoming increasingly expensive, but also because it dealt with a disease which affected a limited number of people in countries with precarious macro-economics. Therefore the international community should take over the funding of research aiming at the relief of disease threatened populations, unable to afford the costs of drugs becoming increasingly expensive due to the new technologies used by the pharmaceutical industry.

The nitrofurazones were then added to the collection of available drugs. In this series Levo-furalta-done appeared, and was very useful for treating patients where trypanosomes had become resistant to melarsoprol. The dose and the length of the treatment restricted the use of nitrofurazones to hospitalized persons only; this was the case, also, for *Nifurtimox* or *Lampit*®.

DFMO or alpha-di-fluoromethyl-ornithine is a new acquisition of particular interest. It is a specific and irreversible inhibitor of ornithine-decarboxylase and ornithine is a precursor for the biosynthesis of essential polyamines. There is very good tolerance to the product which is also active in the second phase. Derivatives which are easier to handle in mass treatment are expected. DFMO (*Eflornithin*®) is moreover active against trypanosoma strains resistant to melarsoprol; it shows also a synergy with pentamidines. Furthermore, it is a prototype drug whose effect inhibits an enzyme essential for trypanosomes, but absent in man. The fact that DFMO was not specifically synthesized to act against trypanosomes is a reminder that research on various active chemical radicals, remains a useful line to follow as it is more and more specifically directed.

Glycosomes could perhaps be another basis on which to develop new active products against parasites.

The need for new products to be used in mass treatment remains very pressing. Whatever the opinions might be, the discovery of a substance meeting the current application criteria will bring victory against sleeping sickness within reach.

STATE OF THE ART

AGENT, VECTOR AND EPIDEMIOLOGY

1. Fundamental elements of the disease

Until the early twentieth century the prevailing ideas about the distribution and origin of *African lethargy* or sleeping sickness were hardly meeting the reality. Knowledge and even perception of the aetiological factors, and epidemiology of the endemic and epidemic forms of sleeping sickness, were rudimentary. The physicians who first observed the disease had not the slightest idea of a chain of transmission. In order to develop the correct treatments and preventive measures, it was necessary to know the pathogen, its vector, the mode of transmission and the peculiarities of the host.

Except the unnoticed febrile phase at the disease's onset, the only known basic information was the cluster of symptoms collated by physicians in a clinical picture, particularly in the West Indies. Later in this chapter the additional elements still to be discovered thanks to the development of more advanced technologies will be mentioned (see pp. 1759 to 1762).

1.1. Pathogenic agent: trypanosomes

Trypanosomes are haemoflagellates whose palaeogenetic origins may have been either plant or animal. They appeared very early in the evolution of species and there is reason to believe that their parasitic role

in vertebrates preceded the development of the hominids (Lambrecht, 1980).

1.1.1. Phylogeny

The protozoa are living beings which are rather difficult to classify, and the trypanosomes are no exception in this composite set. Their history since their discovery by *Gruby* (1843) is rather confused. *Laveran* and *Blanchard* (1895) considered them to be varieties coming from the internal milieu of different hosts. According to *Alexeieff* (1913), the various forms of trypanosomes present in mammals were able to transmute from one to another. *Laveran* and *Mesnil* (1912) divided them into pathogenic and non-pathogenic species. *Roubaud* (1913) and *Duke* (1913) suggested separating them into three groups according to their vector, whereas *Wenyon* (1926) was content with two of the subdivisions.

The current terminology (*Hoare-Wallace*, 1966) recognizes two fundamental genera, one monogenetic and the other digenetic. The monogenetic genus contains all flagellates of the insect-gut transmitted by the faecal route (leptomonas, herpetomonas, blastocrithidia, and crithidia). The digenetic genus passes through two hosts in alternation. Thus, the blood-dwelling trypanosome is characterized by the trypomastigote form, with its kinetoplast situated behind the nucleus in the posterior position. In insects it occurs in the promastigote form, in epimastigote ones with a juxta-nuclear kinetoplast, and amastigote form (characteristic of the genus *Blastocrithidia*; it seems to have originated from monogenetic blastocrithidia of insects, as it passes through a very similar development.

Their degree of kinship, which is based on phylogenetic evidence, makes it possible to understand correctly the natural history of trypanosomes. Detailed identification of their antigenic and enzymatic constitution gives the means to clarify the physiological and biochemical similarities and affinities. These characteristics in turn make it possible to discover diagnostic clues, to screen for sensitivity to drugs and to investigate host parasite relationships. It is therefore sometimes possible to pass around the disconcerting problem of the countless antigenic variants and their endless changes.

The Trypanosomatidae family consists of microscopic living organisms devoid of mineral skeleton; therefore they did not leave any fossilized traces and have no traceable forerunners. We are thus reduced to speculate about their filiation without any possibility to prove it (*Minchin*, 1908). As was to be expected many, often contradictory, opinions have been expressed (*J.R. Baker*, 1963).

The origin of the Trypanosomatidae might be imagined in different ways:

1° they may have originated from biflagellate trypanosomes which were blood-dwelling parasites of cold-blooded animals transmitted by leeches to which transovarian transmission was attributed, while it merely resulted from cannibalism;

2° they may have developed from trypanosomes living in the intestines of vertebrates which entered the blood stream of their hosts, from where haematophagous arthropods took them up, making their intra-intestinal development superfluous;

3° they may have evolved from uniflagellate leptomonas, parasites of the intestine of invertebrates, which might have adapted to the localization in the stinging apparatus.

All three hypotheses have their proponents. Nevertheless, it seems completely logical to conjecture that there might have been a common origin, followed by a division into two groups when the annelids (leeches) and arthropods became separated. This would then reduce the possibilities to two, namely:

- the hypothesis advocated by *Léger* (1904), who affirmed that they stemmed from monogenetic blastocrithidia. According to this theory the blastocrithidia, which first occurred in non-haematophagous insects, adapted to the vertebrates by acquiring the haematophagous potential to feed on blood. This version acquired the most supporters (*Brumpt*, 1908; *Roubaud*, 1909; *Laveran* and *Mesnil*, 1912; *Hoare*, 1925-1967; *Grassé*, 1952; *Taylor*, 1965). *Grassé* regarded this as the most plausible version.

- the other hypothesis defended by *Minchin* (1908), *Mesnil* (1908) and *Lavier* (1943), who were of the opinion that the invertebrate gut was nothing more than a culture medium in which the flagellates made their way to the circulatory system and from there were picked up by blood-sucking insects.

The first scenario places the trypanosomes' ancestral forms in the invertebrates. These forms were thought to have changed their dietary habits by passing from plant juice to animal blood. In the arthropod vector the blood meals contribute to the maturation of fertilized eggs.

This thesis is supported by valid arguments of analogy provided by the Tabanidae, which were originally feeding on plants (phytophagous) and later haematophagous (deriving nourishment from blood), and also by the very closely related Pangonias, which are habitually phytophagous but occasionally haematophagous. So much the more that males of various species have remained phytophagous whereas the

females have become haematophagous. Moreover, the pathogenicity of trypanosomes for vertebrates and the lack of pathogenicity for invertebrates is rather remarkable, even though contact with the latter has been much longer, throughout millions of years.

1.1.2. *Ethological adaptation*

The trypanosomes of interest because of their role in human and animal trypanosomiasis, belong to the genus *Trypanosoma* of the family Trypanosomatidae and the order Kinetoplastidae. These parasites are a very homogeneous group of blood-dwellers whose morphological forms are somewhat similar for animal or human hosts. At the same time, their life cycles are variable and take place in the gut of the invertebrate vector which, in the species under consideration, is an arthropod.

This life cycle enables the various species to be differentiated from each other by the site of their infective forms.

- When they are located in the terminal part of the intestine and thus released by defecation we speak of the **stercoraria** group, referring to transmission via the hindgut. An example is *T. lewisi*, the rat trypanosome.
- In contrast, the infective forms of the tsetse-transmitted African trypanosomes develop in the proboscis or salivary glands, and we speak of forward location. This group is therefore known as the **salivaria** group, referring to transmission via the saliva. This mode of transmission recalls how the leech has previously taken part in the life cycle of the trypanosome.

It is generally accepted that the salivaria group is of more recent origin. The reasons for this are:

- their transformation from the trypanosome stage into crithidia is preceded by division in the midgut;
- these trypanosomes have not yet developed any particular specificity in the choice of hosts for which they are most pathogenic. Since they are not strict parasites of a specific species, they can still adapt to many animals both in nature and in the laboratory;
- finally, their infective power for the vector is usually limited, around a few percent, whereas the infective power of the stercoraria trypanosomes is close to 100%.

The subgenus *Trypanozoon* includes trypanosomes with salivaria-transmission responsible both for human and animal trypanosomiasis. These parasites undergo a two-stage development in the tsetse flies, one in the gut and the other in the salivary gland. This particular feature leads to no morphological differentiation which could allow the members of the species *Trypanosoma brucei* to be distinguished morphologically. These trypanosomes are larger than erythrocytes, but only their short or cylindrical forms (sausage forms) are infectious for the tsetse flies.

This homogenous group has adapted to transmission by tsetse flies. It is subdivided into three types based on the location of the infective form in the proboscis:

- for *Trypanosoma vivax* the transmission is nearest to mechanical transmission;
- for *Trypanosoma congolense* the parasite is located in the mouth of the fly;
- for *T. brucei* the infective form is found in the salivary glands.

The group composed of *T. vivax*, *T. congolense*, and *T. brucei* may be differentiated from the others by a less restricted choice of their host. *T. vivax* is considered the oldest strain, followed by *T. brucei*, *T. congolense* and also *T. gambiense*, which is thought to be the most recent of the species.

The trypanosomes reproduce asexually. Their reproduction occurs in the blood and proceeds with trypanolytic phases, or crises. Genetic mutations or chromosomal abnormalities result in a heterogeneous population. The variants are selected by evolutionary pressure and/or environmental factors. Because hybridization of the zymodemes was present, the existence of sexual reproduction was suspected; that is why *Vickerman* (1986) spoke of the trypanosomes' hidden sex. But the chromosomal division that was observed in the salivary glands was founded on a technical error. However, this hybridization process might explain a number of important problems, such as the idea that these epimastigotes might undergo this process in the tsetse's digestive tract.

The subgenus *Trypanozoon* includes *T. brucei* (pathogenic for cattle), *T. b. rhodesiense* (pathogenic for humans, ungulates and carnivores), and *T. b. gambiense* (pathogenic for humans only). From this very specific choice of hosts arose the question of the means by which their differentiation had occurred.

The simplest hypothesis is based on the presence of *T. b. brucei* across the entire tropical Africa, from east to west. This form may be considered the original one, which did not infect humans and was well tolerated by both game and cattle. Its low pathogenicity may indicate lengthy contact with these species. According to this hypothesis, *T. brucei* might have given rise, in the course of geological periods, to mutants or variants that settled in the two basic biotopes, namely the forest and the savannah, and adapted to the vectors which they found there.

Only when humans appeared on the scene *T. brucei* would have adapted to them – which seems to be what has happened, according to the current opinion, in the savannah where climatic conditions must have been very similar to what they are now. On the basis of this conception, *Lambrecht* (1980) hypothesizes that the

precursors of *Homo sapiens* were exposed to the same infections in the savannah as those found there today. The parasitic diseases whose reservoirs were animals and the vectors winged insects must have jeopardized the very existence of man, while the resulting fatalities would have selected only the most resistant subjects. Similarly, forms of trypanosomiasis that were lethal for animals may have spared only the fittest, so that only the species that acquired tolerance to the trypanosomes survived.

The elimination of receptor animals may have opened ecological niches for the primates when they left their arboreal habitat.

The first contacts in the savannah would probably have taken place at the end of the tertiary period, during the Lower Pleistocene. Before, during the Middle Pleistocene, trypanosomes belonging to the *T. rhodesiense* type would have become parasites of the hominids who slowly took over the forest. During the Upper Pleistocene, *T. rhodesiense* would have persisted in animals and occasionally in savannah-dwelling human beings. In the forests the trypanosomes would have adapted both to the hygrophilic tsetse and to man, becoming transformed into *T. gambiense*. Although not illogical, such an evolutionary path runs against what happens most often.

T. b. gambiense is in any case the variant that adapted to man in the forest biotope, where contact with human beings via members of the *Glossina palpalis* group was occasional but regular. The slow reproduction of this variant and its low virulence are linking up together with high infective power: inoculation with just a few trypanosomes (350 metacyclic forms per tsetse bite or a dozen injected by needle) suffices. The infestation develops slowly in the host because of the small number of parasites in the peripheral blood, which are moreover subject to trypanolysis. In summary, the density of infection is generally low and the development slow, but under certain undefined circumstances the evolution of *T. b. gambiense* may become very active and produce a large number of trypanosomes in the blood.

For *T. b. rhodesiense*, there is an obvious contact with man via the *morsitans* group of tsetse, but it is occasional and irregular; on the other hand receptor animals are always available: the disease is thus a zoonosis. *T. rhodesiense* is found together with *T. brucei*, but not in all the latter's territories. Therefore, hundreds of thousands of people have travelled in caravans from Tabora to Ujiji or by rail from Tabora to Kigoma, which are vast regions infested by *G. morsitans* and *T. brucei*, without one case of *T. rhodesiense* trypanosomiasis occurring among the travellers.

Simultaneous infection with *T. b. brucei* and *T. b. rhodesiense* in the same host is unknown (Gray).

T. b. rhodesiense multiplies rapidly and is usually present in large numbers in the peripheral blood. Like *T. b. gambiense*, it exhibits a wide range of virulence, depending on the circumstances. Both strains also exhibit great variability in their sensitivity to drugs, as shown by resistance to organic arsenic derivatives.

Unlike *T. equiperdum*, which is responsible for dourine in the Equidae, human trypanosomes are not transmitted by sexual intercourse.

1.1.3. Identification of the trypanosome

Trypanosomes were described in the blood of cold-blooded animals as early as 1840 – in a frog by Valentin (at Berne, Switzerland) and in a trout by Gruby (in Paris). *T. lewisi* was found in a rat in 1878. The first pathogenic trypanosome was discovered in 1888 by Griffith Evans in the blood of horses having surra disease and was later named *T. evansi*. The second pathogenic species, *T. equiperdum*, causes dourine (1894). Their discovery was followed shortly thereafter by the discovery in Zululand, by Bruce (1895) of *T. brucei*, the cause of nagana.

Advances in microbiology created the necessary basic conditions to identify pathogens. Despite the sudden spread of sleeping sickness, it was not until 1901 that Forde attributed an antimalarial-drug-resistant fever found in Europeans to a blood-dwelling vermiculate agent. These vermicules were identified by Dutton, in Liverpool (1902), as trypanosomes, which he named *Trypanosoma gambiense*.

Other authors confirmed the discovery. One of them was Aldo Castellani (1903), who on November 12, 1902 revealed their presence in the CSF of a young African with sleeping sickness in Entebbe (Uganda). He made the same observation in several other sleeping sickness patients in the months that followed and named the parasite *Trypanosoma ugandense*. Bruce and Nabarro (1903) detected these same parasites in blood samples from 70% of the observed cases of sleeping sickness and confirmed their absence from the blood of healthy subjects. Nabarro's later research showed that *T. ugandense* and *T. gambiense* were identical. Sleeping sickness was thus merely the last phase of Forde and Dutton's febrile trypanosomiasis.

A few years later, in 1909, another human trypanosome was discovered in another environment. J. Stephens and H. Fantham (1910) noticed a multitude of trypanosomes in the blood of an European prospector named William Armstrong. Armstrong had apparently become infected during an assignment that had taken him from Salisbury (present-day Harare) to

Fort Jameson via the valley of the Luangwa, a sub-tributary of the Zambezi river. These trypanosomes were isolated in Liverpool, where they were found to belong to a different strain of *T. gambiense*, characterized by the posterior position of the nucleus, the ease with which they infected laboratory animals, and their resistance to arsenic compounds. The new strain was named *T. rhodesiense*. These same trypanosomes were found in the Makanga area, in Mozambique, in 1910.

This discovery shows the confusion that reigned between the time of the identification of a micro-organism and the discovery of its role in an infectious disease. The natives had known the disease for generations, but the physicians had chosen to ignore the information they gave.

The origin of Africa's human trypanosomes and their relationships with each other and with related animal trypanosomes are currently under research, which has in many ways a fundamental importance.

It is supposed that over the centuries *T. brucei brucei*, an animal parasite affecting cattle, horses and so on, mutated to produce *T. b. rhodesiense*, a parasite of humans with an animal reservoir, and later *T. b. gambiense*, the more specific human parasite. Lambrecht's hypothesis postulating the inverse chain of events is just as plausible. Anyway, these strains are highly specific. The hundreds of inoculation trials that were carried out to inoculate human volunteers with *T. b. brucei* from either animal blood or laboratory-infected tsetse flies gave only one, rather fleeting, positive result (Van Hoof, 1947). Moreover, Rickman and Robson conducted a Blood Incubation Infectivity Test (BIIT) based on the trypanocidal action of human serum against *T. brucei*, in order to differentiate *T. brucei* from *T. rhodesiense*.

In contrast, *T. rhodesiense* that had been maintained in animals for more than twenty years did not lose its infectiousness for man, that is to say it did not revert to the *T. b. brucei* form (Tinde experiment, strain isolated in man from laboratory-raised *G. morsitans* in October 1934 and maintained by Corson for 23 years, starting in 1936, by 120 cyclic transmissions to sheep (Ashcroft, 1959).

These salivaria parasites reproduce asexually, which gives rise to a heterogeneous population. The eukaryotes must have undergone at one time a sexual reproduction to achieve the level of cellular complexity that they have; yet to maintain this complexity they apparently must rely only on asexual reproduction, although as Vickerman claims, they might have an undetected sexual stage.

- As soon as the trypanosome had been identified as the pathogenic agent of sleeping sickness, physicians

and research workers were intrigued by the *alternation of positive and negative phases of parasitaemia*. Today we know that the sequence of outbreaks and remission is due to the trypanosomes' ability to adapt in the course of its life cycle to very different environments, as well to the intestine, the coeloma and the salivary glands of the tsetse, but also to the lymph and blood of the host. Since the trypanosomes divide very fast (in six hours) the host is quickly submitted to an intolerable overload of parasites without having time to produce antibodies. However, this overload does not reach the same heights for *T. gambiense* and *T. rhodesiense* and, in any event, triggers lysis of the parasite.

- By covering their surfaces with a glycoprotein coat the trypanosomes *protect themselves* from the host's defence mechanisms. The host will then in turn produce the right antibodies to break through the barrier. The fact that trypanosome populations are heterogeneous ensures their survival: from the few trypanosomes that withstand the host's immune defences, a new population will spawn.

The protective mechanism results from the production by the trypanosome of a series of *variable surface antigens* (VSA). The antigenic glyco-proteins are composed of a number of amino-acids, whose sequence is genetically determined, each VSA being coded by a specific gene (Steinert et al., 1987): out of a great number of genes only one is activated when required and induces the synthesis of a unique sequence of amino-acids in the cytoplasm. These variable antigens protect the trypanosomes against the antibodies produced by the host against the previous sequences. The successive populations or types are called variable antigen type (VAT) and, as a whole, a variable antigen repertoire (VAR).

The production of variable antigens stops as soon as the trypanosome enters the tsetse, then this production starts again, but at lower level in the metacyclic form: this fact raises the hope that it will become possible to produce a vaccine.

In the metabolism of the trypanosomes in the bloodstream glucose is the main element. Its favourable effect on the survival *in vitro* had already been known for several years when Yorke et al. (1929) established their method for maintaining pathogenic trypanosomes alive *in vitro* at 37° C for 24hrs. Trypanosome infections in various laboratory animals had been shown to be associated with host hypoglycaemia more particularly in the agonal stage. Later on it will be shown that the glycolytic enzymes are present by the trypomastigotes in glycosomes, cytoplasmatic organelles, and produce pyruvic acid, while the

enzymes of the Krebs cycle are deficient due to an atrophy of the mitochondrion in the trypanosomes. These facts led to a possible connection between hypoglycaemia and the high rate of glycolysis by the parasite in the host and the likelihood that the introduction of glycaemia by insulin in infected animals might be a possible way to control trypanosome infections. Jancsó and Jancsó (1934) tried the therapeutic effect of Synthalin, an antidiabetic agent, on experimental trypanosome infections. Loury and Yorke (1939) confirmed the activity of this aliphatic diguanide and also of several related drugs, which ended up by the selection of aromatic diamidines by King, Lourie and Yorke (1937).

T. gambiense, which has a vast ecological and geographical range, has become firmly installed in areas (or ecotopes, which means an association of various biotopes, e.g. savannah and forest mosaic), with particularly favourable ecological conditions. The clones which have developed in those niches are maintained by a stable natural cycle of asexual reproduction. Consequently there must be vast quantities of different clones and each focus may have its own clone that evolved over very long periods without outside contributions. The identification of such clones by means of iso-enzymatic markers will bring strength to the hypothesis of the variability and persistence of the clones. Molecular analysis of strains from different geographical areas is essential to reach firm conclusions.

The existence of sexual recombination has been advanced by Tait (1980) and confirmed by Jenni et al. (1986), Paindavoine et al. (1986), and Zampetti-Bosseler et al. (1986). If such recombination is proved, one may wonder to what extent the introduction of a new strain might trigger this reproductive mode. It is possible that zymodemes which are detected in human epidemics result from exchanges of genetic material between previously existing zymodemes, which implies a population of parasites belonging to a specific iso-enzymatic type.

1.2. The tsetse flies

For a long time the role of the tsetse flies in African sleeping sickness has remained unknown, even though the part played by this insect in the deadly equine and cattle epidemics had not escaped the notice of the natives. The truth on the parasite's transmission was proven by Taute and Kleine (1909) by analogy with the transmission of *nagana* among horses, as observed by Bruce.

1.2.1. Phylogeny

The oldest tsetse flies are known from fossil impressions discovered at Florissant, southwest of Denver, Colorado (USA), in layers of volcanic ash dating from the Oligocene Era (Tertiary Period) and described by Cockerell (1907-1909). This discovery occurred by chance and fortunately did not go unnoticed. The well-preserved impression of the spread wings allowed their identification. Cockerell was able to distinguish four species, *Glossina oligocena*, *G. osborni*, *G. veterna*, and *G. armatipes*. Doubts that continued to worry certain palaeoentomologists were dispelled when Machado (1959) confirmed the identification.

Various specialists did their best to find the meaning of this discovery, which obliged them to look back farther than the Quaternary Period. Potts (1937) had already tried to explain the existing distribution of tsetse flies based on climatic changes in the Pleistocene. Evens (1953) and Lambrecht (1980) formulated hypotheses based on continental drift. These trials stress how difficult it is to discuss events that occurred some 30 or 40 million years ago (when paraphrasing Buxton, 1955). It would seem prudent to stick to a simple interpretation based on the ideas of Jeannel (1961) and on those of Evens after him (1965).

According to Jeannel, Gondwana – which is, according to certain geologists, a former supercontinent comprising Africa, Madagascar, America, Antarctica, Australia, and the Indian peninsula, but for other geologists a myth comparable to the continental drift from which the Gondwana hypothesis is derived – played the role of an asylum for tsetse flies.

During the Secondary or Mesozoic Era the primitive tsetse lines experienced a Gondwanian drive that lasted until the beginning of the Tertiary Era and spread these lines to the northern hemisphere. According to this hypothesis, the gradual cooling of the northern hemisphere that culminated in the Ice Age was the main cause for the disappearance of the vast majority of species that had emigrated to the north; this Ice Age had pushed Gondwana's fauna to retreat more or less within the boundaries of their original range, tropical Africa.

From this opinion the tsetse flies may be considered to be a Gondwanian line of descent that spread during the Mesozoic Era. Traces of this swarming may have been left in Oligocene fossils, which sedimented in a northern area, whereas the line itself retreated to its original realm in tropical Africa.

It is very possible that similar fossils will be discovered on other continents, including Africa. However, it seems unlikely that they should be found in South America, for the climatological and ecological conditions necessary for the survival of Glossinae were not as favourable as in Africa.

The purely African distribution of tsetse flies is thus the result of a chain of geological phenomena stretching back to the Oligocene, from which period their presence was first ascertained. The tsetses were subjected to great climatic variations, particularly those of Late Pleistocene (Quaternary Period). Africa has known four rainy periods corresponding to the Ice Ages in the northern hemisphere during which the mean temperature was lowered by 5°C and the altitude at which the tsetses could survive declined from 1,500 metres to 700 and even to 500.

1.2.2. Association trypanosomes – tsetse flies

How *Glossina* flies encountered trypanosomes and how the latter adapted to the diptera has also given rise to various hypotheses. It is logical to consider this conjunction in the context of phylogenetic evolution, bearing in mind that the trypanosomes are two-stage protozoa characterized by a specific phase known as tripomastigote (Hoare, 1948).

As mentioned above (pp. 1728-1729), it is generally accepted that the trypanosomes have split into two homogeneous groups at the time the Annelids branched off from the Arthropods. One group was transmitted by leeches and is a parasite of fish, amphibians and water snakes; the other was spread by arthropods and is a parasite of ophidians, crocodilians, birds, and mammals. The stercoraria group, whose development occurs in the digestive tract, is the most numerous one. The salivaria group, which includes the trypanosomes that are pathogenic for man, have an initial intestinal developmental stage, followed by migration to the salivary system, where they complete their transformation into infective forms.

Two hypotheses have been formulated to explain what led the tsetse flies to become the vectors of trypanosomes in the course of their evolution:

- Hoare (1948) claims that the process began with wild ruminants which had been infected with trypanosomes by arthropods, either by the stercorial (or posterior route) or by mechanical inoculation. The bloodsucking tsetse flies should have become infected from these animals while the trypanosomes adapted to their new vectors.

- Baker (1963) hypothesized that the trypanosomes found in aquatic vertebrates were transmitted by leeches (anterior route), who succeeded in colonizing terrestrial vertebrates before the Annelids split from the Arthropods; the trypanosomes would then have separated into two groups, one transmitted by stinging, (which is the anterior route), the other by anal or the posterior route.

Proof of the validity of one or the other of these hypotheses has never been found, but the second seems the more likely. Grassé (1952) elaborated on this as follows: the leech-borne trypanosoma of amphibians, fish and aquatic reptiles could have been ingested by insects that occasionally took blood meals from aquatic vertebrates transmitted by leeches (trypanosome has been discovered in frogs). The parasites managed to adapt to the insect and were eventually transferred to terrestrial mammals or birds during other blood meals. These opinions tie in with the phylogenetic interpretations of Léger (1904) and Minchin (1914).

1.2.3. Ethological adaptation

There are 22 species of tsetses. Their distribution over their area closely matches the favourable macroclimates, for which the essential factors are adequate plant cover, other well-defined macroclimatic features, and a source of blood supplied by vertebrate hosts. Their habitats are determined by very specific microclimatic parameters, such as temperature, humidity, rainfall, wind, light, shade, etc. These conditions are not identical for the pupae, the teneral flies or the adults. The pupae require locations with appropriate substrates, whereas the adults spend a good part of their lives on the vegetation. This set of microclimatic parameters constitutes the prototype of landscape epidemiology.

Some fifteen tsetse species are found in Central Africa. They are divided into three subgenera or groups, namely, *Austenina* or the *Fusca* group, *Nemorhina* or the *Palpalis* group, and *Glossina* stricto sensu or the *Morsitans* group.

The fusca group, which occupies within the virgin forest a larger peripheral area and a few pockets, plays no role in the transmission of sleeping sickness. In contrast, the other two groups or subgenera are involved in the transmission of human trypanosomiasis. They may be subdivided very roughly but usefully on the basis of their bio-ecologies into hygrophilic and xerophilic species (attracted to humid or dry conditions, respectively).

The *hygrophilic* group (*Nemorhina*) includes various subgroups and subdivisions (taxons). Its members live close to water in the mesophilic forest, which means, a forest including undergrowth of bushes and similar dense vegetation. It consists of the following species:

1. *Glossina palpalis*, including the subspecies *G. p. palpalis* that is found in Zaire and lives in mangroves. There is also *G. pallicera*, with the subspecies *G. newsteadi*.

2. *G. fuscipes*, which is represented by three subspecies:

- *G. f. fuscipes*, who live on the edges of the great equatorial forest of the Central Basin and its extensions, in the forest corridors, and along the river banks;
- *G. f. quanzensis*, which is present in the forest corridors in part of the Lower Congo and Kasai;

- *G. f. martinii*, which is found in the Lake Tanganyika area, where it often occurs in banana plantations (Leleup, 1952).

The hygrophilic tsetse require shade and moist warmth: they live close to water and are spread over very great expanses.

The *xerophilic* group comprises five species and three subspecies, two of which are important in Central Africa, namely,

- *G. morsitans*, the tsetse of the wooded savannah, sparsely wooded areas and areas with isolated thickets of trees, and the reforested savannah that is frequented by large game animals; and
- *G. pallidipes*, the tsetse of the thickets, dense vegetation, and copses in which bushbucks and striped antelopes are living.

Between these two extremes there is the whole gamut of transitional forms.

TRYPANOSOMES: Classification of human salivaria trypanosomes

Family	Genus	Subgenus	Species	Subspecies
Trypanosomatidae	Trypanosoma	Trypanozoon	<i>T. brucei</i>	<i>T. b. brucei</i> <i>T. b. rhodesiense</i> <i>T. b. gambiense</i>

GLOSSINA: classification

Family	Genus	Subgenus (Group)	Species	Subspecies
Glossinidae	<i>Glossina</i>			
Hygrophilic		<i>Nemorhina</i> (Palpalis)	<i>G. palpalis</i> <i>G. fuscipes</i>	<i>G. p. palpalis</i> <i>G. f. fuscipes</i> <i>G. f. martinii</i> <i>G. f. quanzensis</i>
Xerophilic		<i>Glossina</i> (Morsitans)	<i>G. pallicera</i> <i>G. morsitans</i> <i>G. pallidipes</i> <i>G. longipalpis</i>	<i>G. p. newsteadi</i> <i>G. m. centralis</i> <i>G. m. submorsitans</i>
		<i>Austenina</i> (Fusca)		

The tsetse flies are able to adapt to changing environmental conditions. Leleup has seen *G. palpalis* var. *martinii* (currently *G. fuscipes martinii*) in Rusizi banana groves, habitats in which they take refuge. The spread of cash-crop farming, intensive deforestation, and other results of economic development have forced the tsetse flies to adapt to what remains of their disturbed habitat. Tsetse flies live close to water, in permanent vegetation, and are scattered over wide areas. Their contacts with human beings are occasional and dependent on human behaviour. *G. tachinoides* lives in the forest corridor galleries and in low vegetation adapted to the heat of the Sahel.

The *palpalis* group transmits *T. gambiense* and the *morsitans* group transmits *T. rhodesiense*. This distinction is not absolute, for *G. fuscipes* has become the vector of *T. rhodesiense* in Uganda because of the new ecological conditions offered by the fenced areas (*rugos*) of *Lantana camara* vegetation around dwellings. Moreover, this possibility was foreshadowed in the laboratory, on account of the fact that climatic factors are prevailing over a specific vegetation.

1.2.4. Identification as vector

In Zululand (northeastern Natal Province) cattle and horses used to be pastured in the uplands, the *Hoogveld*, during the wet summer seasons but returned to the lowlands during the winter. The collapse of the Zulu Kingdom was followed by settlement by the Boers, who did not know that this pattern of bovine and equine seasonal migrations actually prevented epidemics, for the animals remained safe during their stay in the highland, which were free of the tsetse.

The Boers kept their livestock in the lowlands, and the great losses that struck their livestock prompted the authorities to search for the cause of the catastrophe. Bruce was put in charge of the investigation (1893-94); he revealed the role of a haemoflagellate, *T. brucei*, and drew attention to the presence of tsetse flies in the lowland and its role in the transmission.

Also West African herdsmen (Fulani, Peuls, and others) practised a very similar form of transhumance to avoid the danger of serious diseases striking their livestock.

It has highly been suspected that tsetse flies were responsible for infections, since Livingstone's expedition into South Africa and the Zambesi (1875-1858), on the testimony of local people. However, the cause was ascribed to the injection of a lethal poison by the flies. The part played by tsetse flies in trypanosoma infections had already been described by Wiedemann in 1830, and was later recognized by a series of

authors (Bruce, 1895; Nabarro, 1905; Greig, 1905; Brumpt, 1903; and Sambon, 1904). Observers like Roubaud (1908) called attention to the presence of trypanosomes in the tsetse's proboscis and thus to the importance of contacts between man and tsetse; they were forerunners in this field.

Bruce and Nabarro proved in 1903 that *G. palpalis* was transmitting *T. gambiense* when they could infect experimentally a healthy animal by having it stung by contaminated tsetse flies. Roubaud revealed the presence of trypanosomes in the tsetse's salivary glands in 1906, but the credit for elucidating the complete biological cycle in those flies undeniably goes to Kleine and Taute (1909, 1911).

1.2.5. Specific characteristics of the tsetse flies

The specific traits of tsetse flies and their peculiar behaviour were slowly assembled together. Investigators established that the glossinae of both sexes were strictly haematophagous, with the following morphological peculiarities: larger than houseflies, dark-coloured, bearing a visible horizontal proboscis, and having wings folded over themselves like the blades of a pair of scissors; a cleaver-shaped cell (like a butcher's hatchet) between the fourth and fifth wing veins thus making their identification easy. Their common name of tsetse came from the onomatopoeic imitation of the buzzing noise of their flight in Setswana and other African languages.

The tsetse flies are active by day, although *G. morsitans* and *G. pallidipes* may be active on moonlit nights. Besides the males' search for females, their main activity is to search for blood meals. They therefore make successive short flights (30 to 35 minutes each), often at the edge of light and shade, becoming more frequent with increasing light. Their to-and-fro flights are bimodal, limited to the warmest hours in the hot season, whereas during the cold season they have a more uniform activity.

In their flight the tsetse flies keep close to the ground and do not travel long distances. The cruising speed can reach 25 km/hour. In evergreen forests and forest corridors, the flight remains restricted to 200 to 500 metres, but in savannahs the tsetse covers greater distances, although never more than 2 kms. A report of Cuisance and Février (1933) asserts that *G. morsitans* can fly up to 5 kms in one day reaching up to 25 kms in five days. Tsetse flies which were marked are recaptured within a 300 to 500 metre radius. The advance reached may total 15 kilometres a year, but the covered area does not usually exceed 5 to 6 kilometres across. Their daily progression attains about 14 metres/day (13.8 to 15.7 metre). Seasonal movements

are dependent on crops, rice paddies, harvests, fallow periods and their rotations, and according to the seasonal migration of game animals and the life-style of man.

Passive spreading must be added because tsetse flies are inclined to let them be carried by animals and moving vehicles, which can considerably increase the covered distances. They even search for canoes and boats that follow the current of the river. On barges the so-called chicken cages (where passengers seek shelter) are one of their favourite haunts: travellers looking for comfortable shade there run a real risk of being attacked by one or more infective flies. Rapids such as those at Yambuya on the Aruwimi River, Gô on the Itimbiri, Voro on the Bili are a natural barrier for the tsetse flies.

The glossinae feed usually every other day, and possibly every third or fourth day. This means that they must find the right host every 48 hours, or at worst every 96 hours. However the hosts are not chosen randomly: the tsetse flies have a clear preference to feed on the same host. The emerged teneral fly waits one or two days before feeding. The choice of host for the first meal is not without significance, for it will determine the fly's later feeding behaviour. The flies feeding preferences are further discussed below with their anthropophilism (p. 1737). Transmission of trypanosoma by the fly will start one to two weeks after an infective meal, when the cycle in the fly is completed. The glossina will then remain infective its whole life, from three to five months, remaining infective at each feed, at least every two to four days.

The chances that the host is a trypanosome carrier and that it infects the fly are very small (1 to 2%, in rare cases 10%). However if the first meal is infective, the infection rate will be higher than in subsequent infective feeding.

Tsetse flies mate only once and the female stores the sperm in her body. This circumstance has led to the implementation of a tsetse fly control method by releasing a large number of sterilized males. This valid method can only be applied in geographically isolated sites such as islands and small peninsulas.

The inseminated female fly produces a single egg which hatches in the uterus. The fertilized egg matures eight to ten days within the female tsetse which gives birth to never more than six to ten larvae in her lifetime. The tsetse fly is thus larva-viviparous. The larvae have to be deposited in environments that meet very strict requirements. If the larva is deposited on the ground, it must be soft, shaded, relatively dry and, if it is located along a low pathway in a valley, the site must remain above the flood line. The fly sometimes lets the larva fall from its resting place on a tree

branch or trunk. In such a case the pupa will be found at the foot of a tree or dense bush under dead leaves, between roots under a fallen tree, or in a hole, or even in the axil of palm tree leaves (Goutteux, 1984). Puparia of *G. fuscipes* have also been found under coffee bushes, banana-trees, sugar cane, lantana hedges and even under verandas (Okoth, 1986).

Once released, the larva seeks a place where it will bury itself to a depth of 2 to 8 centimetres and rapidly turn into a pupa. The pupation lasts an average of one month at 25°C. During that time the pupa is exposed to predators and infections. The pupal stage is much longer in areas with a wintry period hibernation or where they are more than 1,200 metres above sea level, leaving the larva longer exposed as prey, or open to infection.

When the teneral fly emerges from the puparium it must first spread its wings and wait till they harden and the body becomes firm, so that it takes flight only 30 to 60 minutes after it's hatched. Meanwhile the young fly runs all the inherent risks that threaten an immobile insect in nature. It remains very fragile until its first meal, 24 to 72 hours later, sometimes even longer.

The reproduction dynamics of the tsetse flies fluctuates according to insufficiently known circumstances, which can lead as well to a sudden population increase, as to an intensified potential in a vectorial capacity.

Not all the tsetse flies are trypanosoma vectors, nor do they carry all the six zoonotic species (*T. brucei*, *T. congolense*, *T. uniforme*, *T. simiae*, *T. suis*, and *T. vivax*) or the two species pathogenic for man:

- *T. gambiense* transmitted by *G. palpalis*, *G. fuscipes*, and *G. tachinoides*;
- *T. rhodesiense* transmitted by *G. morsitans*, *G. palpalis*, *G. swynnertoni*, and also *G. fuscipes* of the *palpalis* group.

Purely mechanical transmission to man, as occurs in the case of *Trypanosoma vivax* or *T. simiae*, is unlikely, for when trypanosoma carriers arrived from Africa in tropical America, where sucking and biting insects abound, sleeping sickness did not take root as did *Trypanosoma vivax*.

1.2.6. Ecological requirements

The geographical distribution of tsetse flies by province and locality was mapped especially for Zaïre by Evens (1953), but unhappily the taxonomy he used preceded the revision by Barros-Machado (1954).

In short, the hygrophilic tsetse flies (*G. palpalis*, *G. fuscipes* and *G. tachinoides*) live close to the water but are spread over huge areas. The very strict conditions

governing the tsetse flies' development limit their reproduction. Moreover, their potential as vectors is low because enormous numbers of ingested trypanosomes are wasted before they reach the infective metacyclic form. The life span of the tsetse fly is limited to three to four months. The reproductive rate depends on the total number of living larvae which are deposited by fecundated females; whereby the puparia have to meet very strict environmental conditions and the larvae must survive predation.

Tsetse flies live close to the water, among permanent vegetation spread over vast areas. Cash crops, intensive deforestation, and other consequences of economic development have forced the tsetse flies to comply with the remnants of their disturbed living areas.

Their contacts with human beings are occasional and fleeting, dependent on human behaviour and activities. Close contacts occur, when, for example, women and children go to draw water from the river, or when villages are surrounded with lantana hedges.

Tsetse flies are undisputably archaic insects. Fossil specimens have been found in America but not in tropical Asia. In tropical Africa living fossils, as we may call them, have persisted in only some of the bioenvironmental zones; they nevertheless live over no less than four-fifths of the total surface territory, some 12 million km² between 10° N. lat. and 20° S. lat. Such large regions cover a wide range of climatic areas, answering the range of suitable vegetation and climatic conditions, including an altitude not exceeding 1,500 m. This dispersal area does not keep stable boundaries and is much more extensive than the domains occupied by trypanosomes affecting man. It is striking to see the vigour with which these archaic insects have survived and it had been confirmed by their return to huge stretches of East Africa after being wiped out by the enzootic bovine plague of 1890 which wiped away the cattle and other ungulates in these regions.

Tsetse flies have shown themselves capable of adjusting to such changes and even to exploit new conditions to extend their range, especially when they could find suitable ecological corridors to penetrate the area. They have also shown remarkable adaptability when forced by ecological changes: for example the introduction of cash crops such as coffee, oil palms, sugar cane, and banana plantations, or the use of new shrub plants such as *Lantana camara*, to improve the hedges of the *rugos* or kraals. These hedges have offered the tsetse flies new habitats in which they could easily feed on human or animal blood, thanks to the close contact with habitants and their herds and flocks.

These special niches or refuge habitats are worth drawing our attention. Since the sleeping sickness epidemics of 1885 at the Cuanza coffee plantation various authors have reported foci of the disease due to small *G. palpalis*-infested shrubberies up to 50 to 100 m across. The various vegetation clumps of eleis and raffia palms, bamboos, bushes and climbing plants are sometimes inhabited by large numbers of *G. palpalis*. Some Kwilu villages have three or four such thickets harbouring pigs and the tsetse fly is known in the vicinity as *the pig fly* (Schwetz, 1919).

The introduction of *G. fuscipes* in *Lantana camara* bushes which had been spread to become a major source of firewood surrounding kraals, was followed by *T. rhodesiense*'s adaptation to *G. fuscipes* and the spread of *Lantana camara* thickets, prompted the Alego epidemic in Kenya.

Barlovatz (1934) drew attention to the presence of tsetse flies in the palm groves of Mayumbe. Missionaries have made the same observation. Leleup (1954) noticed, as did also Breur (1910), the presence of *G. fuscipes* var. *martinii* in the banana groves of the Rusizi valley. These were also refuge habitats. This should be kept in mind as well as the observations of Okoth (1986), who found pupae under coffee trees, especially when they were overshadowed by *Ficus* or other shade trees as banana trees, and in *Lantana camara* thickets. The adults rested on the leaves and branches by day. In all these locations *G. fuscipes* was no longer riverine, but could live at a good distance from water courses.

Their infectiveness determined by dissection, is only patchy and is generally low.

However, any upsurge in the tsetse fly population varies according to still insufficiently known circumstances: there may be sudden population explosions with intensified vector capacity and activity.

Considering the very strict conditions governing their biological life cycle, their life span of no more than three to five months, the very small number of larvae that the viviparous female deposits in a very selective setting to develop into pupae, and the risks that the larvae run during their transformation into adult flies, it is amazing that tsetse flies have survived. It is equally surprising that effective tsetse fly control measures have taken so long to develop and that the efficacy of these measures is very local and remaining only partial.

1.2.7. *Anthropophilism of tsetse flies*

The hosts harbouring either human or animal trypanosomes are only involved in transmission if they have contact with the vector tsetse flies. Therefore the

hosts must necessarily live in or come to areas where tsetse flies also live or hunt.

The trophic preferences of these flies, which operate and attack outside the houses, will determine the risks of transmission. Many investigations into the origin of the blood found in tsetse flies have established that their preferences do not follow absolute rules, but meet certain requirements with zoological specificity linked themselves to ecological conditions; however there may be wide fluctuations in the availability of the preferential hosts. In certain shrubberies in which swine and tsetse flies coexist, pigs and river hogs (potamochoerus) may become the flies' sole source of blood.

This rule also applies to the anthropophilic tsetse flies, which might also feed on bovine species, swine, crocodiles, other reptiles and birds.

The indices of anthropophilism vary, depending on the observations and bioecological conditions. They range from 8% to 50% for *G. palpalis* and from 18% to 48% for *G. fuscipes*; the figure goes beyond 50% for *G. tachinoides* and is of the order of 10% for *G. morsitans*. Such differences are explained by the alternation of seasons: during the dry season, for instance, when the antelopes migrate to water holes, the tsetse flies have to rely on man.

The degree of anthropophilism depends on the activities of the inhabitants, their degree of exposure, and the availability of *G. palpalis*, which adapts fairly readily to domestic animals, particularly pigs (see below the animal reservoir, p. 1758).

1.3. The host

1.3.1. The role of humans

The life cycle and ecological requirements of glossinae are not sufficient to explain their distribution. As host-fly contact is essential, the role of humans must not be underestimated. People are constantly changing the ecosystems in which they live, particularly to increase the amount of arable land. Humans clear land, alter geobotanical surroundings, create water sources, plant cassava, introduce new food or cash crops, bring in fallow periods, make use of seasonal fields at a distance from where they live, organize temporary encampments, set up villages near river banks or river mouths, go fishing and so on. All these factors play a large part in promoting contacts between humans and tsetse flies.

The domestication of animals, too, has inevitably brought man into contact with tsetse flies. In herding areas the herdsmen have even had to search for corridors free of tsetse flies along which to drive their herds to other pastures or watering points (see above, p. 1735).

It is fundamental to determine by sophisticated techniques (isoenzyme assays, DNA probes, and others) some undetected infections in people with low levels of parasitaemia (healthy carriers) which so far only the tsetse flies are able to detect. Such undetected infections may provide the reservoir of trypanosomes explaining the persistence of inactive foci during periods where no human illness is found, lasting sometimes twenty years or more.

The tsetse fly population is correlated to a certain extent with the human population density. When the human community exceeds a number of the order of fifty families, tsetse flies persist with difficulty, unless man has made a particularly favourable environment, such as *lantana* hedges. These were set to enclose the *rugos* in which *G. fuscipes* found a favourable residence.

While urban communities really do constitute barriers against the presence of tsetse flies, the chaotic peri-urban spread of people are not as such, but can in some cases bring about the installation of real foci for tsetse flies and sleeping sickness. The pattern of settlement, of human activities or life-style and so on, have a crucial effect on the presence of the vector.

1.3.2. The groups at risk

The early days of colonization remind that some social groups were and are still more exposed than others to the risk of contamination by *T. gambiense*:
- soldiers who, despite medical examinations conducted when they are enlisted, are commonly recruited from all parts of the country and are frequently transferred to take part in field activities run the risk of contamination and can carry over trypanosomes. This was particularly the case when a soldier is discharged at the end of his service, which applies also for his wife and children;

- barge men, engine-room mechanics, pilots and other steamboat crew members, but also canoe paddlers and fishermen who live close to water for very long periods of time (see the shift from endemicity to epidemic conditions, p. 1764);

- personnel cutting wood along the river banks for the steamers, which stopped there and may have introduced various strains of trypanosomes;

- the steamboat, which appeared in Central Africa in the second half of the nineteenth century, has contributed most to the penetration of trypanosomiasis and was increased by the growing interest in raw materials that could be obtained from the hinterland which stepped up contacts with previously isolated areas.

- workers hired for major public works, railway construction, agricultural and industrial estates or mining industry, often recruited from densely populated but

distant and not infected regions. When they are deserting or released at the end of their contracts or for physical disabilities, they are potential trypanosoma-carriers infected during their service period;

- caravan and official porters, usually employed within a limited sector but who could nevertheless play a significant role in spreading trypanosomes. These were often destitutes, as they were selected haphazardly by the village authorities from amongst the undesirables or the poor populations with low resistance, who were doomed to contract all raving infections;
- personnel of the administration and trading posts (clerks, policemen, messengers) or civil servants in an endemic area as well as their families. Prisoners were also at serious risk;
- assistance to abandoned or dying people for humanitarian reasons and slaves bought up during the early colonial period and other destitutes, were sources of contamination for the helpful institution;

These groups were not immutable over time, however any such groups were the most exposed to contract trypanosomiasis and become actual or potential disease carriers.

The organization of medical services and the efficacy of the medical examinations at the hiring or end of employment contracts and also during the service, gradually reduced the risks of the various categories of workers, who were exposed to the bites of tsetse flies. It is well established that never any catastrophic epidemic occurred in a military camp, despite the large number of soldiers found to be infected, thanks to a strict application of the preventive measures against tsetse flies.

The groups that are most exposed today are the inhabitants of known or unknown trypanosomiasis foci and those living in new agricultural areas (near scrubland), fishermen and agricultural labourers on particular types of plantations, agriculture promotion officers and itinerant medical personnel. However, since barely 5% of tsetse flies carry the metacyclic forms of trypanosomes, there is usually no massive influx of parasites, so as to overwhelm completely the immune defences. Reinfection with parasites on a small scale may probably improve an existing tolerance.

In areas of endemic *T. rhodesiense* sleeping sickness the risks were limited to occasional contact during hunting, fishing, group spear-fishing, wild honey collecting, firewood collecting, prospecting, and so on. The settling of the vector glossina in habitats around village dwellings contributed to spread the disease.

1.3.3. *Trypanosome tolerance and healthy carriers*

Individual variations in susceptibility to trypanosomes are due to a combination of immunologi-

cal and nutritional factors, or interactions of infections, and so on. As a result, clinical responses are of variable severity, ranging from the overt disease to mild cases, including healthy trypanosoma carriers. The latter can theoretically become sick after an indefinite lapse of time (although at present nobody can predict whether they will or not become ill). We do not know if the trypanosoma they harbour have the same virulence as those causing the severe disease and hence whether healthy carriers should be treated.

Screening for healthy carriers would provide invaluable epidemiological information, for the proportion of asymptomatic carriers in a population might be used to identify persistent endemic foci. The importance of the problem of healthy carriers is covered below (p. 1756).

1.4. *Non-human foci*

The existence of a non-human trypanosoma reservoir in domestic and wild animals, explaining the survival in tsetse flies, has been a constant worry for all physicians and naturalists engaged in research on African trypanosomiasis. Whereas *T. b. rhodesiense* reservoirs were gradually discovered (Heisch *et al.*, 1958; Onyango, 1971; Geigy, 1971), the same cannot be said of *T. b. gambiense*. This subject has so many epidemiological implications that it will be covered in the proposals for further study (p. 1758).

2. **Endemic foci**

2.1. *Endemicity*

The existence of sleeping sickness foci is linked to the simultaneous presence in the same place of three essential partners: the parasite, the vector and the host.

a) Their encounter led to a situation of coexistence only by consecutive adaptations and variations in a balance which would warrant a chain of transmission and support the appropriate life cycle of the partners.

It seems highly unlikely that this coexistence developed only at one location and at one time (see also pp. 1745 and 1755. Given the fact that the parasite-vector-host triad occurs both in forests and in the savannahs (hygrophilic and xerophilic vectors), it seems certain that adaptations must have been attempted in these different types of vegetation. It would have required too much chance to maintain an adaptation long enough in one single place, before it extended to different areas while by-passing some tsetse flies areas, reaching finally the actual distribution in *T. gambiense* foci.

Concerning *T. rhodesiense*, it is simpler and more logical to consider that, besides occasional deadly infections individual adaptations occurred in several locations and gave rise to endemic foci such as the Luangwa focus where all the basic conditions were found (see below, p. 1746).

b) The other viewpoint favours an epidemiological chain established from a single original focus (see below pp. 1745 and 1755). Assuming that tolerance had developed in a single place would only completely overlook the reactions of vectors and hosts to foreign intruders such as the parasites. These may be destroyed as a result either of susceptibility of the vector, or of the host's innate resistance. On the contrary, the lack of a host reaction will lead to such uncontrolled multiplication of the parasite that their density will kill the infected person. Both eventualities cannot give rise to a sustained parasitism.

The persistence of the trypanosome-vector-host triad requires the trypanosoma survival in the host in such a way that the parasite can be reintroduced into the host-vector cycle. Whereby transmission depends on the presence in a population of individuals who defend themselves but do not kill off the introduced parasites, which must be able to start a new cycle (see p. 1756).

Whatever the situation, the harmony amongst these three elements, has not a stability withstanding all events. The balance may be disrupted by many factors making it difficult to set up limits between endemic and epidemic infections and then transitional stages.

The endemic state is difficult to define and can be difficult to recognize.

a) *T. gambiense* trypanosomiasis is an endemic state known to occur in isolated or sporadic cases, with a trypanosoma developing slowly with lymph nodes often more positive than the blood, at least before the established meningo-encephalitic phase. Once this phase is reached, Muraz's index (the ratio of the number of cases in the secondary phase over the number in the first phase) is greater than one. An isolated episode of lethargy does not draw particular attention in the overall picture of various diseases: the community does not notice the condition, and does not, with exceptions, recognize the vector role of the tsetse flies. Only very occasional flare-ups of the disease drew attention when numerous sick people were present and possibly also when there was an extension of the disease to new areas. The most reliable medicobiological feature to identify endemic or residual foci is the presence of a few healthy carriers in the population, for they are maintaining the focus.

b) In the case of *T. rhodesiense* (East African) trypanosomiasis, three situations might occur:

1) niches of infection have survived the rinderpest panzootic with prevailing conditions similar to the *T. gambiense* foci; however the trypanosoma species is different. The Luangwa focus is a good example (see p. 1746);

2) a second situation corresponds to sporadic infections contracted by those entering a zoonotic cycle, usually wild (hunting, honey gathering, wood collecting, natural parks), but sometimes domestic;

3) a third situation is the spread of trypanosoma infection by tsetse flies which recolonize an area where they disappeared due to the elimination of their blood supply by the rinderpest.

The last two possibilities give rise to acute or extremely acute infections in individuals who have lost or never had produced the necessary specific antibodies.

The normal course taken by a permanent focus is to become the source of new disease outbreaks after unpredictable, sometimes very long, intervals. The rise in the number of infections increases the number of contacts between infected persons and tsetse flies. The glossina population does not have to increase at all: it is in fact often difficult to come across tsetse flies during epidemics.

Before Africa was colonized the conditions that promoted the spread of the gambiense trypanosomiasis were raids, incursions, local conflicts, migrations, famine and importation from other areas as often very clearly recalled by oral tradition.

The initial phase of colonization was characterized by the displacement of large numbers of workers over great distances to carry out major public works or to work in various mining and agricultural concerns, often localized in endemic regions, and simultaneously by increased and faster river and land traffic, military expeditions and famines triggered by climatic conditions, or the outbreak of other epidemics.

During epidemics of trypanosomiasis large amounts of trypanosomes are released into the bloodstream, whereby the transmission potential is increased. Numerous patients are thus in the septicaemic phase and progress quickly over the various clinical stages. The serious nature of the disease and the large number of ill people create a climate of fear that finally causes the survivors to flee. This explains how entire villages disappeared.

As an epidemic wanes the disease becomes sporadic again but the weakened population has difficulty in recovering its vitality, and regeneration is laborious.

The endemo-epidemic state is a transitory situation of resurgence of the disease. It starts with the sporadic transmission mode and culminates in the epidemic mode.

2.2. *The endemic process*

As early as the 18th century endemic sleeping sickness foci were covering huge stretches of tropical Africa. Reports came both from the local population and from explorers, missionaries, medical officers, prospectors and traders who went into the interior and made enquiries about the prevailing diseases.

It should be pointed out that sleeping sickness runs an almost silent course in endemic foci and is all the more easily overlooked or poorly noticed by the local population as only a small number of people are affected. Even today, some foci are detected only by the sudden flare-up of an epidemic.

Endemic foci are the epicentres, from which clinical cases flux and reflux. Some foci remain active, others disappear, depending on as yet unexplained circumstances. Sometimes the foci remain eclipsed for extremely long periods, simulating eradication, to reappear suddenly with surprising vigour. As said, it is often difficult to differentiate clearly endemic from epidemic processes.

An inventory of the foci has been carefully drawn up in all the countries concerned. The geographical distribution of the foci is indicative of the spread of trypanosomiasis. The debate on the location of foci has to ignore political boundaries as these result from negotiations and compromises between colonial powers. To establish the nosogony of sleeping sickness, it is rational and acceptable to consider tropical Africa as a single entity crisscrossed by a large network of waterways either flowing directly into the ocean or being part of major river basins, such as those of the Senegal, Niger, Volta, Benoue, Ogowe, Kwanza (Cuanza), Zaire, and Nile (see map 37 on page 1742).

The simultaneous presence of the parasite, the vector, and the host is a necessity but not a sufficient condition for the existence of a focus. Indeed new foci can arise in territories where the vectors were present, trypanosomiasis can disappear from huge areas without the tsetse flies being eradicated. The sequence of expansion, then disappearance of endemic areas around Lake Tanganyika is a striking example of what remains up till now unexplained.

2.3. *The types of endemic foci*

Endemic sleeping sickness foci are linked to well-defined phyto- and orogeographic biotopes in which trypanosome carriers and vector flies live side by side. Permanent foci, which are usually

well identified within the endemic areas, must be differentiated from secondary foci which result from swarming.

Permanent, primary foci are scattered over vast expanses but are usually limited to one, two or a few villages or hamlets, although some foci cover larger areas. These primary foci will reveal the first signs of a disease recurrence or the risk of an epidemic outbreak.

The secondary or swarming foci shoot out primary foci and are often the result of an epidemic. This swarming pattern is enhanced by ecological changes due, under more, to human activities.

Primary and secondary foci are both residual and make all specific control campaigns difficult.

The apparent recession of secondary foci to their primary focus, stopping the latter's expansion flux, let it suppose that permanent foci share some common factors of an as yet unknown nature. It is of prime importance to try to discover the factors accountable for the permanent foci's persistence and to identify the essential conditions for their survival and the criteria to identify their location. These criteria could be the maximum transmission index for a given period of time and the healthy carrier rate. A related problem is the recognition of one or more mechanisms responsible for triggering the epidemic or non-epidemic expansion of these foci into areas which were previously free of the disease, despite the presence of tsetse flies.

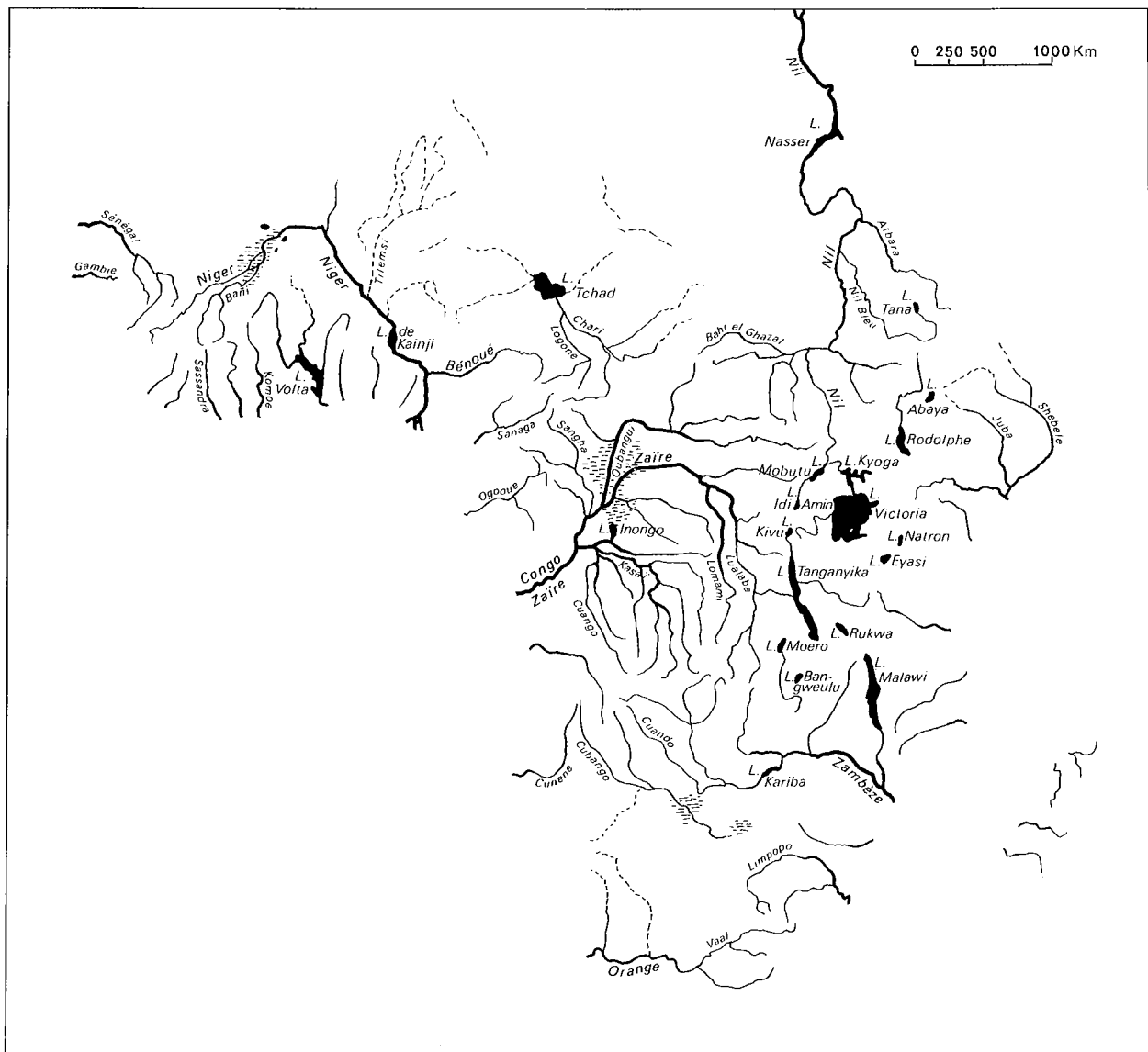
An endemic focus can only exist thanks to specific relationships in one and the same habitat between the essential factors of infection and transmission. These relationships vary according to the trypanosoma species involved, as on other factors as well.

2.3.1. *The T. gambiense endemic foci*

The area of distribution and the characteristics of the *T. gambiense* foci have given rise to some hypotheses about the way sleeping sickness is spread.

2.3.1.1. *Distribution of the T. gambiense foci*

The first foci were scattered over a large part of the huge Western and Central African hinterland, from Senegal in the north to Benguela in the south and from the west coast eastward to the upper stretches of the White Nile and the Great Rift. As soon as the pathogenic agent and vectors of the disease were discovered, all the foci in the hinterland were found to be caused by *T. gambiense* and to occur in a common biotope favourable to the *palpalis* group of flies vectors of the disease (see map 37).



Map 37 – Network of rivers in South-Sahara
(adapted from Stanford, *Geographical establishments*)

- T. Gambiense foci in West Africa

Corré ascertained the presence of sleeping sickness in Joal, Senegal, in 1875. It was identified in Portuguese Guinea in 1855, along the Gulf of Guinea and near the Lake Chad in 1880, where the infection was wrongly attributed to the presence of Haussa soldiers, and finally in Gambia in 1902, but also in most of the countries.

It is remarkable that in the case of the Gambia, *Dutton* and *Todd* indicated on their map published in 1903 the existence of trypanosomiasis along a stretch of approximately 75 kms upstream of the Gambia River. Up to 0.5% of the inhabitants in riverside villages had sleeping sickness and, according to *Hutchinson* (1952), the disease had been known for many generations and centuries, if not millenia. In 1911, the prevalence calculated on 12,000 examinations, reached 0.8%. The disease was of moderate severity and the survival was 30 to 40 months: 12 untreated sleeping sickness patients were still alive seven years later. During the year 1925, ten cases were observed.

In 1940, the situation changed with 1,000 to 2,000 cases observed per year. By 1950, 27% of the population had sleeping sickness and people with enlarged lymph nodes reported spontaneously for treatment. In 1951 the prevalence on 55,000 inhabitants had receded to 0.75%. The course of the disease was only moderately serious and the population was not depleted.

These observations by *Hutchinson* provide some interesting socio-economic informations. The Gambia covers two narrow strips of land on either side of the river, on a distance of 480 km. The river has no tributaries. The coastal area extends up to 50 km inland. It includes many villages. In these lowlands region (between 50 and 200 kilometres further upstream) the water may be brackish and the river is edged with mangroves, the roots of which shelter the puparia. *G. palpalis* abounds in the foliage of the mangroves and oil palms and in the bush that separates the swamps and lowlands from the highlands and villages. These tsetse flies are responsible for the transmission.

Along the river's upper reach (from 200 to 450 kilometres upstream) the banks are high and the water is fresh. The man-fly contacts in this area occur in the creeks during the dry season. The population does not fish and rarely uses dug-out canoes. Contacts with the flies are limited and the population has a great tolerance to the trypanosome. Only newcomers to the area develop serious acute sleeping sickness.

- Central Africa

Sleeping sickness in Central Africa, particularly in the Congo Free State, was known and feared since

1885 in the Cataracts region between Matadi and Leopoldville, and inland around Malebo Pool (at that time Stanley Pool). It had been reported along 150 kilometres of the Congo River as well as along the banks of the Ubangi (see p. 1689).

In his assessment of the health status of Leopoldville Station, *Résumé de l'état sanitaire de la station de Léopoldville*, 1885-1887, *Mense* referred to the presence of sleeping sickness or *ntansi*, which confused the physicians at the time. In 1897, *Mense* gave a more detailed account of his experience, including the macroscopic examination of a brain that did not, however, provide any new clues. The text of his lecture contained only one paragraph about this enigmatic *lalangolo* or *ntansi*. He wrote: "the disease tends to affect young males in the second decade of their life. Its incurability rivals that of leprosy, although it leads in 3 to 4 months to death, with convulsions and complete apathy, at least in the self-observed cases. I managed to open the cranium of a young black with *ntansi* and to examine his brain: the sinuses were found highly filled, with hyperaemia of the brain's envelopes, but there were no haemorrhages in either the white or grey matter and no abscesses anywhere, nor softened areas, nor parasites".

The geographical distribution of the foci in Zaire are well known and described further on. Those of neighbouring countries prompted investigators to search for foci at the border areas which might have affected Zaire's foci. They will be analysed.

- People's Republic of the Congo

Here, an alarming situation prevailed along the caravan routes. It had led to the disappearance of whole villages in 1885 (*Martin et al.*, 1909). The big area to which the disease had spread was defined by *Kermogant* in 1895-1897 (published in 1905). A focus was described on the Niari, at the west of the country, opposite Luozi in Zaire. The Loudima-Jacob focus (Bouenza Territory), in the middle of the country, appeared particularly intractable. While the Malebo Pool's focus had lost its activity, the channel focus notorious since 1887 continued to thrive on both sides of the river at M'Puyo and Tshumbiri, two townships halving the same focus. The Gamboma focus, in the M'Keni, belonged to the group of the Congo River tributaries.

- Central African Republic (CAR)

Its territory that roughly covered the former Ubangi-Chari region was the scene of major military operations, among them Marchand's expedition to the Nile. It included a number of foci:

- The Ouham focus, on the Chadian border, which is still active today;

- The Sangha River which has been surveyed with remarkable regularity for trypanosomiasis from 1904 to the present day;

- the Upper Sangha (Haute-Sangha) focus, in Nola, which showed it was negative around 1950, but is now once more active.

- the Lower Sangha (Basse Sangha) focus, well known in the People's Republic of the Congo, has not disappeared.

- The Lower Bomu foci at Rafai and Zemio, thoroughly studied by *Ouzilleau*, have remained fairly inactive.

- The Upper Bomu focus, known since 1897, attracted little attention because it was located at the sparsely settled southeastern edge of the territory. It was only assumed that the disease had been imported from Sudan. Information is scanty about this area, which used to be the scene of large scale Belgian, French, Anglo-Egyptian, and Mahdi military operations, and continued until the early 20th century. It is clear that the Upper Bomu focus was part of a large infected area straddling the Central African Republic, Sudan and Zaire, but divided artificially by political borders.

- German Cameroon

Under courteous agreement which was concluded with France on December 24, 1885, this territory acquired an outlet to the Congo River (at Bonga, on the Sangha) and to Ubangi province (at Zinga, on the Lobaye River). The two watercourses played a significant role in the spread of sleeping sickness, notably from the focus of Nola. None of the current Cameroonian foci have kept contact with Zaire.

- Sudan

This area had a focus at Tambura-Yambio, in Zande territory at extreme southwest tip. This focus was part of the Upper Bomu and the Yei foci, as a leftover of the Lado enclave focus. It is clear that the foci at Aba and Faradje were not true foci.

- Uganda

Two trypanosomiasis foci of Uganda were in contact with Zaire: near the West Nile and the Semliki-Lake Edward area. Wadelai, Dufili, Mahagi, the Alur and Lugbara lands were cleared of trypanosomiasis. The Busoga area had apparently no links at all with the Zairean foci.

- Angola

To the south, the presence of active foci was attested well before 1870 in Angola at the border of the Congo Free State. These foci included the region of Bas-

Fleuve, Kongo District (São Salvador, or Mbanza Congo), which continues via the Dongo river to the Cuanza District and extends to the neighbourhood of the Kwango (Cuango) river in Malanje District.

This region has paid the price of military campaigns to occupy the interior, and furthermore of a change towards an export economy. This meant farms, coffee plantations, massive recruitment of workers, river navigation (1865), railway construction starting from Luanda harbour (1873), and all the catastrophes which follow the disruption of a patriarchal tribal society. *Mense* (1887) mentioned the presence of many sleeping sickness victims in the São Salvador area. When he expressed the opinion (1906) that sleeping sickness was imported into Angola from the Congo Free State he contradicted the facts (Luanda might already have been affected in the 17th century) and also his own assertions.

- Principe island

Cattle imported the mainland (Rio Muni) tsetse to the island of Principe, in the *Gulf of Guinea*, which was in 1890, at the time of the arrival of workers from Cazengo, the site of an epidemic. *Rio Muni* and the island of *Bioco* were infected in 1898.

- Congo Free State

We have only patchy information about the existence of endemic foci in the *Congo Free State*. It is difficult to recognize an endemic focus when it is at rest. Indeed in a fundamentally steady situation the disease is running a very quiet course; there are symptomless carriers in the communities and at least some members, are resistant to trypanosomes.

This steady state escaped the attention of observers, who were only alerted when the balance was suddenly disturbed: the number of patients rising suddenly and a rapid transmission spreading the disease to tsetse-harbours and to communities without premunition, became quickly a catastrophe. Particular local conditions can speed up the man-fly contacts for example when glossina's resting places are near human dwellings or are frequently visited, which is the case for coffee plantations and banana groves.

Information was often fragmentary because the anecdotal information mentions only those places where unfavourable conditions prompted the outbreak of an epidemic. Moreover, physicians were few and far between in the interior at the end of the 19th century, and microscopic examinations were not yet a routine.

Endemic foci had existed for a very long time (see pp. 1689 to 1711). This was the case at Wathen-Lutele,

at Lukolela on the Congo (Zaire) River, in the Kwango district, which contained small, weakly virulent foci, and in the Semliki Valley. The first medical officers to report the existence of *African lethargy* were those of Mukimbungu-Kibunzi, Bas-Congo Province (Walfridson: in Slade, 1959), of the Equator Province (Dryepondt, 1895), at Leopoldville (Mense, 1888), in the Uele area (Van Campenhout, 1901), along the river at Bolobo (Girling) and at Bolenge (Dye), also in Lower Ubangi (Rodhain). To this must be added the data collected by Boulanger et al. (reproduced by Dutton and Todd, 1904). The foci in the Congo Free State were clearly identified on the map drawn by Dutton and Todd for 1884 (see the map p. 1686).

Other major foci such as the Upper Bomu focus straddling the Central African Republic, Sudan and Zaire could have been missed: for example in the northeast the Doruma-Dungu-Nzoro-Kibali foci, may have been an extension of the great Sudan-focus covering Yambio, Tambura, Obo. The Semliki focus could also have been missed as well as the focus of Gungu (Bandundu) and the focus of the Kasai Province, crossed by the caravan routes linking Pania-Mutombo to Malanje in Angola.

The eastern territories of Zaire, where there was great Afro-Arab activity and no trypanosomiasis, were struck by violent epidemics when the disease did reach them.

2.3.1.2. *Characteristics of the foci and working hypotheses*

Investigations undertaken to understand better the endemic spread and to try to control the disease effectively revealed that sleeping sickness did not cover the entire hinterland, for it was absent above 1,500 ms, where the environment was unsuitable for *G. palpalis*. Neither did the tsetse spread through the vast forests, especially in the equatorial forests of Zaire's Central Basin, except in close proximity to watercourses.

a) *Foci in rural areas and their changing pattern*

Investigations showed that the foci were originally all rural, but could subsequently enter the suburban, even urban, areas of the major towns when these developed rapidly and haphazardly. The observations also unveiled two problems which are now only partly resolved and the subject of further scientific investigation;

- first, the pattern of the foci was found to change in the course of time: they could flare up, remain dormant sometimes for long periods, break out again suddenly and disappear completely. The typical example of a flaring-up rural focus is the microfocus at the

Nzoro-Kibali, and at part of the Doruma-Dungu-Nzoro area in Upper Uele (see p. 1692). The fame of this area stems from an outbreak of the disease that occurred around 1910, and was so devastating that it forced the authorities to close once and for all the Van Kerckovenille post, known as VKville, which was located at the confluence of the Nzoro and Kibali rivers. This focus, which remained fairly passive but persistent for half a century, was monitored by the medical service of Kilo-Moto mines revealing regularly one to three cases per year. No one could unravel the mechanism that maintained this low incidence of *T. gambiense* infection in a *G. fuscipes*-infested area. Obviously such a focus, which produced one case from time to time, was difficult to pinpoint. In this zone epidemics, such as the 1949 Dungu epidemic, broke out suddenly at intervals of one or more decades. Scheitz and van Hoyer (1953) strove to counter the Dungu epidemic.

- the second and relative problem was to discover what was triggering epidemic outbreaks of endemic foci, as well peripherally as in depth, areas which had until then remained completely unaffected by the disease, despite the presence of tsetse flies of the *palpalis* group.

To be sure, attempts were made to find an explanation based on the existence of reservoirs other than human ones. However, all serious investigations in this field led to the same conclusion, namely, that while it was possible to infect some animals with *T. gambiense*, such an infection would be either hyperacute or fleeting, a situation that was incompatible with the maintenance of a *T. gambiense* reservoir.

b) *The single focus theory*

After it had been noticed that several endemic foci, sometimes located at great distances from each other, could all revive at the same time in a broad epidemic outbreak, the hypothesis of an initial focus was formulated: there were indeed no plausible explanations for the phenomenon. The hypothesis was based on the assumption that there was a primary focus of *African lethargy* from where the disease proceeded from spot to spot. Identify a primary focus and its place of origin was acceptable at the start of the 20th century.

In this perspective, a role was also assigned to Senegalese riflemen and to Hausa traders. Senegalese riflemen, who had participated in France's military operations to conquer among others the Lake Chad region, often came from Upper Senegal (today's Mali and Niger) and some of them carried trypanosomes. The Senegalese allegedly should have infected the Hausas, who then according to this theory, should

have spread sleeping sickness in the areas of their trading activities (see p. 1755).

According to the proponents of the single focus hypothesis (Morris, 1963), this would explain how at the time when river traffic started to develop, the disease-infested mouth of the Sangha, a tributary of the Congo (Zaire) river in the People's Republic of the Congo and the Central African Republic, became, with the development of inland navigation, the starting point of the catastrophic spread of African human trypanosomiasis, both southwards to Brazzaville and northwards to Bangui and the Lower Ubangi. The construction of such an epidemiological scenario owed more to fiction than to well-weighed out analysis.

This theory of the invasion from a single initial focus satisfied the underlying desire to find a series of events consistent with the common and familiar patterns of major epidemics, thought at the time to start from a single source.

It is somewhat surprising that *K.R.S. Morris* (1963) subscribed to this theory. He was certainly a sharp observer, but he had gaps in his information. The situation in Cameroon shows clearly that sleeping sickness did not start with Jamot. The Germans had set up a huge sleeping sickness control programme in the area before 1914, using exceptional human and financial resources to this end. Moreover, the World War II operations, with its movements of troops and porters, definitely helped create the right conditions for boosting the foci's activity by spreading parasites along their routes.

Morris's interest in Nigeria and in the Hausas is understandable, but his objectivity is dubious.

c) *Multiple foci infected before the arrival of white men*

Martin, Leboeuf, and Roubaud (1909) upheld a more rational point of view, and first of all differed from Kermorgant (1906), who had put the start of the spread of African lethargy at 1890. They disputed also the idea of a single focus and reported the earlier existence of contaminated settlements in the Lower Congo, on the Niari Plain, and along the river from Brazzaville to the tributaries of the major rivers, including the Lower and Upper Ubangi.

They thought it unlikely that the initial major focus of sleeping sickness would have been in the Lukolela-Irebu-Liranga area. Indeed, the natives of various parts of the Congo-Ubangi river basin and the Niari Plain asserted that the disease had raged in their villages well before the arrival of the white man. *Martin et al.* (1909) added that the maps of the Congo Free State pinpointed foci in many areas neighbouring the French territories of the Lower Congo and the Lower

Ubangi; these authors also underlined that the Congo Free State itself had certainly also several foci. Moreover, the fact that the 1896-1897 extension was detected at both Loango and Liranga simultaneously, ruled out the possible spread of sleeping sickness from Liranga and Lukolela to Loango.

The Loango epidemic (*Gouzien*, 1908) was not necessarily to be explained by a distant origin, since a focus existed on the caravan route, and certainly played a major role in the epidemic's expansion. The confusion seen in the publications covering these events stems from the fact that the date of the discovery of the epidemic was taken to be the date of the onset of sleeping sickness in that area.

It is unnecessary to point out that trypanosomiasis could have been present but unnoticed, given its very slow course. An epidemic outbreak at the contrary would have attracted attention.

2.3.2. *The T. rhodesiense endemic foci*

The first patient who was discovered was a prospector who had been exploring the Luangwa River Valley in Northern Rhodesia (currently the northeast of Zambia; see map 35, p. 1698). For the particular features of the parasites which affected him, they were named *T. rhodesiense*.

Screening for cases or later for foci, showed that in the area of distribution of this human African trypanosomiasis discovered in the Zambezi Valley, the disease transmitted by *G. morsitans*, was limited to the biotope of this particular tsetse fly, the wooded savannahs of eastern tropical Africa. *G. morsitans* can cross treeless expanses for up to one kilometre, and be carried by animals or various man-made means of transportation. In 1910, twelve cases were reported in Nyasaland (Malawi), and 41 in 1911, as part of a small epidemic that totalled 166 cases. A Royal Society commission headed by *Bruce* came to examine the situation in the field.

The *T. rhodesiense* endemic had been spreading northwards along the Great Rift into Tanzania, Uganda, Kenya, Burundi and Rwanda, producing either isolated cases or small epidemics.

This sequence of events requests an explanation.

- One salient fact should immediately attract the attention: the 1890-1896 rinderpest panzootic that practically wiped out ungulated wild game and cattle in East Africa also brought about the disappearance of the tsetse flies, which were deprived of their vital sources of blood. The more limited epizootics which occurred in 1919-1920, 1932 and 1943, confirmed the indirect influence of rinderpest on the decline of the tsetse population.

- The occupation of new territories by trypanosomes and their reoccupation of old territories were linked with the movements of both vectors and hosts. Tsetse flies, which are attracted by moving objects (herds of animals, canoes, cars or lorries and trains), can be carried over sometimes huge distances. Tsetse flies have been gradually recolonizing abandoned territories in successive northward waves. Some tsetse communities escaped the disastrous disappearance of animal life by surviving in particular very localized habitats. The Luangwa focus is a good example of a rather chronic, *T. gambiense*-type evolution.

- Human populations increase and decline in line with epidemics and famine, but the settlement or resettlement of people to cultivated land may have changed the landscape considerably. Populations became much more mobile, a situation which was intermingled with the harmful effects of military campaigns, rebellions and crowds of refugees (often settled in rather inhospitable areas).

2.3.2.1. Distribution area of the foci in Eastern and Southern Africa

In studying the changes in distribution of *T. rhodesiense* trypanosomiasis, starting from the south of the continent to Central Africa, John Ford (1980) pursued the ideas that were fashionable at the time. *T. rhodesiense* had not been seen in East Africa before 1908, and its discovery early in this century had moved the British opinion. Indeed, the British were convinced that this lethal trypanosomiasis had not existed in East Africa and its rapid spread should have been the result of new mobility in formerly sedentary populations. The authorities wanted at all costs to avoid the feeling among British South Africa Company's shareholders that East Africa was threatened by this new scourge.

The prospector William Armstrong was the first foreign traveller to have contracted sleeping sickness (see p. 1730). The second case was that of an African hunter, who must have acquired his infection when travelling through a neighbouring country. The case was thus imported. The third case, a European, could have encountered tsetse flies only in Rhodesia, which he had never left and where trypanosomiasis was advancing from the Rovuma river onto the Mozambique border.

An examination of the local population detected eleven other cases and the local chief explained that the disease had been known for a long time; however, the district medical officer refused to accept this statement.

The cases observed in southern Tanzania in 1911 were attributed to the introduction of the disease from Mozambique, separated from Tanzania only by the

Rovuma river.

Since *G. palpalis* was absent from the Zambezi river basin, it was hypothesized that *T. rhodesiense* might be a form of *T. gambiense* that had adapted, by successive stages, to a new vector, *G. pallidipes* (McLean, 1935; Bourguignon, 1907). To explain the Malawi epidemic, Willet in 1965 continued to maintain this opinion. The great reduced range of *G. pallidipes* by the rinderpest epizootic, both reduced the fly's contacts with man and weakened the human population's immune defences. When *T. rhodesiense* reinvaded the country from its niche it encountered a generation of humans without immune resistance.

- Botswana

Well-documented observations show that sporadic cases of *T. rhodesiense* trypanosomiasis have been present in the northern part of the country, in the Chobe District, bordered on the north by the Chobe River, and in Ngamiland, near the town of Maun probably since 1909 or even earlier. Parasitological confirmation dates from 1935. A control study in 1960 revealed that the course of the disease in Africans tended to be chronic (nine recorded cases), but had a very serious and acute evolution among Europeans. In both groups the infections were occasional spillovers from an animal reservoir. The name *kgotella* given to the disease in the local Setsuwana language means drunkenness, by association with the faltering steps of its victims.

- Southern Rhodesia

Similar observations were made in Southern Rhodesia (Zimbabwe). The discovery of the first case in the north of the country, near the Sengwe River, led to the identification of other cases, including a healthy carrier. Blair (1939) surveyed 21,000 people and found nine chronic cases, two of whom were actually healthy carriers. Other cases were also discovered in the Gokwe District, near the Busi River, in particular among the warehouse workers providing cereals to displaced populations from the Zambezi Valley. The grain depository was located in Mvuranduru Gorge, near resting places of *G. morsitans* and *G. pallidipes*. About twenty people died in a village of those displaced persons.

Some researchers hold these infections to be an intermediate form of trypanosomiasis but parasitological proof of this hypothesis was never produced and this opinion takes no account of the existing focus in Zambia.

- Zambia

As far back as they could remember, the natives of Zambia were aware of a sleeping sickness focus in the

Luangwa Valley. The inhabitants of the valley used to call the disease *chilotera*, which may be translated as "the sickness of dreams". This fact had already been mentioned by *Thorneycroft* in 1911 in his analysis for the *Bulletin of the Sleeping Sickness Commission*, made after a study for the British South African Company. This focus was examined again more recently by *Buyst* (1974).

- Tanzania

According to information gathered by *Koch*, there were no cases of *T. rhodesiense* trypanosomiasis in Tanzania under German colonial rule until the appearance of the Rovuma River focus at the boundary with Mozambique. An initial case diagnosed in 1909 was followed by others, including a small epidemic (twelve people affected) described by *Beck* and *Weck* (1913) and confirmed parasitologically. One of the cases detected was a healthy carrier. In 1914 *Beck* recorded 72 new cases in this focus. Among German soldiers the first cases were detected in 1917, including nine at the Matandu River. They were attributed to an English elephant hunter, who supposedly caught the infection in Mozambique (*Taute* and *Huber*, 1919). This endemic oedema disease, which according to *Dye* was nothing new, would continue in the south. *Dye* witnessed the outbreak of an epidemic triggered by one healthy carrier; the disease killed 44 of the 115 inhabitants of the village of Namabao. He attributed the epidemic to man-tsetse-man transmission (*Dye* 1926, 1927).

When the Tanganyika territory was taken over by the British, the centre and north of the country were free of the disease, but the situation, as detailed by *Fairbairn* (1948), changed quickly. The first serious alert occurred at Maswa, south-east of Lake Victoria in 1920 when, according to *Swynnerton* (1923) an epidemic of initially unknown aetiology with person-to-person transmission broke out. The epidemic was identified as *T. rhodesiense* trypanosomiasis. It died out after 10 years, then resumed in 1958-1959. Meanwhile, the trypanosoma continued to spread out from the Rovuma and Rungwa River foci, first of all towards Tabora; then the epidemic branched north, embracing first Kahama and then Mwanza; it proceeded then west to contaminate the shores of Lake Tangu, the Malagarasi River, and the localities of Kibondo and Kasulu. All of these new foci presented acute infections. Chronic cases were also observed, but the epidemiological surveys were greatly hampered by the proximity of natural game preservations encouraging easily illegal hunting; hunters involved in such forbidden dealing preferred naturally to remain silent.

- Uganda

In Uganda isolated cases coming from Lake Tanganyika area in Tanzania and from the vicinity of Lake Victoria were observed in 1940. The southeastern part of the country was also affected. In 1941 some 100 cases were counted in the Busoga Province, north of Lake Victoria, and this figure rose to 2,500 in 1942. *MacMichan* went on to disclose the role of *G. pallidipes*. The disease continued its northward progress to Lake Kyoga.

- Burundi and Rwanda

These countries were affected by the progressing endemic in 1953, at the same time as its resurgence in Tanganyika Territory. The presence of *T. rhodesiense* in Muhinga, in northeastern Burundi, was discovered by chance. A survey soon revealed 97 cases in this area. The *G. morsitans* fly belt is located in the Ruvubu River area, close to the well-known hunting grounds of Murama Hill. Most of the discovered cases, including an European, were coming from there. The endemic then proceeded northeast towards Rwanda, where cases were imported likewise from the Tanganyika Territory into Migongo (Kibongo Territory) by a point of contact via the Kagera Valley, an area where *G. pallidipes* was also present.

- Kenya

Sleeping sickness due to *T. rhodesiense* was discovered in Kenya in 1942. Two hundred and sixty cases were counted in the Lambwe Valley. The 300 cases identified by screening around Alejo revealed transmission through an unexpected vector, *Glossina fuscipes*, a member of the *palpalis* group.

T. rhodesiense was also identified by laboratory isolation and by scientific identification of strains according to criteria of the time in areas unconnected to the traditional foci. This was the case in *Nigeria* (*Lester*, 1933) and in *northeastern Zaire*, where strain XXIII was isolated in Bambu (Kilo Mines) on April 24th, 1927, by *Van Hoof* and *Lyndhurst Duke* (1928) in a Mubira woman in her forties who had never left the mining area. The diagnosis had been made by *Moncarey* in 1926 (trypanosoma detected in blood and lymph fluid) and confirmed by *Daco* in 1927. This woman was from Nyakunde, a locality close to Irumu, in Ituri, where she had been infected in the proximity of the Semliki Valley and the West Nile. *Van Hoof* (1928) reported also the presence of *G. palpalis* along the Shari (a tributary of the Ituri) and sporadic cases of *T. gambiense* infections.

To those cases must be added the *T. rhodesiense* patients recognized by *Archibald* (1922), then

Archibald and Riding (1926) in Tambura, Bahr-el-Ghazal in the Sudan.

This relatively long-standing fact has acquired new significance since the discovery in March 1967 of a new focus in the area of the Gilo, Akobo and Baro Rivers, which flow in the very isolated southwest Ethiopian region of Gambela, well known to the Italians, the Belgians (1941-1942 expeditionary corps) and the British; during their occupation of the region they saw no cases of trypanosomiasis.

The first two patients belonging to this new Ethiopian focus presented of their own accord at Maji Hospital, near the Gilo and Akobo rivers, where Dr. Marion was working, in 1967. Both had extracted and washed gravel in the Jebba – Dambala goldfields on the Akobo River. At the same time trypanosomes were discovered in blood specimens during a malaria control survey in the Baro River region in northern Gambela; another victim came to Pokwo Hospital. All three patients were Anuak tribesmen. Ten cases followed in their wake: three patients came to Pokwo hospital, four were flown from Gole to the hospital, and another three were discovered through blood sampling for the malaria survey. They were also Anuak. Three foci were identified one after the other on the banks of the rivers Baro and Gilo (clearly the main focus) and in Akobo. The survey also revealed that the inhabitants became aware of the presence of the new disease only after some 50 to 100 cases had occurred. In 1970 there were 322 cases recorded; of these, 210 were from areas bordering the Gilo, four from around the Baro, and 15 from Akobo.

The origin of the infection should be sought in the Anuak and Luer tribes from the Nile area, and in the forest-dwelling Masengo. These were Sudanese refugees who settled along the Gilo river, and any or all three of them could have introduced the trypanosoma to the region. It should also be borne in mind that the health situation along the Ugandan-Sudanese border was not well known; this area used to be the site of unsupervised settlement by Ugandan refugees, who may have included *T. rhodesiense* carriers.

2.3.2.2. Chronic *T. rhodesiense* infections

When the very detailed observations of Ormerod (1961) and Apted et al. (1963) are compared with the body of knowledge gathered on *T. rhodesiense*, its vectors and hosts (human beings, domestic and wild animals), it seems clear that there must be a chronic form of *T. rhodesiense* trypanosomiasis. The infected subjects included apparently healthy trypanosome carriers, and people with acute forms, infected occasion-

ally by penetrating specific biotopes; this is the case for wild-honey gatherers, fishermen, hunters and tourists, who may start small person-to-person transmitted epidemics.

2.3.2.3. The non-human reservoir

The domestication of animals brought necessarily man closer to tsetse flies, which had on the whole rather non-specific dietary preferences; this inevitably raised the issue of the role of domestic or even wild animals as reservoirs of pathogenic trypanosoma for man (see below p. 1758).

3. The epidemic outbreaks

3.1. General remarks

Plague, smallpox, cholera, typhus, exanthematic and yellow fever are prototypes of deadly epidemics. Epidemics usually develop from a known or unknown endemic focus in which balance is upset by a sudden or gradual environmental change.

However, the most destructive epidemics have been seen whenever a new and malignant pathogen was introduced in a population which had no prior contact with the pathogen. When measles was brought in to the Polynesian islands by sailors in the 19th century, the islanders became victims and died of the disease. The effects of a particularly virulent pathogen invading a receptive population also apply to children, immigrants, and immuno-suppressed patients. Hence some viruses, such as those causing measles, chicken pox, poliomyelitis, affect children first and foremost.

An epidemic starts when a transmissible disease spreads at an explosive rate, striking from one to another a large number of people, in a rather short space of time. Some epidemics, such as those of influenza and measles, spread by direct interhuman transmission. Others, such as bubonic plague, yellow fever, dengue fever, and various arbovirus infections, are propagated by arthropods carrying the causal agent. The pathogen may be introduced from outside or become more prevalent by recrudescence of an endemic disease and/or by spreading from a focus in which it was silently maintained in its ecological niche. Such extensions occur in waves of unpredictable periodicity, the cause of which remains a mystery; they may spread from different not contiguous endemic foci, yet exhibiting a more or less simultaneous pattern of resurgence.

Large-scale epidemics are associated with and often result from major disruptions in the human, physical,

biological and ecological environment, which have demographic, economic, sociocultural, and even political consequences.

The factors and mechanisms underlying latent infections, as well as those determining the rapid dispersal of diseases, even over great distances, are insufficiently known (as in the case of influenza and cholera pandemics). The same is true for biological mechanisms ensuring the often hidden survival of a pathogen between epidemics. This survival can sometimes be attributed to an animal reservoir (rats for bubonic plague, pigs for influenza), which merely shifts the persistence of the parasite to zoonoses.

Epidemics subside when conditions for transmission become unfavourable. The population may become tolerant or resistant to the pathogen when they come in contact with it; the people develop either lasting and full immunity to the disease as such is the case for smallpox, or fleeting immunity as in cholera, reducing in a timely manner the number of receptive individuals. The pathogen's virulence weakens as the disease spreads, and its infectiveness decreases in such a way that the pathogens retreat to their normal areas of survival; they may also persist in an extension area of the epidemic when adequate asylum is found, ensuring their survival and transmission. An example of this is *El Tor* cholera in Africa.

3.2. *Sleeping sickness epidemics*

When sleeping sickness epidemics broke out on a large scale in the late 19th century, the dominant sleeping sickness foci scattered at random over vast areas, were not known; whereby the factors governing their establishment and persistence were not understood.

A certain number of apparently unconnected endemic foci were nevertheless present. These foci were clearly linked to environments with specific phylogenetic, geomorphological and hydrographic characteristics; these were subsequently rigorously defined and required the vicinity of trypanosome carriers and vector tsetse flies.

The violence and dispersion of epidemic outbreaks make their detection easy. These outbreaks appear when the equilibrium between one or more links in the trypanosome-tsetse-host chain are disturbed. In endemic areas, even if the foci are well identified, outbreaks are as unpredictable for the public health officer as earthquakes and volcanic eruptions for geophysicists and volcanologists. The impression that the epidemics follow a certain cycle is justified, for they seem to occur at 20 to 30 years intervals and to be triggered almost simultaneously in areas wide apart. This suggests that they share some as yet undefined

climatic, biological, ecological, sociological or other characteristics.

Epidemics flare up like brushfires, starting from an endemic focus or by reawakening of a focus thought to be extinct. Such outbreaks follow the intense competition between the various members of the infection triad, when appropriate conditions to sustain life are secured. Epidemic upsurges quickly deplete the population, members of which either die or leave the area to escape the menace of death. This entails apparently irreversible depopulation. The extension of the disease's territory following all epidemic outbreaks is usually temporary, but may give rise to new foci, which may be either very circumscribed or extensive, temporary or permanent.

3.2.1. *The tsetse's fly belt, adaptability and feeding choice*

The geographic environment in which glossinae live does not explain the disease's epidemic transmission. The vector's distribution commands the possibility of foci. However, there are also extended areas covered by tsetse habitats which escape trypanosomiasis; these areas are nevertheless liable to become infected.

The tsetse's habitat is limited by specific ecological requirements:

- a constant temperature around 20°C;
- an altitude not exceeding 1,500 metres above sea level;
- relative sedentary conditions, for tsetse flies unlike mosquitoes and blackflies, do not move over very wide areas, unless passively conveyed.

Consequently, their hunting grounds, whether strictly fluvial or in the vicinity of lakes, are narrowly circumscribed and reach barely more than 500 ms from watercourses or thalwegs (valley paths). Therefore populations settled on lands which where one or more kilometres from a tsetse fly habitat, like those located on plateaus above 1,500 metres, escape the risk of disease transmission.

Finally, the tsetse fly population density is subject to seasonal fluctuations and drops considerably during the dry season.

a. *Fly belt*

The extent of the fly belt, which is much larger than the distribution area of trypanosomiasis itself, makes possible the spread of the disease from primary foci, and the occurrence of secondary foci. Very early reports and a great number of documents, revealed well before the colonial period the existence of several permanent foci scattered over the whole of Central Africa. On current maps of the disease's distribution these foci roughly correlate with the so-called *historical foci*.

b. Adaptability of tsetse flies

Tsetse flies also present a risk of epidemic outbreaks through their adaptation to another trypanosome or habitat.

- It has been shown when *G. fuscipes* adapted itself to a peridomestic habitat, for example when lantanas are planted around the people's *rugos*, or when glossinae manage to complete their life cycles in banana groves. One wonders whether such adaptations played a part in the still mysterious outbreak of the deadly epidemic that seriously reduced the population of Busoga from 1900 onwards. Referring to the work of Thomas and Scott (1935), J. Ford points out that in 1901 Busoga was still densely populated but offered a vast expanse of banana groves in which the tsetses multiplied. In 1906 the banana groves gave way to a patchwork of forest and grassland populated only by elephants, buffaloes and antelopes; thus human trypanosomiasis died out.
- Tsetse flies can also adapt to another trypanosome. This fact was demonstrated when *G. fuscipes* became in an unusual habitat a vector for *T. rhodesiense*, which is normally transmitted by *G. morsitans* or *G. pallidipes* (Alego epidemic in Central Nyanza).

c. Role in epidemic outbreaks and feeding choice

The tsetses' role in the outbreak of epidemics is usually not very clear. A sudden increase in the number of stinging tsetse flies when ecological or seasonal conditions are favourable leads human beings to avoid contact with them. The proportion of glossinae with infective salivary glands is about 1%. The existence or development of strains with greater vectorial capacity has been hypothesized but not confirmed, but the above-mentioned ability of a tsetse to adapt to a peridomestic habitat is a major hazard. This factor underscores the importance of close man-fly contacts when men, carriers or not, are present, when the fly seeks its first blood meal; indeed the species that supplies the first blood meal will probably determine the fly's subsequent feeding choice, which increases the risk of men to be infected.

It seems safe to argue that the fly belt's ecological conditions alone cannot be used to predict whether transmission will be enhanced.

3.2.2. The human activities

Man lives in close permanent contact with nature and has a permanent need for water. Settlement patterns and population density, the lifestyle, the type of agricultural work, subsistence or cash-crop farming and fallows, remodel the landscape. Contact with tsetse flies, whether remote and occasional or close and reg-

ular, is determined by man's activities, such as hunting, fishing, tree felling, and others, which come into play to reduce or, alternatively, to enhance the risk of contamination, such as the following examples:

- the structure and density of the population; this governs the size and the spatial layout of villages and can influence the distance from sites of infection;
- the stability of the inhabitants or the patterns of their migration;
- the location of fields and water sources;
- the frequency of walking along tsetse-infested pathways;
- women's occupations and children's playgrounds;
- cohabitation, if any, with domestic animals;
- activities performed in the forests or close to water-courses, which can be linked to the development of a new local economy (such as timber depots to supply river steamboats with wood, fisheries, rubber tapping from trees, gathering lianas involving prolonged incursions into the forest, etc.);
- immigration of newcomers when a workforce is recruited from other areas for major civil engineering or buildings projects, and also population movements (porters, refugees, etc.);
- finally, a less obvious factor: the villagers may have reached an economic development enabling to import workers for their plantations, which might be located in an area exposed to infection. The risk is thus shifted to temporary workers, who are neither suspected nor monitored for the disease and can become agents of its transmission.
- Another set of factors should not be overlooked as they are just as important to explain epidemic outbreaks, namely those that undermine the inhabitants' resistance to the disease:
 - The presence of diseases, subclinical infections and healthy carriers, obviously increases the risk;
 - famine, as Africa has regularly been swept by famines and continues to be so today;
 - the degree of immunity or resistance of individuals and the community against trypanosomes;
 - immunodepressive intercurrent endemic or epidemic diseases.

These are only examples of the many conditions that may influence man-fly contact.

3.2.3. Pathogenicity of trypanosomes

Various factors may modify the trypanosomes' pathogenicity:

- the variation of the number of parasites in the blood of contaminated individuals and, according to Dutton and Todd, the density of trypanosomes would enhance the spread of the disease. Van Hoof even defined the

epidemic threshold as being the moment when 75% of the inhabitants carry parasites;

- the infective power of the trypanosome strain involved which is generally higher for *T. rhodesiense* parasites;
- increase in virulence by accelerated transmission or by some unknown manner;
- the appearance of a new variant;
- the importation of a new strain;
- the parasite's introduction into a population free of previous contamination;
- the parasites' possible resistance to certain drugs;
- the poorly elucidated role of healthy carriers and of sporadic and chronic cases with few or no trypanosomes in peripheral blood, so much the more that these parasites do not all have the same degree of infectivity.

4. Advances in knowledge and prospects for epidemiology

After large scale mass campaigns, interest in epidemiology has waned as a result of the considerable progress in cell biology and molecular biology; this knowledge superseded the study of the transmission mechanisms. However, it became apparent after the sixties that great challenges are still to be faced. Even during the years of unfavourable research budgets, due to shortage of funds and other priorities, at least epidemiological studies, focusing on natural sciences should have been encouraged and undertaken.

Indeed naturalists should have relied more on descriptive epidemiology of human sleeping sickness to provide more precise information about the ecosystems in which the endemic foci and epidemic extensions occurred. It is vital to assess correctly the abilities of the parasite-vector-host triad to persist and expand, and to determine the biological and environmental conditions which trigger gradual or sudden extensions of the disease, by the complex interplay of adaptations and interfaces.

The possible contributions of comparative epidemiology should not be overlooked, especially concerning enzootic and epizootic trypanosomiasis, even if their range is rather different from only polymorphic trypanosomes and involves other transmissions modalities than by tsetse flies. There is a need for studies on tolerance to trypanosomes, by facts and experimental details which are easily accessible on animals, while ecological conditions of pastures, the influence of nutrient deficiencies and intercurrent chronic and epidemic diseases can be ascertained.

4.1. Debatable theses

For too long a period the interplay of factors at the origin of the spread of sleeping sickness stemmed from two erroneous epidemiological concepts: extrapolation from other areas or the single initial focus theory.

4.1.1. Extrapolation to Central Africa of the ecological conditions of East Africa and Nigeria

a) East African situation

1) Retrospective analysis of the 1939-1943 Busoga trypanosomiasis epidemic and rinderpest

On his arrival in Uganda in 1944, J. Ford studied the pattern of epidemic outbreaks, their extension, and the variations in the tsetse flies' distribution areas. He started with a retrospective on the course of the second human sleeping sickness epidemic that swept through Busoga in 1939-1943. This epidemic was caused by *T. rhodesiense* and its main vector, *G. pallidipes* and appeared with the development of sugar-cane plantations. The land resettlement scheme carefully drawn up to avoid man-tsetse contact, was rejected by the Soga tribespeople; however the epidemic nevertheless died out by itself.

Ford tried to understand the factors underlying the outbreak of this epidemic. He studied the tsetse fly belts, which led him to consider trypanosomiasis of cattle and changes in herd sizes. Accordingly, the destruction of wild and domestic Bovidae as a result of the 1890-1896 panzootic of rinderpest made clear to him that the tsetse fly is able to change its distribution area when confronting a lack of food. Ninety-five percent of the livestock had been wiped out at the time. The herds gradually recovered and a 1920 count gave 150,000 head of cattle. However, this recovery was slowed down or greatly impeded in 1930-1940 by the resurgence in the fly belt of tsetse flies from a few ecological sanctuaries, such as copses where some small antelopes had managed to survive in the south: these ecological niches had maintained the trypanosomes on the animals that had escaped the toll of death. When the fly belt of tsetse flies went up again, it was followed by the re-establishment of the tsetse-wild animal complex in the northern herds.

However, this migration of tsetse flies was more than a mere reoccupation of their former ranges, for the ecosystem had been altered by the ploughing of previously unfarmed land, while the elephants' annual migrations had been stopped by uncontrolled hunting (the elephant herds were destroying trees in their path). *G. morsitans* was thus able to reinvade the area due to a combination of unusual circumstances that had changed the environment.

Ford considered all these observations with great scientific rigour. He queried on the cause of failure for the apparently rational control measures which had been devised and prescribed by the colonialists. (Ford had become hostile to colonialism and his arguments were taken up by most anticolonialists).

2) *Retrospective analysis of the historical 1900-1908 Busoga trypanosomiasis epidemic*

This led him to examine some well-known historical data, starting with the notorious first sleeping sickness epidemic that depopulated Busoga (1900-1908) in Uganda. This epidemic, which broke out at a time of famine, was caused by *T. gambiense* and transmitted by *G. fuscipes* (palpalis). It had sounded alarm bells among the British, who were worried about their personnel in trading and investment companies. This convinced Ford of the need to engage in basic epidemiological research to pinpoint the real or presumed role of the evacuation of populations, the senselessness of recommendations to exterminate wild animals (Bruce, 1903), or the eradication of the vector insects.

3) *Observations on T. rhodesiense epidemics*

Ford also studied *T. rhodesiense*'s advance in East Africa; and he also tried to determine the causes of the invasion of Ankole (Bahima), Burundi, and Rwanda, by *G. morsitans*.

He examined further the epidemiological patterns in agro-pastoral regions in Sukumaland (between the southern end of Lake Victoria and Tabora) in Tanzania, and also studied the persistence of natural foci such as the Nindo focus.

All these facts were known, but Ford re-examined them and presented them under a new light. With regard to the first epidemic in Busoga he drew attention to the densely populated expanses of banana groves that merged into a patchwork of forest and grassland mosaic where elephants, buffaloes and antelopes lived. To explain the 1957 resurgence he stressed the role played by the modernization of fishing methods which had attracted candidate fishermen and consequently increased the exposure to tsetse flies. This outbreak of *T. rhodesiense* sleeping sickness had *G. fuscipes* as its main vector.

Ford relentlessly continued his quest to understand what had allowed human communities and the trypanosoma-tsetse-animal reservoir complex to coexist during the precolonial period, with no more than mild infections and acquired tolerance.

As to human pathology, he was impressed by measures taken by the local African communities. Chief Mzila had gathered up the entire Ngumi population

along the western border of Mozambique (Swynnerton, 1921). This was a realistic measure that kept the wild animals away and limited the people's exposure to tsetse flies. However the main motivation to congregate the inhabitants arose from raids, road tolls, looting of caravans, and other causes. Ford stressed the extent to which the natives had developed very realistic approaches to limit the tsetse flies' harmful role, whereas the various European strategies had proven to be ineffective.

The villagers' methods were definitely shown worthwhile in dealing with *animal* trypanosomiasis. These diseases, produced by a larger variety of trypanosomes and transmitted by various means, including a broader range of tsetse flies and other biting insects, were well known to the natives; those people had also realized that there was a connection between the disease of animals and the tsetse flies, but ignored their connection with human sleeping sickness. The precautions taken by herdsmen in this regard, whether Zulu, Fulani, Bahema, or Peul, were the fruit of centuries of experience. They included seasonal shifts to highland pastures using routes that skirted around the fly belt, clearing corridors to make them fly-free, as well as the barge transport. The Zulu herdsmen had introduced seasonal transhumance to protect their herds from *nagana*, which sadly depleted the Boers' cattle when they settled in the same lands without applying the transhumance scheme that saved the herds from the tsetses' bites. The Zulus chose corridors through which they could drive their herds without exposing them to tsetses, and their horsemen took care not to enter the tsetse-infested valleys. The natives made no connection between human sleeping sickness and tsetses, and even then with some extremely rare exceptions, they recognized the disease only if there was an epidemic outbreak.

4) *The different control strategies of the colonial time*

Ford expressed his views on the control strategies in a series of publications. When he points out the superiority complex of the colonials, who believed their knowledge enabled them to systematically ignore local people's experiences, we can support him to the extent that the white settlers had overlooked the value of centuries of experience acquired by the savannah populations in avoiding contact with the flies. It is nevertheless unlikely that the clustering of the population in certain areas was the result of a human disease that was still largely unknown. The control strategy proved effective in the savannahs but was impracticable and possibly harmful in forest areas.

Driven by his enthusiasm for this new approach denouncing the colonizers' inconsistencies, Ford eventually reached the point where, by some rather dubious generalizations, he adopted slogans in an Anglo-Saxon way: vector control to avoid the contact with tsetse flies.

However we could not agree with him when he opposes what he calls the *Franco-Belgian strategy* aimed at eliminating the trypanosome by chemotherapy, to the British strategy of concentrating all efforts on vector control. These assertions do not at all match the facts, as *Hutchinson*, among others, showed (1952). Even more serious is that this opinion holds in contempt the rational choices made by officials of the various countries concerning the tailoring of their control programmes to local conditions.

Observation on the tsetse flies and on the changes in their fly belt, as a result of the rinderpest epidemics of 1890 and thereafter, led Ford to stress on the one hand the large part played by the vegetation to maintain the tsetse population and on the other the role of human interventions.

Many of his savannah observations bring arguments which could also be applied to the disease in a rain forest epidemiology, or to its corridors and the remains of the forests. However, despite his efforts to document events and his caution in making his statements, Ford did not manage to come to any conclusions. Knowing only the savannahs, he could not judge correctly the situations specific to the Central African forest with its numerous ecosystems.

Ford's experience is certainly valid with regard to the large strips of land between the forest and the deserts stretching from East Africa westward to Senegal, where geographical conditions are very similar to those of the East African savannahs. However, his reasoning cannot be extrapolated to the forests of Central Africa.

b) Nigerian and Chadian foci

Human sleeping sickness foci are so localized that general conclusions can not yet be drawn while large gaps exist in the epidemiological knowledge, even when limited to the effect of human activities, as was the case in Nigeria.

Duggan's study (1962) in that country dealt with the problem of sleeping sickness in the North of Nigeria and the strategy implemented there by the British. It is particularly interesting because Nigeria had two foci of human trypanosomiasis, each with its own quite different characteristics:

1) The stable *Niger-Benue* focus of the Niger and Benue River valleys and their tributaries. This was a

tropical forest focus made out of small foci linked by streams and rivers, where the disease kept remaining in a quiescent state. This focus was not the origin of devastating epidemics in its centre but has produced deadly epidemic extensions in northern Nigeria, where sleeping sickness had been until then unknown. The only early epidemic extension to have occurred was in Meroe in 1890, when smugglers coming from the Benue area introduced trypanosomes to the region. In twelve years this epidemic practically wiped out the population; only 50 survived out of 4,000.

Epidemic extensions were reported again in 1910 and in 1917-1918. These seem not to have been caused by the numerous troop movements but rather by the transfers of workmen on the railway from Baro to Kano, at a time when roads to the gold and tin mines were being opened and when land earlier abandoned by communities fleeing raiders from the north, was reclaimed by its previous owners. The Kaduna-Zaria-Kano area became the most worrying focus in northern Nigeria. It had no point of contact with the focus in Dahomey (present-day Benin).

2) The focus of the Chad River basin, known since at least 1859, was totally different. It was a usually quiescent savannah focus, periodically shattered by epidemics of such severity that stories about them were recounted from father to son.

Duggan (1970) considered two possible reasons for this pattern:

- either the persistence of the disease in the fishermen's small lakeshore hamlets, where the inhabitants tolerated the presence of the trypanosomes in a state of equilibrium which became upset by the entry of previously unexposed individuals;
- or the persistence of a zoonosis maintained by *G. morsitans*, which might give rise to occasional human infections. It should be borne in mind that *G. morsitans* feeds on human beings only when in dire need. This is also the area where *Lester* (1933) isolated trypanosomes exhibiting the features of *T. rhodesiense*. These occasional human infections are followed by person-to-person transmission by *G. tachinoides* (palpalis group), this shift from zoonosis to interhuman transmission leads to increased virulence.

Careful scrutiny of the situation in Nigeria reveals how diverse the British approaches were. Sleeping sickness control in Nigeria used to be based on drug therapy backed up by tsetse fly control measures, such as the Anchau scheme of intensive deforestation by means of chain-dragging tractors that pulled away all the vegetation. The final success of these schemes was attested by the settlement on the deforested land.

4.1.2. *The single focus theory for Liranga in the People's Republic of the Congo*

Morris (1963), in an attempt to determine the courses by which the epidemic extended into the Congo Free State and the French Equatorial Africa, began by studying the epidemic invasion towards the Upper Nile at the dividing line between Central and East Africa. He concentrated first on the Ubangi and the Bomu areas, in order to detect any possible connection between the epidemics that had broken out along the Upper Nile in Sudan, the Albert Nile, Lake Albert (today's Lake Mobutu), the Semliki River, and in Uganda. He gives an excellent description of the types of landscape related to the endemic disease. He also pointed out that the Congo (Zaire) River and Nile River basins in the Equatoria Province were not clearly separated by a watershed.

Morris choose from the start (see p. 1746) a single focus as being the origin of the epidemic. He supposed that Liranga, along the Sangha River in the eastern part of the People's Republic of the Congo, was the source.

The Hausa hypothesis

He believed that Hausas from the Lake Chad hired by the Congo Free State (EIC) as mercenaries were the trypanosome carriers who had infected this focus. In the presence of Hausa soldiers in the Congo Free State's troops, he had found a decisive argument to back up his single-focus theory. It is correct, but the military brigades also included Krumen, Sierra Leonean, and above all Zanzibarite mercenaries. These soldiers were quickly replaced by the men recruited and trained in the country for the *Force Publique* from 1891 onwards.

The *Hausa hypothesis* as an explanation of the epidemic's origin required clarification: how had the Hausa merchants' caravans travelled down to the Ubangi for centuries without setting off any epidemic whatsoever. This was not to be explained simply by the skillful treatment by local traditional healers. Moreover, the origin of the Hausa, recruited as soldiers could not have been the Lake Chad basin, as the majority of the Hausa people lived in areas where sleeping sickness was still unknown at the beginning of the century. The many Hausa found to be trypanosoma carriers in Nola (on the confluent of the Kadei and Sangha rivers, southwest of the Central African Republic) were political deportees who had most likely been infected on the spot.

Morris also called attention on a number of caravan routes that linked the banks of the Ubangi and the Bomu to the Lake Chad region by means of the Tomé

and Sangha rivers. Hausa, Fulani, and Burnu merchants travelled along these rivers; the distance between Kano and Bambari was covered in three months. French military troops followed these routes to reach Lake Chad; the only exception was the Marchand expedition, which reached Fachoda by following the Ubangi, the Bomu and the Yubu rivers.

Morris referred to two Congo Free State military operations: the Arab campaign and the campaign waged against the Mahdist; but he left out the Franco-Belgian campaign of 1914-1916 which travelled over the Sangha Valley to recover the land corridor that had been granted to Germany to have access to the Ubangi and the Congo.

The rapid spread and exceptional severity of the epidemic are incontestable. However, the assertions that its origin was located in the areas of Liranga, Irebu, and Lukolela, that it descended to Brazzaville because the patients seen there came from the north and that it proceeded towards the coast are not convincing, since a major focus (still active, incidentally) existed along the caravan routes (Loudima, Niari, in the southwestern part of the Republic of the Congo).

Small foci but in small numbers were scattered across both French Equatorial Africa and the Congo Free State (EIC). The spread of the epidemic, from a single focus at Liranga, is difficult to reconcile with the existence of sleeping sickness along the Nile in 1900, yet none was discovered in the Upper Ubangi and Uele before 1906-1912, and its presence along the Semliki River and Lake Edward (Lake Ubanga) in 1896.

It seems simple to accept that several autochthonous foci were present the quiescence of which was disturbed by the overthrow caused by the rapid penetration of the territories thanks to river transport, as Morris in fact points out. Moreover, this spread occurred in communities that had no resistance whatever.

At the time the epidemic began spreading along the river banks it is certain that several foci existed: the foci of the Cataracts, of northern Angola, on the Niari caravan trail, in Kwango, Kasai and Lower Ubangi, a probable focus in Uele, also the Bomu focus (Azande focus) and the one in the Semliki Valley. It is difficult to imagine that there would not have been any autochthonous human sleeping sickness in East Africa before the discovery of *T. rhodesiense* in 1910. This discovery was merely the signal of the territory's reoccupation by a zoonosis that had disappeared after the 1890 rinderpest epidemic, and this reoccupation was launched from niches such as those of Nindo (Ford, 1969) and the Luangwa River in Zambia (see map 35 p. 1698).

It seems unjustified to dwell on the few oversights that slipped into the very interesting and amply documented Morris's overview. The first was due to a failure to differentiate between missionaries (Capuchins, Jesuits, and others present in the Congo in the seventeenth century), and Cardinal Lavigerie's White Fathers, founded in 1868; he also did not differentiate the geographical Kongo district (previous *Kongo kingdom*) on the border between Angola and Zaire, which was the origin of the latest well-documented convoy in 1869, transporting slaves to Martinique. These misunderstandings challenged the findings dating from the eighteenth century on the presence of sleeping sickness. Dutton and Todd established in 1884 a first map on the disease's distribution on the basis of questionnaires (see map 31, p. 1686); the 1905 map added details drawn from local observations of the disease's spread.

4.2. Findings requiring further study

4.2.1. Healthy carriers

It was discovered in small stages that man and human populations who had never been exposed to trypanosomes did not react to infection with this parasite in the same way as those who had lived in endemic areas for a long time (community tolerance?). A previously unexposed population would be very receptive and the overwhelming majority of its members would develop a serious disease characterized by disturbing severe symptoms and a high death rate. Nevertheless, some members of such a population were escaping infection and others were only slightly affected. This explained how in foci which had become endemic, one could find active cases progressing in various stages of development, side by side with healthy carriers and perhaps even naturally recovered subjects.

It should be borne in mind that, just as there are trypanosome-tolerant animals (certain cattle, such as the N'Dama and Mutera breeds), a large proportion of apparently healthy trypanosome carriers can be found.

a) Defence mechanisms, receptiveness and tolerance

Mechanisms of defence, tolerance, and individual and community resistance come into play during the course of an infection. As a result, individuals who have never been exposed to the pathogen or its antigens (both specific and related) will be highly receptive to infection and develop a serious illness. Similarly, in endemic foci some people will remain free of infection, others will develop only a mild illness, while the most receptive will manifest as having the full-blown disease.

The risk of exposure to sleeping sickness varies according to gender and age. The most exposed group consists of individuals between the ages of 15 and 35, especially women. Children are exposed to a variable extent and are the most receptive group. While they generally stand up well to the disease, infantile trypanosomiasis does exist but often goes undetected among other acute illnesses.

The nutritional state and the overall health also influence the course of the disease.

b) Reports on healthy carriers

Various researchers, when comparing their experiences have reported meeting healthy carriers, or spontaneously recovered subjects, but did not publish these observations.

- Healthy carriers of *T. gambiense*

Van Hoof readily recalled that he had found Africans in good health, while their medical records showed them to be trypanosome carriers. *J. Rhodain's* signature on some of these records testified to the fact that they had refused all treatment.

Information and observations were patchy and obtained by chance: *Bruce* (1903) was sent to Uganda by the Royal Society's Sleeping Sickness Commission during the deadly Ugandan epidemic at the start of the century. He found up to 30% healthy carriers even at the height of the epidemic. *Corré* (1876), in Senegal, *Dutton and Todd* in Senegambia (1903), found parasites in perfectly healthy individuals. *Dutton and Todd* estimated the incidence of healthy carriers in the Congo Free State (EIC) at 4.3%. These findings do not mean however that healthy carriers could not become ill. They were simply found side by side with patients in various stages of the disease, with people who had recovered normally and with healthy carriers who had become negative.

The cases of the Bozo fishermen, in Mali, who were found to be carriers of parasites, were readily recalled. *Sentilhes* (in report *Organisation de Coordination pour la lutte contre les Endemies en Afrique Centrale, OCEAC*, Organization for Coordinated Control of Endemic Diseases in Central Africa) monitored a carrier in Upper Volta (Burkina Faso) whose general condition was good, although he remained parasitaemic throughout consecutive quarterly examinations.

Observations from Lake Tumba in the Congo (Zaire) are however of a doubtful validity. Among a group of 4,500 inhabitants of the Central Congo, *Wéry and Burke* (1972), using immunofluorescence found 90 positive reactions or a prevalence of 9%. Fifty-seven of

the positive subjects showed clinical symptoms of the illness, 17 had no symptoms, and three had trypanosomes in their bloodstreams but no abnormal findings in either the lymph or the cerebro-spinal fluid (CSF). Their clinical condition remained unchanged during 18 months of observation.

The case of a woman by the name of Feo de Pagouda, from Togo, is famous. Trypanosomes were found in her blood for more than twenty years, during which time she exhibited no appreciable clinical signs. Moreover, these trypanosomes remained totally indifferent to trypanocides. *Lapeyssonnie* (1960) studied this resistance in detail. The strain was isolated but not identified with certainty.

Dr. Sieniewsky detected a healthy carrier at Lodjo Hospital (Ituri): a man who had not been moving about in an endemic area for more than 17 years. He had been hospitalized by his employer, who wanted to make sure that there was no physical cause for an unexplainable indolence that had suddenly struck this previously excellent worker. The man had harboured the trypanosomes for all those years, providing proof that the absence of symptoms over a long period of time was not a final guarantee that his condition was safe (*Janssens*, 1977).

This case recalls the Senegalese riflemen detected in France by *Baonville*, *Ley* and *Titeca* (1934) as having the infection, as well as the numerous observations made among African soldiers sent back from Indochina to Marseilles for sleeping sickness.

- Healthy carriers of *T. rhodesiense*

More surprising, at first glance, are the healthy carriers of *T. rhodesiense*, since this trypanosome has the reputation to provoke an acute or hyperacute illness. This is perfectly true in the case of new subjects, such as tourists. *Barlow* (1953) mentioned two healthy carriers of *T. rhodesiense* out of a total of 1,850 people he examined; *Graham* (1966) cites two out of 465. *Ormerod* (1961), also *Blair* and coworkers (1965) found a number of healthy carriers in foci with this trypanosoma. A ranger in the Kagera National Park (Rwanda) had peripheral parasitaemia for months before a sterilizing treatment was applied. Once treated he could return to his work without endangering the park's visitors. He had been a healthy carrier who could only be classified as a subject in good health with an uncertain future.

c) The problem of latent infection

The medical physicians, who were intrigued by the up and down pattern of parasitaemia following trypanolytic crises, were completely at a loss when they had to deal with the irritating problem of latency

(*Collomb et al.*, 1956). The many papers and reports on dormant infection reveal some uncertainty about the phase attained by the disease, because some subjects recovered while others developed neurological symptoms after years of latency. This kind of evolution of the disease was to be expected, as trypanosoma carriers may develop an immunity which does not kill all the trypanosomes so that the parasitaemia remains unstable. Intercurrent diseases such as dysentery, pneumonia, viral infections, physical and even mental stress, also famine and any combination of these factors, can bring the period of latency to an unpredictable end.

Close scrutiny of some reports on endemic situations shows that parasitaemia prevalence of 4% or more (often interpreted as a sign of epidemic disease), is devoid of significance in certain foci, such as in the Lake Chad basin (*Duggan*, 1962).

Ginoux and *Frézil's* study (1981) in the historical Ngabé focus, on the right bank of the Congo River opposite Kwamouth (see map 36, p. 1708), is even more interesting. These investigators monitored the population of the focus during five years, by microscopic parasite detection, IF tests and CSF examination. The successively conducted surveys show that 10.3% of the population examined (173 out of 1,678) were seropositive while most of these seropositive patients were without symptoms. The latency period, taken to be five years for the authors, can extend up to 16 years. This period of symbiotic existence with the trypanosome may be disrupted particularly in young people and the elderly. The first warning sign is the development of enlarged cervical lymph nodes, which may be a fleeting phenomenon, but could point to the evolution of a cerebral polarization. This was a very important original and unprecedented finding that we owe to *Ginoux* and *Frézil*.

It is wrong to put those who are only seropositive in the same category as the patients, since they are merely parasite carriers with an unpredictable prognosis. Moreover, in all endemic foci there are chronic trypanosoma carriers who were patients and were treated but not deparasitized; there may be other carriers who refused the treatment but were later discovered to be ill, and finally, carriers devoid of symptoms.

It is normal for the least resistant individuals in an endemic focus to be full prey to the disease. In long-standing foci, this selection of the weakest would lead towards a tolerance for the disease, revealed by quiescent *T. gambiense* infections. Future research will undoubtedly discover the factor or factors which

determine tolerance to trypanosomes, which may be genetic.

Mass sleeping-sickness control campaigns treat all parasite carriers, even those who exhibit no clinical signs including enlarged lymph nodes. Indeed, asymptomatic trypanosome carriers are to be found among those who refuse treatment. But such systematic treatment raises ethical problems. All carriers bear the risk of dissemination, but to what extent does the security brought to the community by treatment of all seropositives offset all risks inherent to the available drugs?

It is therefore very important to screen for asymptomatic carriers by systematic serotesting, bearing however in mind that serotests are not 100% reliable: the field conditions in which samples are taken in rural areas, the existence of multiple parasite carriers and concurrent viral infections request discernment in the interpretation of the tests. The specificity of the antigen also comes into play; the reliability of the CATT test is geographically uneven, because the dominant VAT strains are different from one region to another. It is also very important to be able to identify the most reliable immunological, parasitic, and biological criteria, and herewith the most reliable clinical signs. When current programmes will be adjusted to these new conditions, it will be essential to identify the individuals requiring treatment.

The proportion of asymptomatic carriers in a given population might moreover allow to detect persistent endemic foci, but there is no technique at present that makes possible to identify separately the healthy carriers.

4.2.2. *The animal reservoir*

The existence of a non-human reservoir that might explain the reappearance of *T. gambiense* sleeping sickness in a supposedly extinct focus attracted the attention of many investigators.

a) The role played by pigs drew a particular attention. Indeed some tsetse flies from the *G. palpalis* group are attracted by both domestic and wild pigs, while piglets tend to follow women going to waterpoints or river; those women bring the tsetses with them on their way back to the village.

Greggio (1916) in Inkisi, and Schwetz in Kwango stressed these close relationships between tsetses and swine, due at least in part to the pigs' passiveness (mentioned only in late publications of Schwetz, 1946).

In the thirties and forties *L. Van Hoof* and co-workers (1934, 1947, 1948) examined with great persistence the possibility to infect domestic animals (pigs,

goats, sheep and dogs), and a few wild animals (antelopes and monkeys). These could maintain the trypanosoma transmission cycle through laboratory-raised tsetse flies; in such laboratory infection some animals develop an acute illness, others chronic or fleeting disorders. It is not surprising that a veterinarian caught an acute *T. gambiense* infection from a dog (Denecke, 1941) albeit this is not significant for epidemiology.

The work of *Van Hoof et al.* (1942) on *T. gambiense*'s behaviour with pigs is to some extent completing Greggio's work. The study of the tsetse flies' feeding habits confirmed their preference for pigs; these account for 75% and even more of the glossina's diet, as confirmed by *Laveissière et al.* (1985), who studied the trophic preferences of *G. palpalis* in the forest regions of the Ivory Coast. These researchers were able to detect two different sources of blood in the stomach of the same fly. The origin of the blood was identified by the most sophisticated methods, as complement fixation tests and ELISA, which may be used in series and enables one to identify the blood of successive meals. Human and bovine blood are fairly frequently found together, but other combinations may include human blood and blood from small African antelopes, elephants, or monitor lizards. However, human and porcine blood have never been found together, not even in a series of tests which detected pig blood in 45% of the specimens. This observation might be taken as evidence against the pig's role as a reservoir of West African sleeping sickness in humans. It raises however the question of the role of pig farming: do pigs, perhaps form a protective screen, safeguarding humans from tsetse bites? Such hypothesis finds an echo in the fact that glossinae go for human blood only when herds of pigs have been wiped out by African swine fever (plague of swine). The same holds in forest plantations, where the absence of pigs induces the glossinae to feed almost exclusively on human blood.

b) Another reservoir to be considered in the flies' natural habitat is the bushbuck or harnessed antelope (*Tragelaphus scriptus*), since the flies count on this animal to furnish them with a constant high percentage of their meals (45%). This possibility was put forward by *Duggan* (1962) to explain the persistence of transmission during the quiescence periods in the focus around Lake Chad (see p. 1754).

c) Since the observations of *Heisch et al.* (1958) and *Onyango et al.* (1966), the role of the harnessed antelope and other bovidae has been further demonstrated

in maintaining *T. rhodesiense* sleeping sickness, which is a zoonosis.

More recently, Molyneux (1973), Mehltz (1986), Losos (1972), and Kageruka (1989) in the review of existing data, looked for routinely infected animals. The main difficulties were to identify the isolated polymorphous trypanosomes and to separate the various associated trypanosomes. The use of modern selective diagnostic techniques, such as DNA probes, will probably lead to decisive contribution.

4.3. Research prospects

The complexity of human African trypanosomiasis problems requires an interdisciplinary approach. All disciplines studying mankind have to meet in this research, provided that they should not remain too introspective, as this would lead them either to go completely astray or to achieve only fragmented research.

The clear dissemination of trypanosomiasis in foci requires a thorough study of well-defined foci, the choice and number of which must be representative of their great diversity in Equatorial Africa.

Modern epidemiology will enable to study in depth, all the factors contributing to the existence of a few such endemic foci and to the appearance of epidemic outbreaks, with endemic-epidemic states in between. The programmes must be ambitious in their rigour for analysing the foci chosen to be representative for the whole area. These thorough investigations must circumscribe the true foci without straying into extension areas which represent only temporary phenomena. Such a bio-ecological niche must have clearly limited boundaries to allow a comprehensive study, which should not overlook a single detail of the different pieces in the chain of transmission.

Ecology is all the more important as it determines the areas in which trypanosomiasis may exist or take root. Even if phytosociological studies are difficult, the vegetation in the landscape is an excellent criterion.

Operational research in the area should be reviewed along the country's public health measures and according to the development of a primary health care strategy and the established health centres. It should include an effective surveillance system, preferably based on voluntary participation, so as to create a sentinel type of collaboration. The volunteers must be thoroughly informed about what is expected of them.

Fundamental research is nowadays more vital than ever, for it will provide the key for understanding the complex relationships in the parasite-vector-man triad, to make more reliable diagnoses, and to determine useful areas of pharmacological research.

Fundamental research must cover the triad:

- *man*, as the victim, prime disseminator of the disease and generator of various changes, particularly in the ecology; studies on the relationship between his behaviour and the risk of transmission, with its bio-medical implications;
- the *trypanosoma*, the pathogen and thus the key element; some of its secrets were discovered, but its relationships with both vector and host require further clarification;
- the *tsetse fly*, essential for the transmission, with its not yet fully understood behaviour, could benefit from modern entomology. The natural sciences should provide new explanations.

4.3.1. Research on man

Besides major climatic changes and the disappearance of the tsetse flies' normal food sources, for example when the rinderpest epizootic exterminated the ungulates, it is man who most constantly upsets the balance among the three essential elements of transmission.

The complex relationships which develop as soon as man has been inoculated with trypanosomes include in the human organism local and possibly wider reactions. A latency period of variable length but often of long duration is followed by concentration in tissues. The main factor in this process is immunological, which at the same time involves clinical and biological reactions. The immunity which develops in man is not completely eliminating the parasites and is thus unstable, but sufficient to maintain a state of balance especially in the permanent foci.

The unquestionable presence of *healthy, asymptomatic*, latent or *trypanosoma-tolerant carriers* must be defined quantitatively in various foci, and qualitatively by a better defined infective potential. The potential of these trypanosomes to contaminate other people, must be determined along with the presence of homologous, even heterologous, antibodies. The consequences of trypanosoma elimination from asymptomatic carriers by mass treatment, must also be assessed, particularly regarding their lowering or disappearing of premunition.

The question of spontaneous recoveries require thorough observation on solid ground.

The influence of lifestyle, nutrition (especially the consequences of famines), housing, transportation and occupation, the introduction of various modern agricultural and herding methods, the opening up of forest and botanical reserves, deserve a profound study, but also all factors that may play a part to change existing ecosystems or man-fly contacts:

- in the forest, man is clearing the vegetation to open up new land for agriculture and to allow sunlight to penetrate. This means deforestation, which leads certain tsetse species to disappear, other species to adapt to the secondary vegetation or the introduction of species from other habitats. This process is exacerbated and speeded up by large scale plantations and farming.

- in savannahs, settlement of the grassland is accompanied by the felling of trees in wooded areas to provide timber for the building industry or firewood and by hydraulic engineering projects, mainly for irrigation and rice paddies. These practices will have marked repercussions.

- urbanization is an irreversible process linked to industrialization and technological progress. As in other Third World countries, this trend has gained in Central Africa such momentum that it is getting out of control; projections suggest that 30 to 50% of Central Africa's population will be urban by the end of the century. In theory the towns form an ecological barrier to the tsetse flies, but uncontrolled growth has brought some habitats for these flies within the limits of the city; this is the case in Douala, Bamako, Bujumbura and Kinshasa. Urban sleeping sickness is a common occurrence wherever urban or suburban development maintains micro-environments which favour the life of tsetse flies.

- various means of *transportation* have enabled sleeping sickness to advance; the *caravans* enhanced its progression by their very slowness, broken by numerous pauses; *steamboats* have spread trypanosomiasis via the wood-cutting stations set up along the river to fuel the steamers. Road traffic and railways then took up the relay, transporting not only the tsetses but also men carrying parasites (Ruppel and Kazyumba, 1977). Nowadays modern oil-engined riverboats only occasionally transport tsetse flies. Air transport did not play any role in the spread of the disease.

4.3.2. Research on trypanosomes

a) Species

It is urgent to identify in as much detail as possible the species and subspecies in the various types of foci; to this end enzyme electrophoresis studies will allow differentiation in zymodemes (population of parasites belonging to the same iso-enzymatic type), and the use of DNA probes will enable precise identification. This will provide an objective basis to discern the similarities and differences between the various foci and to determine the origin and the routes followed by the infection.

The first item remains the microscopic diagnosis of the parasite. Blood is not always a first choice because

of the low parasite densities (except for *T. rhodesiense*) and the trypanolytic crises, which commits to repeated examinations. Searching for the parasites in enlarged lymph glands is easier; but in acute cases, the blood analysis might be more reliable. If trypanosomes are present in the CSF, this reveals cerebral extension.

Various biological diagnostic methods have gradually been added to the arsenal of laboratory techniques in use. It is possible to screen blood for the presence of IgM proteins, and to define whether they are monoclonal or polyclonal. For a short period the presence of beta-macroglobulin in the serum was assumed to have a diagnostic value (Mattern *et al.*, 1961), the beta-macroglobulinorachy is a significant element (Mattern, 1962). This does, however, not change the seroproteins. Antigen and antibody tests have led to a range of useful serological techniques: the agglutination test in his most convenient mode: CATT (*Card Agglutination Trypanosomiasis Test*; Magnus *et al.*, 1978) or Tryp Test (Pepin *et al.*, 1986) allows rapid field diagnosis during mass screening for limiting the search for parasite to the seropositive individuals. The Latex Agglutination Test (LAT) is another convenient technique. Complement fixation (CF), immunofluorescence (IF) and ELISA tests lend themselves to automatic processing to be carried out in a laboratory, but this implies the difficulty of finding again the positive individuals during a subsequent visit.

CSF analysis by cytological tests, protein assays, electrophoresis (total IgM and IgM fractions) and serotyping have drawn much attention. Attempts have even been made to seek out significant differences between the lumbar and cerebral fluids by pneumoencephaly (after injecting air as an endeavour to establish the early development of nervous lesions). Suboccipital aspiration, although rather easy to perform, has never been really accepted.

b) Variable virulence

An interesting epidemiological problem is the highly variable virulence of the trypanosomes found in different foci. This variability has been studied in a wide range of laboratory animals, but the significance of the pathogenic role of the various strains has still not been elucidated. A reliable model combining perhaps *in vitro* and *in vivo* methods, would be very welcome. It would also be useful if we were able to define the criteria governing a decline in virulence. However, the changes following the passage in heterologous or unusual tsetse fly are insufficiently known.

The infectiveness of various strains, the infective dose, and the infectivity thresholds of both vector and

host must be determined, as must be the presence, in symptom-free carriers, of infective short forms whose features other than morphological must still be determined.

DNA probes will probably help define the relationships between strains from different foci and thus provide clues as to the ways and means whereby they spread. More specific identification of the strains found in various foci will help in selecting the best variants to serve as the source of antigen for diagnostic tests and the possible production of vaccines. It may also help to determine whether the virulence of strains observed in laboratory animals, is meaningful for the human host. Another so far unanswered question is whether or not specific tissue tropisms exist, such as for instance neurotropism.

c) *Biomolecular research* into the metabolism of blood-dwelling and metacyclic trypanosomes is necessary to define the most specific metabolic axes for the trypanosomes. Such research has already revealed a trypanosome organelle with a major glycolytic function, known as the *glycosome*, present only in Trypanosomidae (see p. 1732). This organelle, which is not found in mammals, is an interesting target for developing a trypanocide that would be harmless for mammals (Oppen- does, 1985). Biomolecular investigations of this kind undoubtedly hold many other surprises.

Basic research will also enable the discovery of the nature of substances which are absent in the host but are able to inhibit irreversibly the trypanosoma enzymes. DFMO (see pages 1727 and 1773) is an example of such a substance, although it was not synthesized for this purpose.

d) *Advanced biotechnology* has brought to light the trypanosomes' ability to cope with host reactions preventing their introduction into the organism. Such selective adaptations are continuously made by trypanosomes thanks to their vast array of variable surface antigens (VSA), each unit of which being coded for by a specific gene; these are activated in a sequential pattern.

A way to control sleeping sickness rests on the discovery and the application of a molecular mechanism specifically designed to block VSAs.

The trypanosomes' adaptability has been displayed by their ability to continue their life cycles in the tsetse, despite an environment with many hindrances. These occur in the digestive system, in the haemocoel and the salivary glands, where the trypanosomes multiply and give rise to their metacyclic forms. The various mechanisms that come into play are still inadequately known.

e) The origin of the trypanosomes that infect man is still a basic problem. Their capacity to develop infective metacyclic forms in the tsetse has probably enabled them to make the switch from life in a cold-blooded insect to life in a warm-blooded vertebrate.

Mechanical transmission of the human trypanosomes is negligible, infections due to transfusions are rare, and congenital infections probably even exceptional.

An animal reservoir is of primordial (but not exclusive) importance in the case of *T. rhodesiense* trypanosomiasis; this is also under consideration in *T. gambiense* sleeping sickness, as it might explain the late or unexpected resurgences of the disease in endemic foci.

4.3.3. *Characteristics of glossinae*

a) More precise *identification* of the glossinae is necessary, based on more sophisticated taxonomic criteria than on morphology alone; the use of biochemical and/or genetic identification techniques such as DNA analysis and monoclonal antibody studies, have become essential to modern epidemiology.

Their systematic classification – revised by Barros Machado in 1954 – divides them into two subgenera, Nemorhina and Glossina, corresponding respectively to the palpalis or hygrophilic group (including *G. p. palpalis*, *G. f. fuscipes*, *G. f. quanzensis*, *G. f. martinii*, and *G. tachinoides*) and the morsitans or xerophilic group (among them *G. m. morsitans* and *G. pallidipes*). Geographic subspecies probably exist and may be of definite epidemiological interest, with particular regard to the characteristics of the tsetse flies theoretically able to transfer the parasite in areas free from sleeping sickness; could they be identical or dissimilar? A study of such a possibility is to be done for various biotopes.

Analysis of the tsetse flies' genetic system (which contains a record of their past) would make it feasible to consider detecting kinship between species, and it would be useful for understanding the origin and the spread of African sleeping sickness.

b) Tsetse flies are known to be poor vectors. It would nevertheless be good to be able to define their *vectorial capacity* in terms of their age and of their genetically-controlled adaptive systems.

c) Despite the difficulties that this would entail, it would be highly desirable to get models to enable the study of the various *adaptations* of tsetse flies under controlled conditions.

The receptiveness of the various species and subspecies should be ascertained. This is important for

grasping the principles that would allow easy laboratory adaptation of various trypanosomes to a range of different tsetse flies. Such adaptations are uncommon in nature, in a few cases they had serious consequences such as giving rise to the adaptation of *T. rhodesiense* to *G. fuscipes* near dwellings, which has completely upset the conditions of transmission.

The vectors certainly have mechanisms enabling them to prevent or limit the entry of trypanosomes into their bodies; these mechanisms are still poorly known.

Much remains to be discovered about the trypanosome's evolution within the vector: its ingestion, the role of the peritrophic membrane, the trypanosome's transit and transformations in the fly, the development of the metacyclic forms and their potential.

d) The flies' feeding preferences, appearing often to be taken at random from one meal to the next, and their different hosts, can be determined thanks to the advances made in biotechnology. The results of these analyses are especially useful to detect the presence of human and non-human blood in the same tsetse fly, thus making it possible to recognize the most probable associated *non-human carriers*. In a studied site in the Ivory Coast, the first findings pointed to the bushbuck (harnessed antelope) and ruled out the pig.

The choice of host is determined by its degree of coexistence and its usual availability. This also applies to the preference for human blood (anthropophilia), which is basically determined by necessity. The most important detail with regard to transmission is the random choice of the first meal, which to a large extent governs the choice of subsequent meals. If the first meal of newly-emerged tsetse flies is taken from a parasite-carrying host, the fly will remain infectious throughout its life.

e) The degree of infectivity, the infective doses, and vector capacity of the various vector tsetse flies must be determined as far as possible, as must be the infectiveness of blood meals taken on hosts who dispose of a non-sterilizing immunity.

f) The dynamics of tsetse fly populations is one of the key epidemiological factors, which depend on a great many factors. These are the fly belt's seasonal movements; the ebb and flow of the tsetse's range and the causes thereof; the density or dispersal of the tsetse flies; the areas of contact or coexistence for two species (with possibility of hybridization); the genesis of tsetse fly swarms, with their significance; daytime or night-time activity; the properties of the

flies' hunting grounds and flight patterns, their active and passive travels, cycles of aggressiveness, day and night resting places; the flies' life span and reproductive cycles. Attention must be drawn to the importance of reappraising naturalists' contributions, for they alone are collecting the ecological and ethological data, whose importance is no longer disregarded by field personnel and managing officers who have to plan selective and as well as effective control programmes.

g) The ecology of the pupae habitats or puparia calls for soil studies. The geomorphology and the physico-chemical characteristics of breeding places determine gaps in the areas over which the flies are distributed. Such studies are even more important when it comes to tsetse flies living in refuge areas.

In nature the connections between *T. gambiense* or *T. rhodesiense* respectively with hygrophilic and xerophilic tsetse flies, appear to be very strict, although in laboratory these relations are easily interchangeable. The transmission of *T. rhodesiense* by *G. f. fuscipes* under new peridomestic ecological conditions was a major change that could be repeated.

h) In Africa there are vast areas harbouring tsetse flies which could theoretically be vectors for human trypanosomes. Such areas are depicted as tsetse infestation without sleeping sickness (Neujean, 1963). On the other hand, despite the persistence of tsetse flies, trypanosomiasis has completely disappeared from areas that used to be swept by particularly virulent epidemics, such as the shores of Lake Tanganyika. The hypothesis that some tsetse flies eliminate the trypanosomes through an anti-trypanosome factor in the haemolymph, is amazing and justifies research into the non-receptiveness of certain tsetse flies; but it does not concur with all the observations.

Tsetse fly control measures maintain all their value. But natural measures can validly be prescribed only if there is a thorough knowledge of the vector's ecology and ethology and provided regular monitoring of their results. The same goes for the use of traps of various designs and colours, for insecticide-soaked screens and ground or aerial spraying of insecticides (with regular control of absence of resistance), or for weed-killers and even for arboricides in infested thickets.

4.3.4. Endemic foci and the outbreak of epidemics

a) Foci

The choice of the term *focus* was a wise one because it signifies as well the gathering of members of a living community as the point from which a fire spreads.

Historical foci (permanent or primary) can be differentiated from secondary or extension foci, and residual foci.

The notion of a focus is convenient, but the full extent of the term's meaning is very complex and does not go beyond describing the ecological circumstances involved, because of both the interdependence of the three components (parasite, vector, and host) and the riverside variants such as lacustrine, suburban, regional stretches or spreadings. It is thus imperative to clarify at least a few of the key elements beyond mere schemata.

Endemic foci are composed of factors which make up a whole and have acquired an equilibrium that is essential for their continuance. They are dispersed within the distribution area of tsetse flies according to as yet unknown heterogeneous patterns which are strictly limited to tropical Africa, with the exception of Madagascar: these foci ignore political boundaries, but have biological boundaries drawn up by the ecosystem.

They are either isolated or occur in small clusters or niches, without any apparent links. These sparse niches are not distributed at random, but result from the convergence of factors of an unknown number and nature. Their classification into localized forest or savannah foci is all too easy, and requires a more thorough study of some well-chosen foci.

The ecosystem in which the focus is located is of prime importance.

b) *Natural and refuge habitats*

Natural habitats are multiple and highly diversified, and correspond to particular vegetations according to the way their type originated. It may be evergreen umbriferous rainforest, secondary forest, thick bushes and scrubland, mangroves, swamp forest, seasonally flooded forest, gallery forest, a forest-savannah patchwork, finally grassy, scrub-covered or wooded savannahs. To describe correctly foci such details are not superfluous. These various types of vegetation have their own specific characteristics which, at a more advanced stage of botanical knowledge, become phytosociological systems. In the savannahs, an experienced eye will immediately pick out areas with the appearance of a focus.

There is also a well-defined relationship between the plant communities and the soil, which is moreover an important element for the choice by the tsetse flies of a site for its pupae.

The descriptions of these biotopes must be supported by climatic data, such as hours of sunshine, temperature, rainfall, atmospheric humidity, wind patterns, seasons and their alternation.

Besides the natural foci there are special biotopes, known as *refuge habitats* which do not follow the usual pattern of host-fly contacts. These consist primarily of palm and banana groves in which human dwellings are implanted, hedges and lantana thickets planted around or near the huts, and camps set up for seasonal labour on the edges of coffee, cocoa or sugar-cane plantations harbouring tsetse flies. Such shelters have been known for a long time but have not received the attention they deserve. Searching for puparia in these biotopes would be very useful.

c) *Epidemic outbreaks*

Epidemiological analysis strives to determine the causes which change the steady state existing in a permanent focus into an appearance of active human disease which sets off an epidemic.

In the tsetse fly area the transmission of human trypanosomes can be the rule in some areas, but be irregular or even temporary, or even absent in others. The identification of a subgenus of trypanosomes became possible by screening for biochemical characteristics (chromatography, large-scale spectrometry of the cuticle) or by enzyme electrophoresis and cytogenetic analyses; such studies could provide useful informations, or if not provide some solutions.

The fact that villages, free from sleeping sickness or resistant to the disease, may be found near similar villages in which the endemic prevails is not a matter of chance, but calls for further research which could make evident some facts useful for reorienting control strategies. The existence, side by side, of active and inactive foci, is the result of adaptations over the centuries or millenia, which helped or impaired the installation of an equilibrium between each of the three partners, thanks to their respective defence mechanisms.

The question of transmission within the same family or the same house or hut, as already regularly mentioned in the first reports on sleeping sickness, but fallen into oblivion, has known a renewed attention since 1975. Such forms of infections seem only occasionally connected to simultaneous infections caused by a tsetse fly whose interrupted infective blood meal on a first host was completed on the next – a rather common occurrence. Other purely mechanical modes of transmission have to be sought for, as such as by mosquitoes (Burke, personal communication).

Epidemiological surveys must pay more attention to the difference between a parasite carrier and a patient, and revise the classical too global indices: the new contamination index (NCI), the circulating virus index (CVI), and the total contamination index (TCI).

It would be useful to check whether the parasitological threshold for an endemic situation set at 25% positive blood specimens and for an epidemic situation at 75% positive blood specimens (Van Hoof, 1928), and an inverse proportion for cerebrospinal fluid tests, correspond to reality.

Clinical examinations should be more thorough in foci or pilot areas, and ascertain if the time at which enlarged lymph nodes develop or revive has a real significance.

Trials to detect why foci are endemic is not merely an intellectual and scientific exercise, but a steadfast search for the factors and circumstances which could upset a steady state to set off an epidemic outbreak.

The direct role ascribed to sleeping sickness as having depopulated certain territories may be a mere coincidence while other factors seem to be real causes. To ascertain this, it is enough to examine the current demographic situation of certain tribes, such as the Alur and the Wanande, again in full increase although they have paid a heavy toll to sleeping sickness in the past. The depopulation threatening certain groups does not often reveal its origin.

4.3.5. *The shift from endemic conditions to epidemics*

a) *The steady state in endemic foci*

The presence of a certain number of endemic foci escaped the attention of the explorers, even that of medical profession, because the local populations were used to the endemic situation with its steady state which prevailed perhaps for centuries.

A patient with symptoms, fitting in the local current picture of diseases, and presenting besides very long periods of quiescence, was perceived by the community as just one common inconvenience among many others. This was the more readily accepted, as the terminal phase included apathy, lethargy and sleepiness, all of which might have another origin.

This steady-state commonly prevails in endemic foci and means that the infection stops at the state of parasitaemia and produces the disease only in a minority of individuals.

b) *Epidemical spreading of the disease*

From such endemic foci sleeping sickness outbreaks spread by leaps and bounds.

Epidemics arise from a destabilized steady-state during which time the focus did not draw attention. Such sudden reawakening of a focus thought to be extinguished, or of a recrudescence in a quiescent focus results from disruptions in the balance between parasite, vector, and host, which may be temporary or lead to extensive expansion of the disease. They spread not nec-

essarily linearly along river systems, giving the impression of having a common origin but also in width by using other penetration ways which suggest an irregular development without excluding a multifocal origin. In this latter case, the disease spreads by leaps rather than continuously.

The mode of infection is no longer sporadic as in endemic foci. The spread includes hosts with poor or even absent defence mechanisms. As a result, parasitaemia reaches very high levels, the biological cycles accelerate and the virulence of the trypanosomes increases.

c) *Promoting conditions for epidemics*

The circumstances promoting an epidemic have been well described by local people, who only became aware of the existence of the disease when a flare up occurs. This came about when ordered and placid existence was disturbed by raids, external or internecine warfare, migrations and famines caused by climatic conditions or locusts. They had also noticed that the epidemic outbreaks at the beginning of colonization were produced by large scale disruption of normal and balanced conditions, but involving more extensive territories. Such epidemics were considered as brought about by the colonizers' military expeditions and pacification campaigns. Interesting examples are described in the Haveaux report (1945), and in those on the Sangha focus.

The fact that outbreaks of particularly widespread and deadly sleeping sickness epidemics occurred at the same time as conquest is undeniable. Medical and public health personnel in tropical areas know that a political change has widespread repercussions, particularly the sudden quickening of a previously rather slow pace of life. The rapid occupation of new economic areas was a requirement of the Berlin Conference and was made possible as well by increased river traffic (barges, boats, caravans of porters) as by the increase of exploratory expeditions. At the same time, wood-cutting posts to provide fuel for the steamboats were set up along the various navigable waterways. This caused infections of the lumberjacks, who changed the ecology through their wood-cutting which produced new habitats for the tsetse flies. These vectors became infected either on site, or on passing parasite carrying travellers.

The increase in overland caravans crossing areas off the traditional tracks of interregional travel, brought about the introduction of the protozoa into new areas by trypanosome-carrying porters.

d) *Changes in ecological conditions*

The distribution of the vectors of human trypanosomes corresponds to specific biogeographical conditions which vary according to the species.

There are two groups of tsetse flies based on macro-ecological criteria: the riparian *palpalis* and the savannah *morsitans* groups; but however practical this division may be it does not allow for any recurring changes. Tsetse flies adapted indeed to new ecological conditions prevailing at the end of the 19th century.

The Cuanza coffee plantations epidemic of 1890 (Kwanza), consequence of the ecological change favouring the development of tsetse flies did not attract the attention it deserved.

The discovery of *G. fuscipes*'s adaptation to the lan-tanas used in Kenya and Uganda as hedges around the kraals which were used to get nearer to the natives' huts, showed the importance of the creation of a peridomestic glossina habitat. The following easier transmission of *T. rhodesiense* provoked the outburst of a serious epidemics. This occurrence revived interest in the presence of tsetse populations in palm tree groves, banana, coffee, cocoa and sugar cane plantations confirming earlier findings.

Such findings gave rise to the reconsideration of the most deadly and least understood sleeping sickness epidemic in Busoga: even after cautiously viewing the extremely high figures, it can not be denied that it was a catastrophe. This re-examination of the past led to the confirmation that *T. gambiense* had been responsible for the epidemic and to formulate a hypothesis about its origin (p. 1753).

As this epidemic broke out at a time when rinderpest had wiped out the domestic and wild ungulates on which the xerophilic tsetse flies were feeding, it led to their disappearance. *J. Ford* (1966) based on the description published by *Thomas and Scott* (1935), and drew attention to the fact that in 1890 southern Busoga was a very densely overgrown area of banana trees, and by 1906 it had become like a patchwork of forest and savannah populated by elephants, water buffaloes, antelopes and wild pigs, while trypanosomiasis had disappeared when the population had been evacuated from the fly belt.

However the epidemic reappeared in 1939-1943, and was attributed to *G. pallidipes* and *T. rhodesiense*, and not once more to *T. gambiense* and *G. palpalis* (re-identified as *G. fuscipes*). *Okoth's* observations (1986) testify to the importance of banana groves in Busoga area.

e) Various hypotheses for the shift to epidemics

1° Some have wondered whether the epidemics had not been preceded by an increase in the number of vectors; experience has proven that this has not always been the case. Intensified man-fly contacts are a determining factor that may arise from various cir-

cumstances. Man plays an important role as a reservoir for disseminating the trypanosomes; at the same time, the tsetse is not an inert carrier, nor is it a *flying syringe*, because mechanical transmission by the tsetse or any other biting or sucking insect if possible, is probably very limited in humans, but it is regular for animal trypanosomes (*T. evansi*, *T. vivax*).

2° Sexual transmission between humans was suspected by analogy with this transmission mode in animal trypanosomiasis (*T. equiperdum*), but if it had been common, the epidemic catastrophes would have attained incredible proportions.

3° Given the conditions presiding over the cycle of the disease, an *isolated carrier* would have minimal importance. Therefore, it is impossible that teneral or young flies might take one or more blood meals from such an isolated parasitaemic host, and later infect another person after the trypanosomes have reached their metacyclic stage, but this lone infection does not appear able to spark off epidemics such as those in the Congo Free State. It seems logical that to produce an epidemic a number of parasitaemic hosts would be required and that several foci would have to develop over a rather short period of time.

4° In their outstanding epidemiological study of the Fankana-Kalakateni focus, in an area neighbouring the historic Kwango-Kwangu focus, *De Muynck* and co-workers (1990) described how a first focus was able to remain unnoticed for five years at Bobo, a fishing village on the banks of the Kwango, not far from Kwamouth. After this initial development the focus expanded into a second one at the adjacent village Mutane, which was then the origin of an explosive spread of the disease to other villages all belonging to the Fankan tribe, probably by just following the roads.

In discussing the episode, the authors reviewed the various possible explanations of the events: a villager could have been infected while fishing or working in an endemic focus; an animal could have become a *T. gambiense* carrier; a former sleeping sickness patient had a relapse; an infective tsetse brought a more virulent subspecies of *T. gambiense* into the area; intensified man-fly contacts occurred or changes in the ecosystem; a lowered resistance of men developed; or there were increased population movements and travelling; or even a natural periodicity for the disease took place, as curiously enough, the 33-year interval in this particular case corresponds to the often quoted a 30 to 40 year periodic cycle.

However, any discussion of these hypotheses tends to be based on a too strictly mathematical analysis.

Computerized data handling can even lead to unconvincing conclusions which do not fit the actual dynamics of sleeping sickness.

The only hypotheses accepted by De Muynck and his fellow workers were the introduction of a trypanosoma carrier, the emergence of a more virulent strain, or the increase of man-fly contacts. The last two hypotheses were not really backed by persuasive arguments, and the first hypothesis did not take into account the occasional very long latent period seen in healthy parasite carriers.

5° But during the same period the recruitment of labour for major deforestation and building embankments required for the Matadi-to-Stanley (Malebo) Pool railway line, or administration and trading posts took place. These projects were labour intensive. As the demand for manpower could not be met locally, it was recruited from densely populated areas. There is no doubt that the village chiefs took advantage of this recruitment to get rid of undesirable subjects, handicapped, and witnesses of bygone raids. When workers were selected, the presence or not of sleeping sickness in their place of origin or destination was not taken into consideration.

The full-time porters employed in traditional interregional trading caravans (more numerous than one might think), travelled on slow journeys, interrupted by the various stops required for trading. This offered many opportunities for infection. Tests performed on porters revealed often high rates of infection as well as on sick men, who were frequently left to a miserable fate.

f) *Soldiers, caravan porters and steamboat depots* alleged as disseminators of trypanosomes pathogenic to man. Such a possibility must be neither underestimated nor overestimated.

1° There is a legend that mercenaries from West Africa, presumed to have been the only source of trypanosomes, contaminated Central Africa; but most of the soldiers arriving in the EIC were in fact from Zanzibar and thus not infected. Hausa recruits, who were generally highly praised, were also brought in and held responsible for the introduction of *T. gambiense* (see p. 1755). However, already in 1890, a local contingent of 20,000 men was recruited from all districts of the Congo Free State for a seven-year tour of duty.

The emphasis with which numerous publications have asserted that military operations had spread the disease or had induced epidemics cannot be ignored. Considering the extreme physical effort required from these soldiers during exploratory expeditions and the

hardships and diseases they endured (such as repeated attacks of blackwater fever (mainly in Europeans) or acute malaria, protracted dysentery and hepatic abscesses which resulted in the killing of many soldiers), it would be incomprehensible that none of the alleged trypanosoma carriers, would have reached the clinical stage of the disease. A number of African infantrymen engaged in military operations in the Vietnam (1946-1954) reached, after the same situation of stress, the clinical stage of sleeping sickness.

Congolese soldiers certainly played a part in the spread of trypanosomiasis. The soldiers benefited during their service of physical training with good medical care. On the other hand, long tours of duty and frequent restationing increased their risk of infection.

In the case of the military population, a large number of men, women and children coming from areas free of sleeping sickness were introduced into active foci and were a prey for the disease. Those who returned after fulfilling their contracts or being released because of illness took their trypanosomes with them, and if they were not discovered by medical screening at the observation posts they introduced their strains into their home regions. This is what happened in Maniema, Kasai, Sankuru, Lomami, and the northern Katanga.

2° In addition to the riverine firewood posts with up to 50% infected lumber men, camps were set up in rubber and copal harvesting areas, as well as in clearings of forest areas, made to establish plantations. The tsetse readily settled in such ecological niches.

3° The economic changes, the collapse of the social structure, the recruitment of soldiers and manpower required for major public works, and for mining industries or plantations all were considered blessings of civilization. The superiority complex of the colonizers was an insurmountable obstacle for gathering information about the reactions and problems of the socio-cultural displaced persons. These fundamental changes may have been an underlying cause of the shift to an epidemic course of sleeping sickness.

g) The question as to whether or not other diseases contributed to the development of epidemic sleeping sickness was often raised. There does not seem to be any connection with smallpox. Similarly, a link with meningitis is merely related to a superinfection occurring in the terminal phase of sleeping sickness. Malaria, which is omnipresent, even in sleeping sickness-free areas, does not seem to have had any particular role. If bacillar dysentery was the main cause of death during the Berghe-Sainte-Marie epidemic, this probably reflected a deteriorating state of health among this group of children.

Nothing is known about the possible role of arbovirus infections. Such infections accompany all deforestation operations, since tree-felling is the cause on the ground of all the arthropods maintaining the viral cycles in the canopy. A few studies have shown a certain synergy between the presence of these viruses and that of trypanosomes.

As already said, the role ascribed to sleeping sickness in depopulating certain territories rests on the simultaneous presence of other unidentified factors. The causes of the decline in population that continues to threaten certain groups remain a mystery.

Conclusion

Analysis of the facts shows that there are fundamental differences between foci which do not result only from the existence of different pathogens and vectors.

a) There are *different interpretations for well-established facts*, due to a too narrow field of investigations.

Many specialists (medical officers, biologists, entomologists and epidemiologists) tried to draw general conclusions from a sometimes very broad but geographically limited experience. The same pitfall awaits the new partners in the extended study of trypanosomiasis: the geographers, demographers, sociologists, anthropologists, historians, etc.

1° There is a world of difference between the great stability enjoyed by the hygrophilic tsetse flies in their forest environments, and the changes which xerophilic tsetse flies share with their customary hosts threatened of extermination by epizootics. In the case of rinderpest, except for the flies sheltered in a rinderpest-free pocket, the tsetse flies have each time reinfested the territories.

2° Equatorial forest, sub-Saharan savannah, and East African grassland – to mention only the most typical environments in which the trypanosomes' life cycles take place – differ so much that it would be paradoxical if a naturalist discovered a great number of common determining factors in them!

3° It should be stressed with strong conviction that it is unjustified and fallacious to stick to an approach dominated by a contrast *T. b. gambiense* – *T. b. rhodesiense* as being predictable for the clinical pattern mild and chronic or acute and short-lived. Examples of acute, fast deadly Gambian and chronic Rhodesian trypanosomiasis abound.

In the tropical forest *T. gambiense* infections are indeed proceeding cautiously and under conditions of equilibrium with periods of latency and of a particular tolerance to the pathogen.

In the East African savannahs *T. rhodesiense* persists as a zoonosis producing sporadic or limited epidemic superacute infections, but also permanent foci of subacute or chronic disease. The fact that a wide range of intermediate situations exists between these extremes matches the laws of nature. However, one must remember the very disturbing fact that there are huge areas infested with tsetse flies in which human trypanosomiasis has never gained a foothold.

4° Current and past economic geography must not be overlooked. The important contributory role of the caravan routes and their points of intersection, the long distance travelling to acquire essential commodities deserve further investigation in many territories. The importation of *mondjo*, iron with special properties deemed indispensable by the Kakae who were accustomed to get from the Pahouin, or *benge*, another type of iron that the Zande used to bring back from other distant lands.

The patterns of migration, their routes and the periods which ethnolinguists are just starting to discover, should be carefully listed.

5° Finally, it should be recalled that sleeping sickness control campaigns have completely eradicated the disease from territories swept by huge epidemics despite the persistence of vectors, whereas such campaigns have failed to produce the same effect in other foci. Another unsolved problem is the reappearance of an old focus, decades after it was thought to have been eliminated.

To clarify scientifically the problem of the waxing and waning of sleeping sickness foci it is necessary to conduct more epidemiological surveys in particular situations, such as those carried out by Frézil (1983), Ginoux et al. (1981), and De Muynck et al. (1990). Any new surveys should cover the various topographic macro- and microvariants of the foci. Priority should be given to foci located at the edge of great equatorial forests and savannahs, especially the forest corridors and forest fringes in various types of grasslands, and to foci at the watersheds of the major river basins, or on promontories next to the confluents of major rivers and in mangroves, and similar mosaics.

b) *Other puzzling not yet elucidated problems* have to be solved by the most performing biological and biomolecular techniques currently available:

1° As regards *trypanosomes*, attention should focus on variations in virulence, on infective doses, infectivity thresholds, changes in resistance to arsenic compounds, tissue tropisms and, above all, on more precise identification of the strains responsible for the disease in its various

foci. Such findings will aid to identify the origin and to follow the epidemiological extension of the disease.

2° Concerning *tsetse flies*, it seems possible to decipher the secrets of their physiology, their natural history and ecology; it is vital to better understand the interactions between the vector-parasite and the vector-host systems. To gain knowledge of the fly's vectorial capacity in various foci, further details must be collected on receptiveness, as well as on successive feeding preferences, on the lifespan of tsetse flies and on the length and steps of the parasitic cycle. Tsetse flies, usually very poor vectors of human parasites, are very good vectors of animal parasites; for what reason, remains a mystery.

Accurate field observations carried out by experienced naturalists should enable one to fill many gaps in our knowledge. These naturalists should pay attention to the type of habitats, the active and passive dispersion, the patterns and periods of aggressiveness,

the resting places during day and night, the vector density, the sudden multiplications without influencing the endemic situation; as well as on the well-known difficulty to find tsetse flies in active foci. The development of peridomestic habitats calls both for closer attention and for practical means to eradicate them.

This long recollection over time, through progressing scientific knowledge associated with day-to-day realities, has all along been accompanied by the hope to see a light dawning and enabling the discovery of some connection between scattered truths to establish their common core. Such light has not yet appeared, but the new generation of researchers will be better equipped to reach this goal and consequently to find alternative means of controlling effectively this persistent state of great misery, to the greater benefit of the peoples of Africa.

P.G. Janssens

THE DISEASE AND ITS TREATMENT

1. Infectious agent

The trypanosoma is a unicellular flagellate protozoon, detected for the first time by *Ford* and *Dutton* in 1902 in blood of a man affected by trypanosomiasis. It is a parasite both of man (in whom it causes sleeping sickness) and cattle (causing *nagana*).

Trypanosoma in fresh blood preparations are seen under the microscope as tiny, constantly twisting, mobile, fusiform, unicellular micro-organisms occurring in two forms, a long slender one (25 to 35 μ) that multiplies in the bloodstream and interstitial fluid of the vertebrate host, and a short stumpy shape (14 to 22 μ) that survives and reproduces in the tsetse's midgut.

Stained blood preparations reveal an elongated fusiform unicellular cytoplasmic body, usually incurved by the fixing process, in which one can see a single large round or oval central nucleus containing an achromatic karyosome and chromatin. A small, round, intensely stained kinetoplast will be seen at the posterior end of the nucleus. The flagellum is a thin filament that arises from a parabasal body close to the kinetoplast, runs along the border of the cell's membrane, and emerges as a free flagellum at the anterior end of the cell. It gives the parasite its mobility. The undulating membrane is a fold of the cell wall linking the flagellum to the cytoplasmic body.

The trypanosomes live in the host's tissue fluids, the bone marrow, blood stream, lymphatic system, and

cerebro-spinal fluid (CSF), but are never intracellular. The parasite multiplies by longitudinal binary fission, starting from the blepharoplast (kinetoplast), which duplicates first, followed by the cell nucleus.

2. Vectors

These belong to two distinct groups of tsetse flies: the hygrophilic glossinae (found in forests, forest corridors, isolated forest patches, lake shores) and the xerophilic glossinae. The following species are implicated in the spread of trypanosomiasis in Central Africa (see also p. 1734):

- hygrophilic: *G. palpalis palpalis*
G. fuscipes fuscipes
G. fuscipes quanzensis
G. fuscipes martinii

- xerophylic: *G. morsitans* and *G. pallidipes*

Tsetse flies are medium-sized (8 to 10 millimetres), the male being smaller than the female. They have a stinging proboscis easy to see and long wings that fold over one another at rest. Both sexes can bite.

The larvae develop to term in 8 to 12 days within the fertilized female's body. The female deposits only one larva at a time, making a total of six to ten larvae in her lifetime.

The larvae are deposited on loose and relatively dry soil in the shade. The larva burrows into the

ground and rapidly changes into a pupa. Pupation lasts an average of one month at a mean temperature of 25°C. The adult cannot start to fly until a half an hour to an hour after emerging from its puparium, as it must first allow the hardening of its wings and body cover. It takes its first meal after one or two days. The first host which is selected will determine the fly's habitual dietary preferences. The infection rate will also be higher if the infection occurs at the first bloodmeal than if it happens at a later meal. The fly is infected for a period of two to three months of its life, during which time it will feed every three to four days, preferentially on warm-blooded animals, without excluding reptiles and other cold-blooded animals. Some tsetse flies have a preference for human blood and are said to be anthropophilic.

Tsetse flies mate only once, and the female stores the sperm in her spermatheca. Tsetse control by releasing sterilized males exploit this particular behaviour.

Transmission of infection is due almost exclusively to tsetse flies carrying the short, stumpy, infective metacyclic trypanosomes in their salivary glands. Mechanical transmission by direct transmission from the mouthparts of deerflies or tsetse flies infected during a blood meal, is possible but rare. However one and the same tsetse fly, can contaminate two or more individuals if it is disturbed during a meal. The doses required to infect a host are estimated at ten parasites for mechanical transmission and 350 for the inoculation by metacyclic forms.

The number of infected tsetse flies does not exceed 5% of any population. Furthermore, the species that transmit human trypanosomes are very specific: *T. gambiense* is not transmitted by *G. morsitans*, which is the main vector of *T. rhodesiense*; however this last parasite has become adapted to the *G. palpalis fuscipes*, when this glossina became peridomestic in Uganda, during the epidemic of Alego.

Tsetse flies do not cover a great distance, but have a tendency to let themselves be carried by water and road traffic.

The possibility of animal reservoirs has been recognized for a long time. Wild game animals and domestic cattle are the reservoir for Rhodesian sleeping sickness, while the *Gambiense* form may be harboured by pigs, dogs, sheep, etc. However, the infections are chronic and occasional, and do not play any role in sustaining epidemics. In *gambiense* trypanosomiasis, men (not only patients, but also asymptomatic *healthy carriers*) are a source of infection for tsetse.

3. Transmission

Transmission from the vector to a vertebrate host is linked to the life cycle of the trypanosomes in the fly, ending with the presence of infective metacyclic forms in the hypopharynx.

The short, stumpy forms ingested by the tsetse reach the tsetse fly's midgut, where they develop and reproduce by binary fission.

Long, slender forms appear between the 10th and 15th day and migrate through the peritrophic membrane and intestinal wall to the proventricle, where they metamorphose into crithidia (short form in which the kinetoplast is anterior to the large nucleus). The crithidia then migrate to the pharynx and salivary glands, where they metamorphose back into elongated metacyclic forms, the only stage that can infect the vertebrate host.

The entire life cycle in the fly lasts 18 to 25 days. An infective tsetse remains infective for its entire life (around 120 days).

4. The disease

When a tsetse, carrying trypanosomes in its salivary glands, stings the host to have a blood meal, it inoculates its prey with saliva containing the so-called metacyclic trypanosomes.

After a 10 to 21 day latency period (invasion or incubation phase), a rather painful, itchy reaction develops. This lesion, called a *trypanosomal chancre*, takes the form of a central vesicle surrounded by an infiltrated plaque with whitish desquamation of the skin. This swelling goes often unseen as the physical examination of the individual is not complete, or are masked by other pathological skin conditions.

This initial stage lasts from a few days to about a week and is followed by the spread of the disease throughout the body; invasion of the lymphatic and the reticulo-endothelial system is accompanied by the first symptoms: persistent headaches, intermittent bouts of fever, shivering and weakness.

This parasitic invasion is followed with production of antibodies by the host which bring about lysis of most of the initial trypanosoma population, causing febrile trypanolytic attacks. Polyconal gammaglobulins, particularly IgM, are produced in a high amount. However, some of the trypanosomes will survive the action of the antibodies, on account of the diversity of antigens in the population and will multiply, thus engendering a second invasion of parasites with a different antigenic system. These will, in turn, be attacked by a new generation of sensitized antibodies, and the

process will continue until only a few trypanosomes remain in the blood; at the same time the host's immune defences will be exhausted (see the chapter Serological aspects of parasitic and infectious diseases in Volume I, p. 266).

In some fairly rare cases host and parasite may coexist in biological balance, a phenomenon known as *trypanotolerance*.

The functional and general clinical signs of the disease become overt. Reactions of the mononuclear phagocyte system also appear, including the characteristic Winterbottom's sign (enlarged lymph glands in the posterior triangle of the neck). Other lymph glands may also become swollen. Hepatosplenomegaly is less common.

The visible signs are a typical swelling (oedema) of the face (Heckenroth's *Japanese face*) or a red-coloured rash on the thorax (trypanids), together with itching, which gives rise to the characteristic *aimless scratching*.

The triad headache-oedema-pruritus calls for the diagnosis of trypanosomiasis. Other milder cardiovascular symptoms also occur, such as persistent tachycardia, unrelated to bouts of fever, extrasystoles, and abnormal ECG tracings.

At this stage the first symptoms of impaired CNS regulation of body functions begin to appear: behavioural disorders such as altered personality, insomnia, sensitivity to cold, anorexia, sensory disorders (deep hyperaesthesia, known as Kerandel's sign), and hyperaesthesia of the joints observed by Heckenroth. Neurological examination will reveal specific, cheilo-oral and peri-oral skin reflexes, called cerebral trunk reflexes.

The duration of the haemo-lymphatic stage varies, and may even be interrupted by periods of remission. However, the disease will continue its progress to the stage of CNS invasion, on average after 6 to 12 months.

The clinical picture during this central nervous system invasion is dominated by neurological symptoms: disturbed sleep pattern (insomnia by night, somnolence by day), disrupted thermal regulation (shivering in full sunlight), disordered instincts (anorexia, frigidity), and disorganized endocrine functions (myxoedematous appearance, amenorrhoea, etc.). Motor impairment and impaired equilibrium are striking: awkward gait, legs set apart like an infant, uncoordinated and involuntary movements, shaking of the extremities, and typical tremors of the lips and facial muscles. The picture is fulfilled by sensory disorders, such as pruritus or cramps, and mental disorders: indifference, apathy and depression, sometimes alternating with euphoria and extreme agitation.

The terminal phase is that of a cachectic, emaciated, demented, bedridden patient swept by nervous twitchings. If terminal cardiac or renal complications do not shorten the course of the disease, the patient will eventually slip into a deep coma and die. Brain involvement was already described by Van Bogaert in 1936 (see ref.)

Trypanosomiasis in young children and possible congenital trypanosomiasis deserve to be specially mentioned. In infants and young children the disease progresses very rapidly to the stage of cerebral invasion. The clinical picture may take one of two forms: either a sudden convulsive attack with fever, followed by a coma with characteristic hypertonia of the limbs, or evidence of psychomotor retardation. A lumbar puncture will almost always provide a correct diagnosis.

The incidence of congenital trypanosomiasis is much higher than it would seem from the statistics. This type of the disease follows also a very rapid course, and is invariably recognized at the stage of CNS involvement. Hydrocephaly at birth is a frequent complication.

5. Parasitological and biological diagnosis

As soon as there is any possibility that trypanosomiasis could be present, attempts should be made to detect the parasites by using all available laboratory procedures. Confirming a diagnosis of *T. gambiense* sleeping sickness is difficult, as roughly 40% of infected individuals go undetected in direct examination of blood or lymph; therefore the constraints inherent in mass screening campaigns have emphasized the enormous advantages but also the budgetary constraints of serological screening. Increasingly sensitive serotests have been developed over the past twenty years and were adapted for field use.

This virtually immediate detection of the suspects has still to be confirmed by detecting the parasite itself. Parasite enrichment techniques for body fluid specimens have also benefited from advances in research. The risks involved in treatment with trypanocidal drugs are a good enough reason to ensure that current diagnostic tests give the correct answer.

5.1. Parasitological diagnosis

5.1.1. Direct examination

- Lymph gland juice: Puncture of the enlarged glands (usually the posterior cervical nodes) and examination of the lymph under the microscope (10 × 40);

- Blood: Examination between slide and cover slip of a drop of fresh blood, taken from a finger;
- Bone marrow: Puncture of the sternum or iliac crest to examine a small fragment of bone marrow on a covered slide.
- CSF: spinal fluid, obtained by lumbar or suboccipital puncture, is examined by cell-count (Nageotte, Fuchs-Rosenthal cell), when possible after centrifuging at 2,500 rpm for 10 minutes.

5.1.2. Stained preparations

Thick drops and blood film preparations should be stained with May-Grünwald or Giemsa stain.

5.1.3. Concentration methods

a. Blood

- Triple centrifugation: this consists in centrifuging 10 ml of whole citrated venous blood, then again the supernatant, and finally the second supernatant at 1,500 rpm for 10 minutes before searching for the trypanosomes in the last aliquot.
- Micro-haematocrit centrifuging test (MHCT) in a heparinized capillary tube (Buffy coat technique or Woo's method): the blood is collected in a haematocrit or capillary tube and centrifuged. The tube is then broken at the boundary between the precipitated RBC's and the buffy coat (WBC layer). The trypanosomes are concentrated in the terminal portion of the buffy coat, and a drop of fluid from this separation layer is then examined on a covered slide.
- Anion-exchange filtration technique (Miniature Anion Exchange Centrifugation Technique, MAECT, or Lanham and Godfrey's technique): the trypanosomes are separated from the blood's cellular constituents by running over a DEAE-cellulose anion exchanger column in a buffer solution. The separation is possible by the difference in electric charges of the red blood cells (retained by the cellulose in the column) and of the trypanosomes, which run through with the buffer. The filtrate at the bottom of the collecting minitube or the aliquot obtained by centrifugation is examined under the microscope.
- Blood culture is performed on GLSH medium (glucose, lactalbumin, serum, haemoglobin) or on Tobie's blood agar, a refined version of Brutsaert-Henrard's medium.
- Inoculation of laboratory animals: mainly *Cricetomys gambianus* or the Gambian rat, although Guinea pigs or *Mastomys natalensis*, (house rat) can also be used.

b. CSF

- Single centrifugation (10 minutes at 2,000 rpm).
- Double centrifugation: the sediment of the first centrifugation is recentrifuged in a haematocrit tube with

one end sealed; the sealed end of the tube is examined under a microscope.

5.2. Biological tests

5.2.1. Serological screening tests

- Immunoglobulin determinations (IgM or β 2-macroglobulins) by Mattern's double diffusion technique and Cunningham's droplets or *confetti* technique; these methods have been replaced by more sensitive techniques;
- Indirect fluorescent antibody test (IFAT): the principle of the reaction is based on the formation of an antigen/antibody/anti-antibody complex. The antigen (smear of *T. brucei*-infected rat blood or a suspension of purified and lyophilized parasites) reacts with the antibodies in the human serum, a drop of which is placed on the slide (a droplet eluate can also be used). After a prolonged contact (30 to 60 minutes) and washing, the slide is covered with a fluorescent human antibody conjugate, left for another 30 minutes, then washed and examined under a microscope and ultraviolet light. The trypanosomes will be seen as bright fluorescent green areas if the reaction is positive. The intensity of fluorescence is graded +, ++ and +++.
- Indirect haemagglutination (IHA): the principle is based on binding purified trypanosoma antigens to the red blood cells. The antigen-labelled, lyophilized erythrocytes placed in suspension at the time the test is carried out, are mixed with the test serum. If the serum is positive the RBCs will agglutinate and form a thin film on the bottom of the test tube, if negative they will form a sediment at the bottom of the tube. This test is not as popular as the direct haemagglutination tests.
- Direct haemagglutination (*CATT-tryptest*®, Card Agglutination Trypanosome Test): given its simplicity and rapidity, is by far the preferred method for *T. gambiense* trypanosomiasis screening. The principle is based on agglutination of the antigen present on fixed and stained trypanosomes by antibodies in the patient's serum. A drop of reagent deposited on a plastic card is mixed with a drop of heparinized whole blood in a capillary tube and shaken for five minutes. The agglutination reaction is easily detected in the form of large clumps visible to the naked eye (see the chapter Serological Aspects of parasitic and infectious diseases, p. 269).

5.2.2. Blood analyses

- Cell count (anaemia and hyperleukocytosis), high sedimentation rate;
- Protein electrophoresis (elevated IgM).

5.2.3. Examination of cerebrospinal fluid

- Cell counts: lymphocytes per mm³ (using the Nageotte or Fuchs-Rosenthal cell: normal count 3 lymphocytes per mm³);
- Presence of morular or Mott cells (large modified plasma cells with eosinophilic inclusions);
- Rachialbuminimetry: total proteins, mg per 100 ml (Sicard and Cantaloube's method: 25 mg per dl is normal; Dye Binding Protein Assay with sulfosalicylic: 37 mg per 100 ml is normal);
- IgM and antibody titres.

6. Treatment

The search for an effective drug to treat sleeping sickness has given rise to the production and trial of countless chemical substances. The dyes such as *trypan red* that were so popular in the 19th century marked the beginning of this trend. They were followed by arsenical compounds. More recently, work has focused on nitrofurane derivatives and polyamine synthesis inhibitors such as alpha-DFMO (see below p. 1773).

Drugs that were commonly used in the early days of colonization – the *heroic period* – such as *Atoxyl*®, a cyclic pentavalent arsenical compound (Thomas, 1905) and tartar emetic (J. Rodhain), are no longer used. Others, such as Tryparsamide (used by Louise Pearce, Leopoldville, May 1920) are no longer manufactured – maybe not judiciously. Tryparsamide, for instance, was a good CSF modifier and was active against neurological lesions.

There used to be a great deal of experimentation with trivalent arsenicals (arsphenamine, *Salvarsan*®, neoarsphenamine, *Neosalvarsan*®) on infected small animals. These compounds were very effective against spirochaetes and inaugurated a *therapia sterilisans magna* (Dubois and Van den Berghe, 1947). Trivalent arsenicals (including *Arsobal*®) were later used to cure sleeping sickness (see below 6.3).

Trypanarsyl® (glyphenarsin or *tryparsamide*), given alone or in conjunction with potassium emetic and *Suramin*® (*Bayer 205*®), was the standard treatment for sleeping sickness from 1926 to 1956. Unfortunately, the arsenicals had the serious drawback that they could involve optic nerve damage such as amaurosis and blindness. By the late thirties resistance to tryparsamide was becoming an increasing problem.

Today the drugs are traditionally classified as those acting at the early (haemolymphatic involvement) stage but without any proven effect on the late meningo-encephalo-myelitic stage (such as *Suramin*®,

diamidines, diaminazene), and those acting at all stages of the disease, and are both peripheral trypanocides and active against deep nerve damage (these are melarsoprol, melarsonyl, nitrofuranes and DFMO). However, it is important not to ignore the possible synergistic actions of the combination of available drugs such as *Suramin*®, *Lomidine*®, melarsoprol, levofuraltadone and so on.

6.1. *Bayer 205*®, *Suramin*®, *Germanin*®, *Belganyl*®, *Moranyl*®, *Antrypol*®):

These are today's trypanocides. *Suramin*®, a white crystalline powder, is a complex urea derivative, inhibiting numerous enzymes including RNA polymerase. It is available in phials or multiple-dose bottles to be diluted to a 10% solution in distilled water for IV injection once or twice a week until the curative dose of 200 mg per kg body weight is reached. In practice, an adult will receive six to nine injections of 1 to 1.5 gramme each.

Suramin® may cause fever and attack the skin with mucous membrane reactions, and almost always sets off transient albuminuria. Cases of fatal toxicity have been reported.

6.2. The diamidines (*propamidine* and *pentamidine*)

Pentamidine, an aromatic diamidine, was the most widely used of the diamidines on the market (dichlorohydrate, 4-4-diamidino-diphenoxypentane isothionate or dimethane-sulfonate (*Lomidine*®)). Among other possible modes of action are kinetoplast fragmentation and inhibition of glycolysis (Sands et al., 1985). In practice, the commercially available 4% *Lomidine*® solution was used.

Lomidine solution is to be given by strictly IM daily injections of a basic 3 mg per kg for 10 consecutive days. The injection is administered preferably in the vastus lateralis muscle (lateral part of the thigh), to avoid possible diffusion towards the sciatic nerve. In the early stage of the disease a combination of *Suramin*® and *Lomidine*® appeared to be the treatment of choice.

Diminazene aceturate or *Berenil*® used against trypanosomiasis in livestock, has been used successfully against both *T. gambiense* and *T. rhodesiense* in men. It blocks selectively DNA kinetoplast synthesis.

It is sold as yellow powder in sealed packets of 1.05 g and is a product soluble in distilled water. A 7% solution of the drug is administered three times by intramuscular injection on alternate days, at a dose of 7 mg per kg body weight, or 1 ml of solution per 10 kg body weight.

6.3. Trivalent arsenicals

- Melarsoprol or *Mel B*® (Friedheim), *Arsobal*®

The trivalent arsenical derivative of melarsenoxide combined with dimercaptosuccinic acid (British Anti-Lewisite or BAL) which shows an irreversible binding to trypanothione. Trypanothione is N¹, N⁸-bis (glutathionyl) spermidine, uniquely found in trypanosomatids and is absent from mammalian cells. As trypanothione synthesis is also reduced by eflornithine, a synergistic effect is possible (Fairlamb et al., 1989). Melarsoprol is dissolved in propylene glycol, an oily, colourless liquid. The solution contains 3.6% active substance and is sold in 6 ml (216 mg) phials. The drug must be given only intravenously, with much care to avoid leakage. It is given in doses of 3.6 mg per kg body weight, or 1 ml per 10 kg, with a maximum of 200 to 250 mg.

The procedure for treatment with melarsoprol as devised by Neujean in 1951 was based on the degree of CSF deterioration and is still applicable: the basic design consists of a course of three to four injections, one per day. One to four courses of injections are prescribed, at one-week interval between each course depending on whether the CSF contains 0 to 3, 3 to 20, 20 to 100, or more than 100 cells per mm³.

However, the emergence of arsenic-resistant strains of the parasite has obliged the staff to start treatment with the maximum recommended doses.

Treatment with *Arsobal*® is dangerous, requiring several precautions. The patient's general physical condition must be improved and a close follow-up after each injection carried out. One must be able to recognize and treat Jarisch-Herxheimer type anaphylactic reactions, early incidents of arsenic toxicity (offset by administering BAL, British Anti-Lewisite), and the always serious reactive encephalopathy, for which the only treatment is a slow perfusion with corticoids, with little chances of recovery.

- Melarsonyl or *Mel W*® (Friedheim), *Trimelarsen*®

This melarsoprol analogue is a trivalent arsenical molecule and a derivative of phenylarsenoxide combined with potassium salt of dimercaptosuccinic acid. The white powder product was available in 200 mg doses to be diluted immediately before use in 3 ml of distilled water. It is given by subcutaneous or intramuscular injection.

The dose is 5 mg per kg body weight, or 0.75 ml per 10 kg, up to a maximum of 200 to 250 mg. One or two courses of three to four injections (one per day), with a seven-day interval between courses, was prescribed, depending on the degree of the CSF deterioration.

Melarsonyl had no advantages over melarsoprol other than its ease of administration to young children or as an alternative if melarsoprol had obstructed the veins. It was withdrawn from the market in 1984.

6.4. 5-nitrofurans (*heterocyclic nitrogen compounds*)

Nitrofurans inhibit trypanothione reductase. Trypanothione (see above) is the major polyamine-containing analog of glutathione in trypanosomes (Fairlamb, 1990).

- *Furacin*®, *Nitrofur*®

The Nitrofurans cross the meningeal barrier and are taken orally. The average dose is 30 mg/kg body weight per day, taken in three divided doses for 7 to 10 days, as well for *T. gambiense* as for *T. rhodesiense* trypanosomiasis.

- These derivatives, for which patient tolerance is low, have been replaced by the L-isomer levofuraltadone, which is much better tolerated. Levofuraltadone (NF 602) and its chlorohydrate (NF 902) *Altafur*®

The 200 mg tablets are given orally at an average dose of 30 to 50 mg per kg body weight per day in three divided doses during 15 to 21 days.

This trypanocide has yielded spectacular recoveries in case of melarsoprol-resistant relapses. The drug has also been shown to be highly effective and practically free of harmful side-effects in treating very early stages of both *T. gambiense* and *T. rhodesiense* trypanosomiasis. Unfortunately it has been used only in clinical trials and has never been released on the market.

- Nifurtimox, *Bayer 2502*®, *Lampit*®

This product, which was synthesized for treatment of Chagas' disease, has on the whole yielded very good results in clinical trials for patients with *T. gambiense* or *T. rhodesiense* trypanosomiasis. It may cause also DNA breaks. It is active against arsenic-resistant trypanosomes.

The recommended dosage is 10 to 20 mg per kg body weight per day in three divided doses *per os* for 60 to 120 days.

6.5. Eflornithine DFMO, alpha-difluoromethyl-ornithine, *Ornidyl*®

Eflornithine is a selective and irreversible inhibitor of ornithine decarboxylase, the enzyme that catalyzes the conversion of ornithine to putrescine, the first step in the synthesis of putrescine and of spermine and spermidine, polyamines essential for the growth and the multiplication of eucaryotic cells (see p. 1727).

Since eflornithine has an excellent CSF penetration (Taelman et al., 1987), it has been successful in treating patients at an advanced stage of CNS invasion and doomed because of the parasite's resistance to traditional trypanocides.

The treatment is arduous because it requires administration at regular intervals: either intravenously, 100 mg/kg four times a day for 14 days, followed by oral administrations of 75 mg/kg four times daily for 21-30 days, or by mouth 200 to 400 mg per kg and per day (up to an average of 15 g), given in divided doses every six hours for five to six weeks. There is hope that more effective derivatives (methyl-esters of the monofluoride) will soon be available.

6.6. Combination of drugs

From the early days, combination or consecutive use of trypanocidal drugs have been used routinely. A possible synergetic mode of action was expected, but the aim was more distinctly to resolve the day by day difficulties: side effects, relapses, the crossing of the meningeal barrier, resistance. Current associations have been atoxyl and tartar emetic, suramin and trypanamide. During the period that followed, the preference was given to single drugs: pentamidine and melarsoprol, but diminazene, nitrofurans and suramin posed again the problem of synergetic combinations. Nowadays the main target is the trypanothione oxydoreduction system in the trypanosome. Eflornithine blocks the pathways and melarsoprol combines with the trypanothione and removes it from the system, as a result melarsoprol can be used in minimal doses. In addition, antimony compounds surfaced again as the combination Eflornithine Astiban® (*Fouadine*®) was extremely active in experimental models. Pentamidine, suramine, nitrofurantoin will also enable the medical personnel to reduce the dosage of Eflornithine.

7. Prevention and control

The aim of trypanosomiasis control is to break the chain of trypanosoma transmission by acting on the human reservoir, by controlling the vector and protecting receptive hosts. These measures are implemented either alone or in combination.

7.1. Reducing the human reservoir

This approach is useful only in cases of endemic infection by *T. gambiense*. It consists of active, careful detection of any infection, followed by mandatory treatment. This method has been imposed wherever

the disease occurs. These operations are carried out in endemic foci by specialized mobile units visiting the villages and hamlets; the aim of these teams is to reduce the size of the human reservoir by eradicating the trypanosomes from the patients' blood as soon as possible after infection.

Since mobile teams have to examine an average of 200 to 300 individuals a day, screening must be based on easily detectable and as reliable as possible symptoms. Up to date palpation of the cervical lymph nodes has been the technique contributing best to detecting the infection. However, the relative unreliability of this symptom results in a large proportion of infected individuals overlooked in the detection. The scarcity or periodic disappearance of the trypanosomes in the blood is also limiting the detection rate from 60 or 80% of the cases. The development of fast and simple methods has noticeably improved the yield of serological detection. Suspected seropositives require however parasitological evidence before carrying on treatment. For the time being the cost of serological tests, which is still prohibitively high, precludes their routine use in mass screening.

7.2. Vector control

a) Large-scale spraying of insecticides against flies of the *Glossina palpalis* group is very difficult to achieve and extremely costly, although micronized aerial spraying (ultra low volume spray of roughly 140 l per km² of endosulfan or decamethrin) has yielded encouraging results. The *Glossina morsitans* group, living in the savannah is more vulnerable to aerial spraying.

b) Trapping and capturing tsetse flies is an easy, inexpensive solution. The cow-shaped Harris trap, the dog-shaped Morris trap, and the Lewillon trap with human profile, were the simple forerunners of today's biconical trap designed by *Challier* and *Laveysières* out of blue-and-white cloth impregnated with an insecticide (pyrethroid) and attractant products. Such traps can reduce the tsetse population by 90%.

c) Prophylactic destruction of the vector's preferred biotope by selective clearing should also not be forgotten.

7.3. Protecting receptive hosts

Prophylactic administration of *Suramin*® or *Lomidine*® brings healthy subjects incomplete protection for only a short, uncertain period. Moreover, such chemoprophylaxis carries the risk of masking an infection, detection being delayed until the disease has already reached the stage of CNS invasion.

8. Epidemiological characteristics

The sleeping sickness foci are scattered over extremely vast areas (see pp. 1689 to 1712, and pp. 1741 to 1749). This distribution is definitely not due to chance, but is linked to very specific geomorphological and phytogeographical conditions, as well as to the frequency of man-fly contacts.

As a result, epidemiological analysis cannot remain limited to the description of the surroundings of natural or permanent foci, and to the collection of parasite indices in the vectors and in the population. Very thorough analysis of the **human factor** is required (see p. 1751).

This involves a knowledge of: 1) the structure of the population and its density; 2) the spatial distribution of the inhabitants with the size of the village and their location in relation to the possible sites of infection; 3) the stability or migratory movements of individuals; 4) the location of fields and sources of water; 5) the commonly-used trails; 6) children's games and activities; 7) the family infections; 8) the frequent mixed infections of domestic animals; 9) the social and economic conditions, such as villagers hiring out their labour to subcontractors or an outside workforce, etc. A multidisciplinary approach is thus essential.

Considering that sleeping sickness can spread dramatically through a region, it is hard to believe that the **tsetse fly** is in fact not a good vector. The tsetse fly is indeed a fragile haematophagous Diptera whose survival and reproductive ability depend on very strict climatic conditions. It is vulnerable to predators and parasitic insects and is forced constantly to find hosts on which to feed. The larva deposition and their transformation into pupae are also dependent on a very narrow range of conditions. The parasite must complete a long, complicated life cycle in the vector and it quickly loses its virulence if the infective blood meal does not take place soon after the fly hatches. The fly covers only short distances, unless it can find a means of transport.

The reason for ongoing epidemics lies largely, although not entirely, in the close **contact** between humans and tsetse flies without which the disease cannot propagate. In forest areas, where sources of food (antelopes, reptiles) are plentiful, the tsetse flies will not look for human blood. It follows that areas swarming with those flies do not necessarily correspond to endemic foci. In transition zones between forest and savannah, where sources of water become scarcer in the dry season, man and tsetse fly come together more easily. In such areas man will be exposed to a greater risk of infection, especially since his presence will scare away the game and other potential hosts.

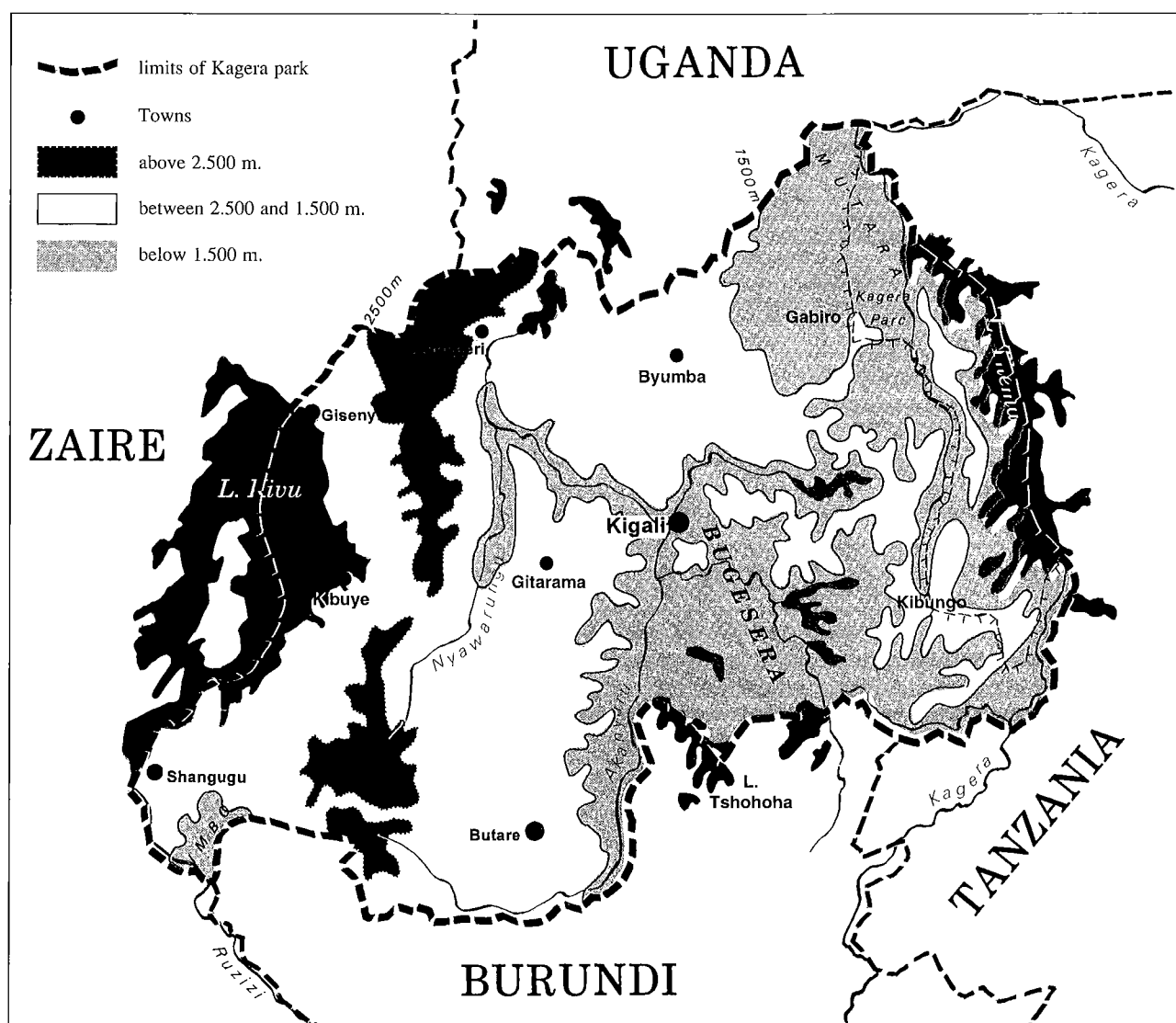
Given the vector's propensity to fix on a preferential host (determined by its first blood meal), the epidemic will spread out, like oil on water, following the tsetse fly infested areas and places where men gather to bathe, to ret cassava (soaked in water) or to fish.

In the course of the infection the parasite's antigenic determinants are forced to change in response to the host's immune defences; the successive antigenic variants progressively lose their virulence. However, the parasite's cyclic evolution in the vector will restore its initial virulence. This indicates that the tsetse is not just a vector, but will allow to restore the parasite's original antigenic structure.

The environment plays an important role in the epidemiology of trypanosomiasis. Over the centuries of the disease's existence some sort of balance had to be developed among the various links of the epidemiological chain. If the strain was too virulent or the vectors too numerous, man either died or moved on, taking the disease with him. In other places, where the strain was less virulent, resistance better, and host-vector contacts less frequent, a hypo-endemic state persisted, enabling the community to survive without too many losses. This equilibrium was maintained by the populations' sedentary traditions, which in the pre-colonial era were seldom upset. During the colonial era this equilibrium was also restored through constant monitoring of the foci and strict control over the movements of people from endemic areas. At the dawn of the new era of independence of African States, after a long history of successive outbreaks and periods of calm, there remained only a few residual foci.

The existence of sleeping sickness in Central Africa was discovered some 600 years ago. From then on, it seems to have spread in successive waves, mainly after the second half of the 19th century. This phenomenon has gone hand in hand with the penetration of Europeans into the heart of the continent, coupled with the attendant extension of trade, the introduction of industry and transport, and the pursuit of military campaigns. The slave trade with the slavers' caravans, the exploratory expeditions – Stanley's among others –, the construction of railway tracks or telegraph lines, all of these formed so many paths which sleeping sickness followed.

Experiences over the last few decades have confirmed that population movements, migrations and uncontrolled massive use of various means of transportation, are some factors responsible for upsetting the balance of residual foci. Population movements can facilitate contacts with the vector, either through



Map 38 – Rwanda, Burundi and Rusizi valley
(adapted from Meheus *et al.*, 1982).

the influx of individuals into a *new* residual focus or through contact with a strain of trypanosomes against which the population has not built up any immunological barriers.

This brings to consider the so-called *new old-foci*. Practically 20 years after the resumption of the control of the sleeping sickness in Zaire, following a period of inactivity resulting from the independence of the country, the disturbing epidemiological problem of resurgence in former extinguished foci, had to be faced despite regular control. The often very long interval (25 years) between the foci's extinction and the new outbreak justifies to adopt the paradoxical term *new old-foci*. None of the attempts to explain the phenomenon is completely satisfactory: neither the accidental introduction of a new strain brought by a sick individual or by infected tsetse flies introduced into the focus, nor the persistence of an animal reservoir or a reservoir in healthy carriers, nor even possible flaws in the patients' detection, nor treatments which did not achieve the intended parasitological sterilization.

All of these conditions were either present or were to be found in combination at any time: ubiquitous tsetse flies, favourable biotope, permanent game and domestic animal populations, resident hosts with their travelling activities, visitors, and sick individuals detected in nearby settlements. The reawakening of a focus after 20 or more years of quiescence is difficult to explain. It seems that all of these epidemiological parameters converged suddenly upon a given site.

What is more, the disease seems to revert in cycles, like an epidemic wave running across the historical evolution of the endemic. The first epidemic wave in Africa struck around 1885, to be followed by a second wave around 1920, which stabilized in 1940. The third wave, whose ripples do not yet seem to have disappeared, began its surge around 1960.

One plausible hypothesis is that this phenomenon is based on a mechanism of acquired or inborn immunity among the individuals living in the foci.

On the basis of this hypothesis, the following scenario could be imagined:

- at a certain time the focus encounters one of its epidemic outbreaks and becomes hyperendemic, with an elevated new contamination index (NCI). A large proportion of the population is sick and others will become infected. Some will succumb to the disease while others will be treated successfully or develop a chronic infection with relapses. Cured patients will be reinfected, while a few rare individuals will remain healthy carriers of the trypanosome;
- after a few years, during which time most of the population will have been infected, the focus will die out as a result of control measures and gradually become residual, while at this stage the community enjoys a high degree of immunity; the focus will eventually be completely destroyed, but with time the protected generation will disappear, while a new, unprotected generation takes its place, as little by little, the immunological protection is becoming patchy, to disappear completely once the new generation has replaced the old one; this change seems to occur after an average of thirty years;
- throughout the period starting from the extinction of the focus up to the progressive disappearance of the generation with immunity, the introduction of a new trypanosoma will not find a favourable field for a new epidemic, although a few sporadic cases might be seen;
- when the proportion of protected individuals in the community has fallen below a critical threshold, conditions will once again be suitable for the accidental introduction of trypanosomes pathogenic to man; those trypanosomes will infect a certain number of unprotected individuals, causing the disease to flare up once again.

J. Burke

SLEEPING SICKNESS IN RWANDA AND BURUNDI

(translation from Burke, in: Meheus *et al.*, 1982)

Concerning epidemiology, Rwanda belongs to the Interlake Region between the Western Great Rift Valley and Lake Victoria. This area has been submitted to the advance of the fly belt of tsetse flies and the development of *T. rhodesiense* sleeping sickness (Ford, 1971).

Rwanda is on a north-south mountainous plateau averaging an altitude of 2,000 metres above sea level, which is a watershed between the Zaire and the Upper

Nile basins. This particular topography explains why the *country of the thousand hills* was for long relatively spared from the scourge of sleeping sickness whilst the epidemic was raging on all sides, while about a quarter of the country's areas are lower than 1,500 metres and are infested with tsetse flies.

Burundi has about the same profile, except for a part of the Rusizi valley shared with Zaire, and the Tanganyika shore (see map 38).

LOCAL HISTORICAL BACKGROUND

The situation of sleeping sickness in Rwanda must be considered and studied within the broad epidemiological framework, going beyond its own borders.

The history of sleeping sickness in Southern and East Africa was determined as much, if not more, by animal trypanosomiasis as by human sleeping sickness. Indeed there was in those countries an ancestral equation of livestock with wealth, a natural abundance of game, and a potential for animal husbandry. Furthermore Rwanda is located at a unique epidemiological crossroad in Africa. Indeed, all efforts to control this endemo-epidemic in Rwanda have been dictated by ecological factors.

The chronic *T. gambiense* West African endemic sleeping sickness reached the shores of Lake Victoria and was definitely spread by *Glossina pallidipes*, whereas the *T. rhodesiense* zoonosis, in its spread to the north, was carried by *G. palpalis fuscipes*. The human *T. rhodesiense* epidemic reached the habitats of the local populations.

The first explorers who reached the Interlake Region hardly ever mentioned either the tsetse fly, or sleeping sickness. There was a good reason: the disease was absent from this area in 1861, when *Speke* and *Grant* crossed the Karagwe marshes between the Akagera river and Lake Victoria, or when *Stanley* crossed the area in 1876, and even in 1889, when Stanley returned with Emin Pacha. In 1891, when Emin Pacha and the eminent biologist *Stuhlman*, together with *Captain Luggard* crossed the Karagwe area again, Luggard could not escape witnessing the terrible havoc brought by the rinderpest panzootic, which severely depleted the herds of game and cattle throughout Southern and East Africa between 1890 and 1896.

The presence of *G. morsitans* was observed for the first time by *Meyer* in 1911 in the Ihangiro depression and around the Low Akagera. If tsetse flies had been present, it would have been impossible for scientists such as *Stuhlmann* not to have come across those flies during their travels. A German official in Gatsibu, in the Mutara area, reported around the same time the presence of the first signs of sleeping sickness in cattle (1911). If we can accept that *G. morsitans*'s invasion of the Interlake Region was a new event after 1891, we must nevertheless bear in mind that the local populations were not unaware of the animal trypanosomiasis (*omurasho*) which existed in a few circumscribed foci, for they did recognize three foci of infection, namely, Buhamira, Nsongezi (on the lower Kagitumba river), and Kikagati (on the Akagera river). The first eye-witness account of a horse that died of *omurasho* near the Akagera river dates back to 1903.

It was *Bruce* who set up the epidemiological and clinical bases of animal sleeping sickness, during his mission to Ubombo (in Zululand), between 1894 and 1897. According to *Livingstone*'s accounts and local oral tradition, Bruce became convinced that cattle bitten by tsetse flies were to perish from *nagana*; this was apparently due to injection of poison by the fly. Bruce's investigations came to isolate the parasite and he could conclude that *nagana* was caused by a trypanosoma parasite transmitted by the tsetse fly. He also discovered that wild ungulates were the reservoir of the disease, and the tsetse flies' reproductive behaviour. Hereby Bruce laid the foundations for all subsequent investigations on tsetse-fly transmitted trypanosomiasis in Africa (*Duggan*, 1977).

MAJOR CHALLENGES

1. *T. gambiense* sleeping sickness

The first major sleeping sickness epidemic started, around 1885, and came from West Africa. It began somewhere near the convergence of the Zaire and Ubangi river tributaries and spread around 1898 to Lake Mobutu Sese Seko (previously Lake Albert), between Zaire and Uganda and around 1905 and 1907 respectively, further southwards to the shores of Lake Tanganyika then along the Luapula River, on the border with Zambia.

A terrible epidemic broke out in the Busoga region of Uganda at the end of the 19th century. According to H.H. Scott, it may have started when Stanley's car-

vans left the shores of the Lualaba in 1887 to go to Emin Pacha's aid. Later on, indeed, the soldiers and porters and their families settled in the fertile areas of the Busoga and elsewhere in Uganda.

In 1902 the British Royal Society sent a commission to investigate the epidemic. It was composed of Low, Christy, and Castellani, joined later by Bruce. During his work *Castellani* discovered trypanosoma in the cerebrospinal fluid, and *Bruce* concluded that the epidemic was due to a tsetse-transmitted trypanosoma. Finally, *Adams* (1903) suggested that trypanosomal fever and sleeping sickness were one and the same disease (International Colloquium on the Control of Major Endemics, 1970).

It is at the turn of the century that *T. gambiense* sleeping sickness appeared on the shores of Lake Tanganyika and the Rusizi Plain, in Burundi (see p. 1702). It was the expansion of the epidemic which was spreading from West Africa right across the Sudano-Guinean climatic zone. The epidemic was in full swing around 1911 and entire villages had been ravaged and abandoned by the survivors. In his book *Die Barundi*, Hans Meyer established that during this period the number of sleeping sickness cases in the area was around 200,000.

Entire populations were evacuated to upland areas free of tsetse flies. The FOREAMI (Queen Elisabeth Foundation), see p. 126) carried out a systematic control of the disease and an eradication campaign from 1932 to 1934. At that time the new case index (NCI) was around 2.5%. Thanks to the prophylactic administration of *Bayer 205*® the epidemic was checked so that the population returned to their villages in 1937, making it easier to clear the tsetse fly infested thickets. A low profile endemicity persisted at the northern tip of the lake and in the Rusizi Valley (NCI 0.15% in 1938). A new flare-up occurred around 1942 with a NCI greater than 1% in 1946. Thereafter a campaign of preventive administration of pentamidine stopped the endemic: only one case was observed in 1960, from 1960 to 1965 another seven cases, but by 1965 the disease could be considered as being eradicated.

Rwanda's endemic focus was limited to the small strip of land on the east bank of the Rusizi River, the Bugarama, at the north-eastern part of the Rusizi Plain.

2. *T. rhodesiense* sleeping sickness

Across all of East Africa, including Rwanda and Burundi, as far as Ethiopia, *T. rhodesiense* sleeping sickness spread from Ngamiland, in northern Botswana, over the Zambezi Basin, and the east of Angola. The endemic covered some 4.5 million square kilometres of land where some 50 million people were exposed to the risk. Unlike the *T. gambiense* endemic, this trypanosomiasis was basically a zoonosis, and only some 1,000 to 1,500 cases occurred each year in the entire area exposed to the disease (Apted, 1970). Some reports seem to indicate that the disease existed for years in the Luangwa Valley (Zambia) and in Ngamiland (Botswana) before it was discovered.

2.1. Origin of the disease

Epidemiologists are not always in agreement as to its origins: some believe that a virulent strain of *T. gambiense* was brought into an area with *G. morsitans*

by travellers from Zaire, and that this strain of *T. gambiense* managed to maintain itself by turning its attention to game (Willett, 1965); others believe that *T. rhodesiense* had existed for a very long time and was the result of an adaptation of *T. brucei* to a non-immunized human host.

T. rhodesiense trypanosomiasis was thought to have been eliminated for the time being by the rinderpest epizootic that swept across Africa at the end of the 19th century, killing off most of the animals on which the tsetse fed and thus reducing considerably the fly belts. Since then, both the herds and tsetses have recovered in the area (Buyst, 1977).

In any case, the first case of human sleeping sickness discovered in October 1908 in Malawi near Nkata Bay was detected in an African who had travelled along the Zaire River, and it seems to have been an imported case of *T. gambiense* sleeping sickness. In 1909 an Englishman, W. Armstrong, contracted a virulent febrile infection after travelling through the Luangwa Valley in Zambia. The parasite that was isolated from this patient was described by Stephens and Fantham in 1910 as *T. rhodesiense*; it differed from *T. gambiense*, as a morphological study showed, and on account of its virulence in rodents.

In 1909 Kleine and Taute established the trypanosome's life cycle in the tsetse and in 1913 Kinghorn and Yorke established the cycle of *T. rhodesiense*'s transmission in *G. morsitans*. Before that time the common belief had been that *G. morsitans* was unable to transmit sleeping sickness. By the end of 1909, twelve cases of *T. rhodesiense* sleeping sickness had been diagnosed. Bruce was sent to Malawi in 1912 to investigate this *Rhodesian* form of sleeping sickness. He very quickly concluded that the infection was identical to *nagana* and that *T. rhodesiense* was actually a camouflaged *T. brucei* and named it *T. brucei vel rhodesiense* (Duggan, 1977).

2.2. Extension and outbreaks

It is well known that this sleeping sickness had spread subsequently further northwards. In 1911 an isolated focus was present close to the Rovuma River, on the border between Mozambique and Tanzania. *T. rhodesiense* spread out from this focus in the twenties and thirties across all of western Tanzania in epidemic waves.

The first serious study of a typical *T. rhodesiense* epidemic outbreak was conducted during the year 1922, from an outbreak in Maswaland (Tanzania). The epidemic spread to the Ikoma District in Tanzania in 1925, to reach its full extent in 1932 (Fairbairn, 1948).

South-Eastern Uganda was struck by a severe outbreak transmitted by *G. pallidipes*. It was imported in

the historic region of Busoga by immigrating farmers from Tanzania during the year 1940. This outbreak became responsible for the introduction and the spread of *T. rhodesiense* sleeping sickness, around 1942 to the north-eastern shores of Lake Victoria, in Kenya (Apted, 1970).

The first westward-moving wave of the epidemic from the Ngara District, Tanzania, reached Burundi in 1954 and went on from there to reach Rwanda around 1955.

In March 1954 Marneffe reported the first focus of *T. rhodesiense* infection in Muhinga territory, in the north-eastern part of Burundi. The region located at some 1,200 to 1,800 metres above sea level, has a typical environment for *G. morsitans*, which can be found as high up as 1,500 metres. In that area Burundi, Tanzania and Rwanda have common borders. Ninety-seven cases of sleeping sickness were very quickly detected in the 7,500 inhabitants of Busoni and the surrounding hills. The new case index at the epidemic's centre reached 14% (Marneffe, 1955). The epidemic then spread from Muhinga south to Ruyigi in 1955 and to Rutana in 1957. After the focus had broken out the epidemic had ups and downs over the years: new outbreaks occurred in 1966 and 1970, at the same time as outbreaks in the bordering foci of Tanzania and Rwanda. A total of 1,126 cases were diagnosed from 1954 right through to 1975.

To the north, Rwanda was reached in 1955 after the epidemic had pervaded into the Bugesera, near Lake Tsoho, via a narrow strip of the savannah along the Akagera River Valley. Fortunately, the disease did not take on dramatic epidemic proportions, for only 83 sleeping sickness patients were detected between 1955 and 1960. A subsequent inactive period was followed by a small outbreak in 1966, when another 45 cases of sleeping sickness were discovered in the area between 1966 and 1968.

The actions taken by ISAR (*Institut des Sciences Agronomiques du Rwanda*) in 1968-1969 to eradicate the tsetse from its permanent habitat proved very successful. These actions led to the complete eradication of *G. morsitans* from the entire Bugesera with the immediate disappearance of sleeping sickness in both the human population and, a little later, in the herds of cattle (Van Minnebruggen, 1970).

A second focus, at Migongo in south-eastern Rwanda, appeared in 1958 with 14 patients affected by sleeping sickness. A few more cases were diagnosed in 1960, but the focus did not become epidemic until some ten years later, when 133 people were found to have sleeping sickness from 1969 through 1975. The Mutara region and the Akagera National Park were also affected around 1969 and sleeping sickness continued to develop, albeit sluggishly from

1969 until 1979: 44 cases of human sleeping sickness were reported in this area.

Outside Rwanda the endemic continued to spread around Lake Victoria in Uganda. The explosive onset of a focus at Alego (Nyanza District in Central Rwanda) gave the first typical example of a *T. rhodesiense* epidemic transmitted by *G. palpalis fuscipes*. Close to 3,000 new cases were reported in Tanzania between 1961 and 1966, mainly in the Lambwe Valley and the West Lake region at the Rwandan-Tanzanian border. As early as 1958 *T. rhodesiense* was isolated from a patient at the Murchinson Falls (Uganda); in 1966 the disease was reported in Teso District, Northern Uganda, probably brought in by fish merchants from Busoga. Finally, in 1967, it was tracked down close to Gambela, in Ethiopia, along the 8th parallel of the Northern Hemisphere, the actual limit of its expansion.

2.3. Epidemiology

It will chiefly be the epidemiology of *T. rhodiense* sleeping sickness in Rwanda that will be taken in consideration. Its particular feature is that it is actually a zoonosis having wild game as intermediate vertebrate host and the parasite's reservoir. Man is merely an incidental host. There are in fact four links in the transmission: the parasite, game, tsetse fly, and man; these components determine the epidemiology of the disease by their mutual interactions within their particular environment.

2.3.1. Parasite-vector-host relationships

It is commonly believed that the early man had contacts with *G. fusca* and *G. morsitans* only by chance and that whereby humans in the forest and the wooded savannah, were not receptive for *T. brucei* strains, whilst frequent contact between man and *G. palpalis* maintained the human sensitivity to *T. brucei* strains in a river ecosystem (Buyst, 1977). The more or less acute *T. rhodesiense* infection is indisputably the result of genetic human immunity, acquired or not through centuries of man's contacts with tsetse flies.

In the laboratory, sleeping sickness infections were attributed to *T. rhodesiense* infection; infectious material inoculated to animals produced after a one to ten-day incubation period a high parasitaemia that failed to respond to trypanamide. The diagnosis was considered confirmed, even though the clinical symptoms were not very obvious. The diagnosis was supposed not to be influenced by the type of vector in the focus. The vector was especially *G. palpalis fuscipes*, traditionally believed to transmit only *T. gambiense*; but as any tsetse of the *G. morsitans* group it was also

capable of transmitting *T. rhodesiense*. However, the opinion that *T. gambiense* acquires enhanced virulence when it is transmitted by *G. pallidipes* and behaves in several respects like *T. rhodesiense*, is not sustainable. Similarly, *T. rhodesiense*'s passage in *G. palpalis* does not, in any way, diminish the parasite's virulence (Onyango, 1969).

The predominant transmission chain passes from game to tsetse and then to man. Man is infected when he enters an endemic area, and an epidemic outbreak will occur when a man-tsetse-man transmission becomes possible, in the case of carriers on the move, lack of treatment, anthropophilic tsetse flies, etc. These epidemic outbreaks are usually located in areas where the infection was previously unknown and where, as a rule, everyone is receptive to the parasite. The first cases usually go undetected, but their number increases until the number of deaths reaches an alarming level so that the community becomes aware that an epidemic has broken out.

Man is responsible for the epidemic dissemination of *T. rhodesiense* in new foci. Except during outbreaks, the generally very low incidence of *G. morsitans* infected with *T. rhodesiense* allows one to suspect that game animals harbour a large amount of infectious material and are a reservoir (Leach, 1969). In recent years epidemiological investigations conducted in East Africa have led to important discoveries. Carnivores such as lions and hyenas are reservoirs for *T. rhodesiense*; they are probably infected by contamination of wounds in or around the mouth when they devour an infected prey (WHO, 1979).

As early as 1958 Heisch and his colleagues isolated, from a bushbuck or harnessed antelope, (*Tragelaphus scriptus*) an extremely virulent trypanosoma for man. During a relapse of trypanosomiasis on Lake Victoria's northern shores in 1940 MacKishan and later Jackson discovered *G. pallidipes* to be the vector. This tsetse feeds on the harnessed antelope and lives in certain thickets of the wooded savannah. Later, Lumsden also isolated a virulent strain of trypanosoma from *G. pallidipes* from a place where these flies fed preferentially on the harnessed antelope. Various experiments also proved that the trypanosomes isolated from human beings lost none of their infectivity for man, even after the parasites had been transmitted to animals by savannah tsetse flies, in their normal cycle.

All the research conducted in East Africa has proven that human sleeping sickness is a zoonosis and that the clinical symptoms reflect the degree of the parasite's adaptation to man. It is generally accepted that *G. pallidipes* is the animal-to-man vector and *G. morsitans* the man-to-man vector, in so far as such transmission has time to occur, in the rapid and debilitating course

of the disease in man up to a fatal outcome. In contrast, *T. rhodesiense* infections in the bushbuck take a chronic course, and this well explains the persistence of a parasite reservoir.

2.3.2. Fly belt movements

The regression of *G. morsitans* fly belts as a determining factor in the epidemiology of East African sleeping sickness after the 1890-1896 rinderpest epidemic, tends to be overlooked. Such a shrinking of the habitat was well known in the Zambezi Basin, where certain foci had remained for many years outside the tsetse flies' normal range, before resuming their eastward spread. The same shrinking had occurred in Tanzania. Resettlement followed by expansion of these ranges was rapid until the mid-1920s, then slowed down, and finally stabilized around 1946 (Ford, 1965).

G. morsitans came from Zimbabwe around 1910 and its relentless northward march in savannahs below 1,500 metre could be followed by a date on the map (Van den Berghe, 1964). The eastern Rwandan fly belt of these tsetse flies is a continuation of the huge Tanzanian fly belt following the Ruvubu Valley. The infested area in Rwanda covers about 570,000 hectares, or a quarter of the country's surface. It is bound to the north by the Kizinga and Kagitumba rivers, to the east by the swampy lakes feeding into the Akagera River, to the south by the Akagera River with its marshlands, and to the west by the plateaus surrounding Mohasi Lake and the line of escarpments stretching up to the Byumba highlands.

This tsetse fly offensive broke through Rwanda's defences at two places:

a) The right wing crossed the Akagera river at Rusumo Falls, probably in the late 1920's, and headed west along the river until it reached the Nyabarongo river's marshes, then north to invade the Akagera Park. It is most likely that the tsetse flies were accidentally carried across the Akagera river by people and their livestock fleeing the threat of famine when climatic conditions, the vegetation and game were suitable for the extension of tsetse colonies.

The fly belt's movement towards the Akagera National Park must have occurred after 1928 and was partly caused, in the wake of the great famine, by the exodus of the inhabitants from the Migongo area on the edge of the lakes, towards Karagwe in Tanzania. This depopulation was reinforced later by tribal disputes. As a result, once the lands had been abandoned, the flora and fauna were free to take over and proliferate to finally favour the settlement of tsetse flies. The southern part of the Park was infested later by

tsetse flies passing through the very narrow bottleneck between mountains higher than 1,700 metres and the Rwampanga Lake.

b) The left wing started to invade part of Burundi between 1945 and 1950. It fanned out from the shores of the Rugwera Lake to reach the Busoni area in Burundi and the Bugesera in Rwanda (Van Minnebruggen, 1973).

Lake Victoria owes its origin to a huge depression in the earth's crust dating back to the Miocene. Before this period, Africa's vast central basin, which would later become the Zaire river Basin, extended further eastwards from a watershed ridge in Kenya and Tanzania. The depression trapped the upper reaches of the rivers and their currents reversed in their middle stretches. The Akagera waters were stopped by a ridge rising upwards, as a counterpart to the depression of Lake Victoria. This river became the main tributary of Lake Victoria and its flow was reversed eastwards, to spill into a series of lakes and marshes before entering Lake Victoria through a narrow gorge in the ridge, which would appear to be the passage followed by *G. morsitans* between 1907 and 1913. During their westward progress (Ford, 1971), these flies invaded the Mutara area in north-eastern Rwanda.

Other authors state that the tsetse flies were seen for the first time in the Ngarama area of the Akagera valley in 1913 (Buyst, 1977). Veterinarians had already reported the presence of animal sleeping sickness near the Kagitumba river in 1912. This was an *omurasho* (sleeping sickness) focus maintained by *G. pallidipes*.

The invasion of the Mutara by *G. morsitans* was confirmed in 1925. This was later corroborated by inhabitants of the Bugoye area, who stated that the tsetse had appeared around 1925 in the Akagera valley by entering from the northern Ankole area. From there, it would have spread out, along the Akagera valley via the acacia savannah of the Mutara area.

2.3.3. The vector and its geographic range

a) Three main types of vector were recognised, as follows (Henrard, 1951; Evens et al., 1957):

- *Palpalis* group: *G. fuscipes martinii*, found only in the Rusizi Valley below 1000 metres;
- *morsitans* group: *G. pallidipes*, found in the north-east, in Mutara area and the Akagera National Park; *G. morsitans morsitans*, present throughout eastern Rwanda from Mutara area, the Akagera National Park, the lakes region, and the Bugesera up to Rusumo Falls.

Two other vectors have since been identified in Akagera National Park (Hanotier, 1977): *G. swynner-toni* (*morsitans* group) and *G. brevipalpis* (*fusca* group).

b) Three fly belts can be defined, even if only two of these are currently relevant in Rwanda. They are:

- the north-eastern fly belt, which covers the Ndorwa area, the Mutara region and the edge of Akagera National Park. The part of the fly belt along the Akagera river, at the intersection of Rwanda, Uganda, and Tanzania, is of particular epidemiological importance. The great swampy valley along the Tanzanian border could not spread westward because a strip of the highland was running from north to south through the middle of the Akagera Park.

- the central fly belt (covering the lakes and the Bugesera area), is a huge depression of wooded savannah below 1,500 metres with abundant game, offering a favoured area for *G. morsitans*. The advance of this fly belt is limited by the 1,500 metres contour line, a natural barrier almost never crossed by *G. morsitans*; indeed, Mohasi Lake is free of tsetse flies. The central region communicates with the other fly belts only through the narrow gorge of the Akagera river. In the last few decades, game has become scarce and the tsetse flies were victims of eradication campaigns.

- the Rusizi fly belt, which is negligible in Rwanda, as it covers a very small southwestern area. In the case of Burundi, the Rusizi valley used to be an important focus of *T. gambiense* (see pp. 1702 and 1778, and also p. 126). In the past *G. palpalis* and *G. fuscipes martinii* were also present.

2.4. Sleeping sickness control

Sleeping sickness control measures concentrate mainly on the vector. The exposed populations are protected by epidemiological screening and surveillance techniques.

The factors enhancing the tsetse flies responsible for epidemic outbreaks, to progress in an area, are extraordinarily diverse and their importance varies according to the circumstances. The relationships and interplay between the various parameters in a given ecosystem are extremely complex. The tsetse's progression depended first and foremost, at one hand on the natural regeneration of the fauna (the natural host) after the great epizootics, and on the other hand on the fleeing of human populations from the infected area, which allowed an immediate new growth of wild bush.

Climatic changes definitely played a role which is not easy to evaluate. The resettlement of abandoned land after the tsetse invasion, during the years 1920 to

1930, led to the clash of two opposing ecosystems: one being the fauna and vector in their natural environment of wooded savannah, the other being man and his livestock settled down to farming or ranching (Ford, 1965). In recent years the risk of epidemic outbreaks has increased considerably because of economic development projects which have attracted huge numbers of people into tsetse-fly-infested areas.

The Akagera National Park presents a special problem because it is part of the country's cultural heritage, an important potential tourist attraction and, from time to time, a source of food. Sleeping sickness control in the park must take the conservation of nature and the environment into account, at the same time as it must protect tourists, guard rangers and neighbouring populations. The latter must be protected from possible occasional infective contact, as well as from importing exogenous sleeping sickness parasites into the park. Poachers by their unpredictable and difficult to control incursions are one of the causes of *T. rhodesiense* sleeping sickness's persistence in the Akagera Park.

Theoretically, *T. rhodesiense* sleeping sickness is relatively easy to control. The vector is destroyed by insecticide applications in its preferential habitat; selective clearing of the bush and installation of tsetse fly barriers, and also constant medical and environmental surveillance are the means used to control the disease. In practice, this is hampered by ecologists' demands and budgetary or technical constraints making it difficult to set up and maintain tsetse fly hindrances. Therefore medical and public health surveillance with destruction of the vector by non-polluting traps have acquired great value (Hanotier, 1977).

In *T. rhodesiense* areas preventive chemo-prophylaxis is not to be recommended, for the known drugs protect man completely for only a short and variable period. What is more, the use of chemo-prophylaxis entails the great risk of masking the disease at its early stage, just when it is the easiest to cure.

2.5. Development of control measures

Tsetse fly eradication was planned for the ranching project in the Mutara area. A campaign was to be organized in some parts of the Akagera Park along the procedures conducted so successfully by ISAR in the Bugesera area and by the International Association for Rural Development (AIDR, *Association Internationale pour le développement rural*) in the Rusumo area.

It emerged from several studies that total eradication of the tsetse fly from Rwanda was technically feasible (Van Minnenbruggen, 1970) and could be carried out in three phases:

- the first phase was undertaken to drain the Bugesera marsh and was completed successfully in 1969;
- the second phase or draining of the Kibungo-Rusumo area, had a number of setbacks after its initial success in 1972;
- the third phase would be to eradicate the tsetses from the entire Akagera Park and Mutara region.

The current state of knowledge indicates that the eradication of sleeping sickness from Kibungo Prefecture should be carried out at the same time as the tsetse fly control measures in the Park, for the flies follow a south-north line. It would appear that the Akagera river forms a natural barrier for the tsetse flies, and that it would only be necessary to obliterate a few accesses for the tsetse flies.

When resuming the second phase, new protective corridors had to be considered, as well as cleaning out the flies' residual habitats. The progress of the campaign to eradicate the flies must be followed by resettlement of the land; in addition, tsetse-free corridors must be maintained at the same time as health control barriers.

3. Animal sleeping sickness

Animal sleeping sickness cannot be ignored in a country of 700,000 heads of cattle, of which 20,000 on average, must be treated each year for *T. congolense* or *T. vivax* infections.

The history of this control goes back to the time of the first World War, during which the veterinarian René Van Saceghem, by his studies and work in Rwanda, laid the foundations of a therapeutic and immunological approach to animal sleeping sickness. His well-known works, dated from 1914, deal with the classification of pathogenic trypanosomes in Africa and describe his experiments with therapeutic action of trypanflavine on animal sleeping sicknesses. By 1923 he was considering the trypanosomes' influence on the immune system of infected animals and in 1936 he tested various immunization procedures on livestock (Van Saceghem, 1921a; 1921b; 1922a; 1922b; 1934). More than half a century has elapsed since then, and immunity experiments continue to stimulate research.

The various trypanosoma species, *T. brucei*, *T. congolense*, *T. vivax* and *T. simiae*, have all been reported in Rwanda, where they infect cattle, swine and dogs. It should be mentioned that cows are liable to be infected by *T. rhodesiense* and thus form a potential reservoir of the parasite, which should be treated whenever possible.

4. Developments of the disease and research

T. rhodesiense sleeping sickness is not an alarming public health problem, unlike chronic West African sleeping sickness. Indeed, the number of patients is small: as a rule a few dozen cases here and there, or cases clustered in small foci. The recovery prospect has brightened considerably since the use of powerful trypanocides and medications to counter the drugs' side effects.

A low population density prevents any insidious spread of the disease. Nevertheless, when there is person-to-person transmission through close man-fly contact, serious acute cases can occur.

Therefore it is thus important to consider carefully the various economic, social, climatic and other factors promoting man-fly contacts in endemic areas. Indeed all tsetse fly control strategies are closely linked to the development of farmland and pastures and to the settlement of populations in these areas.

Research should be geared first of all to destroy the vector by highly effective non-polluting means such as the new synthetic pyrethroids. The only way to achieve lasting tsetse flies' eradication without having to resort to costly tsetse fly barriers, is to make their habitat hostile. Other methods are under consideration such as insect growth inhibitors, biological destruction by predators, bacteria, viruses,

fungi, or by releasing sterile males and trapping the flies with various attracting means such as pheromones (Goodwin, 1978). Further field research is obviously vital if eradication methods are to be continued.

Conclusions

Over the past century sleeping sickness has proved to be a real economic hindrance to the development of sub-Saharan Africa. It remains a permanent handicap for animal husbandry and ranching, as with the population increase people cannot settle in several areas, while the land is increasingly scarce. Sleeping sickness is also a danger for tourists, who are a valuable source of hard currency. Finally the disease calls for the establishment of a permanent medical and public health surveillance. While it is certain that current research will give rise to new, less expensive control methods, an effort must nevertheless be made to consider sleeping sickness in a broader context as one of the many biological, economic and social factors that affect life in tropical Africa.

This means improving besides the overall health of the people, the use of the land as well in order to achieve a full eradication programme (Ford, 1952).

J. Burke

ROLE OF BELGIUM IN CONTROLLING SLEEPING SICKNESS IN THE CONGO (ZAIRE)

When, towards the end of the 19th century, the first physicians were recruited for Africa by the Upper Congo Study Committee (*Comité d'Etudes du Haut-Congo*) of the International Association of the Congo, it was already known both from the explorers' accounts and from African oral tradition that sleeping sickness, far from being a new disease, had been endemic for centuries. The endemic belt stretched along the west coast of the continent and into the heart of Central Africa. It soon became obvious that this dreadful enemy for man and animal was a major obstacle for the development of these large areas.

The disease expanded during the last quarter of the 19th century, a period that Buxton called "the dark history" of sleeping sickness. The extension of this disease is explained by the population movements which were, at that time, brought about by trade, European expeditions and Arab slave raids. Medical officers, engineers and military officers in increasing numbers were sent on exploration, trade routes were

opened, work forces were recruited, and soldiers were enlisted. The infection followed all existing roads. Diseases previously confined to a single clan began to move outwards, bridging entire areas that had until then remained untouched. The socio-economic changes altered the previous host-parasite equilibrium, so that a violent epidemic of sleeping sickness was soon raging across Africa.

1. The first rudiments of a health service

The first Belgian physician recruited by the *Comité d'Etudes du Haut-Congo* was Jean-Baptiste Allart, who arrived in October 1882. His assignment was to set up the rudiments of a health service at Boma and Vivi. One month later Theodore Van den Heuvel arrived to set up the first medical extension at Leopoldville. He was followed by Leslie, then by Jean Nilis in May 1884. Allart was replaced by Karl

Mense, who disembarked at Boma on May 29, 1885, the year that the Congo Free State (*Etat Indépendant du Congo*) was proclaimed.

Mense was the first to relate in his Report on Health Conditions at Leopoldville from November 1885 to March 1887, his personal observations on Africans struck by sleeping sickness in the Lower Congo (Bas-Congo). That was all he could do, as he did not even have a microscope.

Some missionary posts were severely hit and even wiped off the map by this disease (see pp. 1689, 1709 and 1717):

- Berghe-Sainte Marie, near Kwamouth at the confluence of the Kwa and the Congo (Zaire) River, had to be totally evacuated in 1898. Its survivors were sent to other missions, such as the one at Nouvelle-Anvers, where they all died, having maybe in the interim inadvertently transmitted the disease to those around them;
- a death rate of 70% due to sleeping sickness was reported at the Tumba mission, along the railway line linking Matadi and Leopoldville, between 1900 and 1905;
- it was impossible to count the number of dead at the St Trudo Mission, near Lusambo, in the Kasai Province, in 1903;
- the St Donatien de Bruges Mission, in Maniema District, saw four-fifths of its residents wiped out by the disease;
- starting from Kasongo, the epidemic went on to hit the missionary posts of the White Fathers in the eastern part of the country.

The natives were not far off the mark when they accused Baron Dhanis of having sheltered what they called *usingizi* (sleep) in the telegraph lines being laid at this time. In 1905, Mpala mission escaped extermination thanks to the first measures taken against the tsetse fly, namely by clearing the bush. The mission at Lusenda, near Baraka, on the other hand, had to be evacuated.

Two physicians, *Gustave Dryepondt* and *Jean-Emile Van Campenhout* had reached the Congo in 1890. No specific scientific mission had been given to them. Dryepondt has written a work entitled *Guide pratique hygiénique et médical des voyageurs au Congo* (A Practical Health and Medical Guide for Travellers in the Congo). This vade mecum devoted no more than a few lines to the terrible scourge, including the laconic statement that this *bizarre affliction* was found only among the populations of the Lower Congo and in the Equateur Province. The outcome remained fatal for lack of appropriate therapy.

Henri de Marbaix, founder of the first bacteriological laboratory in the Congo, arrived at Boma in 1894. Unfortunately, the young scientist, ravaged by malarial

fever, had to be repatriated only two years later and died before he could publish the results of his research.

2. Research in laboratories

Van Campenhout, after having accompanied the Van Kerckhoven expedition to the sources of the Nile in 1890 and after some years as a bush practitioner in the Uele basin, was the first to introduce real medical research to the Congo. In 1889 he began his scientific research at the Leopoldville laboratory, concentrating on sleeping sickness. He did not hesitate to spend several months in one of the deadliest foci, the Berghe-Sainte-Marie Mission, to observe the patients. Together with *Dryepondt* he published a very important report on the work of the Leopoldville medical laboratory in 1899-1900. Even though he did not succeed in discovering the trypanosoma itself, his observations of the victims' enlarged cervical lymph glands would prove later on to be of great value. In August 1900 Van Campenhout handed over the laboratory to *Dr. Alphonse Broden*.

The 19th century closed without the discovery of any link between trypanosoma and sleeping sickness. Even after *Ford* discovered in 1901, the parasite in the blood of a sea captain suffering from bouts of fever at Bathurst (Gambia), and after *Dutton* identified it that same year as *Trypanosoma gambiense*, the connection was still not made.

After taking over the Leopoldville laboratory in 1901, *Broden* lost no time in starting to rearrange the trypanosomiasis research activities.

On June 10, 1902, the London Royal Society sent a Sleeping Sickness Commission composed of *Low*, *Castellani* and *Christy* to Uganda to study a violent epidemic of sleeping sickness that had broken out on the northern bank of Lake Victoria. As the investigators were not in agreement as to the approach to the problem, the Royal Society sent *Bruce* to settle the discordances. Bruce arrived in Entebbe on March 16, 1903, 48 hours after the discovery by Castellani of trypanosomes in the cerebro-spinal fluid of a patient with sleeping sickness. Broden was the first to prove that Europeans could also contract the feared disease.

At the request of King Leopold II of Belgium, a scientific team from the Liverpool School of Tropical Medicine began, in 1904, a comprehensive study of tropical diseases in the Congo Free State, especially human and animal trypanosomal infections, and it showed this report on health conditions at the various stations through which it passed (see pp. 1687, 1690, 1717).

The investigation was conducted by physicians *Dutton* and *Todd*, joined later by *Christy*. It stopped at Boma in order to study the numerous records there on trypanosoma cases. At Leopoldville the investigators were able to examine fifty or so hospitalized victims of the disease. Continuing up the Congo River, they learned at Coquilhatville that their colleagues *Greig* and *Gray*, in Uganda, had preceded them by discovering trypanosomes in the lymph of the cervical lymph nodes, an invaluable means of diagnosis. The expedition reached Kasongo, where Dutton died in February 1905. At the end of 1905, Todd returned via Lusambo, crossing areas that had been ravaged by sleeping sickness, to go back to Europe.

The expedition's two-and-a-half years of research and observation in the Congo contributed greatly to the diagnosis and the treatment of sleeping sickness; health measures to stop the spread of the disease were also formulated.

In 1905 *Thomas* demonstrated the effectiveness of *Atoxyl*®, the first active, but relatively dangerous trypanocide.

By a Belgian Royal Decree of June 3, 1906, a prize of 200,000 gold francs was to be awarded to the first person, irrespective of nationality, to discover a cure for sleeping sickness. At the same time an appropriation of 300,000 francs was set aside to finance and promote research into the treatment and prevention of sleeping sickness. It was also at that time that King Leopold II decided to set up a School of Tropical Medicine in Brussels, along the lines of the one in Liverpool.

Van Camphenout had already instituted a preparatory course in Brussels in 1906 for medical officers to be sent to the tropics, but the School of Tropical Medicine as such was not founded officially until the Royal Decree (*Arrêté Royal*) of September 30, 1910, created this institution.

Also in 1906, at the instigation of the medical officers working at the Leopoldville laboratory, the administration announced a number of measures to coordinate the control of the disease. The chief measure was to detect actively the infected individuals, thanks to lymph node puncture and to keep patients in isolation lazarets (lazarets) specifically designed for this purpose with quarantine for suspect cases. The worst-affected areas were encircled by observation posts where each traveller was compelled to undergo a medical examination and, depending on the outcome, to remain there if trypanosomiasis was suspected or to be taken to the nearest lazaret if he was found to be ill. These measures were supplemented by clearing all scrub land around steamers' landing stages along the waterways, the land around wells and springs, and other areas harbouring tsetse flies.

Jerome Rodhain, who was to become one of the leading authorities on tropical diseases, arrived at Leopoldville in 1907 to help Broden in his work at the laboratory. The scientific contributions of Broden and Rodhain in the Congo constitute an important chapter in the history of Tropical Medicine. It is summarized in the well-known Report on the Work Performed at the Laboratory of the Belgian Society of Colonial Studies in the Hospital for Blacks and the Lazaret for Trypanosomal Patients in Leopoldville (Broden, 1906).

One of their practical contributions, of the utmost importance, was to introduce systematic lumbar puncture as a means of recognizing the stage reached by the disease; this was necessary in order to prescribe the proper treatment. They also recommended to centrifuge a blood sample to improve the detection in blood. As to therapy, Rodhain advised administering tartar emetic as an adjunct to *Atoxyl*®.

Meanwhile *Kleine* and *Taute*, working in East Africa in 1909, brought to light the way in which the trypanosome's life cycle developed in the tsetse fly. *Fantham* and *Stephens*, working in Nyasaland, identified *Trypanosoma rhodesiense* in 1910 and demonstrated the role of *G. morsitans* as its vector in 1912.

3. Epidemiology, mobile teams and dispensaries to control the disease

Once the epidemiological sequence trypanosome-tsetse-man had been brought to light, it became clear that the disease's transmission could be stopped only by eliminating the parasite or the vector.

Belgium took over the Congo Free State in 1908. In 1910 King Albert I decided to set aside one million gold francs annually for the control of sleeping sickness. Queen Elisabeth, for her part, launched a campaign to sell an item of jewellery known as the *Queen's Edelweiss* for setting up a lazaret in Leopoldville.

Physicians from the Leopoldville laboratory visited regularly the wood storage depots for steamboats, along the river, administering a sterilizing arsenophenylglycine injection every three months to both the sick and the suspected carriers. This was most probably a first and an initial chemoprophylactic step towards mobile medical treatment. The idea for this procedure occurred at about the same time to a number of medical officers confronted with the problem of endemic sleeping sickness; and in the final analysis it was not important to know who was the first to implement it. More important was the new method pioneered by *Lejeune*, a young Belgian physician assigned to the lazaret (isolation lazaret) at Kiambi, in Katanga: it consisted of a medical census

of the entire population of a given area, and a follow-up treatment of the detected sick, by auxiliary personnel in centres set up in the villages themselves. This is how the first rural dispensaries came into being and how all subsequent campaigns were organized.

The following years were extremely eventful. In 1911 *Broden* and *Rodhain* handed the laboratory of Leopoldville over to *Mouchet* and *Dubois*, who ran it until the arrival in 1913 of its new director *Van den Branden*.

Fifty-one medical officers assisted by twelve heads of mobile medical units were in the front line of the fight against sleeping sickness when the First World War broke out. Reports on the situation and work have been rather rare, and the conduct of the war hampered the implementation of the necessary health measures. *Broden's* research conducted in the Leopoldville laboratory from 1909 to 1911, which was recorded in a lengthy paper on the diagnosis of sleeping sickness, could not be submitted until May 1920 at an Anglo-Belgian Congress on *Coloniale Hygiène* (Public Health) held in Brussels and published in 1921.

The Belgian Minister Lippens sent out the first medical and administrative expedition to the Congo after World War I. The aim was to resume the census and the treatment for all those with the disease who lived to the Kwango and Uele river basins. The expedition, led by *Schwetz*, spent three years (from 1920 to 1923) in the Kwango basin, examining more than half a million inhabitants spread over 80,000 km² and treating 84,740 victims of sleeping sickness, including 62,479 new cases. Similar work was carried out in the Uele basin, where new cases accounted for 9% of the population.

Other expeditions set off for the Middle Congo and the Bangala district, as well as for Mayumbe, in the Lower Congo (*Bas-Congo*). The industrial companies located in endemic areas did not remain inactive. For example the *Forminière* Company, in the Kasai Province, entrusted *Fourche* with the task of eradicating the disease from the areas around Tshikapa, where 2,000 victims of sleeping sickness and as many suspects were discovered. A special task force operated on the stretch of the river between Bukama and Kindu from 1923 to 1925.

4. New drugs and activities of FOREAMI

Miss *Louise Pearce*, accompanied by *W. H. Brown*, arrived at Leopoldville in May 1920. She was sent by the Rockefeller Institute to test tryparsamide, a new drug against sleeping sickness. Miss Pearce's report, containing the observations made in collaboration with *Van den Branden* and *Van Hoof* on 77 patients, concluded that

tryparsamide had an excellent effect against sleeping sickness caused by *Trypanosoma gambiense*.

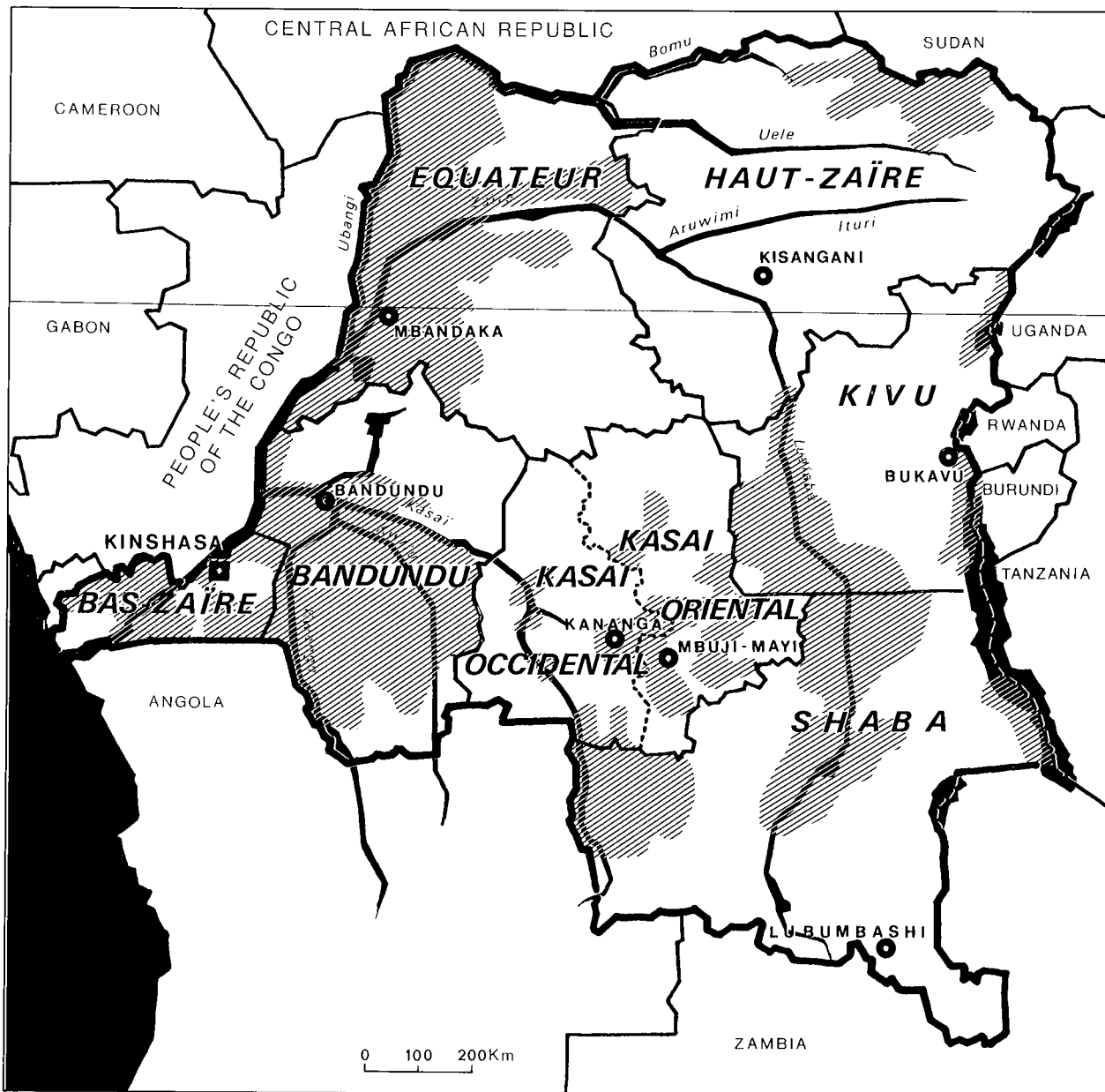
At about the same time (1921-22), *Professor Kleine* was experimenting in East Africa with a new product – *Bayer 205*[®] – whose formula was temporarily kept secret. This substance had no effect on patients in the lethargic stage, but was very active during the early stage of invasion of the blood and the lymphatic system, and particularly against *Trypanosoma rhodesiense*. *Bayer 205*[®] was also a protection for some time against infection, thanks to its slow rate of elimination. This property led *Van den Branden* to use it from 1926 as a means of prevention in endemic foci where, despite treatment, contamination persisted. *Van Hoof* later resorted to this preventive *Bayerization* with great success.

The Queen Elisabeth Fund (FOREAMI, see p. 119), an autonomous semi-public institution set up after the royal couple's visit to the Congo in 1928, began operating in the Lower Congo in 1931, under the direction of *Dupuy*. Without any doubt, it contributed greatly to the control of sleeping sickness in the Lower Congo. Its success was due to an organized approach relying on scrupulous population censuses and subdivision of the areas in circles of about 30,000 inhabitants, each with a full medical staff made up of a physician responsible for the special staff *agents sanitaires*, and an appropriate number of medical auxiliaries, nurses and nursing-aids. The prevalence rate was brought down to around two per thousand.

Individual medical examinations were followed by the issue of a medical passport or a treatment card and the development of a network of health centres operating over a radius of no more than 15 kilometres; a systematic six-month screening of the population, improved environmental sanitation in the villages, clearing the ground around springs and areas where cassava was soaked in water or where bathing occurred, helped in the control of the disease. A more thorough understanding of the factors favouring the spread of transmissible diseases and finally the effective monitoring of sleeping sickness foci, enabled the prevalence of infection to drop to about two per thousand.

5. Further control of the epidemic

The year 1930 was an important turning point. With 33,502 new cases of sleeping sickness recognized, the epidemic had probably peaked. The assignment of the special teams (*Missions Médicales*; see p. 1725) came to an end and their duties were taken over by the Public Health Department. The number of new cases of sleeping sickness dropped steadily between 1930 and 1940, while the size of the monitored population increased



Map 39 – Distribution of sleeping sickness in the Congo in 1952
(from Neujean, 1952)

steadily from three million to five million. The new case index (NCI) also dropped steadily, falling from 1.20% in 1930 to 0.45% in 1935 and 0.24% in 1940.

Under the impetus of *Van Hoof* new plans had been drawn up for the construction of a new, modern laboratory in Leopoldville. Building progressed well, and in 1937 *Brutsaert*, the laboratory director, and his deputy, *Neujean*, were able to leave the historic Leopold II Laboratory and take up quarters in the new Institute of Tropical Medicine named after Princess Astrid (*Institut de Médecine tropicale Princesse Astrid*).

The resumption of normal activities after the Second World War coincided with the release on the world market of a new drug, diamidine. Its effectiveness had been proven by *Lourie* and *Yorke* in 1939. It was hoped that injections of diamidine given to all of the inhabitants of an endemic area would protect them against inoculation by the tsetse fly for a sufficient period of time, so that the disease would disappear on its own.

The most spectacular progress in the control of sleeping sickness came after *Van Hoof* and his colleagues instituted mass prophylaxis with pentamidine (diamidine).

The Tsetse-Fly and Trypanosomiasis Inter-African Bureau, TFTPI (*Bureau Permanent Interafricain pour les Tsé-tsés et la Trypanosomiase – BPITT*) was set up in Leopoldville in 1948 to serve as a clearing house for all scientific information related to sleeping sickness. The services of this information office had become imperative, but unfortunately, it was closed down in July 1960. Its archives were transferred to Brazzaville, where the World Health Organization took over its work.

Despite scientific advances, the disease would not accept defeat. Regular inspection of the protected areas soon revealed the first signs of disease among villagers who had earlier received preventive injections of *Lomidine*® (another diamidine).

Between 1951 and 1955 approximately 70% of the entire population underwent annual medical examinations in the areas struck by the disease (see map 39). The contamination index dropped, in a sort of mathematical progression, from 0.09% to 0.06% to a bottom line of 0.03% (see graph 1 p. 1791).

Unfortunately, the parasite gradually developed resistance to the drugs being used, and it was high time for new products to replace Miss Pearce's trypanamide. Fortunately, Friedheim's new arsenic derivative, *Arsobal*® became available.

Meanwhile the prize offered by King Leopold II to the first person to find a cure for sleeping sickness had never been awarded; a decision taken by King Baudouin and dated July 13, 1953, confirmed King Leopold II's earlier decree and raised the prize money to one million francs. The elected jury wisely decided

to award the prize to different research-workers: Miss Pearce, physicians Brown, Jacobs and Heidelberger, who had discovered trypanamide, and Thomas, who had pioneered the use of *Atoxyl*®.

In 1959, on the eve of the Belgian Congo's independence, only 1,100 new cases of sleeping sickness were diagnosed out of a total population of 13 million people. The contamination index had fallen to 0.01% and, with the exception of a few residual foci, all danger seemed to have disappeared.

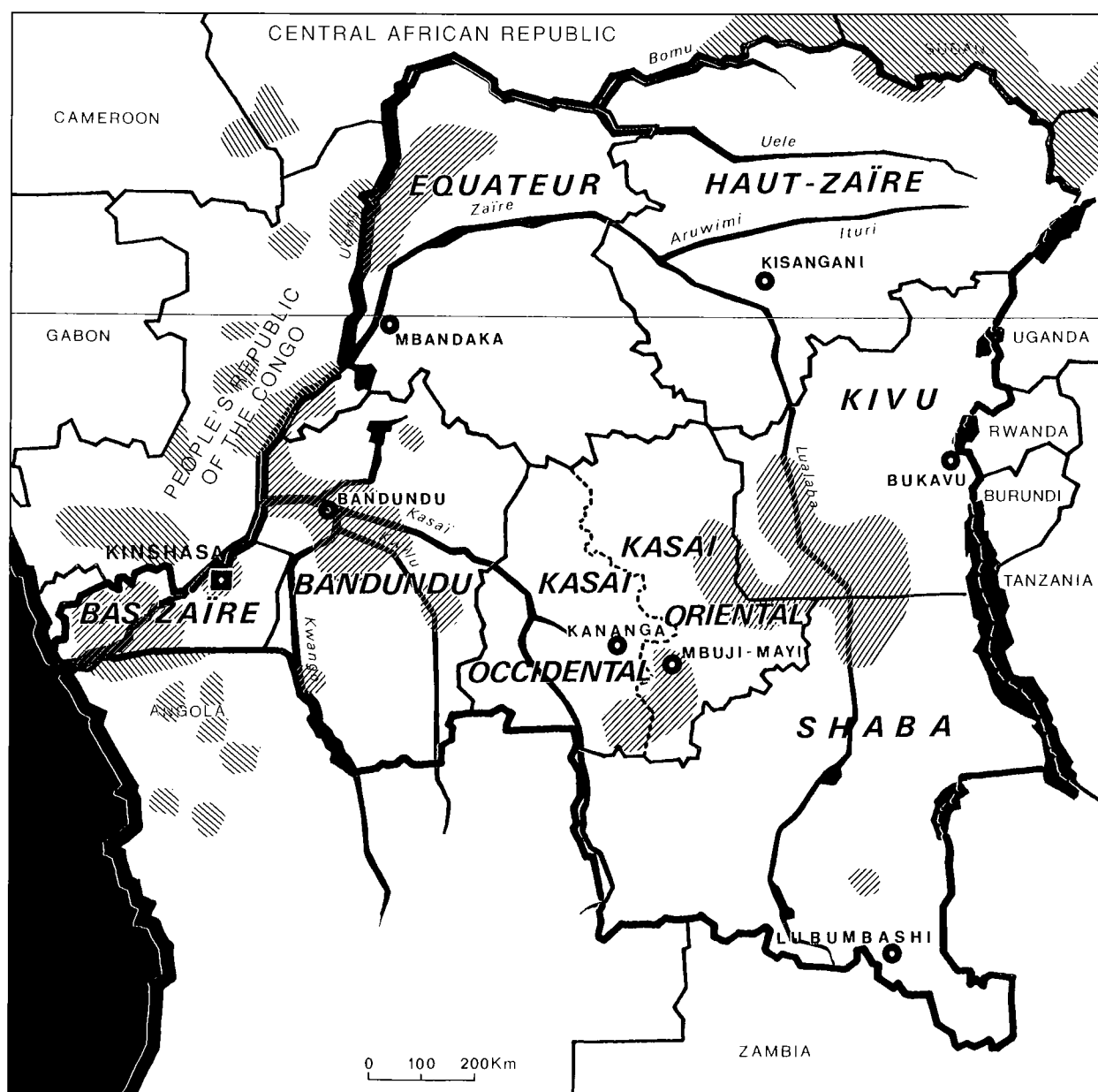
At the end of the colonial period, after an unrelenting 40-year struggle to overcome the disease, 150 million examinations had been conducted, more than 500,000 cases of sleeping sickness had been detected and 1½ million courses of treatment had been given.

6. New extension after the end of the fifties

Unfortunately, between 1958 and 1962, when more than 20 African states with endemic sleeping sickness foci gained their independence, what we might call, according to Professor P.G. Janssens, the "politico-epidemic extension of sleeping sickness", quickly plunged these countries again into the epidemic situation of 1930.

The FOMETRO (*Fonds Médical Tropical* or Tropical Medical Fund, see p. 144), created at the initiative of some Belgian physicians immediately after the Congo became independent, in order to restore a normality of the destabilized medical and health situation, managed to draw up a plan in 1963 for systematic action along the lines of the mobile teams (*Missions médicales*) of former times. From 1964 onwards the FOMETRO's small number of mobile units travelled over the country, trying to make up for the lack of adequate medical care. As more and more teams were trained and acquired experience it became possible to cover all of the known endemic areas and set up centres for screening, treating, and monitoring the cases of sleeping sickness. FOMETRO, supported by Belgium and the Presidential Office in Zaire, has set up the Central Office for Sleeping Sickness; from there it has been possible to send out instructions to control the endemic throughout the Republic of Zaire.

Despite a well-planned operation, drawing on the lessons of the past and on new technical possibilities, it had to be acknowledged that it was once again extremely difficult to bring the sleeping sickness epidemic under control. After independence, it reached its peak in 1970, at which time 30 mobile units were working to control the endemic. It then took a further five years – until 1975-1976 – to lower the contamination index from 8.4% to 2.8% and thereby reduce the spread of the disease by two-thirds.



Map 40 – Distribution of sleeping sickness in Zaire, 1985
(from Burke, 1985, and WHO, 1991)

The situation in 1985 is illustrated by the endemic zones shown on Map 40. The situation in the future will depend on proper control of the disease, especially in the regions bordering the foci.

The incidence of new cases and the yearly rate of new infections from 1930 or 1960 to 1990 are shown on graphs 1 and 2, next page.

In 1961, Belgium was the first country to sound the alarm against African trypanosomiasis and to organize a model strategy for the eradication of sleeping

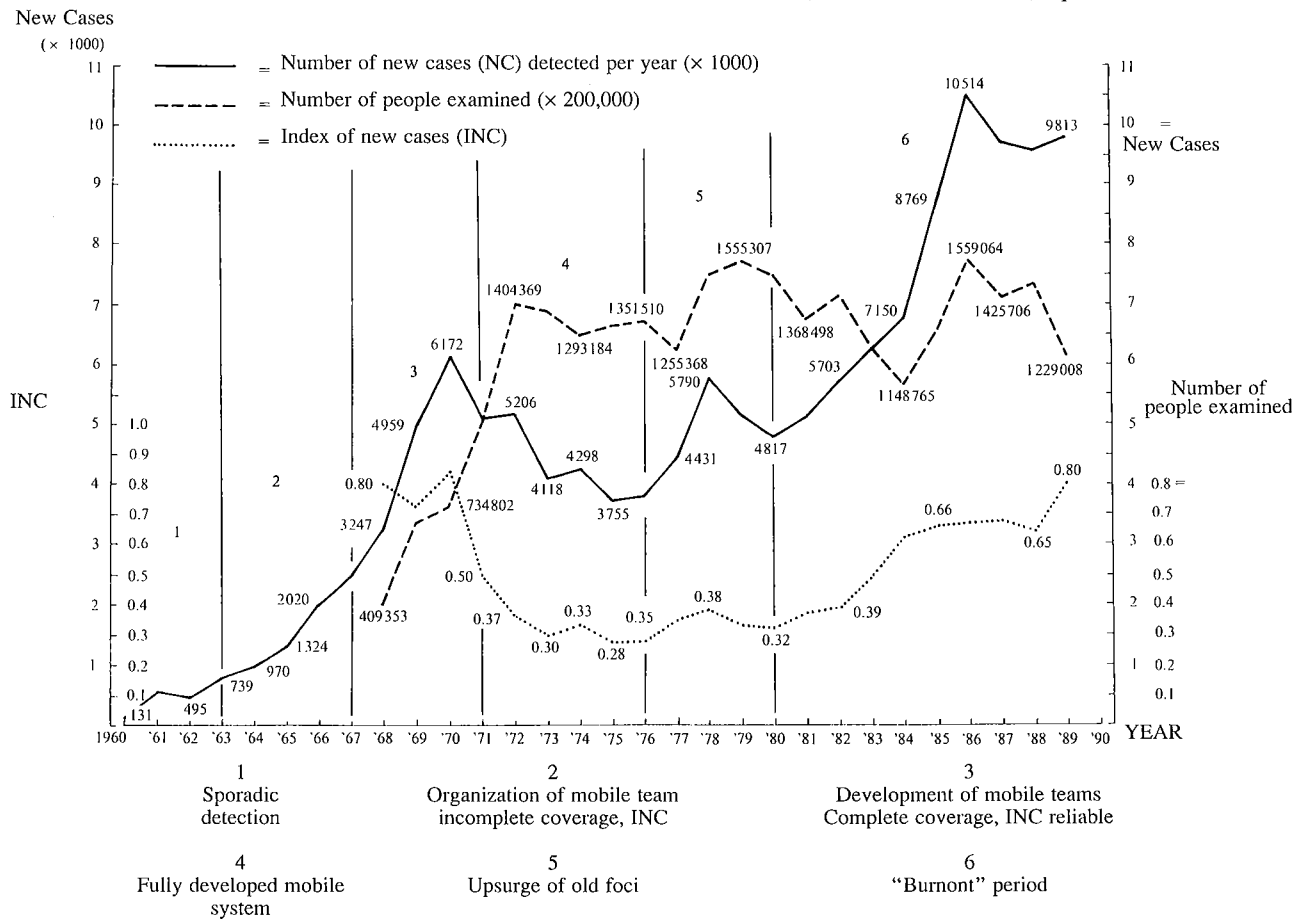
sickness. Technical staff was trained to provide an efficient means for the control of the disease in Africa.

A review of this campaign shows that 15 million persons have been examined by a handful of Belgian and Zairean *agents sanitaires* and a few physicians; by December 31, 1984 65,000 new cases of sleeping sickness had been diagnosed and 150,000 patients (both new and old cases) had been treated.

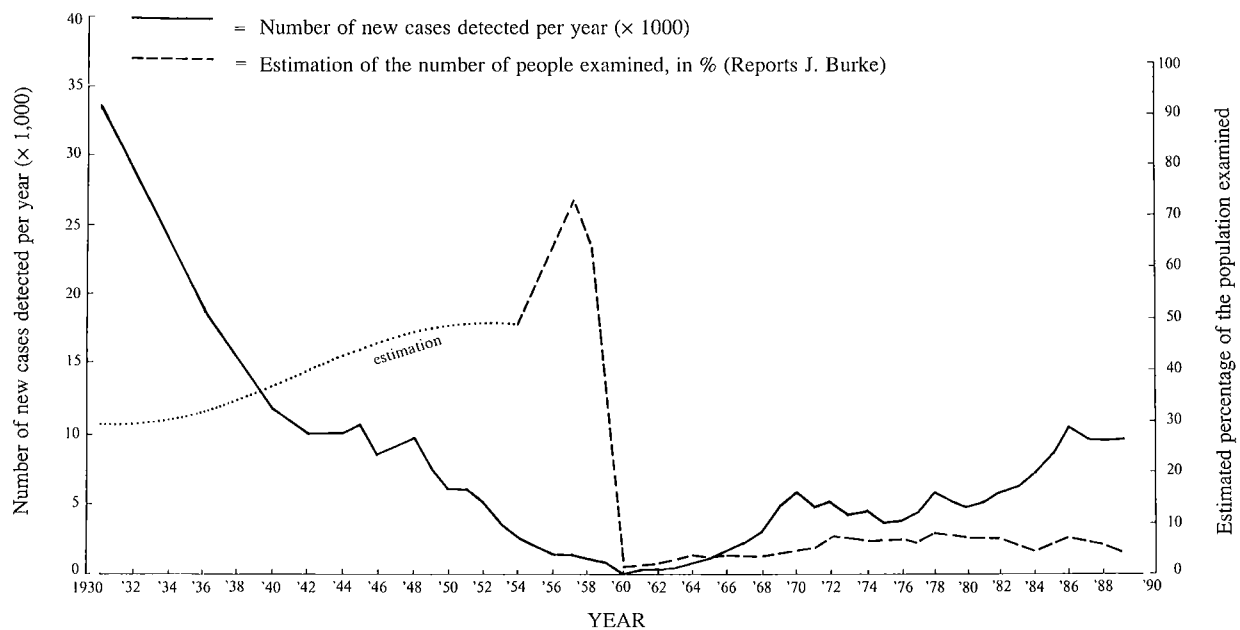
J. Burke

HUMAN AFRICAN TRYPANOSOMIASIS

EVOLUTION OF SLEEPING SICKNESS IN ZAIRE OVER SIX PERIODS, from 1960 to 1990 (Reports J. Burke)



INCIDENCE OF THE TRYPANOSOMIASIS IN THE BELGIAN CONGO AND ZAIRE, 1930-1990



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